



supercurriculum
year10

History – Basic Supercurriculum (OCR Board)

Why Supercurriculum Matters in History

Supercurriculum activities in History help you move beyond memorising facts to *thinking like a historian*.

They allow you to explore causes and consequences, analyse different perspectives, and connect past events to today's world. By engaging independently, you strengthen your skills in research, evaluation, and argument — essential not only for exams but also for informed citizenship.

Reading – Breadth and Comprehension

Deepen your understanding of key historical eras and debates.

Read: *A Short History of the World* by H. G. Wells (for broad context).

Explore BBC Bitesize IGCSE History topics

(<https://www.bbc.co.uk/bitesize/subjects/zj26n39>).

Read historian articles on *Spartacus Educational* (<https://spartacus-educational.com/>).

Keep a "History Reading Journal" summarising one new fact and one question after each article.

Writing – Argument and Precision

Practice constructing clear and persuasive arguments.

Write a 300-word essay answering: "Was the Treaty of Versailles fair?"

Practice PEEL paragraph structure (Point, Evidence, Explain, Link).

Edit a paragraph to improve historical vocabulary (e.g., *source*, *bias*, *reliability*).

Try "history micro-essays": 50-word answers to "What caused the Cold War?" or "Why did women gain the vote?"

Listening – Interpretation and Empathy

Engage with historians' discussions and firsthand accounts.

Listen to BBC Radio 4's *In Our Time* –

History (<https://www.bbc.co.uk/programmes/p01dh5yg>).

Watch or listen to *History Extra* podcasts from *BBC History*

Magazine (<https://www.historyextra.com/podcast/>).

Note one key argument from each episode and discuss it in class.

Watching – Visual and Applied Understanding

Analyse how media presents the past.

Watch *The World at War* (1973) documentary series.

Watch *13 Minutes to the Moon* (BBC) or *The Crown* (Netflix) and note factual vs. fictionalised moments.

Reflect: "How do TV or films shape public understanding of history?"

Researching – Independent Enquiry

Develop curiosity through personal investigation.

Choose a topic like “Propaganda in World War II” or “Civil Rights Movement 1954–68.”

Use <https://www.nationalarchives.gov.uk/education/> for primary sources.

Summarise findings in 3 slides or a short report using correct source citation.

Student-Led Tasks – Collaboration and Leadership

Lead historical exploration in class or online.

Create a mini-quiz for your peers using key dates and events.

Present a short “History Minute” every week about a major historical figure.

Organise a group debate: *Which had a greater impact — WWI or WWII?*

Creative Tasks – Innovation and Self-Expression

Use creative expression to bring the past alive.

Write a diary entry from the perspective of a soldier, suffragette, or factory worker.

Create a historical comic strip illustrating a key turning point.

Design a museum exhibit poster for your chosen topic.

Reflection – Metacognition and Growth Mindset

Think critically about your learning journey.

Keep a “History Reflection Log” after each unit: What surprised me? What did I find difficult?

Set a SMART goal for improving essay structure or source analysis.

Reflect on how historical understanding helps you interpret current events.

Interdisciplinary Connections – Synthesis and Transfer

Link History to other subjects.

Geography: Migration patterns after WWII.

Literature: Context of *Animal Farm* or *To Kill a Mockingbird*.

Economics: Impact of the Great Depression on society.

Politics: Comparing past and modern democracies.

Digital & Career Literacy

Understand how history connects to real-world paths.

Research careers using historical skills: journalism, law, museum curation, diplomacy.

Learn to evaluate digital sources for reliability and bias.

Use online archives and databases (Google Scholar, JSTOR) responsibly.

History – Extended Supercurriculum (OCR Board)

Reading – Breadth, Depth, and Historiography

Objective: Develop a sophisticated understanding of multiple perspectives and historical debates.

Suggested Advanced Readings:

Primary sources:

The Nuremberg Trials transcripts (The Avalon Project – https://avalon.law.yale.edu/subject_menus/imt.asp)

Letters from the Trenches – Imperial War Museum collection.

Secondary readings:

Richard J. Evans, *In Defence of History*

Niall Ferguson, *The War of the World*

Mary Beard, *SPQR: A History of Ancient Rome*

Eric Hobsbawm, *Age of Extremes: The Short Twentieth Century*

Extension Task:

Compare how two historians interpret the causes of World War I. Write a 500-word review evaluating evidence, tone, and bias.

Reflect: *Can history ever be objective?*

Writing – Analytical Precision and Academic Argument

Objective: Construct complex arguments supported by evidence and historiography.

Activities:

Produce a **mini dissertation (1,200 words)** on a self-chosen question, e.g.

“To what extent was the League of Nations a success?”

“How far did the Cold War shape post-colonial Africa?”

Integrate at least three different historians’ views and evaluate their arguments.

Use Harvard or MLA citation style.

Attend to analytical transitions, topic sentences, and evaluative conclusions.

Challenge Exercise:

Write a “synoptic essay” connecting two historical periods (e.g., the 19th-century revolutions and modern social movements).

Listening – Scholarly Discourse and Evidence Interpretation

Objective: Engage with advanced historical debate and oral argumentation.

Resources:

BBC Radio 4 – In Our Time (History) – analyse one episode weekly.

The British Museum Podcast – artefacts as historical voices.

Dan Snow’s History Hit – critical discussions on modern historiography.

Task:

Listen to two podcast episodes presenting opposing views on the same topic (e.g., “Causes of the Cold War”).

Create a comparative chart summarising each historian's thesis, evidence, and interpretive stance.

Watching – Critical Analysis of Historical Representation

Objective: Evaluate how visual media interpret and distort historical events.

Suggested Materials:

Documentary: *The Fog of War* (2003) – analysis of decision-making.

Film: *Schindler's List* – moral interpretation of genocide.

Documentary: *The Vietnam War* (Ken Burns) – narrative techniques.

Historical fiction: *Darkest Hour* – leadership portrayal.

Analytical Task:

Write a 600-word film critique discussing:

“How does the director's use of imagery and dialogue influence the viewer's understanding of historical truth?”

Researching – Independent Historical Enquiry

Objective: Formulate and investigate an original historical question.

Process Framework:

Define your question: e.g., “How did propaganda shape public opinion during the Cuban Missile Crisis?”

Gather evidence: National Archives, JSTOR, Britannica Academic.

Evaluate reliability: Author credentials, publication date, potential bias.

Synthesize findings: Create a digital research poster or Google Sites mini-project.

Reflect: What limitations did your research encounter?

Extension:

Enter a national essay or research competition such as the *Historical Association Young Historian Awards*.

Student-Led Tasks – Leadership, Debate, and Collaboration

Objective: Build initiative and communication skills through shared inquiry.

Suggested Activities:

Organise a **Historical Debate Society** in school.

Example motions:

“The dropping of the atomic bomb was justified.”

“Empires did more harm than good.”

Lead a weekly ‘**History in the News**’ briefing linking modern events to historical contexts.

Mentor younger students (Year 8–9) by running short workshops on source analysis.

Extension:

Host a virtual panel discussion with classmates acting as historical figures in a simulated conference.

Creative Tasks – Reconstruction and Perspective

Objective: Use imagination responsibly to deepen empathy and historical accuracy.

Activities:

Create a **historical short film** (2–3 minutes) recreating a past event, integrating period-accurate dialogue and sources.

Design a **virtual museum exhibit** using Canva, Google Arts & Culture, or ThingLink, showcasing a key historical theme (e.g. “Propaganda through the Ages”).

Write a **first-person historical fiction story** incorporating at least three verified historical facts per page.

Extension:

Submit creative writing to the *Historical Association Young Historian Creative Award*.

Reflection – Metacognitive Growth

Objective: Foster self-awareness and intellectual maturity.

Activities:

Keep a “**Historian’s Journal**” where you:

Record evolving opinions on moral questions (e.g., war, empire, resistance).

Reflect monthly: *What kind of historian am I becoming?*

Write a “Learning Manifesto” on how history has reshaped your worldview.

Extension:

Present a reflective talk (TED-style, 3–5 min) titled “Why History Still Matters in the 21st Century.”

Interdisciplinary Connections – Synthesis and Cross-Curricular Thinking

Objective: Integrate history with other disciplines to generate new insights.

Links:

Economics: Industrial Revolution and capitalism’s growth.

Psychology: Mass persuasion and leadership personality theory.

Geography: Environmental consequences of empire and trade.

Science: Technological innovation during wartime.

Art & Literature: Political propaganda through visual culture and novels.

Task:

Produce a **concept map** linking one historical era to at least three other subjects. Annotate the connections with explanatory sentences.

Digital & Career Literacy – Historical Skills in the Real World

Objective: Apply historical thinking to modern digital and professional contexts.

Activities:

Practice analysing digital archives using the **British Library’s “Learning” platform** (<https://www.bl.uk/learning/>).

Use AI tools (e.g., ChatGPT, Perplexity, or Google NotebookLM) to summarise long primary sources — while critically checking for bias.

Explore digital storytelling or podcast creation tools (Audacity, Anchor, Canva Video).

Research degree and career pathways in:

- Law and political analysis

- Journalism and historical consultancy

- Heritage, curation, and museum management

- Policy and diplomacy

Extension:

Create a personal “Portfolio of Historical Inquiry” — combining essays, creative artefacts, and research reflections — for inclusion in future university or scholarship applications.

Recommended Long-Term Project

Title: *“Power, Memory, and Truth: How History Shapes Identity”*

Develop over one term.

Combine at least three supercurriculum dimensions (reading, research, creative).

Present at a school exhibition or publish digitally.

History – Complete Student Guide & Extended Supercurriculum (OCR Board Style)

PART A: Understanding the Exam

Overview of Cambridge IGCSE History

Paper	Focus	Type	Skills Tested	Duration	Weight
Paper 1	Core Content Questions	Structured Source & Essay	Knowledge, causation, change, continuity	2 hours	40%
Paper 2	Source Paper	Source Evaluation	Interpretation, reliability, comparison	2 hours	33%
Paper 4 (Alternative to Coursework)	Depth Study	Essay	Extended argument, evaluation	1 hour	27%

Note: Some centres offer **Paper 3 (Coursework)** instead of Paper 4. In that case, you submit a 2,000-word internally marked investigation.

Assessment Objectives

Code	Objective	What It Means	Examples
AO1	Recall, selection, and communication of knowledge	Use factual detail to show understanding	Explain causes of WWI using accurate examples
AO2	Causation, consequence, change, continuity	Analyse patterns and relationships	Compare effects of WWI and WWII
AO3	Interpretation and analysis of sources	Evaluate bias, reliability, purpose	Assess whether propaganda posters are trustworthy
AO4	Organisation, clarity, argumentation	Structure answers logically and persuasively	Write essays with strong introductions and conclusions

Key Themes and Depth Studies

Common Cambridge IGCSE/OCR topics include:

Core Content:

The 19th and 20th Centuries: International Relations (1919–1990)

The League of Nations, WWII, the Cold War, the UN, Vietnam, etc.

Depth Studies (choose one):

Germany, 1918–1945

Russia, 1905–1941

USA, 1919–1941

China, 1945–1990

South Africa, 1948–1994

Each study requires *factual precision + interpretive skill + analytical reasoning*.

PART B: Study Skills and Strategies

The Historian's Toolkit

Causation: Why did events happen? (short-term vs long-term causes)

Consequence: What changed as a result?

Continuity: What stayed the same?

Significance: Why was this event or person important?

Interpretation: How do historians disagree — and why?

Mnemonic for essay planning:

□ C-A-T-S (Causes, Actions, Turning points, Significance)

Exam Technique Tips

Paper 1 – Structured Questions

Identify the command word (*Describe / Explain / Evaluate*).

Support every point with **specific evidence** (date, treaty, leader, event).

Plan essays quickly (5 min max): intro, 3 paragraphs, conclusion.

Aim for balanced argument — always show both sides.

Paper 2 – Source Paper

OPCVL approach:

Origin – Who created it?

Purpose – Why was it created?

Content – What does it say?

Value – What can it tell us?

Limitation – What can't it tell us?

Always cross-reference sources — never treat one as absolute truth.

Use short, precise quotations or visual evidence (e.g., “poster slogan says...”).

Paper 4 – Depth Study Essay

Directly answer the question, not the topic.

Link every paragraph back to the question.

End with a judgement: “Although X was important, Y had a greater impact because...”

Revision Techniques

Method	Description	How to Apply
Flashcards	Key dates, treaties, definitions	Use Quizlet for timed recall
Timelines	Visual sequence of causes/effects	Create colour-coded war timelines
Mind Maps	Connect themes and events	Compare revolutions or alliances
Practice Papers	Real exam exposure	Use Cambridge past papers (https://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-igcse-history-0470/)
Self-Assessment Grid	Track AO1–AO4 progress	Rate yourself 1–5 per objective

PART C: Independent Enrichment (Extended Supercurriculum)

These go **beyond the syllabus** — developing university-level historical thinking.

Reading for Depth

A People's History of the United States – Howard Zinn

The Origins of the Second World War – A. J. P. Taylor

The Cold War: A New History – John Lewis Gaddis

Gulag Archipelago – Aleksandr Solzhenitsyn

Task:

Compare two historians' accounts of the same event. What explains the differences in interpretation (nationality, ideology, time period)?

Research Projects

Choose one of the following:

“How did propaganda affect the outcome of WWII?”

“How did women's roles change during total war?”

“To what extent was the League of Nations doomed from the start?”

Expected Output:

1,200–1,500-word research report + bibliography + reflection paragraph.

Digital Exploration

Explore **BBC Archive: Voices of History** (<https://www.bbc.co.uk/archive>).

Use **Google Earth Historical Maps** to visualise empire changes.

Use AI (ChatGPT / NotebookLM) to summarise primary source sets and test for bias.

Reflection Journal Prompts

Write a short entry each week:

“What event this week reshaped my view of humanity?”

“What parallels can I see between history and modern politics?”

“Which skill improved most — evidence use, analysis, or empathy?”

PART D: Assessment and Progress Tracking

Weekly Skill Tracker

Week	Unit	AO1 Knowledge	AO2 Analysis	AO3 Sources	AO4 Writing	Comments
1	Treaty of Versailles	4/5	3/5	2/5	3/5	Improve use of factual evidence
2	League of Nations	5/5	4/5	4/5	4/5	Excellent explanation
...	...	...	...	...	...	...

Self-Evaluation Rubric

Level	Description
Emerging	Recalls events but lacks explanation
Developing	Begins to link causes and consequences
Proficient	Balances knowledge with analysis
Advanced	Integrates multiple perspectives
Expert	Challenges interpretations with confidence and evidence

PART E: Examination Preparation Calendar

12-Week Revision Plan (Example)

Week	Focus	Core Actions
1–2	Overview of 20th Century	Create visual timeline
3–4	WWI and Versailles	Essay: “Was the Treaty fair?”
5–6	League of Nations	Practice Paper 1 Question 4
7–8	Hitler’s Rise to Power	Watch documentary + note causes
9–10	Cold War Origins	Source comparison exercise
11	Practice Mock	Timed paper under exam conditions
12	Review & Reflect	Revise weak topics, final mind maps

PART F: Long-Term Project (Capstone)

Title: *“Power, Memory, and Truth: The Historian’s Responsibility in the Modern Age”*

Research + creative synthesis + oral presentation.

Present in school exhibition or publish digitally.

Include primary and secondary sources, annotated bibliography, and reflection on ethical interpretation.

PART G: Tools and Resources

Past Papers: Cambridge International (0470) site

BBC Bitesize: <https://www.bbc.co.uk/bitesize/subjects/zj26n39>

National Archives: <https://www.nationalarchives.gov.uk/education/>

Spartacus Educational: <https://spartacus-educational.com/>

Historical Association: <https://www.history.org.uk/>

Podcast: *In Our Time – History*

Digital Tools: Canva, Quizlet, Notion, Google Sites for projects

Final Reflection

“The past is not dead. It is not even past.” – William Faulkner

To master History is to learn how to question, to empathise, and to think critically.

Use this supercurriculum not just to prepare for exams, but to understand the power of ideas — and the responsibility of truth.

Geography – Basic Supercurriculum (Cambridge Board)

Why Supercurriculum Matters in Geography

Supercurriculum Geography helps you *see the planet as an interconnected system*.

By exploring physical landscapes, human settlements, and global issues beyond the textbook, you build curiosity, critical thinking, and awareness of how your choices shape the world.

It turns “learning facts” into *understanding change* — from rivers and rainforests to megacities and migration.

Reading – Breadth and Comprehension

Strengthen your understanding of places, processes, and patterns.

Suggested Activities:

Read BBC Bitesize: *Geography*

IGCSE (<https://www.bbc.co.uk/bitesize/subjects/zkv76sg>).

Explore National Geographic articles on climate and sustainability (<https://www.nationalgeographic.com>).

Read *Geography Review* (school library or online issues).

Keep a “Geography Fact Journal”: one global issue, one new term, one map per week.

Extension:

Choose one country per month and make a short “Country Profile” including population, economy, and major environmental challenges.

Writing – Argument and Precision

Practice clear geographical explanations and structured reasoning.

Activities:

Write a 300-word mini-report: “*Causes and Effects of Deforestation in the Amazon.*”

Use the **PEE method** (Point, Evidence, Explain).

Create a paragraph comparing rural–urban migration in two countries.

Write one “Geographical Opinion Piece” per term — for example:

“Should all countries aim for net zero emissions by 2050?”

Listening – Interpretation and Empathy

Learn from global voices and real-world stories.

Activities:

Listen to *BBC Earth Podcast* (<https://www.bbcearth.com/podcast>).

Follow *Geographical Association* talks or webinars.

Watch “Planet Earth” or “Our Planet” and describe how geography explains what you see.

Discuss one episode in class, identifying physical and human geography themes.

Watching – Visual and Applied Understanding

See geography in action through visuals and documentaries.

Resources:

Our Planet (Netflix) or *Seven Worlds, One Planet* (BBC).

Explained: The World’s Water Crisis (Netflix).

YouTube Channel: *Geography Now!* – country-by-country overviews.

DW Planet A – short videos on climate solutions.

Task:

Summarise one documentary and link it to an IGCSE topic (e.g., coasts, urbanisation, population, industry).

Researching – Independent Inquiry

Investigate the world using data, maps, and first-hand evidence.

Ideas:

Conduct a mini-fieldwork project:

“How does land use vary around my school?”

Take photos, sketch maps, and record observations.

Use Google Earth to explore natural features and city growth.

Use UN data (<https://data.un.org/>) to compare two countries’ population structures.

Present findings in slides or a short infographic.

Student-Led Tasks – Collaboration and Leadership

Take initiative and lead your peers in exploring big ideas.

Activities:

Create a short quiz on countries, capitals, or climates.

Lead a “World News Minute” presentation each week.

Form a “Green Team” to monitor your school’s environmental impact.

Work in pairs to model a river system or volcanic structure.

Creative Tasks – Innovation and Self-Expression

Show understanding through visual or creative expression.

Activities:

Design an illustrated map of your favourite ecosystem.

Make a model volcano or coastal landform using everyday materials.

Create a digital poster: “10 Facts About Climate Change Everyone Should Know.”

Write a travel blog or brochure about a country you’d love to visit, explaining its geography.

Reflection – Metacognition and Growth Mindset

Think about how your learning connects to your life and values.

Prompts:

“What surprised me most about Earth this week?”

“Which environmental issue do I feel most passionate about?”

“How does geography influence where and how people live?”

Keep a monthly reflection log with personal insights and goals.

Interdisciplinary Connections – Synthesis and Critical Transfer

Link Geography with other school subjects to build full understanding.

Subject	Connection Example
Science	Climate, weather, ecosystems
Economics	Development, trade, globalisation
History	Colonisation, migration, urban growth
Maths	Data interpretation, graphs, statistics
Art	Landscape drawing, environmental design

Task:

Create a poster linking geography, economics, and sustainability — e.g. “The Geography of Energy Use.”

Digital & Career Literacy

Learn how geographers use technology and how skills transfer to real careers.

Activities:

Explore careers: GIS specialist, urban planner, environmental scientist, cartographer, climatologist.

Use free tools like **Google Earth** or **ArcGIS StoryMaps** to visualise data.
Follow a sustainability campaign (UNEP, WWF, Greenpeace).
Research a real job ad for a “geospatial analyst” — list the skills required and how you can start building them now.

Final Reflection

Geography is not just about maps and facts — it’s about *making sense of our changing world*.

By exploring beyond the classroom, you become a curious global citizen — able to see connections, question assumptions, and imagine solutions.

Geography – Extended Supercurriculum (Cambridge Board)

Why Extended Geography Matters

Geography connects science, society, and ethics.

The extended supercurriculum takes you beyond textbook learning — helping you to question *how environments function, why inequalities exist, and what sustainable futures might look like.*

You'll read like a geographer, think like a researcher, and act like a global citizen.

Reading – Breadth, Depth, and Global Understanding

Objective: Explore geographical theory and debate.

Recommended Readings:

Prisoners of Geography – Tim Marshall

Factfulness – Hans Rosling

The Sixth Extinction – Elizabeth Kolbert

Why Geography Matters – Harm de Blij

UN World Urbanisation Prospects reports (<https://population.un.org/wup/>)

Advanced Tasks:

Compare two authors' explanations for global inequality.

Write a 400-word reflection: "How do physical geography and geopolitics influence each other?"

Keep a *Reading Log* with three columns — "New Knowledge", "Questions Raised", "Connections to Class Topics."

Writing – Analytical Argument and Precision

Objective: Express complex geographical ideas clearly and convincingly.

Tasks:

Write an **extended report (1,000–1,200 words)** on a current issue, e.g.:

"To what extent can renewable energy solve the climate crisis?"

"How far can cities ever be sustainable?"

Include maps, data tables, and citations (Harvard style).

Practice **evaluative writing** using connectives

like *however, whereas, nonetheless, in contrast, and ultimately.*

Challenge:

Write a comparative essay linking two global case studies (e.g., coastal management in the UK vs Bangladesh).

Listening – Global Perspectives and Debate

Objective: Engage with professional and policy voices in geography.

Suggested Sources:

The Geographical Podcast (Royal Geographical Society)

BBC Costing the Earth – human impact on the environment

TED Talks: The Future of Cities / Water / Food Security

Activity:

Create a short podcast review: identify the issue, summarise perspectives, and reflect on bias or regional focus.

Extension:

Invite a class discussion: “Is economic growth compatible with environmental protection?”

Watching – Visual Literacy and Critical Analysis

Objective: Learn to interpret and critique visual geographical narratives.

Resources:

Before the Flood (National Geographic, 2016)

Don’t Panic – The Truth About Population (Hans Rosling)

Years of Living Dangerously (documentary series)

The True Cost (fashion and globalisation)

Task:

Analyse how film producers use visuals, statistics, and emotional appeal to shape understanding of global issues.

Write a 500-word review connecting the documentary to an IGCSE topic.

Researching – Independent Geographical Enquiry

Objective: Apply real geographical methods.

Extended Fieldwork Ideas:

Urban Geography: “How does land use and pedestrian flow vary across my local town centre?”

Environmental Geography: “How clean is the local river or coastline?”

Human Geography: “What factors influence local commuting choices?”

Steps:

Form a hypothesis (e.g., “Environmental quality decreases with distance from the town centre”).

Collect data: counts, surveys, photos, secondary data (e.g., local council GIS).

Present data using maps, graphs, and annotated photos.

Analyse patterns and anomalies.

Conclude and evaluate reliability.

Extension:

Create a 5-slide digital presentation of your fieldwork with commentary — ready for a Geography Fair or competition.

Student-Led Tasks – Leadership and Collaboration

Objective: Lead enquiry, present findings, and mentor others.

Ideas:

Form a **School Geography Society** — organise debates, film nights, and model-UN-style simulations.

Lead a class activity using live data (e.g., *earthquake tracker* or *World Air Quality Index*).

Run a weekly “Geography News Roundup” — summarise three global events with maps.

Mentor younger students on how to analyse choropleth or population pyramids.

Extension:

Plan a one-day *Model COP Conference* where students act as nations negotiating climate policy.

Creative Tasks – Synthesis and Communication

Objective: Use art, media, or design to express geographical meaning.

Projects:

Create an **interactive StoryMap** using ArcGIS or Google Earth.

Design a campaign poster: “Protect Our Coastlines.”

Produce a 2-minute digital video titled “*A Day in the Life of a Megacity.*”

Write a *poem or diary* from the viewpoint of a glacier, rainforest, or urban migrant.

Extension:

Submit creative work to *The Young Geographer of the Year* competition (Royal Geographical Society).

Reflection – Metacognition and Self-Evaluation

Objective: Think critically about what you’ve learned and why it matters.

Weekly Prompts:

“Which concept this week changed the way I see the world?”

“How do my personal choices affect global systems?”

“What kind of geographer do I want to become?”

Activity:

Keep a “Geo-Journal” with sections for *observations*, *questions*, and *personal connections*.

Term Reflection:

Write a one-page reflection on how studying Geography has influenced your view of citizenship and responsibility.

Interdisciplinary Connections – Systems Thinking

Objective: Link geography to other disciplines to create deeper understanding.

Discipline	Example Connection
Biology	Ecosystems, biodiversity, conservation
Chemistry	Carbon cycle, atmospheric changes
Physics	Renewable energy, water flow
Economics	Development, trade, inequality
Politics	Migration policy, resource conflicts
Art & Media	Infographics, landscape design

Challenge Task:

Create a “Systems Map” linking climate, economy, and population using arrows and annotations to show feedback loops.

Digital & Career Literacy – Geography for the Future

Objective: Explore real-world applications of geography.

Activities:

Learn basic GIS mapping via ArcGIS or QGIS tutorials.

Follow *RGS-IBG Career Resources* (<https://www.rgs.org/schools/>).

Explore *World Bank Data Portal* for real development indicators.

Research how geography informs work in:

- Urban planning
- Disaster management
- Environmental law
- Tourism and logistics
- Sustainable design

Extension:

Create a digital portfolio — a professional online showcase of your research, projects, and reflections.

Advanced Long-Term Project (Capstone)

Title: “*Mapping the Future: Human Impact on the Planet*”

Duration: 6–8 weeks

Components:

Select a global issue (e.g., plastic pollution, urban sprawl, desertification).

Collect and analyse data (primary, secondary, and GIS).

Write a 1,500-word report including visuals and evaluation.

Present to peers or upload as a digital StoryMap.

Outcome:

Students demonstrate analytical depth, creativity, and awareness of global systems — ideal preparation for A-Level or IB Geography.

Recommended Digital & Print Resources

Type	Source	Link
Learning platform	BBC Bitesize	https://www.bbc.co.uk/bitesize/subjects/zkv76sg
Case studies	Cool Geography	https://www.coolgeography.co.uk/
Data visualisation	Our World in Data	https://ourworldindata.org/
Interactive maps	National Geographic MapMaker	https://mapmaker.nationalgeographic.org/
Podcasts	Costing the Earth	https://www.bbc.co.uk/programmes/boo6r4wn
Competitions	Royal Geographical Society	https://www.rgs.org/schools/competitions

Final Reflection

“Geography is the subject which holds the key to our future.” — Michael Palin
Through the Extended Supercurriculum, you develop not just academic skill but *global vision*: the ability to read landscapes, interpret change, and imagine sustainable futures.

The more you observe, question, and connect, the more you become a true geographer — one who understands both people and planet.

Geography – Complete Study Guide & Extended Supercurriculum (Cambridge Board)

PART A: Understanding the IGCSE Geography Exam Overview of the Exam Structure

Paper	Title	Focus	Format	Duration	Weight
Paper 1	<i>Geographical Themes</i>	Core content: population, migration, settlements, natural environments, economies	Structured written exam	1 hour 45 mins	45%
Paper 2	<i>Geographical Skills</i>	Map skills, graphs, data interpretation, decision-making	Structured written exam	1 hour 30 mins	27.5%
Paper 4	<i>Alternative to Coursework (Fieldwork)</i>	Data collection, analysis, conclusion, evaluation	Structured written exam	1 hour 30 mins	27.5%

(Some schools may offer Paper 3 – Coursework instead of Paper 4.)

Assessment Objectives (AO1–AO3)

Code	Objective	Description	Example Skills
AO1	<i>Knowledge with Understanding</i>	Recall places, environments, processes	Define and explain key terms (e.g. erosion, urbanisation)
AO2	<i>Interpretation and Application</i>	Analyse relationships, interpret data, explain processes	Explain how deforestation affects climate and population
AO3	<i>Analysis, Evaluation, Decision-Making</i>	Judge evidence, form conclusions, evaluate sustainability	Evaluate whether a management strategy is effective

Core Units of Study

Theme	Key Topics
Population and Settlement	Population growth, migration, urbanisation, settlement hierarchy
The Natural Environment	Volcanoes, rivers, coasts, weather, climate, ecosystems
Economic Development	Agriculture, industry, energy, trade, globalisation
Fieldwork	Hypothesis, data collection, analysis, evaluation

PART B: Study and Revision Skills

The Geographer's Toolkit

To succeed in Geography, you must *think spatially* and *reason systematically*.

Core Skills to Practice:

Map reading and interpretation (scale, grid, contour, direction)

Analysing graphs and climate data

Interpreting aerial photos

Constructing reasoned, evidence-based arguments

Evaluating human and environmental interactions

Mnemonic for essay planning:

□ **D.E.E.P.** = *Describe* → *Explain* → *Evaluate* → *Predict*.

Exam Technique Tips

Paper 1 – Geographical Themes

Identify **command words** (*describe, explain, compare, evaluate*).

Always use **specific case studies** (named countries, data, facts).

Structure long answers with clear paragraphing.

Example: *Explain two causes of urbanisation in LEDCs (4 marks)*

→ 2 causes, each with evidence and explanation.

Paper 2 – Map and Graph Skills

Practice reading **OS maps** and **topographical maps**.

Use the **4- and 6-figure grid reference system** confidently.

Identify contour lines, patterns of settlement, and river features.

Learn how to draw and interpret graphs: line, bar, pie, scatter, choropleth.

Annotate diagrams clearly — clarity counts in marks!

Paper 4 – Alternative to Coursework

Read questions carefully — they follow **fieldwork enquiry stages**:

Identify the question/hypothesis.

Describe data collection methods.

Present data in correct format.

Analyse and identify trends.

Evaluate reliability and suggest improvements.

Practice interpreting unfamiliar data in graphs, tables, and maps.

Revision Strategies

Method	How It Works	Example
Case Study Cards	One card per topic (country, cause, impact, solution)	"Flooding in Bangladesh – Causes / Effects / Management"
Map Practice	Label features, draw climate graphs	Use past-paper map extracts
Mind Maps	Show connections between physical and human factors	Link population growth ↔ resource demand
Quizzes & Flashcards	Self-test on definitions and examples	Quizlet: "IGCSE Geography Terms"
Past Papers	Practice under timed conditions	Use Cambridge past-paper archive
Peer Teaching	Explain one topic to a friend	"Teach Back" = stronger memory

Effective Case Study Management

Guidelines:

Learn 2–3 detailed examples for each topic.

Use **facts, dates, data, and place names**.

Structure every case study using **C.A.R.E.**

C – Context

A – Aims / Actions taken

R – Results

E – Evaluation (successes & problems)

Example:

Case Study: Volcano – Mount St. Helens (1980)

Context: USA, composite volcano, erupted 1980.

Actions: Evacuation zones, monitoring systems.

Results: 57 deaths, new research led to improved preparedness.

Evaluation: Effective warning but severe environmental loss.

PART C: Independent Enrichment (Extended Supercurriculum)

Academic Reading

The Power of Geography – Tim Marshall

Collapse – Jared Diamond

There Is No Planet B – Mike Berners-Lee

An Atlas of Impossible Longing – for literary geography context

Challenge:

Compare how two texts approach global sustainability and human impact.

Fieldwork and Data Collection

Practice Local Enquiry:

Question: “How does traffic congestion vary near the school?”

Method: Tally counts, noise surveys, environmental quality index.

Presentation: Graphs, bar charts, annotated map.

Conclusion: State pattern, possible causes, reliability limits.

Online Tools:

Google Earth, MapChart, ArcGIS StoryMaps, Excel for graphs.

Record GPS data via smartphone or app (e.g., *Map My Walk*).

Digital & Global Literacy

Explore global datasets:

Our World in Data: <https://ourworldindata.org/>

World Bank Data: <https://data.worldbank.org/>

NASA Earth Observatory: <https://earthobservatory.nasa.gov/>

UN Environment Live: <https://environmentlive.unep.org/>

Task:

Investigate one global indicator (e.g., deforestation rate or urban growth).

Create a short infographic comparing two regions.

Reflection and Growth

Geo-Journal Prompts:

“Which global issue do I care about most, and why?”

“How do my daily actions connect to distant places?”

“What kind of future geographer do I want to be?”

Extension:

End-of-term essay: “*The World in 2050: My Vision for a Sustainable Planet.*”

PART D: Progress and Self-Evaluation

Weekly Progress Tracker

Week	Topic	AO1	AO2	AO3	Comments
1	Population & Migration	4/5	3/5	2/5	Need to add more case study detail
2	Coasts & Erosion	5/5	4/5	4/5	Great diagrams; revise processes
3	Industry	3/5	3/5	3/5	Focus on examples

Self-Evaluation Rubric

Level	Descriptor
Emerging	Basic knowledge, limited use of examples
Developing	Some understanding, needs stronger structure
Proficient	Balanced explanation, clear examples
Advanced	Detailed knowledge, independent thinking
Expert	Evaluates data and draws nuanced conclusions

PART E: 12-Week Revision Plan

Week	Focus	Core Actions
1-2	Population & Settlement	Mind map + quiz + 1 case study
3-4	Natural Environments	Diagrams + definitions + videos
5-6	Economic Development	Case study cards + essay
7-8	Map & Graph Skills	Practice Paper 2 tasks
9-10	Fieldwork	Create and analyse sample dataset
11	Mock Paper 1 & 2	Timed conditions + feedback
12	Final Review	Summary sheets + weak topic revision

PART F: Long-Term Project (Capstone)

Title: “Mapping the Future: Geography, Humanity, and Change”

Objective: Synthesize physical and human geography insights.

Duration: 6–8 weeks.

Deliverable: 1,500-word research report or multimedia presentation.

Assessment Criteria:

Data accuracy and interpretation

Analytical depth

Visual presentation quality

Reflective insight

PART G: Recommended Resources

Category	Resource	Link
Study Support	BBC Bitesize Geography	https://www.bbc.co.uk/bitesize/subjects/zkv76sg
Data & Maps	Our World in Data	https://ourworldindata.org/
Case Studies	Cool Geography	https://www.coolgeography.co.uk/
Fieldwork	National Geographic Education	https://education.nationalgeographic.org/
Revision Practice	Save My Exams – IGCSE Geography	https://www.savemyexams.com/igcse/geography/
Professional Body	Royal Geographical Society	https://www.rgs.org/schools/

Final Reflection

“Geography illuminates the past, explains the present, and prepares us for the future.” — Michael Palin

Through this guide, you have the tools to master IGCSE Geography not only as an academic subject, but as a lens through which to view humanity and the planet.

Global Perspectives – Basic Supercurriculum (Cambridge Board)

Why Supercurriculum Matters in Global Perspectives

Global Perspectives encourages you to see the world from *multiple viewpoints*. It helps you question assumptions, analyse real-world problems, and collaborate across cultures.

The supercurriculum extends classroom learning into *active citizenship*, allowing you to connect knowledge to real human challenges — from climate change and technology to equality and peace.

Reading – Breadth and Comprehension

Develop awareness of current global issues and diverse perspectives.

Suggested Activities:

Read *BBC Future Planet*, *The Guardian Global Development*, or *National Geographic Global Issues* sections.

Explore UN Sustainable Development Goals (<https://sdgs.un.org/goals>).

Keep a “Global Issues Journal” — summarise one news article per week and write your opinion in 100 words.

Read *UNICEF Voices of Youth* stories to understand how young people worldwide experience the same challenges differently.

Writing – Argument and Clarity

Practice communicating your ideas with structure and balance.

Tasks:

Write a short essay (300 words): “*Is technology making the world more connected or more divided?*”

Use **PEEL structure** (Point, Evidence, Explain, Link).

Write a letter to a local newspaper expressing your view on a global issue (e.g., water scarcity, human rights).

Keep a “Position Log”: note whether your opinion on issues changes after reading or discussions.

Listening – Interpretation and Empathy

Learn to understand other people’s views through active listening.

Activities:

Watch TED Talks such as “The Danger of a Single Story” by Chimamanda Ngozi Adichie.

Listen to *The Guardian Global Development Podcast* or *UN News Audio*.

Summarise each talk in one paragraph, highlighting new perspectives you hadn’t considered.

Discuss in class: *What surprised you about another country’s experience of the same issue?*

Watching – Visual and Applied Understanding

See global themes come alive through film and media.

Recommended:

Before the Flood (environment & responsibility)

He Named Me Malala (education & gender equality)

The Social Dilemma (technology & ethics)

Explained: Global Inequality (Netflix series)

Task:

After watching, answer: “What’s the main issue? What are the causes? How might different people view this differently?”

Researching – Independent Inquiry

Learn to find, analyse, and compare sources.

Activities:

Choose one global topic (e.g., poverty, climate change, migration).

Find two articles with *different perspectives*.

Summarise each viewpoint in your own words.

Create a simple table comparing **causes, impacts, and proposed solutions**.

Use websites like the *World Bank Data Portal* or *Our World in Data* to find reliable evidence.

Student-Led Tasks – Collaboration and Leadership

Develop teamwork and project planning.

Ideas:

Lead a small group presentation on a Sustainable Development Goal (e.g., SDG 5: Gender Equality).

Create a class debate on a controversial topic: “Is globalisation good for everyone?”

Form an “Eco or Equality Committee” to promote awareness in school.

Design a short survey on student opinions about a global issue and share the results visually.

Creative Tasks – Expression and Impact

Show your understanding through imagination and empathy.

Examples:

Create a short awareness poster for World Water Day or Earth Day.

Write a diary entry from the perspective of someone affected by a global issue.

Produce a short video (1–2 minutes) with your opinion on a global challenge.

Write a poem or short story titled “*A Day in a Global Citizen’s Life*.”

Reflection – Metacognition and Growth Mindset

Think about your own learning and values.

Prompts:

“What global issue do I care about most, and why?”

“How can I make a difference locally?”

“Which skill have I improved most — collaboration, research, or communication?”

Keep a weekly reflection log to record how your views evolve through discussion.

Interdisciplinary Connections – Global Synthesis

Subject	Connection Example
Science	Climate change, health, sustainability
Economics	Globalisation, inequality, trade
History	Conflict, colonialism, migration
Sociology	Social justice, gender, community
ICT	Cybersecurity, AI, communication ethics
Languages	Cross-cultural communication

Activity:

Design a visual “Global Web” linking five subjects and showing how they all affect one world issue (e.g., poverty or food security).

Digital & Career Literacy

Learn to use global data responsibly and explore careers linked to social impact.

Activities:

Research jobs in NGOs, policy, journalism, diplomacy, or environmental science.

Explore how data is visualised at *Gapminder* or *Our World in Data*.

Practice writing an email to a real organisation (e.g., UNICEF, WWF) requesting information for a school project.

Create a small “Career Map” showing how Global Perspectives connects to your interests and strengths.

Final Reflection

Global Perspectives teaches you to think critically, care deeply, and act wisely.

Through reading, debate, and creative exploration, you become not just a learner — but a global citizen ready to make positive change.

Global Perspectives – Extended Supercurriculum (Cambridge Board)

Why Extended Global Perspectives Matters

Global Perspectives challenges you to *investigate real-world issues through multiple lenses*.

It builds the habits of mind that define true global citizens: **critical inquiry, ethical reasoning, collaboration, and action**.

In this extended version, you move from *awareness* to *analysis* — researching, debating, and proposing solutions grounded in evidence and empathy.

Reading – Depth, Comparison, and Source Evaluation

Objective: Read critically to understand how knowledge and bias shape our worldview.

Advanced Readings:

Factfulness – Hans Rosling

The World is Flat – Thomas Friedman

Half the Sky – Nicholas Kristof & Sheryl WuDunn

Invisible Women: Data Bias in a World Designed for Men – Caroline Criado Perez

UN Human Development Reports (<https://hdr.undp.org>)

Chatham House Global Trends and *The Economist – Global Agenda*.

Tasks:

Choose one current global issue (e.g., inequality, technology, migration).

Read **two contrasting perspectives** — one from the Global North, one from the Global South.

Write a **400-word comparative commentary** highlighting differences in argument, evidence, and assumptions.

Keep a **Critical Reading Log** noting:

- Key claim

- Supporting evidence

- Type of bias (cultural, political, economic)

- Your reflection

Writing – Persuasion, Synthesis, and Ethical Reasoning

Objective: Develop academic writing that balances emotion, logic, and ethics.

Tasks:

Write a **mini research essay (1,000–1,200 words)**:

“To what extent should countries prioritise economic growth over environmental sustainability?”

“Is technology improving equality or widening the gap?”

“Can individuals make a meaningful difference in global challenges?”

Include **3–5 sources**, correctly cited (Harvard or MLA style).

End with a **personal reflection**: how your perspective evolved during research.

Extension:

Write an **op-ed (opinion article)** aimed at a youth audience — clear, engaging, 600 words max. Submit to your school newsletter or a student magazine.

Listening – Debate and Global Discourse

Objective: Engage with international voices and evaluate real-world dialogue.

Resources:

The Economist Podcasts: The Intelligence / Money Talks

BBC World Service – The Documentary

TED Global (e.g., “The Case for Optimism on Climate Change” – Al Gore)

World Economic Forum debates on YouTube

Tasks:

Listen to two podcasts presenting different perspectives on one global issue (e.g., AI ethics).

Create a **two-column summary** comparing arguments and evidence.

Present orally: “Which speaker was more persuasive, and why?”

Extension:

Join or start a **Model United Nations (MUN)** team and represent a country’s perspective on global issues.

Watching – Visual Evidence and Media Literacy

Objective: Analyse how documentaries and media shape opinion and identity.

Recommended:

The True Cost (fashion & globalisation)

13th (race & justice)

Minimalism: A Documentary About the Important Things

An Inconvenient Sequel: Truth to Power

Inside Bill’s Brain (innovation & leadership)

Task:

Choose one film, and complete a **media critique** including:

Main argument

Tone and emotional appeal

Evidence used (data, testimony, imagery)

Missing perspectives

Overall reliability (1–5 scale)

Extension:

Create a short **documentary review podcast** (2 minutes) with personal reflection.

Researching – Independent and Comparative Inquiry

Objective: Conduct structured investigation into global issues.

Research Framework (adaptable to IGCSE Component 1):

Define your question (e.g., “What are the main barriers to gender equality in education?”).

Plan your research – decide on keywords, regions, and perspectives.

Gather sources – at least 4 from contrasting viewpoints.

Analyse reliability using the **CRAP test** (Currency, Relevance, Authority, Purpose).

Synthesize findings – what patterns or contradictions appear?

Draw conclusions – how do perspectives vary, and why?

Output:

Create a 3–4 page **mini research portfolio** with summary tables and an analytical reflection.

Extension:

Enter your work in the *Cambridge Schools Global Perspectives Research Challenge* or *Young Reporter for the Environment* competition.

Student-Led Tasks – Leadership and Collaboration

Objective: Build teamwork, communication, and decision-making skills.

Projects:

Organise a **Global Issues Fair** at school — each team presents one Sustainable Development Goal (SDG).

Conduct a **peer survey** on a chosen issue (e.g., social media use, consumption, recycling habits) and display results in charts.

Lead a **debate series**:

“Should rich countries accept more refugees?”

“Is surveillance necessary for security?”

Collaborate with students abroad via eTwinning or virtual exchange.

Extension:

Complete a **Team Project Plan** (mirroring IGCSE Component 2):

Define shared goal

Assign responsibilities

Record decisions and reflections

Produce a joint outcome (poster, campaign, short film)

Creative Tasks – Innovation and Storytelling

Objective: Express complex ideas through imaginative, human-centred formats.

Projects:

Create a **digital story or short film**: “My Vision of a Better World.”

Design an **interactive infographic** comparing human development across countries.

Write a **monologue or diary** of someone living through a global issue (e.g., refugee, scientist, activist).

Compose a **photo essay** titled “*Inequality in Everyday Life.*”

Extension:

Submit entries to the *UNICEF Voices of Youth* or *Royal Commonwealth Essay Competition*.

Reflection – Metacognition and Global Mindset

Objective: Develop awareness of your own values and intellectual growth.

Activities:

Keep a **Global Perspectives Journal**:

Record weekly reflections (What did I learn? What changed my mind?)

Note your emotional reactions to injustice or inequality.

Reflect on cultural assumptions you may have challenged.

End each term with a **Self-Reflection Report** on your collaboration, communication, and research skills.

Extension:

Deliver a **TED-style talk** in class: “How studying Global Perspectives changed the way I see the world.”

Interdisciplinary Connections – Systems and Synthesis

Objective: Link local and global issues across multiple subjects.

Discipline	Connection Example
Science	Climate change, pandemics, bioethics
Economics	Development, inequality, sustainability
History	Conflict, human rights, revolution
Sociology	Gender, identity, media, culture
ICT	Digital ethics, privacy, AI
Art & Design	Visual activism, poster campaigns

Challenge Task:

Build a **Global Systems Map** showing how one issue (e.g., climate migration) links science, politics, economics, and ethics.

Digital & Career Literacy – Responsible Global Citizenship

Objective: Learn how digital tools shape global discourse and future opportunities.

Activities:

Practice **fact-checking** using *Reuters Fact Check* or *Snopes*.

Learn **data visualisation** through Canva, Flourish, or Infogram.

Explore careers in:

- International relations

- Humanitarian work

- Journalism & media

- Environmental policy

- Education & advocacy

Extension:

Create a **LinkedIn student profile** showcasing your global research and teamwork achievements.

Advanced Long-Term Project (Capstone)

Title: “A World in Dialogue: Understanding Perspectives on a Shared Issue”

Duration: 6–8 weeks

Structure:

Choose one global issue (e.g., education access, food insecurity, AI ethics).

Research at least **three perspectives** (cultural, national, economic).

Collect data (interviews, surveys, online sources).

Write a **1,200–1,500 word report** with visuals and critical reflection.

Create a public outcome (poster, campaign video, article).

Present your findings to an audience.

Assessment Criteria:

- Depth of research and balance of perspectives

- Analytical evaluation and ethical awareness

- Creativity and presentation quality

- Reflection on personal learning and action

Recommended Resources

Type	Resource	Link
Official Syllabus	Cambridge IGCSE Global Perspectives 0457	https://www.cambridgeinternational.org/programme/s-and-qualifications/cambridge-igcse-global-perspectives-0457/
Data & Analysis	Our World in Data	https://ourworldindata.org/
Debates & News	The Economist	https://www.economist.com/
UN Framework	Sustainable Development Goals	https://sdgs.un.org/goals
Youth Voices	UNICEF Voices of Youth	https://www.voicesofyouth.org/
Academic Skills	BBC Ethics Guides	https://www.bbc.co.uk/ethics
Competitions	Royal Commonwealth Essay / Young Reporter	https://www.thercs.org/youth/competition/

Final Reflection

“Global Perspectives is not about the right answer — it’s about the right questions.”

Through this extended journey, you evolve from a student into an *inquirer and changemaker*.

You learn to analyse complexity, communicate respectfully, and act responsibly — skills that define true leadership in a connected world.

Global Perspectives – Complete Study Guide & Extended Supercurriculum (Cambridge Board)

PART A: Understanding the Cambridge IGCSE Global Perspectives Course

Overview of the Course

The Cambridge IGCSE **Global Perspectives (0457)** course develops transferable skills in **critical thinking, research, communication, and collaboration**.

It encourages you to explore complex global issues from personal, local, and international viewpoints.

Component	Focus	Format	Weight
Component 1 – Written Examination	Individual critical response to global issues	1 hour 15 minutes	35%
Component 2 – Individual Report	Independent research on a global issue	1,500–2,000 words	30%
Component 3 – Team Project	Collaborative investigation with reflective paper	Group presentation + individual reflection	35%

Assessment Objectives (AO1–AO3)

Code	Objective	Description	Demonstrated Through
AO1	<i>Research, Analysis, and Evaluation</i>	Gather, analyse, and compare information from a range of perspectives	Individual Report & Exam
AO2	<i>Reflection</i>	Evaluate your own and others' perspectives	Written Paper & Reflection Components
AO3	<i>Communication and Collaboration</i>	Express ideas clearly and work effectively with others	Team Project & Presentation

Key Global Themes

Students select and investigate issues within these broad themes:

Environment – climate, biodiversity, water, energy

Technology – AI, privacy, data, automation

Education – access, gender, literacy, inclusion

Health – pandemics, inequality, wellbeing

Conflict & Peace – migration, human rights, international law

Culture & Identity – gender, diversity, language, heritage

Politics & Governance – democracy, corruption, globalisation

PART B: Study Skills and Approaches

Thinking Like a Global Citizen

Global citizenship means recognising interdependence, questioning bias, and seeking solutions that balance people, planet, and prosperity.

Core Skills to Master:

Identifying bias and reliability

Comparing perspectives (local vs global)

Evaluating ethical implications

Synthesising evidence into coherent arguments

Reflecting on your own worldview

Exam Preparation – Component 1 (Written Paper)

Structure:

Section A: Analyse short sources (identify perspectives, evaluate evidence)

Section B: Compare perspectives and develop your own response

Tips for Success:

Read questions carefully — highlight *command words*: *Explain, Compare, Evaluate, Assess*.

Always **reference evidence** from the source.

Use **balanced argument**: “Some may argue... however...”

Support your own opinion with facts or examples from current affairs.

Write clear introductions and concise conclusions.

Practice Resource:

Cambridge Past Papers & Specimen Papers:

□ <https://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-igcse-global-perspectives-0457/>

Component 2 – Individual Report (IR)

Purpose:

A 1,500–2,000-word report investigating a *global issue* of your choice, exploring **different perspectives** and proposing **solutions**.

Framework:

Stage	Description	Example
1. Topic Selection	Choose a current, real-world issue that connects personal, local, and global dimensions	“How far does fast fashion harm the environment and workers’ rights?”
2. Research	Use diverse sources (academic, media, statistics, interviews)	UN data, NGO reports, news analysis
3. Perspectives	Identify at least three viewpoints: personal, local/national, global	Local designer, government, NGO
4. Analysis	Compare arguments, evaluate evidence	Reliability, bias, author purpose
5. Reflection	How did your view change through research?	“Initially I thought... now I realise...”
6. Conclusion	Suggest practical and ethical solutions	Sustainable production & fair trade
7. Bibliography	Cite sources using Harvard or MLA style	Consistent referencing mandatory

Tip:

Use **colour-coded mind maps** to visualise perspectives and impacts.

Component 3 – Team Project

Purpose:

Work collaboratively to investigate a global issue and create a **shared outcome** (poster, video, campaign, article).

Each student also writes a **200–300-word reflection** on collaboration and learning.

Process:

Form a team (3–5 members).

Agree on topic and goal.

Plan roles and tasks (researcher, designer, editor, speaker).

Gather data from multiple countries/perspectives.

Create outcome – e.g. campaign video, website, or community initiative.

Reflect individually on teamwork and personal growth.

Assessment Criteria:

Quality of research and synthesis

Balance of perspectives

Clarity of outcome and communication

Depth of reflection on collaboration

Example Topics:

“How can schools reduce plastic use?”

“Does social media harm mental health?”

“Can AI be used ethically in education?”

PART C: Revision and Independent Study**Analytical Tools for Global Thinking**

Tool	Purpose	How to Use
OPVL	Evaluate reliability of sources	Origin, Purpose, Value, Limitation
CRAP Test	Assess online information	Currency, Relevance, Authority, Purpose
PEEL Writing	Structure arguments	Point, Evidence, Explain, Link
Three-Level Thinking	Explore causes and effects	What? → So what? → Now what?

Weekly Revision Strategy

Week	Focus	Core Tasks
1	Review global issues	Mind map + short summaries
2	Source analysis	Practice OPVL and bias detection
3	Perspective comparison	Create a table of 3 viewpoints
4	Practice Paper (Component 1)	Timed mock exam
5–6	Individual Report draft	Research, analysis, bibliography
7–8	Reflection writing	Review growth and skills
9	Team Project prep	Plan topic + roles
10–12	Presentation and peer feedback	Final review and polish

Independent Enrichment (Extended Supercurriculum Highlights)**Read:**

Factfulness – Hans Rosling

There Is No Planet B – Mike Berners-Lee

The Power of Geography – Tim Marshall

Listen/Watch:

BBC The Documentary

The Economist – *Global Issues Podcast*

TED Global Talks

Before the Flood, *13th*, *The True Cost*

Explore:

UN SDG Tracker – <https://sdg-tracker.org/>

Our World in Data – <https://ourworldindata.org/>

Global Citizen Campaigns – <https://www.globalcitizen.org/>

Create:

Short blog on a global issue

Infographic summarising key statistics
 Poster promoting one UN Sustainable Development Goal

Reflection and Growth Tracker

Date	Topic	What I Learned	How My View Changed	Next Step
Week 3	Climate Change	Local actions matter	Realised importance of political cooperation	Research Greta Thunberg's influence
Week 6	AI & Society	Tech inequality exists	Became more critical of automation	Read UN AI ethics paper

PART D: Progress and Evaluation Self-Evaluation Rubric

Level	Descriptor
Emerging	Identifies an issue but limited perspectives
Developing	Describes perspectives with some evidence
Proficient	Compares perspectives using analysis
Advanced	Evaluates perspectives and reflects insightfully
Expert	Synthesises ideas, challenges assumptions, proposes solutions

Weekly Skill Development Tracker

Skill	Description	Self-Assessment (1-5)	Teacher Comment
Research	Locating and evaluating information	4	Excellent use of sources
Critical Thinking	Analysing arguments and bias	3	Needs deeper evaluation
Reflection	Linking learning to self-awareness	5	Mature reflection
Communication	Clarity and structure	4	Very strong writing
Collaboration	Leadership and teamwork	5	Reliable and creative team member

PART E: Capstone Project

Title: “Connecting Perspectives: My Voice in a Global Conversation”

Duration: 8 weeks

Output: Multimedia presentation + 1,500-word analytical essay

Process:

Choose a global issue (climate, poverty, education, migration).

Research three perspectives (individual, national, international).

Conduct interviews or surveys if possible.

Create a visual product (poster, video, digital article).

Reflect on learning — how your view evolved.

Evaluation Criteria:

Research depth

Balance of perspectives

Quality of reflection

Creativity and communication

PART F: Recommended Resources

Category	Source	Link
Official Syllabus	Cambridge IGCSE Global Perspectives 0457	https://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-igcse-global-perspectives-0457/
SDG Framework	UN Sustainable Development Goals	https://sdgs.un.org/goals
Youth Voices	UNICEF Voices of Youth	https://www.voicesofyouth.org/
Statistics	Our World in Data	https://ourworldindata.org/
Academic Analysis	Chatham House	https://www.chathamhouse.org/
Ethics Resources	BBC Ethics Guides	https://www.bbc.co.uk/ethics
Debate Platform	Intelligence Squared	https://www.intelligencesquared.com/
Competitions	Royal Commonwealth Essay / Young Reporter	https://www.thercs.org/youth/competition/

PART G: Final Reflection

“The test of a first-rate intelligence is the ability to hold two opposing ideas in mind at the same time and still function.” — F. Scott Fitzgerald

Global Perspectives teaches you not *what to think*, but *how to think*: to listen, evaluate, empathise, and reason in a complex world.

Through research, reflection, and collaboration, you become a problem-solver — one capable of connecting ideas and inspiring change across cultures.

Business – Basic Supercurriculum (Cambridge Board)

Why Supercurriculum Matters in Business

Business isn't just theory—it's decision-making under uncertainty.

Supercurriculum tasks help you apply concepts (marketing, finance, people, operations) to real firms, sharpening analysis, judgement, and communication for top exam performance and real-world readiness.

Reading — Breadth and Comprehension

Read one short business article per week (e.g., BBC Business, FT School edition if available). Keep a log: **What happened? Why? Who's affected?**

Browse company pages (About/Investors) for two competing brands; list **mission, products, target market**.

Create a glossary (aim for 10 new terms/week): *stakeholder, revenue, break-even, market share, added value...*

Writing — Argument and Precision

250–300 words: "Should Firm X focus on product or price to grow market share?" Use **PEEL** (Point–Evidence–Explain–Link).

Draft a **business email** to a supplier negotiating payment terms (tone, clarity, next steps).

Convert a messy paragraph into bullet-pointed recommendations for a manager.

Listening — Interpretation and Empathy

Listen to a business news brief or founder interview. Note: **opportunity spotted, risks faced, decisions taken**.

Share a 60-second class recap: "One insight I'd apply if I ran a start-up."

Watching — Visual and Applied Understanding

Watch a short factory/warehouse/process video; sketch the **operations flow** and identify two potential bottlenecks.

Watch a pitch episode (e.g., Dragons' Den clips) and judge: product need, market size, margins, founder credibility.

Researching — Independent Enquiry

Mini-case: pick a local business. Find: customer segment, key costs, main competitors, one risk.

Use a supermarket or e-shop to compare **pricing & positioning** for three rival products; infer strategies.

Student-Led Tasks — Collaboration and Leadership

In pairs, run a **10-question customer survey** (Google Forms) on a new snack/drink idea; present findings with one chart.

Lead a 5-minute "This Week in Business" briefing—one global story, one local story, and a takeaway for managers.

Creative Tasks — Innovation and Communication

Design a one-page **lean canvas** for a school-friendly business idea (problem, solution, key metrics, costs/revenue).

Create a simple **logo & tagline**; write a 50-word brand story.

Make a product mock-up photo or sketch and a shelf talker (USP + price).

Reflection — Metacognition and Growth Mindset

Weekly prompt: “What decision would I reverse if I were this CEO, and why?”

Set one SMART target (e.g., “Improve finance calculations—5 break-even questions by Friday”).

Interdisciplinary Connections — Synthesis and Transfer

Maths: Percentages, ratios, break-even, contribution.

Geography: Location, logistics, emerging markets.

ICT: Spreadsheets for cash flow; presentation design.

Psychology: Consumer behaviour, motivation at work.

Economics: Supply/demand, elasticity, externalities.

Digital & Career Literacy

Build a simple spreadsheet for **cash inflows/outflows** and plot a mini cash-flow line.

Explore roles: marketing assistant, operations coordinator, junior analyst, HR assistant—note **skills required** and one way to start building each.

Core Topic Map (to align with lessons)

Business Activity & Objectives (enterprise, stakeholders, aims)

Types of Organisation (sole trader, partnership, company, franchises)

People in Business (recruitment, training, motivation, leadership styles)

Marketing (market research, segmentation, 4Ps, e-commerce)

Operations (productivity, quality, lean ideas, location)

Finance (costs, revenue, profit, break-even, cash flow, sources of finance)

External Influences (economic, legal, ethical, environmental, technological, globalisation)

Quick Weekly Routine (30–40 mins total)

Read one article + 5-line summary.

One calculation set (e.g., break-even or margin).

One mini-write-up (150–200 words) arguing a decision.

Update glossary + one reflection line.

Business Studies – Complete Study Guide & Extended Supercurriculum (Cambridge Board)

PART A: Understanding the IGCSE Business Studies Course

Overview of the Course

The Cambridge IGCSE Business Studies course develops understanding of how organisations operate and how managers make decisions about **people, finance, marketing, and production** in a changing global environment.

Paper	Title	Focus	Duration	Weight
Paper 1	<i>Short-Answer and Structured Questions</i>	Core theory & definitions	1 hr 30 min	50%
Paper 2	<i>Case Study</i>	Application, analysis, and evaluation	1 hr 30 min	50%

Assessment Objectives

AO	Objective	What It Tests	Typical Question Words
AO1	<i>Knowledge and Understanding</i>	Key terms, definitions, examples	Define, State, Identify, Outline
AO2	<i>Application</i>	Apply concepts to a business scenario	Explain, Describe how..., Give an example
AO3	<i>Analysis</i>	Explore cause-effect, consequences	Analyse, Why, How, Advantages/Disadvantages
AO4	<i>Evaluation</i>	Make judgements with justification	Recommend, Evaluate, To what extent

Main Units of Study

Unit	Core Topics
1. Business Activity	Purpose of business, enterprise, objectives, stakeholder groups
2. Types of Organisation	Sole trader, partnership, company, franchise, public sector
3. People in Business	Recruitment, training, motivation, communication, leadership
4. Marketing	Market research, segmentation, product life cycle, pricing, promotion, distribution, digital marketing
5. Operations Management	Productivity, quality control, lean production, location decisions
6. Finance	Sources of finance, costs/revenue/profit, break-even, cash flow, accounts
7. External Environment	Economic influences, ethics, legal framework, globalisation, exchange rates

PART B: Study Skills and Exam Strategy

The Business Toolkit

To succeed, you must connect **theory** → **data** → **decision** → **evaluation**.

Core Skills

Define precisely (AO₁)

Apply to a named business (AO₂)

Use logic and linking words for analysis (AO₃)

Weigh up and justify recommendations (AO₄)

Mnemonic for structuring long answers:

□ **D.A.R.E.** = *Define* → *Apply* → *Reason* → *Evaluate*

Paper 1 Tips (Short & Structured Questions)

Read the mark allocation: 2 marks = definition + example.

Keep answers concise but accurate — avoid narrative.

Use your own case examples where possible.

Example: “State two stakeholder groups of a school and explain why each is interested.”

Paper 2 Tips (Case Study Analysis)

Read the case twice: highlight key numbers, dates, and strategic problems.

Always use the **company name** in your answer (AO₂ marks!).

For 6–8 mark evaluative questions:

Compare at least two options.

End with a clear recommendation and reason (“Therefore, X is better because...”).

Quote or paraphrase data to support reasoning.

Revision and Memory Techniques

Method	Description	Example
Glossary Cards	Write definition on one side, example on the other	“Added Value – Difference between selling price and cost of inputs”
Topic Mind Maps	Link concepts visually	Marketing mix branches
Formula Practice	Daily 10-min finance drill	Break-even = Fixed Costs ÷ (Price–Variable Cost)
Timed Practice	Do one 8-mark question under 10 min	Mark using examiner report

PART C: Extended Supercurriculum Activities

Reading – Depth & Application

Shoe Dog – Phil Knight (Nike): entrepreneurship & resilience

The Everything Store – Brad Stone (Amazon): strategy & innovation

Zero to One – Peter Thiel: innovation & monopoly thinking

The Undercover Economist – Tim Harford: pricing & markets

Articles: *Financial Times for Schools*, *BBC Business Daily*, *The Economist Espresso*.

Challenge:

Read one case article weekly and write 100-word reflections: “What key business decision was made? Why did it succeed or fail?”

Writing – Business Argument & Decision-Making

Produce a 1,000-word **analytical report** on a chosen firm:

“How does Tesla maintain competitive advantage through innovation?”

Include SWOT table + evaluation of risks.

Practice **boardroom memos**: short persuasive summaries for a decision.

Write an **evaluation essay** using data (profit %, costs, market share) to justify a recommendation.

Listening – Business Communication and Leadership

Podcasts:

BBC Business Daily

The Economist Money Talks

Harvard Business Review IdeaCast

How I Built This (NPR)

Task:

After each episode, summarise one leadership lesson and one strategic insight in 50 words.

Watching – Real-World Business in Action

Documentaries & Series:

Inside the Factory – operations

The Billion Dollar Code – innovation & IP

The Social Network – entrepreneurship & ethics

Dragons’ Den / Shark Tank – investor evaluation

Task:

Analyse one pitch or company:

What is the business model?

What was the investor’s decision and why?

What risk factors did they identify?

Researching – Independent Business Investigation

Sample Enquiry Framework:

Choose a business issue (e.g., “Should McDonald’s go fully plant-based?”).

Collect data (articles, company reports, surveys).

Compare **two stakeholder perspectives** (investors vs consumers).

Draw conclusion with reasoned recommendation.

Extension:

Write a **mini-consulting report** (800–1,000 words) including:

Executive summary
Analysis (data + chart)
Recommendation
Reflection on learning

Student-Led Tasks – Collaboration and Leadership

Form a **mini-enterprise** (e.g., school merchandise, snack kiosk).
Track revenue, costs, break-even, and reflection.
Lead a “Boardroom Debate”: “Should firms prioritise profit or ethics?”
Create an **Advertising Pitch** (poster + tagline + 60-second presentation).

Creative Tasks – Innovation and Self-Expression

Design a **business plan poster**: product, market, USP, budget, risks.
Create a **marketing video or advert** for a sustainable product.
Build a digital **brand storybook** on Canva or Google Slides.
Prototype a product model with materials or 3D software.

Reflection – Metacognitive Growth

Prompts:

“Which skill have I strengthened most—analysis or creativity?”
“What type of leader am I becoming?”
“What business decision this week made me think differently?”

Task:

Keep a *Business Growth Journal* with three sections: Knowledge, Skills, Reflection.

Interdisciplinary Connections

Subject	Example of Integration
Economics	Inflation, interest rates, supply/demand
Maths	Percentages, ratio analysis, financial formulas
Geography	Location, resources, trade routes
ICT	E-commerce, spreadsheets, digital marketing
Psychology	Motivation, leadership, consumer behaviour

Digital & Career Literacy

Use Excel or Google Sheets to simulate profit/loss and cash flow.
Explore jobs on LinkedIn: marketing intern, financial analyst, operations trainee.
Research local entrepreneurship incubators or youth enterprise contests.
Follow business influencers (Simon Sinek, Indra Nooyi, Elon Musk) — analyse leadership style.

PART D: Assessment and Progress

Weekly Skills Tracker

Week	Topic	AO1	AO2	AO3	AO4	Comment
1	Business Objectives	4	4	3	2	Add more real examples
2	Marketing Mix	5	5	4	3	Good balance of evidence
3	Finance & Break-Even	5	5	5	4	Excellent calculations

Self-Evaluation Rubric

Level	Descriptor
Emerging	Recalls terms, little application
Developing	Applies ideas with some analysis
Proficient	Analyses and draws reasoned conclusions
Advanced	Evaluates effectively with clear justification
Expert	Integrates data, insight, and creativity in solutions

PART E: 12-Week Revision Plan

Week	Focus	Key Task
1	Business Activity	Define key terms + case summary
2	Objectives & Stakeholders	Diagram + example questions
3	People in Business	Create motivation summary table
4	Marketing Mix	Product life cycle diagram + essay
5	Operations	Sketch production flow + lean tools
6	Finance	20 break-even & cash-flow questions
7	External Environment	News analysis + pros/cons table
8	Case Study Practice	Mock Paper 2 (timed 90 min)
9	Weak Topic Review	Flashcards & 8-mark drills
10	Mock Paper 1	Review under exam conditions
11	Evaluation Writing	"To what extent" essay practice
12	Consolidation	Final summaries + reflection journal

PART F: Capstone Project – “Enterprise in Action”

Title: *“Design, Launch, and Evaluate a Sustainable Mini-Enterprise.”*

Duration: 6–8 weeks

Steps:

Form 3–5-person team.

Brainstorm product/service idea.

Conduct basic market research (survey 20+ people).

Set budget, calculate break-even point.

Create promotional materials.

Present business pitch and financial projection.

Reflect: What went well? What would you change?

Assessment Focus: Creativity, feasibility, teamwork, financial accuracy, ethical awareness.

PART G: Recommended Resources

Category	Source	Link
Official Syllabus	Cambridge IGCSE Business 0450	https://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-igcse-business-studies-0450/
Case Studies	Tutor2u	https://www.tutor2u.net/business
Calculations Practice	BBC Bitesize	https://www.bbc.co.uk/bitesize/subjects/zpsvr82
Global Data	World Bank	https://data.worldbank.org/
Entrepreneurship	Young Enterprise UK	https://www.young-enterprise.org.uk/
Ethics & CSR	UN Global Compact	https://www.unglobalcompact.org/
Revision Notes	Save My Exams	https://www.savemyexams.com/igcse/business-studies/

PART H: Final Reflection

“Business opportunities are like buses — there’s always another one coming.” —

Richard Branson

By exploring both the analytical and creative sides of business, you develop the mindset of an entrepreneur and strategist.

The IGCSE Business Supercurriculum transforms theory into *practical intelligence*: the ability to read markets, lead people, and make balanced, ethical decisions.

English Literature – Basic Supercurriculum (Cambridge Board)

Why Supercurriculum Matters in Literature

Literature trains close reading, empathy, and argument. Going beyond class texts builds range, interpretive agility, and exam-ready precision (quotation, analysis, evaluation).

Reading — Breadth and Comprehension

Read one poem, one short story, and 10 pages of a play/novel weekly.

Keep a **Reading Journal**: summary (2–3 lines), one *writer's method* (e.g., imagery, structure), and one *theme* connection.

Sample short fiction: Poe, Mansfield, Dahl, Saki, Lahiri.

Poetry tasters: Keats, Duffy, Heaney, Grace Nichols, Maya Angelou.

Writing — Argument and Precision

250–300 words: respond to a character/theme question using **PEEL** (Point–Evidence–Explain–Link).

Practice an **alert thesis**: a debatable claim in one sentence.

Write one paragraph each week using a **terminology target** (e.g., caesura, anaphora, free indirect discourse).

Listening — Interpretation and Empathy

Hear poems performed (Poetry Archive / BBC clips). Note how voice alters meaning.

Record yourself reading a monologue; annotate where tone or pace shifts.

Watching — Visual and Applied Understanding

Watch a stage scene (Shakespeare or set play). Bullet: setting, proxemics, lighting, gesture → *how do these shape theme?*

Compare one filmed soliloquy by two actors; note contrasting interpretations.

Researching — Independent Inquiry

Author snapshot: 10-slide mini-biography linking life to themes.

Context cards: 5 facts per set text (era, social issues, genre conventions).

Create a **Quotation Bank** (minimum 25 quotes across texts), grouped by theme.

Student-Led Tasks — Collaboration and Leadership

Run a 5-minute “**Close-Read Spotlight**” on one stanza or paragraph.

Lead a mini-seminar: three open questions that provoke debate (no yes/no).

Curate a class pad of alternative interpretations with citations to critics/productions.

Creative Tasks — Innovation and Self-Expression

Rewrite a key scene from another character's viewpoint.

Compose a found poem using only words from a chapter; add commentary on tone/imagery.

Storyboard six frames for a pivotal scene with captions showing subtext.

Reflection — Metacognition and Growth Mindset

Weekly prompt: “Which line stayed with me and why?”

Set a SMART goal (e.g., “Embed quotations fluently; no quotation longer than 8 words”).

End-of-unit reflection: how did my interpretation change?

Interdisciplinary Connections — Synthesis and Transfer

History: contexts (war, empire, civil rights).

Art/Film: visual symbolism, adaptation choices.

Psychology: motivation, cognition, unreliable narration.

Ethics/Philosophy: justice, identity, freedom.

Digital & Career Literacy

Build a simple **Lit Portfolio** (Docs/Slides/Notion) for essays, reflections, and quote banks.

Explore roles: publishing, journalism, law, advertising, teaching, editing—map which Lit skills transfer (analysis, rhetoric, empathy).

Core Text Map (choose per your syllabus set)

Shakespeare: *Macbeth*, *Romeo and Juliet*, *Merchant of Venice*, *Julius Caesar*.

Drama (20th/21st C): Miller (*A View from the Bridge*), Russell (*Blood Brothers*), Hansberry (*A Raisin in the Sun*), Priestley (*An Inspector Calls*).

Prose: Steinbeck (*Of Mice and Men*), Orwell (*Animal Farm*), Achebe (*Things Fall Apart*), Hosseini (*The Kite Runner*).

Poetry Anthologies: selection of pre-1914 and contemporary poems (love, nature, conflict, identity).

Weekly Micro-Routine (30–40 mins)

Read: poem + 10 pages prose/play.

Journal: 6 lines (summary, method, theme).

One PEEL paragraph with a short quotation (≤ 8 words).

Add two quotations to your bank (tag by theme + device).

English Literature – Extended Supercurriculum (Cambridge Board 0475)

Why Extended English Literature Matters

Studying literature is not only about reading texts; it's about reading *human experience*.

Through poetry, prose, and drama, you explore universal themes — love, power, conflict, identity, morality — while developing advanced skills in analysis, empathy, and written argument.

This extended supercurriculum helps you engage as a literary critic: questioning authorial choices, exploring interpretations, and connecting literature to history, art, and ethics.

Reading – Depth, Breadth, and Comparative Study

Objective: Expand reading range across genres, periods, and cultures. Learn to compare style, structure, and theme.

Core Extension Activities:

Read **one supplementary text** each term linked by theme to your set work.

Examples:

If studying “Macbeth” → read *Othello* or *Dr Jekyll and Mr Hyde* (power, guilt, duality).

If studying “Of Mice and Men” → read *Animal Farm* or *To Kill a Mockingbird* (justice, morality, equality).

Keep a **Comparative Reading Journal**: note setting, conflict, theme, tone, symbols, and social context.

Read at least **two critical essays** (e.g., JSTOR, Poetry Foundation, SparkNotes “In Depth”) and summarise arguments in your own words.

Read aloud once a week: tone and pacing improve analytical reading and exam performance.

Challenge:

Create a *Literary Family Tree* — map connections between your texts and other works from the same period or genre (e.g., Romanticism, Modernism).

Writing – Critical Argument and Evaluative Clarity

Objective: Write analytically, persuasively, and precisely — every paragraph driven by insight, not summary.

Tasks:

Write a **comparative essay**:

“How do two authors explore the idea of moral choice?”

Use clear thesis, balanced structure, embedded quotations, and evaluative conclusion.

Rewrite one of your exam essays, focusing on tightening thesis and textual evidence (≤8 words per quotation).

Practice writing **critical introductions** (2 sentences summarising theme + technique + tone).

Produce one **literary article** (600–800 words): “Why [Author] Still Matters Today.”

Extension:

Create a **criticism notebook** — collect striking phrases from your own essays, peers, or critics, and refine them for future use.

Listening – Interpretation and Aural Appreciation

Objective: Deepen understanding of language rhythm, tone, and emotion through spoken performance.

Activities:

Listen to **Poetry Archive recordings** (Heaney, Duffy, Agard, Angelou).

Annotate vocal choices and rhythm.

Analyse **actors’ interpretations** (RSC Shakespeare scenes): how pacing and gesture alter meaning.

Record yourself performing a monologue or poem; reflect on mood, stress, and diction.

Listen to *BBC In Our Time – Literature* podcasts or *Guardian Books* interviews and note interpretive differences.

Challenge:

Write a 200-word comparison of two performances of the same extract, analysing tone and subtext.

Watching – Visual Literacy and Performance Analysis

Objective: Interpret staging, cinematography, and adaptation choices critically.

Recommended Visuals:

National Theatre Live productions

Baz Luhrmann’s Romeo + Juliet (film language vs Shakespearean text)

An Inspector Calls (1992 film vs stage set design)

BBC Shakespeare Animated Tales for stylistic variation

Tasks:

Compare how setting or lighting symbolises theme.

Design a **director’s concept sketch** for a chosen scene.

Create a **shot-by-shot breakdown** of a dramatic moment, linking cinematic technique to meaning.

Researching – Literary Inquiry and Contextual Understanding

Objective: Investigate historical, cultural, and philosophical influences on literature.

Research Framework:

Define a question: “How does industrialisation shape Victorian literature?”

Gather primary & secondary sources (articles, interviews, historical documents).

Evaluate reliability — use the CRAP test (Currency, Relevance, Authority, Purpose).

Synthesize findings into 3–4 analytical points.

Reflect on how context changes your reading.

Project Ideas:

“The Great Depression in Literature” – comparing *Of Mice and Men* and a poem by Auden.

“Women’s Voices in Modern Poetry” – comparing Duffy, Nichols, and Angelou.

“Fear and Guilt in Tragedy” – Shakespeare vs Miller.

Extension:

Compile a mini “Context Dossier” for your set text (3 pages, with visuals and key dates).

Student-Led Tasks – Collaboration and Academic Leadership

Objective: Develop discussion leadership, peer feedback, and interpretive dialogue.

Activities:

Lead a **seminar**: choose one theme or symbol (e.g., fate, identity, memory).

Prepare 3 critical extracts and discussion questions.

Create a **peer-marking rubric** for essay evaluation.

Organise a **Literary Debate**:

“Is Macbeth a villain or a victim of ambition?”

“Does Orwell offer hope or despair in his fiction?”

Host a **Class Symposium** — each student presents a 5-minute “insight talk” on a favourite author.

Extension:

Write a “Seminar Reflection” summarising what perspectives changed your understanding of the text.

Creative Tasks – Reinterpretation and Imaginative Response

Objective: Explore literature creatively, showing insight into authorial style and tone.

Ideas:

Rewrite a scene from another character’s point of view.

Compose a **found poem** using only words from your novel.

Design a **book cover and blurb** reflecting tone and symbolism.

Write a **monologue** revealing a character’s inner conflict not shown in the text.

Create a **modern adaptation proposal** (200-word synopsis + design concept).

Extension:

Submit work to a literary competition (e.g., *Foyle Young Poets* or *Commonwealth Essay Competition*).

Reflection – Critical Self-Awareness and Metacognition

Objective: Cultivate reflective habits about your growth as a reader and writer.

Prompts:

“Which text challenged my empathy the most — and why?”

“What interpretive risk did I take this term?”

“How has my understanding of authorial purpose changed?”

“Which essay shows my clearest critical voice?”

Activity:

Maintain a **Literary Reflection Journal** — one entry per fortnight including a favourite quotation, a new interpretive insight, and a goal for next essay.

Interdisciplinary Connections – Literature in Context

Discipline	Connection Example
History	War poetry, empire, industrialisation
Psychology	Motivation, repression, identity
Sociology	Class, gender, ideology
Philosophy	Morality, free will, existentialism
Art / Music	Visual symbolism, tone, aesthetic movements
Media Studies	Adaptation, narrative structure, audience effect

Challenge Task:

Create a visual “Interdisciplinary Web” connecting your set text to three other disciplines — label each with a 1-line analytical link.

Digital & Career Literacy – Communication and Cultural Skills

Objective: Use digital tools for research and presentation while recognising transferable skills.

Activities:

Build a **digital portfolio** (Google Slides / Notion) containing essays, creative tasks, and reflections.

Learn to cite sources correctly using EasyBib or Zotero.

Record a **short podcast episode** discussing a literary theme.

Explore careers linked to literature: journalism, publishing, law, advertising, education, screenwriting, and diplomacy.

Analyse one real job description — identify which “literary” skills (analysis, empathy, communication) are relevant.

Capstone Project – “The Power of Words”

Duration: 6–8 weeks

Output: 1,200–1,500-word analytical essay + creative response

Structure:

Choose a theme (*Freedom, Justice, Identity, Love, Ambition*).

Select two texts — one set text + one wider reading text.

Research historical and social background.

Write a comparative analysis exploring how each author conveys the theme.
Create a short creative piece inspired by that theme (poem, monologue, or journal entry).

Reflect on how creative writing deepened your understanding.

Assessment Focus:

Insightful interpretation

Precision of language and structure

Balanced comparative evaluation

Reflection and originality

Recommended Resources

Type	Resource	Link
Official Syllabus	Cambridge IGCSE English Literature (0475)	https://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-igcse-english-literature-0475/
Study Support	SparkNotes / Shmoop / LitCharts	https://www.sparknotes.com/
Critical Essays	Poetry Foundation / JSTOR Daily	https://www.poetryfoundation.org/
Performances	National Theatre Live / Globe Player	https://globeplayer.tv/
Competitions	Foyle Young Poets / RCS Essay Competition	https://foyleyoungpoets.org/ / https://www.thercs.org/youth/competition/

Final Reflection

“Literature is the question minus the answer.” — Roland Barthes

Through the Extended Supercurriculum, you become not just a reader, but a *thinker* — analysing, questioning, and creating.

Each text becomes a mirror for your own curiosity about people, language, and the world.

English Literature (0475) – Study Guide & Extended Supercurriculum

Course Overview

Cambridge IGCSE English Literature develops understanding of how writers use **language, form, and structure** to shape meaning.

You will analyse **poetry, prose, and drama**, exploring both set texts and unseen material.

The course balances close reading with independent interpretation.

Component	Focus	Format	Weight
Paper 1	Poetry & Prose (Set Texts)	2 essays (1 poetry + 1 prose)	50 %
Paper 2 or 3	Drama (Set Texts) or Coursework Portfolio	1 drama essay or 2 coursework pieces	50 %

Schools may choose either Paper 2 (exam) or Paper 3 (coursework).

Assessment Objectives

Code	Objective	Description	Demonstrated Through
AO1	Knowledge & Understanding	Show detailed knowledge of text, theme, character, context	Accurate quotation + commentary
AO2	Analysis of Language / Form / Structure	Explain how writers achieve effects	Literary terminology + close analysis
AO3	Personal Response & Evaluation	Give informed, relevant, original views	Judgement, comparison, insight
AO4	(For Coursework) Organisation & Accuracy	Structure argument & use coherent English	Fluent academic style

Understanding Each Paper

Paper 1 – Poetry & Prose

Section A: one question on a poem / poems from the anthology or set text.

Section B: one question on a prose text.

1 hour 30 min = 2 essays (~45 min each).

Choose either (a) *essay* question or (b) *passage-based* question per section.

Top tips

Identify *what* the question asks (e.g. feelings, contrast, relationships).

Integrate short quotations (≤ 8 words).

Comment on **method** (not just meaning): tone, imagery, rhythm, punctuation.

Keep balance: 70 % analysis / 30 % narrative context.

Paper 2 – Drama (Set Texts)

45-minute essay on one play (Macbeth, An Inspector Calls, A View from the Bridge, etc.).

Either essay or extract question.

Focus: character, theme, dramatic methods (stagecraft, tension, dialogue).

Approach

Underline key phrases in the question.

Build a clear thesis responding to that focus.

Organise essay by ideas (not chronology).

Conclude with effect on audience and relevance today.

Paper 3 – Coursework Portfolio (*if selected*)

Two assignments (600–1000 words each):

Close analysis of an extract or poem.

Comparative essay on two texts linked by theme or technique.

Marked internally, moderated by Cambridge.

Analytical Frameworks for Essay Writing

PEEL Structure

P – Point | **E** – Evidence | **E** – Effect | **L** – Link

Example: Shakespeare presents Macbeth's guilt through the imagery of blood (**P**). After killing Duncan he cries "Will all great Neptune's ocean wash this blood clean from my hand?" (**E**). The hyperbole reveals his moral panic and inner corruption (**E**). Thus Shakespeare turns a physical stain into a psychological burden (**L**).

Critical Method Checklist

Identify tone (angry, tender, satirical ...).

Examine form (sonnet, monologue, tragedy).

Comment on structure (turning point, contrast).

Analyse language (imagery, metaphor, sound pattern).

Explore context and audience response.

Unseen Poetry & Prose Strategy

Read twice – first for meaning, second for technique.

Annotate tone shifts, imagery, structure.

Plan quickly: intro + 3 paragraphs (idea / device / effect).

Avoid summary; focus on "how" not "what".

End with a 2-sentence evaluation of overall impact.

Context & Critical Perspective

Literary Contexts

Period	Features	Example Authors
Renaissance / Elizabethan	ambition, fate, moral conflict	Shakespeare, Marlowe
Victorian	realism, social morality	Dickens, Hardy
Modernist	fragmentation, inner voice	Woolf, Eliot
Post-colonial / Contemporary	identity, race, voice	Achebe, Atwood, Agard

Critical Lenses for Extension

Feminist reading – representation of gender and power.

Marxist – class conflict and economic control.

Psychoanalytic – motivation and subconscious.

Post-colonial – voice of the “other”.

Eco-critical – human–nature relationship.

Students can choose one lens to frame an optional independent essay or presentation.

Supercurriculum Extensions

Category	Advanced Tasks
Independent Reading	Read an extra novel or play linked by theme (e.g. “Ambition”, “Justice”). Keep a comparative journal.
Critical Research	Summarise two scholarly articles and evaluate their arguments in 150 words each.
Creative Response	Rewrite a scene or poem in modern context + 200-word commentary explaining choices.
Performance Task	Direct and record a short scene; analyse staging and actor choices.
Seminar Leadership	Prepare a 5-min presentation on a key symbol or motif and lead peer discussion.

Interdisciplinary Connections

Subject	Link to Literature
History	Context of war, empire, industrial change
Psychology	Character motivation, dreams, guilt
Philosophy	Morality, existentialism, free will
Art	Symbolism & visual imagery
Music	Rhythm & emotion in poetic form
Politics / Sociology	Censorship, gender roles, social class

Practical Study Habits

Keep a **Quote Bank** for each text (20–25 short quotations with themes).

Revise one theme each week and build a “theme map”.

Use colour coding: yellow = language, green = structure, blue = context.
 Read aloud daily for 5 minutes to train interpretive tone.
 Swap essays with a peer and assess AO performance using rubric below.

Self-Evaluation Rubric

Level	Descriptor
Emerging	Summarises plot with minimal analysis
Developing	Uses quotations but limited comment on effect
Proficient	Integrates language and form analysis confidently
Advanced	Evaluates interpretations and draws insightful links
Expert	Writes with critical independence and flair across texts

12-Week Revision Planner

Week	Focus	Core Tasks
1	Poetic Form & Terminology	Annotate 3 poems for form, tone, structure
2	Characterisation	Write 2 PEEL paragraphs on character change
3	Themes & Motifs	Create a theme grid for each text
4	Language Analysis	10 mini-exercises on imagery and syntax
5	Context Research	Half-page summary per author
6	Unseen Poetry Practice	2 timed responses (45 min each)
7	Comparative Essay	Plan and write 1 essay (2 texts + evaluation)
8	Drama Techniques	Diagram 1 scene for stagecraft
9	Exam Paper 1 Mock	Full timed practice + teacher feedback
10	Refine Introductions & Conclusions	Write 3 sample intros
11	AO Checklist Review	Highlight strengths & targets
12	Final Timed Mock & Reflection	Paper 1 or 2 under exam conditions

Recommended Resources

Type	Resource	Link
Official Syllabus	Cambridge IGCSE English Literature (0475)	https://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-igcse-english-literature-0475/
Study Support	Cambridge Learner Guide & Past Papers	https://www.cambridgeinternational.org/support-and-training-for-schools/
Critical Essays / Poems	Poetry Foundation / The British Library Discovering Literature	https://www.poetryfoundation.org/ / https://www.bl.uk/discovering-literature
Performances	Globe Player / National Theatre Collection	https://globeplayer.tv
Revision Sites	BBC Bitesize / Save My Exams	https://www.bbc.co.uk/bitesize/subjects/zckwzhv/ / https://www.savemyexams.com
Competitions	Foyle Young Poets / RCS Essay Competition	https://foyleyoungpoets.org/ / https://www.thercs.org/youth/competition/

Final Reflection

“Reading makes a full man, conference a ready man, and writing an exact man.” — Francis Bacon

The Literature Supercurriculum guides you from *reader* to *critic*.

By analysing structure, evaluating viewpoints, and expressing ideas with precision, you develop both intellectual depth and creative empathy — the twin aims of literary study.

English Literature (0475) – Drama Supercurriculum & Study Guide

Why Drama Matters

Drama is literature brought to life.

It combines **language, movement, and emotion** to explore human behaviour under pressure.

Studying plays trains you to interpret **speech, silence, stage direction, and audience reaction** — skills essential for both analysis and empathy.

The Drama component rewards students who can:

Recognise how playwrights create tension and character.

Analyse dramatic methods — **stagecraft, structure, and dialogue**.

Evaluate moral, political, and emotional impact on audiences.

Exam Overview (Paper 2 – Drama Texts)

Section	Type	Duration	Task	Marks	AO Focus
Paper 2	Drama Set Text	45 min	One essay (extract or essay question)	25	AO1–AO3
Coursework Option 2 (Paper 3)	assignments	Internal	Analytical or creative essay	50	AO1–AO4

Assessment Objectives (Drama Focus)

AO	Skill	Description	Example Application
AO1	Knowledge & Understanding	Accurate reference to play, theme, structure	Explains character arc, identifies key acts
AO2	Analysis of Dramatic Methods	Explains how playwright uses language & stagecraft	Discusses lighting, tension, irony, pacing
AO3	Informed Personal Response	Evaluates ideas, audience impact	“The ending is both tragic and hopeful because...”
AO4	(Coursework only) Coherent Expression	Clarity, organisation, accuracy	Fluent academic writing

Core Analytical Framework: S.T.A.G.E.

Letter	Meaning	What to Look For
S	Structure	Scene order, turning points, rhythm of acts
T	Tension	Conflict, silence, stage directions, pauses
A	Audience	Response, emotion, irony, empathy
G	Gesture	Movement, tone, body language
E	Effect	Overall message, emotional outcome, symbolism

Approaching the Drama Question

Step 1: Read the question carefully

Underline focus words: *How does the playwright...? / Explore the theme of power through...*

Step 2: Choose your route

- (a) Essay: broad thematic discussion (e.g. guilt, justice, class).
- (b) Extract: close reading of given scene (analyse line-by-line).

Step 3: Structure

Introduction: set context + answer question directly.

Body: 3 analytical paragraphs (method + effect).

Conclusion: link message, audience, and relevance.

Step 4: Technique

Embed quotations (short phrases, no long copying).

Comment on stage directions (movement, light, tone).

Consider audience emotion — *shock, pity, humour, irony*.

Link to whole play and author's purpose.

Writing Paragraphs – Model Framework

PETER Paragraph

Point — key idea related to question

Evidence — short quotation

Technique — dramatic method used

Effect — emotional/psychological impact

Reader/Viewer Response — your evaluative insight

Example:

Priestley builds tension through overlapping dialogue. When Mrs Birling interrupts Sheila, it shows hierarchy breaking down, increasing the scene's urgency. The audience senses both chaos and justice converging — a hallmark of the play's social message.

Drama Supercurriculum Extensions

Area	Advanced Activities
Performance Study	Watch two versions of the same scene (stage vs film) and write a 300-word comparison of acting and direction.
Stage Design	Create a sketch or digital mock-up of set & lighting for one key act. Label mood, symbolism, and movement paths.
Director's Notebook	Choose one character. Plan gestures, tone, and pacing for their three most important scenes.
Comparative Study	Compare how two playwrights handle moral responsibility (e.g., <i>An Inspector Calls</i> vs <i>The Crucible</i>).
Research Task	Study theatre movements (Greek tragedy, realism, expressionism). Present how your set text reflects or challenges these forms.
Critic's Voice	Write a short "review" of a performance from the audience's viewpoint — style, message, relevance.

Key Dramatic Devices to Master

Device	Definition	Purpose
Aside	Private speech to audience	Creates irony / reveals secrets
Dramatic Irony	Audience knows more than character	Builds tension, humour
Symbolism	Object/action represents theme	The letter in <i>A View from the Bridge</i> , the inspector in <i>An Inspector Calls</i>
Lighting / Sound	Stagecraft to mark emotion or change	Red wash = guilt, thunder = foreshadowing
Pacing / Pause	Rhythm of dialogue	Emphasises discomfort or realisation
Contrast / Juxtaposition	Opposite moods or characters	Highlights moral or social divide
Stage Directions	Playwright's guidance on movement/tone	Reveal hidden emotion, mood shifts

Contextual Insight (for Depth and Evaluation)

Playwright	Contextual Lens	Key Themes
Shakespeare	Renaissance moral order, kingship, fate	Ambition, guilt, conscience
Priestley	Post-WWII socialism, responsibility	Social class, moral awakening
Miller	McCarthyism, American Dream	Guilt, justice, identity
Hansberry	Civil Rights, African-American family life	Race, hope, generational conflict
Wilde / Shaw	Victorian hypocrisy, wit	Morality, gender, class
Sophocles	Greek tragedy, fate vs free will	Hubris, destiny, moral order

Reflection & Interpretation Prompts

“What does this scene reveal about human weakness or power?”

“How might an audience today react differently from the original audience?”

“How does silence convey meaning here?”

“What moral choice defines this character’s tragedy?”

Use these prompts to deepen AO3 (personal response).

Interdisciplinary Links

Subject	Application
History	War, industrial class, revolution context
Sociology	Gender roles, hierarchy, family structure
Psychology	Motivation, guilt, cognitive dissonance
Philosophy	Fate, moral law, utilitarianism
Art & Music	Lighting tone, sound motif, mood painting

Study and Revision Strategies

Create a **Scene Map** for each act — summarise key events, emotional shifts, symbols.

Keep **Character Logs** (goal, obstacle, revelation, transformation).

Practice one timed extract essay per fortnight (45 minutes).

Annotate scripts with colours:

☐ Language & tone

☐ Stagecraft

☐ Audience effect

Record yourself performing a scene; note emotional rhythm and diction.

Self-Evaluation Rubric (Drama Essays)

Level	Description
Emerging	Retells plot with limited method analysis
Developing	Identifies some dramatic devices, basic interpretation
Proficient	Integrates close analysis with audience awareness
Advanced	Evaluates impact and author’s purpose confidently
Expert	Writes with flair, insight, and awareness of performance possibilities

Sample 12-Week Drama Revision Plan

Week	Focus	Key Task
1	Scene-by-scene summary	Create visual map of each act
2	Character arcs	Track changes + key lines
3	Themes	Build theme sheets (quotes + methods)
4	Language techniques	Identify 10 key devices
5	Stagecraft	Draw or describe set & lighting
6	Performance study	Watch one filmed version + compare
7	Essay planning	Draft 2 outlines using PETER method
8	Context	Write 200-word context summary
9	Extract practice	Timed response under 45 min
10	Peer review	Swap essays + feedback with AO grid
11	Full mock	Paper 2 timed essay
12	Reflection	Final evaluation + performance showcase

Recommended Digital Resources

Type	Resource	Link
Official Syllabus	Cambridge IGCSE English Literature 0475	https://www.cambridgeinternational.org/programmes-and-qualifications/cambridge-igcse-english-literature-0475/
Scripts & Notes	BBC Bitesize / SparkNotes / Drama Online	https://www.bbc.co.uk/bitesize/subjects/zckw2hv
Performances	National Theatre / Globe Player / Digital Theatre+	https://www.nationaltheatre.org.uk/learning
Revision	Save My Exams / Teachit English	https://www.savemyexams.com
Criticism	British Library Discovering Literature	https://www.bl.uk/discovering-literature
Youth Theatre	NT Connections / RSC Education	https://www.rsc.org.uk/education

Final Reflection

“Theatre is the most immediate way in which a human being can share with another the sense of what it is to be human.” — Oscar Wilde

Through the Drama Supercurriculum, you move beyond reading plays to **living them intellectually**: analysing structure, feeling character dilemmas, and understanding the moral heartbeat of each story.

By integrating analysis, performance, and empathy, you become both critic and creative interpreter — the true goal of studying Drama in Literature.

English Literature (0475) – Extended Supercurriculum (Drama)

Why the Extended Drama Supercurriculum Matters

Drama transforms reading into human experience.

It unites language, movement, and morality — teaching us to interpret tension, silence, and empathy.

Through this extended framework, students progress from reading plays to *inhabiting* them: analysing the playwright's craft, staging concepts, and emotional design while linking performance to history, psychology, and society.

Reading – Depth, Context, and Comparison

Objective: Read widely across dramatic forms to understand evolution of theatre and authorial technique.

Advanced Reading Tasks:

Read **one additional play each term** from a different period or style:

Classical: Sophocles – *Antigone* (fate vs. conscience)

Elizabethan: Shakespeare – *Macbeth* or *Othello* (ambition, jealousy)

Modern: Miller – *The Crucible*, Priestley – *An Inspector Calls* (morality, responsibility)

Contemporary: Hansberry – *A Raisin in the Sun*, Shaffer – *Amadeus*, Friel – *Translations* (identity, race, art, language)

Keep a **Comparative Reading Journal:**

How is conflict presented?

What emotions are central?

What techniques recur (irony, foreshadowing, monologue)?

How does performance differ between eras?

Challenge Extension:

Read critical introductions from *The Cambridge Companion to Theatre Studies* or *Modern Drama Journal* and summarise how theatre has reflected political and social change.

Writing – Analytical and Reflective Drama Criticism

Objective: Move from observation to interpretation, from commentary to evaluation.

Extended Writing Tasks:

Compose a **Director's Commentary (800–1,000 words)** on how you would stage one key scene — explain lighting, tone, actor movement, and symbolism.

Write a **Comparative Essay:**

“How do two playwrights use conflict to expose social injustice?” (e.g., Priestley vs. Hansberry).

Produce a **Critical Review** (600 words) after watching a performance (live or filmed). Discuss acting, staging, and whether the director remained faithful or subversive.

Keep a **Theatrical Vocabulary Log** — 40+ terms (blocking, aside, soliloquy, subtext, catharsis, proxemics, diegetic sound).

Extension:

Publish your review or directorial essay in a school magazine or blog; respond to one peer review to develop critical dialogue.

Listening – Performance Literacy and Aural Interpretation

Objective: Understand how sound, pace, and silence create meaning.

Activities:

Listen to *BBC Radio Drama* performances (e.g., *Macbeth*, *An Inspector Calls*).

Annotate how voice modulation conveys power, guilt, irony.

Watch **two contrasting performances** of the same play (stage vs. film; classical vs. modern setting). Note differences in delivery, energy, and rhythm. Record your own 1-minute reading of a monologue; experiment with tone and tempo to reveal subtext.

Analyse **score and sound**: what emotional colour do music and pauses add to a scene?

Challenge:

Create a 2-minute podcast episode comparing “voice as weapon” in two characters from different plays.

Watching – Visual Literacy and Directorial Analysis

Objective: Develop visual and cinematic understanding of theatrical meaning.

Advanced Viewing List:

National Theatre Live: Macbeth (Cumberbatch, 2021)

An Inspector Calls (Stephen Daldry, 1992 film and 2015 stage revival)

A View from the Bridge (Young Vic, Mark Strong, 2014)

Oedipus Rex (BBC Greek Drama adaptation)

A Raisin in the Sun (Lorraine Hansberry, 1961 and 2014 productions)

Extended Tasks:

Write a **Performance Comparison Report (500–700 words)**:

Focus on one scene; compare stage blocking, lighting, costuming, and tone.

Design a **Storyboard** or digital mood board illustrating your interpretation of one act (include colour palette, texture, symbolism).

Analyse **camera vs. stage perspective**: how does intimacy or distance affect tension?

Extension:

Create a “Director’s Journal” — one visual entry per week with sketches or screenshots and brief commentary on aesthetic impact.

Researching – Theatre History and Contextual Inquiry

Objective: Investigate drama as a mirror of its time and society.

Advanced Research Projects:

“How did post-war theatre challenge moral complacency?” (*Priestley, Miller, Beckett*)

“The evolution of tragedy: from ancient Greece to modern realism.”

“Women playwrights and feminist theatre in the 20th century.”

“Political theatre and protest: Brecht, Boal, and documentary drama.”

Research Process:

Select one topic from above or propose your own.

Gather 4–6 sources (plays, articles, critical essays, interviews).

Use the **CRAP test** (Currency, Relevance, Authority, Purpose).

Write a 1,000-word synthesis highlighting key shifts in theatrical method and message.

Reflect on how this research deepened your appreciation of your set text.

Extension:

Collaborate with peers to create a short presentation — “From Stage to Society: How Drama Drives Change.”

Student-Led Tasks – Collaboration, Leadership, and Discussion

Objective: Transform group reading into collective interpretation.

Advanced Group Activities:

Scene Reconstruction: Divide into teams to reinterpret a scene using a different tone or context (e.g., *An Inspector Calls* set in 2025).

Socratic Discussion: One student poses a critical question (e.g., “Is Macbeth’s downfall fate or free will?”) and moderates debate.

Round Table Critique: Each student brings a 100-word comment from a critic and defends or refutes it.

Theatre Club: Once a month, host a lunchtime viewing of a short play extract followed by analysis of theme and audience impact.

Leadership Extension:

Plan a mini “School Theatre Festival” — students perform key scenes and introduce their interpretive choices.

Creative Tasks – Performance, Design, and Adaptation

Objective: Develop artistic expression alongside textual understanding.

Creative Projects:

Write a **10-minute original scene** inspired by your set text’s theme (e.g., greed, power, equality).

Compose a **character diary** — 5 entries revealing off-stage thoughts or unseen moments.

Create a **costume design portfolio** linking colour, texture, and symbolism to emotion.

Record a **modern monologue** adaptation (e.g., Macbeth as a corporate CEO).

Collaborate on a **short devised performance** combining movement, soundscape, and dialogue from multiple plays.

Extension:

Submit your creative work to youth festivals (*RSC Shakespeare Schools, National Theatre Connections, or Young Writers Competition*).

Reflection – Critical and Emotional Awareness

Objective: Build meta-cognitive understanding of your interpretive process.

Reflective Prompts:

“What emotion drives the climax of the play, and how did I feel it personally?”

“Which directorial choice would I change to strengthen the theme?”

“How do power, silence, and empathy interact in this scene?”

“What truth about human nature did this play make me confront?”

Advanced Task:

Keep a **Drama Reflection Journal** (one page per week):

A quotation or stage direction that resonated.

Your personal response and insight.

How your perspective evolved through discussion or performance.

Interdisciplinary Connections – Theatre as a Living Art

Discipline	Connection Example
History	Political revolutions, social reform, war influence on theatre
Psychology	Character motivation, moral dissonance, group behaviour
Sociology	Class struggle, gender roles, cultural identity
Philosophy	Ethics, existentialism, fate vs free will
Art & Architecture	Set design, spatial composition, lighting aesthetics
Music	Underscoring, leitmotif, emotional tempo

Interdisciplinary Challenge:

Design a “**Theatre Web of Knowledge**” linking your set text to at least three other disciplines. Annotate each with a one-sentence interpretive insight.

Digital & Career Literacy – Theatre in the Modern World

Objective: Connect literary study to digital innovation and professional pathways.

Advanced Digital Tasks:

Use Canva or Padlet to build a **digital performance portfolio** (images, video links, critical notes).

Record a **vlog reflection** on your directorial process.

Research careers in: stage design, dramaturgy, theatre criticism, media production, scriptwriting, education.

Write a 300-word professional profile summarising transferable skills: analysis, empathy, collaboration, leadership.

Extension:

Interview a local director, actor, or set designer about how they transform text into performance — summarise findings in a reflective report.

Capstone Project – “Reinventing the Stage”

Duration: 6–8 weeks

Goal: Integrate research, creativity, and critical analysis.

Project Options:

Directorial Vision Portfolio – choose a play, design its full staging (concept, set, costume, lighting, sound, audience impact).

Comparative Essay (1,200 words) – “How do two playwrights use the stage to question authority?”

Devised Performance – create a 10-minute original piece blending text, sound, and movement; submit with a 500-word reflection on theme and method.

Assessment Focus:

Depth of textual understanding

Creativity and conceptual clarity

Awareness of dramatic form

Reflection and self-evaluation

Recommended Resources

Type	Resource	Link
Performances	National Theatre Live / Digital Theatre+	https://www.digitaltheatreplus.com
Scripts & Notes	BBC Bitesize / SparkNotes / Drama Online	https://www.bbc.co.uk/bitesize/subjects/zckwzhv
Research	Modern Drama Journal / JSTOR / Cambridge Companions	https://www.cambridge.org/uk/education
Competitions	RSC Shakespeare Schools Festival / NT Connections	https://www.rsc.org.uk/shakespeare-schools-festival
Context	British Library – Discovering Literature (Drama)	https://www.bl.uk/discovering-literature
Podcasts	The National Theatre Podcast / Royal Court Playwrights’ Podcast	Spotify & Apple Podcasts

Final Reflection

“Theatre is the art of looking at ourselves.” — Augusto Boal

The Extended Drama Supercurriculum invites you to analyse, perform, and question — to see how language and gesture build human truth.

By combining scholarship, creativity, and empathy, you move beyond the page and into the living heartbeat of literature.

English Literature (0475) — Drama Study Guide

What you're assessed on (Drama)

Component	What you do	Timing/Weight	Notes
Paper 2: Drama	1 essay on your set play (choose extract or essay question)	~45 min (half of a 1h30 paper if paired) / 50% if school selects Paper 2 overall	Close analysis + whole-play knowledge
Paper 3: Coursework (option)	2 assignments (600–1000 words each)	Internally assessed / 50%	One close analysis + one comparative task

Assessment Objectives (mapped to Drama)

AO1 Knowledge/Understanding: precise reference to plot, character, themes, structure.

AO2 Writer's Methods: dramatic technique (dialogue, pause, irony, stage directions, lighting, sound, proxemics).

AO3 Personal Response: your judicious, text-based judgement (audience effect, evaluation).

AO4 (coursework only): organisation, clarity, accuracy.

What the questions look like

You'll usually face two options on your set play:

(a) Extract question — a printed passage from the play: *Analyse how the playwright presents X in this extract; refer to the rest of the play as appropriate.*

(b) Essay question — thematic/character question on the **whole play**.

Choose one. If you're strong at close reading, pick the extract; if you have excellent whole-play command, the essay can be ideal.

Fast frameworks that score

The S.T.A.G.E. lens (for Drama-specific AO2)

Structure (scene placement, turning points)

Tension (conflict, silence, pace)

Audience (what we feel/expect/realise)

Gesture (movement, tone, proxemics)

Effect (message, symbolism, moral/emotional impact)

PETER paragraph (tight and reliable)

Point → Evidence (short quote) → Technique (dramatic method) → Effect (on audience/meaning) → Reasoned link (to Q/whole play)

*Priestley heightens **tension** via overlapping dialogue (T). When Sheila interrupts—"Please be quiet, mother"—(E) the role reversal shocks the audience (E), exposing a moral awakening that undercuts class hierarchy (R).*

How to answer an Extract question (step-by-step, 45 mins)

0–3 min: Read the rubric & **underline the focus** (e.g., power/guilt/justice).

3–7 min: Second read + annotate **three clusters** (language, stagecraft, shifts).

Mark one **link to elsewhere** in the play for each cluster.

Plan (1–2 bullet points per paragraph):

Para 1: Immediate context + central tension

Para 2–3: Two strongest moments (device → effect)

Para 4: A third moment or structural turn (before/after)

Micro-links to **another scene** each time

Write (30–33 min):

Intro (2–3 sentences): argument that answers the focus.

3 body paragraphs using PETER + S.T.A.G.E.

Mini comparative link to a different point in the play within each paragraph (secures whole-text reach).

Conclusion (2 sentences): overall effect + author's purpose.

Final minute: Circle key words in the Q; check you actually answered them.

Golden rule: avoid re-telling; do **how** not **what**.

How to answer a Whole-play Essay (45 mins)

2-minute plan:

Thesis: a **one-sentence answer** with stance (e.g. "Priestley argues responsibility is collective, staged through Inspector's control of pace and lighting.")

3 ideas (ordered for logic, not chronology)

1 counter/complication (nuance = AO3 maturity)

Template

Intro: text + task + stance (no plot summary).

Para 1–3: Idea → quoted evidence (≤8 words) → method (dialogue/irony/light) → effect on audience → link to Q.

Para 4 (complication): where the idea is challenged/complicated; evaluate.

Conclusion: "So what?" for audience/ethics/theme.

Stagecraft toolkit (micro-glossary to deploy)

Proxemics (spacing =

power/intimacy), **levels**, **blocking**, **gesture**, **pause/silence**, **overlap**, **sticho**

mythia (rapid exchange), **dramatic irony**, **aside**, **soliloquy**, **lighting**

cue, **sound motif**, **symbolic props**.

Use **one** stagecraft term per paragraph with purpose explained.

High-value context (one-liners you can wield)

Shakespearean tragedy: moral order + vaulting ambition → disorder (*Macbeth*).

Priestley 1945: post-war social conscience; interrogates class/individualism (*An Inspector Calls*).

Miller/McCarthyism: hysteria/exclusion; public guilt vs private fear (*The Crucible*).

Hansberry/Civil Rights: dignity, aspiration, systemic barriers (*A Raisin in the Sun*).

Keep context **relevant** and **brief**; tie it to a stage choice.

Model intros & topic sentences (plug-and-play)

Intro (extract)

In this extract Priestley intensifies moral pressure through accelerated turn-taking and symbolic lighting changes. By compressing status reversals into a few volatile lines, he compels the audience to confront responsibility as a shared, not private, duty.

Topic sentence (essay)

Shakespeare frames Macbeth's rise as a collapse of ethical language; euphemism and fractured rhythms reveal a conscience that speaks only in evasion.

Common pitfalls — and fixes

Pitfall	Quick Fix
Plot retell	Start each paragraph with method not event.
Overlong quotes	Use ≤8 words ; zoom on a telling fragment.
Ignoring stage directions	Quote and explain [Aside] [Pause] [A loud ring].
One-sided argument	Add a counter-reading in one paragraph.
No links to whole play (in extracts)	Add 1–2 micro-links per paragraph.

Coursework (if your centre selects Paper 3)

Assignment 1: close analysis of a short extract (600–1000 words).

Assignment 2: comparative essay (two texts or two scenes).

Tips: keep a bibliography; signpost AO2; paragraph topic sentences; conclude with evaluation, not summary.

12-Week Drama Revision Plan (compact)

Week	Focus	Output
1	Scene map (acts & turning points)	1-page diagram
2	Character arcs	3 mini-profiles (goal/obstacle/revelation)
3	Themes & motifs	Theme grid (quotes + method)
4	Stagecraft	Annotate one scene for 5 devices
5	Language	10 micro-commentaries on key lines
6	Extract drill #1	Full timed response

7	Whole-play essay #1	Planned + written essay
8	Context bites	4 bullet-point context links
9	Performance study	300-word comparison (stage/film)
10	Extract drill #2	Timed + teacher/peer marking
11	Whole-play essay #2	Add counter-argument paragraph
12	Mock + reflection	Paper 2 essay + targets sheet

Self-assessment grid (use after each practice)

Level	AO1	AO2	AO3
Emerging	Basic story	Spots devices rarely	Limited personal view
Developing	Some references	Describes effects	Simple judgement
Proficient	Accurate detail	Explains how method shapes meaning	Balanced, text-based evaluation
Advanced	Wide command of play	Integrates stagecraft with language	Nuanced, considers alternatives
Expert	Selective precision	Method drives every paragraph	Insightful, original, audience-aware

Last-minute checklist (2 minutes before you finish)

Have I **answered the exact focus** of the question?

Do I have **3+ short quotations** used surgically?

Did I name **at least three dramatic methods**?

Is there **one link** to elsewhere in the play in each body paragraph (for extracts)?

Is my **conclusion evaluative**, not a retell?

Final thought

*Great drama criticism shows **how** the stage makes us feel what we feel.*

Keep your eyes on **method** → **effect** → **meaning** and your essays will land consistently in the top bands.

English First Language – Basic Supercurriculum (Cambridge Board 0500/0990)

Why a supercurriculum for EFL?

First Language English is all about *precision and purpose*: reading closely, selecting evidence, writing clearly for a specific audience, and controlling tone, structure, and grammar. These activities take you beyond classwork to build fluent, confident, and accurate communication.

Reading — Breadth & Selection

Goal: read widely; Practice *selecting and synthesising* key points.

One non-fiction article per week (science, history, culture, tech).

→ Highlight 5 key facts + 3 writer techniques (tone, bias, structure).

One opinion/column per week.

→ Note viewpoint, counter-view, and rhetorical devices.

Keep a **Reading Log**: source, 2-line gist, 3 bullet points of evidence, 1 sentence on tone.

Mini-task: 120-word *objective* summary of an article (no opinion, no quotes).

Writing — Clarity, Audience, Purpose

Goal: write to *inform, explain, argue, persuade* with control.

200–250 words: **letter or email** (formal). Focus: clear structure + polite register.

250–300 words: **article or speech** (persuasive). Use triads, contrast, rhetorical question.

300 words: **report** (neutral tone; subheadings; data references).

PEAR paragraph (for argument): **P**oint – **E**vidence – **A**alysis – **R**eaders effect.

Composition — Descriptive & Narrative

Goal: craft imagery, control pace, and end well.

Descriptive snapshot (180–220 words): one place, 5 vivid details (sight/sound/smell/touch/movement).

Narrative moment (220–300 words): one turning point; show don't tell; limit characters.

Build a **Sensory Bank** (original verbs, precise nouns, fresh metaphors).

Micro-skills: varied sentence openings, paragraphing for pace, accurate tense control.

Listening / Speaking — Voice & Audience

Goal: speak with structure; listen for argument.

90-second mini-talk: problem → cause → solution (note cards only).

After a podcast or talk, jot a **30-word précis** (main claim + reason).

Watching — How writers persuade

Goal: spot technique in the wild.

Watch a short editorial/brief talk.

Identify: **claim**, **evidence**, **appeal** (logic/emotion/credibility).

Rewrite the same message for a *different* audience (teen readers → parents; formal ↔ informal).

Research — Finding & Using Sources

Goal: select reliable material and integrate it.

Pick a current topic (e.g., social media and sleep). Find **two contrasting** articles.

→ Create a **Comparison Table**: viewpoint, 3 key points, tone, best evidence.

Write a **blended summary** (120–150 words) combining both sources, neutrally.

Student-Led — Lead, share, refine

Weekly **Language Spot**: one student explains a device (e.g., anaphora) with a 2-sentence example.

Pair-swap edits: highlight (i) strongest verb choices, (ii) one sentence to tighten.

Creative Tasks — Style & Voice

Headline flip: take a sensational headline; rewrite neutrally (and vice versa).

Tone palette: write the same 80-word paragraph in *formal*, *friendly*, and *urgent* tones.

Hook lab: craft 3 opening sentences for a narrative; keep the best.

Reflection — Accuracy & Growth

End-of-week **Skills Tally**: ideas, organisation, vocabulary, sentence variety, accuracy (✓/–).

Set a **micro-target** (e.g., “one semicolon used correctly per task” / “cut 20 redundant words”).

Interdisciplinary Links

History/Geography/Science: Practice concise, bias-free summaries of factual texts.

Global Perspectives: balanced argument, counter-argument, evaluating sources.

Business/ICT: emails, reports, and proposals with clear layout and headings.

Digital & Career Literacy

Build a simple **portfolio** (Docs/Slides/Notion): best summary, best article, best description, target list.

Grammar tools: keep your own **error log** (comma splices, agreement, homophones).

Real-world writing: a CV profile (60–80 words) and a professional email request (120–150 words).

Core Topic Map (aligns with class)

Reading & Analysis: main ideas, inference, writer's effect, bias, structure.

Directed Writing: letter, speech, article, report; register & purpose.

Summary: select, condense, *objective* tone.

Composition: descriptive techniques; narrative structure & viewpoint.

Accuracy: punctuation, sentence variety, paragraphing, spelling, vocabulary precision.

Quick Weekly Routine (30–40 mins total)

Read one article; write a **120-word summary**.

One **directed writing** piece (200–250 words).

One **composition** mini-task (180–220 words).

Update **error log** + 1 reflection line.

Handy Mini-Checklists

Directed Writing (exam-style)

Audience & purpose clear?

Intro that frames *what/why*; logical paragraphs with connectives.

Varied sentences; accurate punctuation; decisive ending.

Summary

Only relevant points; your own words; objective tone; within word limit.

Descriptive/Narrative

Concrete nouns, strong verbs, sensory detail; controlled paragraphing; satisfying close.

Micro-Drills (5 minutes each)

Cut the clutter: remove filler words from 5 sentences.

Elevate verbs: swap weak verb + adverb for one precise verb.

Comma clinic: fix three comma splices.

Synonym sense: replace 5 vague words with exact ones (thing, nice, very...).

Final Thought

Clear thinking shows up as clear writing.

Build small, consistent habits: read with purpose, summarise cleanly, write for a real audience, and polish accuracy. That's the foundation for top grades in English First Language — and for confident communication everywhere.

Είπες:

English First Language extended

English First Language (0500/0990) Extended Supercurriculum

Why the Extended Supercurriculum Matters

English First Language is more than exam technique — it is mastery of communication itself.

The extended programme develops your **critical reading, rhetorical control, and precision of style**, preparing you to write, speak, and think like an independent scholar and communicator.

You move from *comprehension* to *interpretation*, from *writing clearly* to *writing with impact*.

Reading – Depth, Breadth, and Critical Evaluation

Objective: read beyond the set extracts, exploring form, bias, and purpose across global English texts.

Advanced Reading Tasks

Read one **editorial or long-form essay** weekly (The Guardian, National Geographic, Aeon, The Atlantic).

→ Summarise in 150 words: main argument, supporting evidence, tone, and rhetorical stance.

Read one **speech** per fortnight (e.g., Malala Yousafzai, Churchill, Mandela, Greta Thunberg).

→ Annotate: appeal to ethos (credibility), logos (logic), pathos (emotion).

Create a **Writer's Toolkit Table** for each text:

Device	Example	Purpose	Effect
Repetition	"We shall fight..."	Builds rhythm	Emphasises resolve

Extension

Compare how two writers shape tone for opposite audiences (e.g., *advocacy article* vs *corporate report*). Write a 200-word evaluation: who is more effective, and why?

Writing – Voice, Structure, and Precision

Objective: master adaptable, controlled writing styles for a variety of audiences and purposes.

Advanced Writing Tasks

Directed Writing (Argumentative / Discursive)

300–350 words: *"Should schools replace homework with independent projects?"*

Plan: 3 main points (for / against), counterargument, decisive conclusion.

Embed rhetorical questions, balanced tone, and controlled paragraphing.

Informative Writing (Report / Article / Letter)

250–300 words: write a *press report* on a fictional event; maintain formal register and factual clarity.

Persuasive Writing (Speech / Campaign)

Write for a **defined audience** (students / local council / online readers).

Use inclusive pronouns, emotional logic, and memorable slogans.

Include one **short personal anecdote** for authenticity.

Reflective Writing

350–400 words: “*The moment that changed how I see success.*”

Focus on introspection, tone, and rhythm of sentences.

Extension Challenge

Rewrite a persuasive article *from the opposite viewpoint*, keeping tone consistent.

Keep a **Personal Style Notebook** – 10 openings, 10 transitions, 10 endings you admire (from fiction & non-fiction).

Composition – Narrative & Descriptive Mastery

Objective: write with imagery, rhythm, and emotional logic, not just vocabulary.

Advanced Composition Tasks

Descriptive: “*The station at dawn*” – structure around 3 sensory moments (light, sound, movement).

Narrative: “*The message that never reached me*” – focus on inner change, not twists.

Hybrid task: blend narrative with reflection (e.g., a journey that changes perspective).

Tone variation exercise: rewrite the same description in three tones: peaceful / tense / nostalgic.

Skill Focus

Figurative economy (metaphor sparingly used).

Punctuation rhythm: semicolons for connection, em dashes for emphasis.

Cohesion markers: however, meanwhile, although, yet, in contrast.

Extension

Analyse your own paragraph: highlight **5 verbs** and replace 3 with more specific alternatives.

→ Example: “*walked slowly*” → “*drifted*” or “*trudged*”.

Research & Synthesis – Independent Inquiry

Objective: find, combine, and evaluate information from multiple reliable sources.

Extended Task

Choose a topical issue (e.g., AI, climate change, privacy, media bias).

Locate **two contrasting articles**.

Identify main ideas, tone, and evidence.

Write a **neutral synthesis summary** (150 words).

Follow with a **personal commentary** (150 words) evaluating techniques and reliability.

Keep a **Bias & Rhetoric Log**:

| Text | Writer's goal | Type of bias | Evidence | Your evaluation |

Extension:

Present your synthesis orally in 90 seconds, focusing on clarity and neutrality (ideal practice for Paper 1 summaries).

Listening & Speaking – Argument and Empathy

Objective: refine tone, clarity, and audience adaptation through structured oral work.

Tasks

2-minute persuasive talk: begin with a striking hook and end with a quotation or call to action.

Podcast-style discussion: record a pair dialogue on an ethical issue (scripted intro + spontaneous debate).

Reflective monologue: describe an event from two viewpoints (younger vs older self).

Extension:

Transcribe 30 seconds of your own speech; analyse filler words, pauses, and intonation. Rewrite for maximum clarity and impact.

Watching & Listening – Language in the Real World

Objective: observe professional rhetoric, register, and structure.

Watch 3 TED Talks: identify (i) structure, (ii) key phrase repetition, (iii) emotional turning point.

Watch contrasting newscasts (BBC / Al Jazeera / DW): evaluate tone and objectivity.

Listen to short story podcasts (*The Moth*, *BBC 4 Short Works*) — note rhythm and natural dialogue.

Extension:

Write a 150-word critique of one talk: “How effectively does the speaker use language to persuade?”

Student-Led Tasks – Collaboration and Leadership

Objective: become both peer editor and rhetorical coach.

Form a **mini editorial team**: one student edits for structure, one for tone, one for grammar.

Host “**Write & Swap**” sessions: give anonymous feedback using a rubric (clarity, coherence, effect).

Conduct a **Micro-Workshop**: 5-minute demo on one skill (e.g., “how to open with contrast”).

Lead a “Sentence Surgery” activity — collectively rework an overly long paragraph into three concise sentences.

Extension:

Maintain a **Peer Feedback Log** – for each peer’s advice, note one change you actually applied.

Reflection – Metacognition and Language Awareness

Objective: build self-awareness as a communicator.

Reflection Prompts

“Which of my writing voices feels most authentic?”

“When I read non-fiction, how quickly can I detect bias?”

“What techniques make my arguments persuasive yet respectful?”

“Where do my sentences lose energy — and why?”

Extended Task:

At the end of each month, select one paragraph you are proud of and one you’d revise; annotate both, explaining decisions.

Interdisciplinary Connections

Discipline	Application
History / Geography	Objective summaries of factual data; clarity in report writing
Global Perspectives	Evaluating sources, balanced argument, reflective conclusions
Economics / Business	Concise formal tone in proposals and briefs
Literature	Analysing figurative language and narrative voice
Science / ICT	Translating complex information into accessible language

Extension:

Choose a topic from another subject and write both an **academic summary** (informative) and a **magazine-style article** (engaging) about it.

Digital & Career Literacy

Objective: connect academic writing with real-world communication.

Tasks

Build a **digital writing portfolio** (Docs / Notion / Canva): best summaries, letters, articles, and narratives.

Practice **email professionalism**: 3 models — formal request, complaint, appreciation.

Research 2 careers (journalism, law, marketing, diplomacy) — write 100 words on how language skills apply.

Use Grammarly or Hemingway Editor once a week — not for correction, but to **analyse tone and sentence rhythm**.

Extension:

Write a **LinkedIn-style bio** (80–100 words) — informative yet personable.

Capstone Project – “The Power of the Word”

Duration: 4–6 weeks

Goal: demonstrate synthesis, voice, and adaptability.

Project Options

Investigative Feature Article (700–900 words)

Topic: a social, scientific, or cultural issue of personal interest.

Combine interviews, research, and reflection; include at least one contrasting viewpoint.

Personal Essay / Memoir (600–800 words)

Reflect on identity, experience, or growth — modelled after writers like Orwell or Didion.

Rhetorical Portfolio

Three short pieces for different audiences (formal report, persuasive speech, creative narrative).

Add a 300-word reflection on style and audience adaptation.

Assessment Focus

Range of tone and structure

Evidence of synthesis and evaluation

Original voice and stylistic control

Accuracy, coherence, and register consistency

Recommended Resources

Type	Resource	Link
Official syllabus	Cambridge IGCSE English First Language 0500 / 0990	https://www.cambridgeinternational.org/programmes-and-qualifications/
Advanced reading	The Guardian Opinion, Aeon Essays, The Atlantic	https://www.theguardian.com/commentisfree / https://aeon.co / https://www.theatlantic.com
Writing skill	Purdue OWL / BBC Bitesize English Language	https://owl.purdue.edu / https://www.bbc.co.uk/bitesize
Listening	TED Talks / BBC 4 Short Works / The Moth	https://www.ted.com / https://www.bbc.co.uk/sounds
Editing & grammar	Hemingway / Grammarly / Quillbot	https://hemingwayapp.com
Academic argument	Cambridge University “Thinking Writing” resources	https://www.cctl.cam.ac.uk/thinkingwriting

Final Reflection

“To write well, one must think clearly; to think clearly, one must read deeply.”

The Extended Supercurriculum trains the *writer, thinker, and communicator* together.

Every summary, paragraph, and reflection is a rehearsal for leadership — because precision in language is power in the real world.

Would you like me to continue next with the **English First Language – Full Study Guide** (exam papers, AO grids, assessment strategies, 12-week plan, and model answers)?

Είπες:

yes

To ChatGPT είπες:

Excellent — here is the **Year 10 Cambridge IGCSE English First Language (0500/0990) – Full Study Guide**, written in the same structure and tone as your Literature and Business guides.

It covers: syllabus overview, assessment objectives, paper structure, key strategies, analytical frameworks, writing templates, and a 12-week skill-based revision plan — everything needed for top-band exam performance.

Year 10 Cambridge IGCSE English First Language (0500/0990) — Study Guide

Course Overview

Cambridge IGCSE English First Language tests the ability to:

Read with understanding and evaluation, Select, organise, and synthesise ideas, and

Write accurately, effectively, and for purpose.

The focus is on clear communication, structure, and style — not literary analysis.

Paper	Title	Focus	Weight
Paper 1	Reading	Reading for meaning, summary, and writer's effect	50%
Paper 2	Directed Writing & Composition	Writing for audience/purpose; narrative/descriptive composition	50%
(Alternative 0990)	1 & 2 (same)	For international series	same

Assessment Objectives

AO	Description	What It Means
AO1	Demonstrate understanding of explicit and implicit meanings	Inference and evidence
AO2	Analyse how writers use language and structure	Comment on tone, word choice, imagery
AO3	Select, organise and use relevant details	Concise, accurate summaries
AO4	Communicate clearly, effectively and imaginatively	Structure and voice for audience
AO5	Use accurate spelling, punctuation, and grammar	Technical control

Paper 1 – Reading (2 hours)

Section A – Reading for Meaning (Questions 1–3)

You'll read two passages and answer comprehension, analysis, and summary questions.

Skills Tested

Selecting relevant details

Explaining writer's effects

Writing concisely in your own words

Question Types

Comprehension: short answers testing understanding.

Writer's effect: analyse language, imagery, and tone.

Summary: 120–150 words — *in your own words, no comment, no quotes.*

Strategy

Annotate while reading: underline ideas, tone shifts, key verbs/adjectives.

Label paragraphs: *main idea* of each.

For writer's effect:

Choose 3–4 phrases → explain how *sound*, *connotation*, or *image* supports tone.

For summary:

Bullet 12–15 points.

Cut repetition.

Rewrite neutrally, formally, and concisely.

Language Analysis Template (for Question 2)

Device	Quotation	What it shows	Effect on reader
Metaphor	“The storm clawed at the windows”	Suggests violence of nature	Creates tension and fear

Section B – Directed Writing

Respond to both passages with your own argument (e.g., report, article, letter).
You’ll need to *synthesise* both texts and write persuasively.

Tip

Use information from both passages, not imagination.

Keep register formal, tone balanced.

Include evidence, connectives, and paragraphing.

Paragraph Framework (for 250–300 words)

Introduction — clear purpose.

Point 1 — evidence + explanation.

Point 2 — comparison or contrast.

Point 3 — solution / recommendation.

Conclusion — evaluative summary.

Paper 2 – Directed Writing & Composition (2 hours)

Section A – Directed Writing

Task: respond to one text (article, speech, letter) with your own viewpoint or summary.

Key Skills

Select key points.

Adopt appropriate register.

Write clearly and persuasively.

Plan:

Identify *purpose* (inform, argue, advise).

Identify *audience* (youth magazine, council, readers).

Annotate the source: tone, 5–7 ideas.

Plan structure: 4 paragraphs (intro–3 points–close).

Use linkers: *moreover, however, ultimately, in contrast*.

Section B – Composition

Choose **one descriptive or narrative** task.

Type	Key Features	Checklist
Descriptive	Mood, setting, atmosphere	5 senses, paragraph focus, figurative language
Narrative	Story structure, character, climax	Opening hook, rising tension, resolution
Reflective (some syllabuses)	Inner thought, tone shift	Authentic voice, emotional journey

Composition Scoring (AO4–AO5)

Content: originality, coherence, tone.

Structure: paragraphs, transitions, flow.

Style: varied vocabulary and syntax.

Accuracy: grammar, punctuation, spelling.

Writing Frameworks

Descriptive Writing – 4-Part Flow

Zoom-in Opening – detail the smallest image.

Movement – shift focus or time.

Contrast – change in tone or weather.

Echo Ending – return to image or phrase from opening.

Sentence Rhythm Example:

Long sentence = build image.

Short = jolt reader.

Fragment = emphasis.

Narrative Writing – 5-Stage Arc

Exposition – introduce setting/character quickly.

Rising tension – conflict or discovery.

Climax – turning point.

Falling action – reaction or consequence.

Resolution – closure or reflection.

Keep 1 character, 1 event, 1 theme. Depth > complexity.

Directed Writing (Report/Letter/Article)

Paragraph	Function	Example opener
1	Purpose + hook	“Recent reports suggest...”
2	Main point	“One major concern is...”
3	Supporting evidence	“For example...”
4	Balanced counterpoint	“On the other hand...”
5	Conclusion / appeal	“It is clear that...”

Register Reminder:

Formal = no contractions, precise vocabulary, calm tone.

Informal = contractions allowed, conversational flow.

Advanced Writing Techniques (for Top Grades)

Skill	How to Use	Example
Syntactic variety	Vary sentence length & openings	"Under the old streetlight, I waited. My hands trembled."
Parallel structure	Emphasise rhythm	"We came, we saw, we built."
Rhetorical devices	Triads, contrast, direct address	"You've seen the adverts. You've felt the pressure. You can make the change."
Transitions	Logical flow	"Initially / subsequently / conversely / therefore."
Lexical precision	Replace vague words	big → vast / huge → immense

Editing & Reflection Process**The 3-Pass Edit**

Content pass – sense, structure, audience.

Style pass – sentence rhythm, vocabulary.

Accuracy pass – punctuation, spelling, agreement.

Personal Error Log

Keep a small table after each task:

Error	Correction	Rule
their / there	there (place)	homophones
comma splice	fix with semicolon	punctuation

Language Devices Bank (for Writer's Effect Qs)

Device	Function
Simile / Metaphor	Comparison, imagery
Personification	Adds emotion to description
Alliteration / Assonance	Rhythm, emphasis
Repetition	Reinforces theme
Juxtaposition	Creates contrast
Sentence variety	Controls pace and tone
Dialogue	Adds immediacy
Tone shifts	Reveal attitude changes

Use one quotation → identify → explain → link effect to reader.

Common Mistakes & Fixes

Problem	Example	Correction
Retelling in summary	"The writer says that..."	Paraphrase in your own words
Over-describing in narrative	Too many adjectives	Focus on verbs & imagery
No paragraph control	One long block	New idea = new paragraph
Flat openings	"I woke up and..."	Start mid-action or with sensory detail
Misusing register	"Hey there council members!"	Match tone to task

12-Week Revision Plan

Week	Focus	Tasks
1	Reading Paper overview	Read 2 short non-fiction texts; identify bias & tone
2	Summary writing	Practice 150-word summary (factual)
3	Writer's effect	Analyse 3 metaphors + 2 tone shifts
4	Directed writing	Write a report (250 words)
5	Argumentative writing	Speech to school council
6	Descriptive composition	"The city at night"
7	Narrative composition	"The mistake"
8	Editing & grammar	Build personal error log
9	Timed Paper 1 extract practice	2h mock
10	Timed Paper 2 composition	2h mock
11	Review AO targets	Identify weakest area
12	Final full mock & reflection	Whole-paper simulation

Self-Assessment Rubric

Level	Reading (P1)	Writing (P2)
Emerging	Basic comprehension	Limited control of tone and grammar
Developing	Some analysis, uneven structure	Clear ideas, simple vocabulary
Proficient	Balanced interpretation	Varied sentences, good accuracy
Advanced	Analytical precision, secure synthesis	Controlled tone, strong cohesion
Expert	Evaluative insight and concise summaries	Sophisticated, fluent, original voice

Recommended Resources

Type	Resource	Link
Official Syllabus	Cambridge IGCSE English First Language (0500 / 0990)	Cambridge International
Reading	The Guardian, The Atlantic, BBC Future	www.theguardian.com / www.theatlantic.com
Exam Practice	Save My Exams / GCE Guide / XtremePapers	www.savemyexams.com
Writing Support	BBC Bitesize, Grammarly, Hemingway App	www.bbc.co.uk/bitesize
Listening & Speaking	TED Talks / BBC Sounds / NPR	www.ted.com / www.bbc.co.uk/sounds

Final Reflection

“Precision is not just about grammar — it’s about truth.”

The English First Language Study Guide builds mastery through **clarity, evidence, and voice**.

Every reading builds judgment; every sentence builds control.

Your goal is not only to pass an exam but to *command language with purpose* — the single most powerful lifelong skill.

English as a Second Language (0510/0511/0991)
Basic Supercurriculum (Cambridge Board)

Why Supercurriculum Matters in English as a Second Language

Language is not just a tool for passing exams — it's the gateway to understanding, collaboration, and confidence in every subject.

The Supercurriculum helps you move from classroom English to *global English*: reading authentic texts, writing with purpose, listening actively, and speaking clearly in real-life situations.

Reading — Understanding and Response

Goal: read accurately, find key information, and understand the writer's message and tone.

Core Tasks

Read one **short article or report** each week (BBC Learning English, National Geographic Kids, News in Levels).

→ Highlight *main idea*, *supporting details*, and *writer's opinion*.

Complete a **“headline challenge”**: summarise an article in **one sentence** (max 15 words).

Learn **10 new words per text**. Write a sentence for each.

Mini Skill Focus

Skimming for main idea.

Scanning for facts/numbers/names.

Inference (reading between the lines).

Writing — Accuracy and Organisation

Goal: write clear, correct, and well-organised texts for different audiences.

Core Tasks

Email / Letter (formal + informal)

Start and end appropriately (Dear Sir/Madam – Yours faithfully / Dear Maria – Best wishes).

Structure: greeting → purpose → details → closing.

Article / Blog

4 paragraphs: introduction, two points with examples, conclusion.

Add one rhetorical question or short personal example.

Report

Use subheadings, bullet points, and a neutral tone.

Focus Skills

Skill	Example
Linking words	however, moreover, on the other hand, therefore
Paragraph openers	Firstly, In addition, As a result, Finally
Accuracy	Correct tenses, punctuation, subject–verb agreement

Mini Drill: write 80–100 words every 2 days — one clear paragraph with accurate grammar.

Listening — Comprehension and Note-Taking

Goal: understand spoken English in different accents and speeds.

Core Tasks

Listen to one **short news report or dialogue** per week (BBC Learning English / ESLPod).

→ Write 5 bullet points of what you understood.

Repeat key phrases aloud to Practice pronunciation and rhythm.

Identify **signal words** (e.g., *firstly, however, in conclusion*).

Extension

Dictation challenge: listen once and write a short summary (60–80 words) from memory.

Speaking — Fluency and Confidence

Goal: speak naturally and accurately on everyday and academic topics.

Core Tasks

2-minute talk each week on a familiar topic: *My favourite hobby, How to stay healthy, Technology and life.*

Pair work: short debates (agree/disagree tasks).

Practice **clarity**: stress key words, avoid fillers (um, you know).

Prepare **5 conversation questions** before each topic.

Mini Drill: record your answer to one question each week; listen and note one pronunciation to improve.

Watching — Real English in Use

Goal: learn vocabulary, tone, and cultural context from real situations.

Activities

Watch **news clips, short documentaries, or film scenes** (with English subtitles).

Write a short reaction:

What was it about?

What did you learn?

One new phrase you liked.

Researching — Independent Vocabulary Growth

Goal: build word banks through curiosity.

Core Tasks

Choose one theme per week: e.g. *Food, Travel, Education, Environment.*

→ Find **10 new words or idioms** and make a mini dictionary (word, part of speech, example).

Use **Quizlet** or **Memrise** for digital flashcards.

Test yourself: 5 random words every Friday.

Student-Led Tasks — Collaboration and Sharing

Goal: learn from peers, Practice communication.

Word of the Week: one student presents a new expression and example.

Mini Debate: 3 minutes on a topic (“Is social media helpful or harmful?”).

Peer Correction: swap short paragraphs and underline grammar/vocabulary to improve.

Creative Tasks — Expression and Confidence

Goal: use English creatively and personally.

Write a **short story (150 words)** — 3 paragraphs: beginning, problem, ending.

Write a **dialogue** (8–10 lines) between two people solving a problem.

Record a **short video** presenting a topic of your choice.

Keep a **personal diary** in English — 3–4 sentences per day.

Reflection — Learning to Self-Correct

Goal: take control of your learning.

Keep a **language log**:

Date	Mistake	Correction	Rule
1 Oct	“He go”	“He goes”	3rd person -s

End of week: write one sentence about your **biggest improvement** and one **goal** for next week.

Interdisciplinary Connections

Subject	Application
Geography / Science	Summarise information clearly in reports
ICT / Business	Write emails, instructions, short presentations
Global Perspectives	Express opinion politely and give balanced arguments
Literature / Media	Understand tone, theme, and audience

Digital & Career Literacy

Goal: use English in real-world contexts.

Write a **simple CV paragraph** (50 words) introducing yourself.

Practice **formal emails**: enquiries, applications, complaints.

Create a **digital portfolio** (Google Docs / Slides): best paragraph, report, and talk script.

Use grammar tools (Grammarly / Reverso) for checking but **edit manually first**.

Core Topic Map (aligned to syllabus)

Skill Area	Main Objectives
Reading	Identify main ideas, inference, writer’s attitude
Writing	Organise, use register, control grammar

Listening	Extract gist and detail, predict content
Speaking	Express opinions, compare ideas, use intonation
Use of English	Vocabulary, verb forms, linking words, prepositions

Weekly Routine (30-40 mins total)

Read 1 article → summarise in 3 sentences.

Write 1 paragraph (80-120 words) → check grammar.

Watch/listen → list 5 new words.

Speak 2 minutes → record yourself once.

Reflect → 1 sentence goal for next week.

Final Reflection

“To learn a language is to gain another way of seeing the world.”

Through steady reading, writing, listening, and speaking, your English becomes not only a subject but a skill for life.

Every paragraph, every conversation, every reflection builds fluency, confidence, and connection.

Chemistry (o62o/o971)

Basic Supercurriculum

Why the Supercurriculum Matters in Chemistry

Chemistry explains how the world works — from the air we breathe to the batteries in our phones.

The Supercurriculum develops scientific curiosity and methodical thinking. It helps you connect classroom facts with real-life materials, reactions, and discoveries.

Reading – Building Scientific Curiosity

Goal: Understand chemistry's real-world impact.

Read short science features weekly (BBC Bitesize, Chem4Kids, Science News for Students).

Try:

“What is an atom?” (BBC Bitesize)

“How plastic is made and recycled.”

“Why salt dissolves in water.”

Activity:

After each article:

Write 3 new terms and define them.

Write 1 question you still have.

Summarise the main idea in 2 sentences.

Writing – Recording Experiments Clearly

Goal: Communicate results and ideas with accuracy.

When doing or watching experiments, Practice writing **lab reports** using this structure:

Section	What to Include
Aim	What are you trying to find out?
Hypothesis	What do you predict will happen?
Method	Steps written clearly and safely.
Results	Table with headings and units.
Conclusion	Compare results with your prediction.

Mini Task:

Write a 100-word “experiment diary” after a class demo. Focus on **observation words** (colour change, fizzing, smell, temperature).

Watching – Chemistry in Action

Goal: Visual learning and observation.

Recommended Channels / Clips

Fuse School Chemistry (YouTube – animated explainers).

Periodic Videos (University of Nottingham).

BBC Bitesize: GCSE Chemistry experiments.

Activity:

After each video, complete:

3 facts you learned.

1 process or reaction shown.

1 “what if” question to explore further.

Experimenting – Safe Curiosity

If you have access to school lab time or home-friendly setups:

Simple safe experiments (with supervision):

Acid + metal carbonate → gas test (limewater → CO₂).

Evaporation of salt solution → crystals.

Chromatography with felt pens → separation by solubility.

Record:

What did you observe?

What does it show about reactions or mixtures?

Listening – Science Communication

Goal: Understand scientists’ voices and ideas.

Activities

Listen to short science podcasts (e.g. *BBC Science Focus*, *Chemistry for Your Life*).

Note down:

The topic discussed

3 key facts

1 new word (and definition)

Extension:

Practice explaining the same idea in simple English to someone younger — if they understand it, *you* understand it.

Researching – Extending Curiosity

Goal: Connect Chemistry with other sciences and everyday life.

Mini projects (choose one):

“The chemistry of baking bread.”

“How soap cleans.”

“Why iron rusts and how to stop it.”

“What’s inside a rechargeable battery?”

“The chemistry of the human body (water, salts, oxygen).”

Steps:

Find one reliable article or short video.

Take notes (key elements, reactions, everyday examples).

Present findings in 5–6 sentences or a small poster.

Student-Led Tasks – Collaboration

Goal: Learn together, teach others.

Ideas:

Create a **Periodic Table quiz** for your classmates.

Lead a “3-minute explainer” on one element (name, symbol, use).

Make a short slideshow titled “*Where we use chemistry every day.*”

Creative Tasks – Linking Science and Art

Goal: See beauty and design in chemistry.

Projects:

Design a **poster** of your favourite element — include symbol, atomic number, uses, and colour scheme that fits its personality.

Draw a **reaction comic strip** (e.g., acid meeting base = neutralisation!).

Build a **model molecule** (plasticine, LEGO, or 3D app like MolView).

Reflection – Becoming a Scientific Thinker

Goal: Notice your learning progress.

After each topic:

Write one thing you now understand clearly.

Write one thing you want to research further.

Write one “connection” (e.g., “This links to biology because...”)

Interdisciplinary Connections

Subject	How It Connects
Physics	Atoms, energy, and states of matter
Biology	Enzymes, respiration, and photosynthesis reactions
Geography	Rock formation, atmosphere, pollution
Maths	Balancing equations, ratios, concentration, data graphs
Design & Technology	Materials science, corrosion, fuels

Digital & Career Literacy

Goal: Understand real-world chemistry roles.

Tasks:

Research 3 jobs that use chemistry (e.g., pharmacist, chemical engineer, forensic scientist).

Write 50 words about what each one does and what skills it needs.

Explore online simulators (PhET Reactions & Balancing, ChemCollective Labs).

Core Topic Map (Cambridge alignment)

Unit	Theme	Example Concept
1	States of Matter	Particle theory, changes of state
2	Atoms, Elements & Compounds	Atomic structure, bonding
3	Stoichiometry	Formulae, equations

4	Electrolysis	Movement of ions
5	Acids, Bases & Salts	Indicators, neutralisation
6	Chemical Energetics	Exothermic/endothermic
7	Rates of Reaction	Collision theory
8	Periodic Table	Groups & trends
9	Metals	Reactivity series, extraction
10	Organic Chemistry	Alkanes, alkenes, alcohols

Weekly Routine (30–40 minutes total)

Day	Activity	Outcome
Monday	Read 1 short science article	New vocab + 2-sentence summary
Tuesday	Watch 1 chemistry video	Write 3 facts
Wednesday	Mini experiment / observation	1 photo or sketch
Thursday	Quiz or flashcards	10 terms revised
Friday	Reflection log	3 sentences on what you learned

Final Thought

“Chemistry is the study of change — in matter and in ourselves.”

Every observation, question, and experiment sharpens your understanding of the invisible patterns that build our world.

Start simple. Stay curious. Think like a scientist.

Chemistry (0620 / 0971)

Extended Supercurriculum

Why the Extended Supercurriculum Matters

Chemistry at this level goes beyond memorising formulae.

It is about understanding *why* reactions occur, *how* structures determine properties, and *where* chemistry shapes our future — from renewable energy to nanotechnology.

The extended programme builds analytical thinking, data literacy, and creativity in scientific reasoning.

Reading – Deepening Conceptual Understanding

Objective: Read scientifically, not just factually — identifying logic, argument, and application.

Advanced Reading Tasks

Weekly article from *Chemistry World*, *Scientific American*, or *Royal Society of Chemistry Education*.

→ Write a 150-word “summary + significance” paragraph: what was learned and why it matters.

Read one **case study** per month (e.g. Haber Process, Polymer Recycling, Green Chemistry).

→ Identify: input materials, reaction type, social/environmental effect.

Critical Reading Log

Article	Concept	Chemical Principle	Real-World Application
“Ammonia and the planet”	Industrial catalysis	Equilibrium, pressure	Fertilisers, food supply

Extension: Read short extracts from university resources like *Chemguide* or *Khan Academy* and explain each equation in your own words.

Writing – Scientific Communication & Precision

Objective: Write reports that show clear reasoning, variable control, and accurate language.

Extended Tasks

Experimental Design (200 words)

Describe aim, apparatus, variable control, risk assessment, expected results.

Example: *Investigate how concentration affects rate of reaction between acid and metal.*

Data Commentary (150 words)

Given a results table, describe trends, anomalies, and suggest causes.

Extended Explanation (100–150 words)

Explain why ionic compounds conduct electricity when molten but not solid.

Style Targets

Use *third-person passive voice*: “The solution was heated until...”

Quantify observations (“blue solution faded to pale green after 20 seconds”).

Always include **units, conditions, variables, and uncertainty**.

Experimenting – Advanced Laboratory Thinking

Objective: Plan, perform, and evaluate experiments like a chemist.

Design Challenges

Titration simulation: standardise an acid/base using known concentration.

Calculate molarity.

Electrolysis of copper(II) sulfate: measure mass change of electrodes, discuss purity.

Rate investigation: effect of temperature on sodium thiosulfate reaction.

Evaluation Framework

Step	Question
Accuracy	Were readings taken precisely?
Reliability	Were repeats consistent?
Validity	Did it test what it claimed to?
Improvement	How could error be reduced?

Safety Focus: personal protective equipment, safe disposal of chemicals, awareness of exothermic reactions.

Quantitative Chemistry – Thinking in Numbers

Objective: Move from simple mole counting to logical, ratio-based problem solving.

Core Concepts

Mole triangle: $n = m/M_r$ $m = nM_r$

Gas volume relation: $V = n \times 24 \text{ dm}^3$ $n = V/24 \text{ dm}^3$ (at r.t.p.)

Concentration: $C = n/V$ $n = CV$ (mol/dm³)

Advanced Tasks

Calculate mass of CO₂ from 10 g of CaCO₃ decomposed.

Determine empirical formula from percent composition.

Create your own question involving a **limiting reagent** and solve it.

Extension: Research how stoichiometry applies to industrial chemistry — e.g. fuel combustion efficiency.

Interpreting Data – Graphs, Rates, and Patterns

Objective: Analyse experimental trends and reason about reaction dynamics.

Practical Analysis Exercises

Draw rate graphs (volume vs. time). Identify where rate is fastest.

Compare **exothermic** vs **endothermic** profiles.

Analyse **activation energy** diagrams.

Explain “collision theory” using real data.

Extension:

Use simulation tools (PhET *Reactions & Rates* or *Energy Changes*) to visualise collisions at molecular level and summarise what changes temperature or concentration do.

Watching – Applied Chemistry and Innovation

Objective: Observe chemistry’s role in solving world problems.

Recommended media

Periodic Videos – advanced episodes (catalysis, nanomaterials).

BBC Horizons – “Plastic Planet” or “Climate Chemistry.”

TED-Ed: “How batteries work,” “The chemistry of soap.”

Viewing Tasks

Identify main chemical principle.

Note one industrial process mentioned.

Summarise environmental impact (positive/negative).

Researching – Inquiry and Interdisciplinary Links

Objective: Connect chemistry to wider scientific, economic, and ethical questions.

Suggested Projects

Green Chemistry: design safer, less wasteful reactions.

Renewable Energy Materials: investigate lithium-ion batteries or hydrogen cells.

Medicinal Chemistry: how molecular design affects drug action.

Atmospheric Chemistry: ozone, greenhouse gases, acid rain.

Mini Research Report Structure (300–400 words)

Problem background.

Scientific explanation (key reactions or data).

Implications (economic, social, environmental).

Evaluation and personal reflection.

Student-Led Tasks – Collaboration and Leadership

Objective: Become confident in explaining and defending chemical reasoning.

Team Roles

Researcher – finds sources and data.

Presenter – explains experiment or concept to class.

Technician – builds models or visuals.

Analyst – checks data accuracy and conclusions.

Group Challenges

Debate: *Should plastic be banned or reinvented?*

Create a model Periodic Table with images of real-life uses for each element.

Host a “mini science fair” showing one experiment’s principle.

Creative Tasks – Science Beyond the Lab

Objective: Use imagination to express chemistry through art, design, or storytelling.

Ideas

Write a short story titled *“The Life of a Molecule.”*

Design an infographic: *“10 reactions that changed the world.”*

Compose a poster comparing renewable vs fossil fuels.

Create a digital periodic table with QR codes linking to real examples.

Reflection – Thinking Like a Chemist

Objective: Build metacognitive awareness — not just what you know, but how you reason.

Weekly Prompts

“Which experiment changed how I think about matter?”

“What mistake taught me the most?”

“Where did I apply a concept from another subject?”

“What chemical process do I see in daily life?”

Keep a **Science Journal**: 1 paragraph per week connecting learning with observation.

Interdisciplinary & Career Connections

Field	Chemical Focus	Example Applications
Physics	Energy changes, quantum structure	Spectroscopy, semiconductor design
Biology	Biochemical reactions, enzymes	Drug design, nutrition
Engineering	Materials, fuels, corrosion	Alloys, batteries, polymers
Environmental Science	Pollution, recycling	Water treatment, green fuels
Economics / Policy	Production cost vs. sustainability	Carbon trading, waste management

Career Investigation Task:

Research 3 chemistry-based careers. For each, write:

What knowledge is required.

What global issue it helps to solve.

Which personal skills are most important.

Digital & Data Literacy

Objective: Use digital tools to simulate and communicate chemistry.

Explore **PhET**, **ChemCollective**, and **MolView** to visualise molecules.

Create digital lab logs (Google Sheets for data tables, Canva for lab posters).

Use **Excel** or **Desmos** to plot reaction rate graphs and interpret gradients.

Record a 1-minute explainer video on a chosen topic (e.g. “Why metals conduct electricity”).

Capstone Project – “Chemistry for a Sustainable Future”

Duration: 4–6 weeks

Options:

Research Presentation: *How can chemistry help reduce plastic waste?*

Experimental Report: *Comparing efficiency of natural catalysts (yeast, enzymes, etc.).*

Innovation Proposal: *Design a new product using recycled materials.*

Deliverables:

Written summary (500–700 words).

3-slide visual presentation.

150-word personal reflection (how chemistry links to real life).

Assessment Focus:

Scientific accuracy

Application of principles

Creativity & relevance

Communication clarity

12-Week Extended Planner

Week	Focus	Output
1	Atomic structure & isotopes	Model or infographic
2	Bonding types	Table: ionic vs covalent vs metallic
3	Formulae & equations	Balanced examples + worksheet
4	Moles & stoichiometry	3 solved problems + explanations
5	Energy changes	Reaction profile sketch
6	Rates & catalysis	Data table + graph
7	Equilibrium	Real-life application summary
8	Acids & bases	Neutralisation experiment reflection
9	Electrolysis	Diagram + half-equations
10	Metals & reactivity	Series + extraction notes
11	Organic chemistry	Hydrocarbon chart
12	Environmental chemistry	Presentation: “Chemistry & the Earth”

Suggested Resources

Type	Resource	Link
Revision	BBC Bitesize GCSE Chemistry	https://www.bbc.co.uk/bitesize/subjects/zs6h34j
Enrichment	Royal Society of Chemistry – Learn Chemistry	https://edu.rsc.org
Video	Periodic Videos /	YouTube

	FuseSchool / TED-Ed Chemistry	
Simulation	PhET Reactions & Rates / ChemCollective Labs	https://phet.colorado.edu
Reading	Chemguide (A-Level prep)	https://www.chemguide.co.uk
Interactive	MolView 3D Molecule Builder	https://molview.org

Final Reflection

“To study chemistry is to study connection — between atoms, between ideas, between people.”

The Extended Supercurriculum transforms curiosity into scientific literacy. You'll learn to observe, measure, reason, and imagine — exactly the mindset of a modern chemist, environmental innovator, or researcher.

Cambridge IGCSE Chemistry (0620 / 0971) – Study Guide
(Updated for the current syllabus)

Course Overview

Aims:

To develop scientific knowledge, experimental skills, and analytical reasoning across the three main domains:

Physical Chemistry (structure, bonding, energetics, rates)

Inorganic Chemistry (periodic trends, metals, reactions)

Organic Chemistry (hydrocarbons, alcohols, macromolecules)

Paper	Title	Duration	Weight	Assessment Focus
Paper 2	Multiple Choice (Core/Extended)	45 min	30%	Factual recall, calculation, reasoning
Paper 4	Theory (Extended)	1 h 15 min	50%	Structured, short + long answers
Paper 6	Alternative to Practical	1 h	20%	Experimental design, analysis, accuracy

Assessment Objectives

AO	Description	What It Tests
AO1	Knowledge & understanding	Recall, define, state, describe
AO2	Handling information & problem solving	Apply ideas, interpret data, use equations
AO3	Experimental skills & investigations	Plan, analyse, evaluate experiments

Paper 2 – Multiple Choice (45 min)

Structure

40 questions (A–D)

Mix of knowledge, data, and calculation

Negative marking does *not* apply → always attempt every question

Strategy

Read question first, then look at data/diagram.

Eliminate two wrong answers quickly.

Use **units** and **prefixes** to reason logically.

Watch for “NOT” or “INCORRECT” wording.

Estimate before you calculate.

Example:

Which of the following describes an endothermic reaction?

- A) Temperature rises
- B) Heat released
- C) Bonds broken absorb energy ✓
- D) Reaction feels warm

Tip: underline the keyword (e.g., *endothermic* = absorbs heat).

Paper 4 – Theory (1 h 15 min)

Question Types

Short answers (define, describe, state)

Calculations (moles, energy, titration)

Structured reasoning (steps, patterns, explanation)

Extended writing (why/how questions)

Marking Focus

Skill	Weight	What Examiners Look For
Knowledge	40%	Correct facts, definitions
Application	40%	Logical reasoning, patterns, reactions
Communication	20%	Clear sequence, correct units, labelled diagrams

Essential Techniques

Use labelled diagrams:

Particle model for states of matter.

Bonding models: ionic lattice, covalent molecule.

Apparatus for experiments (titration, gas collection).

Command Words

Word	What to Do
Define	Give a short, precise statement.
Describe	Say what happens.
Explain	Say <i>why</i> it happens.
Deduce	Use data to reach a conclusion.
Suggest	Logical prediction using knowledge.
Calculate	Show full working and correct units.

Calculation Method (always show steps):

Write known data.

Select formula.

Substitute values.

Show unit conversions.

Round to correct significant figures.

Paper 6 – Alternative to Practical (1 h)

Skills Tested

Recognising apparatus.

Interpreting tables and graphs.

Drawing conclusions from data.

Identifying sources of error.

Describing safety and improvements.

Practical-style Questions

Titration Data: calculate mean, concentration, identify anomalies.

Qualitative Analysis: test for ions, gases, and conclusions.

Rate Experiments: describe trend, sketch graph, suggest improvements.

Separation: chromatography / distillation diagrams.

Common Ions & Tests

Ion	Test	Result
Cu^{2+}	NaOH(aq)	Blue precipitate
Fe^{2+}	NaOH(aq)	Green precipitate
Fe^{3+}	NaOH(aq)	Brown precipitate
Cl^-	AgNO_3	White ppt
CO_3^{2-}	Acid $\rightarrow$ limewater	Effervescence, turns milky
SO_4^{2-}	BaCl_2	White ppt (insoluble in acid)

Flame Tests

Ion	Colour
Li^+	Red
Na^+	Yellow
K^+	Lilac
Cu^{2+}	Blue-green
Ca^{2+}	Orange-red

Key Equations & Formulae

Concept	Equation	Units
Moles	$n = \frac{m}{M_r}$ $n = \frac{m}{M_r}$	mol, g, g/mol
Gas volume	$V = n \times 24$ $V = n \times 24$ (r.t.p.)	dm^3
Concentration	$C = \frac{n}{V}$ $C = \frac{n}{V}$	mol/dm^3
Energy change	$Q = mc\Delta T$ $Q = mc\Delta T$	J
% Yield	$\frac{\text{actual}}{\text{theoretical}} \times 100$ $\frac{\text{actual}}{\text{theoretical}} \times 100$	%
Atom economy	$\frac{M_r(\text{useful})}{M_r(\text{total})} \times 100$ $\frac{M_r(\text{useful})}{M_r(\text{total})} \times 100$	%
Rate of reaction	$\frac{\text{change}}{\text{time}}$ $\frac{\text{change}}{\text{time}}$	g/s or cm^3/s

Key Topics Breakdown

Unit	Extended Understanding
States of Matter	Kinetic theory, diffusion, evaporation vs boiling
Atomic Structure	Isotopes, electron configuration, ions
Bonding	Ionic, covalent, metallic; link to conductivity
Stoichiometry	Empirical formula, moles, limiting reagent
Electrolysis	Half-equations, electrode products
Energetics	Exo/endo reactions, bond energy calculations
Rates of Reaction	Collision theory, catalysts, graphs
Equilibria	Reversible reactions, effect of conditions
Acids & Bases	Neutralisation, titration, indicators
Periodic Trends	Group I, VII, VIII behaviour, reactivity
Metals	Extraction, reactivity series, rust prevention
Organic Chemistry	Alkanes, alkenes, alcohols, polymers

Environmental Chemistry	Pollution, greenhouse gases, water treatment
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Data Interpretation Skills

When given tables/graphs:

Identify **trend** (increase/decrease/constant).

Quote **numerical evidence** ("as temp increases from 20°C to 50°C, rate doubles").

Explain **why** (collision frequency / energy / particle movement).

Label axes with units, draw smooth curve, not sketchy lines.

Evaluation Practice:

"The rate graph flattens because reactants are used up."

"The yield decreases with temperature since equilibrium shifts backward (endothermic)."

Experiment Design Framework

Step	Include
Aim	Clear statement (e.g. To investigate...)
Hypothesis	Prediction with reason
Variables	Independent, dependent, controlled
Method	Sequential, safe, measurable
Results	Table with headings and units
Analysis	Compare with theory
Evaluation	Errors, improvements, reliability

Error Sources

Misreading meniscus, inconsistent heating, gas loss, impure reagents.

Improvements

Repeat, average, precise instruments, controlled variables.

12-Week Study Plan

Week	Topic	Focus Task
1	States of Matter	Kinetic theory diagram + short quiz
2	Atomic Structure	Draw 3 atom models (Na, Mg, Cl)
3	Bonding	Venn diagram: ionic vs covalent vs metallic
4	Stoichiometry	3 mole calculations + check with teacher
5	Energetics	Draw energy profiles, label exo/endo
6	Electrolysis	Predict electrode products + write equations
7	Rates of Reaction	Graph + analysis paragraph
8	Equilibrium	Explain effect of pressure on Haber process
9	Acids, Bases, Salts	Neutralisation summary + titration table
10	Metals	Reactivity series summary chart
11	Organic Chemistry	Flow chart of homologous series
12	Revision & Mock	Full Paper 2 + reflection log

Self-Assessment Rubric

Level	Knowledge	Application	Communication
Emerging	Basic recall	Limited use of formulae	Simple sentences
Developing	Some connections	Partial reasoning	Clear layout
Proficient	Accurate & detailed	Logical explanation	Correct units
Advanced	Evaluative & analytical	Quantitative mastery	Clear, concise structure
Expert	Integrates ideas across topics	Creative application	Precision in data + argument

Exam Day Checklist

- ✓ Calculator (check batteries)
- ✓ Ruler, pencil, sharpener
- ✓ Pen (blue/black only)
- ✓ Periodic Table (school issue)
- ✓ No correction fluid
- ✓ Show working clearly — even if wrong number
- ✓ Use correct units (marks often awarded for these)

Top Revision Tips

Concept maps — link topics (bonding ↔ conductivity ↔ structure).

Equation flashcards — one side formula, other side sample question.

Practice every calculation twice — no guessing.

Explain aloud — if you can teach it, you know it.

Balance revision — theory + practice papers weekly.

Recommended Resources

Resource	Type	Link
BBC Bitesize Chemistry	Revision	https://www.bbc.co.uk/bitesize/subjects/zs6h34j
Chemguide	Advanced explanations	https://www.chemguide.co.uk
Royal Society of Chemistry	Enrichment	https://edu.rsc.org
SaveMyExams	Practice papers	https://www.savemyexams.com
PhET Interactive Simulations	Virtual labs	https://phet.colorado.edu
Chemistry Olympiad (Intro)	Challenge prep	https://edu.rsc.org/chemistry-olympiad

Final Reflection

“Chemistry is problem-solving in the language of atoms.”

Mastering IGCSE Chemistry means learning to balance equations, not just in your notes — but between logic and curiosity.

By observing, quantifying, and explaining clearly, you learn the habits of real scientists: precision, curiosity, and evidence-based reasoning.

Year 10 Cambridge IGCSE Physics (0625 / 0972) Basic Supercurriculum

Why the Supercurriculum Matters in Physics

Physics explains how everything moves, shines, and interacts — from the way your phone battery works to why the sky changes colour.

This Supercurriculum builds curiosity and critical thinking, showing how classroom physics connects to real life, technology, and the universe.

Reading — Curiosity About the Physical World

Goal: Develop understanding through observation and imagination.

Weekly Habit: Read 1 short article or watch 1 short video about physics in action.

(Examples: *BBC Bitesize Physics*, *Physics World*, *HowStuffWorks*, *NASA Kids*.)

After each article/video:

Write 3 new facts.

Define 2 new scientific words.

Ask 1 “why” question to research later.

Suggested Topics:

Why does the moon look different every night?

How do magnets work?

What makes sound travel through walls?

What is electricity made of?

Writing — Explaining Clearly Like a Scientist

Goal: Learn to describe and record ideas accurately.

Mini Tasks:

Experiment diary: 100–120 words after each lab activity — what you did, what you saw, and what you learned.

Simple report template:

Section	Example
Aim	To investigate how the length of a pendulum affects time period
Apparatus	String, stopwatch, ruler, mass
Method	Measure time for 10 swings, repeat for different lengths
Result	Table + graph
Conclusion	Longer pendulum = longer period ($T^2 \propto L$)

Writing tip: Use short, factual sentences — no storytelling, no “I think”.
Just *evidence + logic*.

Watching — Seeing Physics in Motion

Goal: Visualise the laws that govern everyday life.

Suggested Channels / Clips:

BBC Teach Physics

Veritasium – real-world explanations

MinutePhysics – quick animations

Physics Girl – demonstrations

Activity:

After each video, make a 4-line note:

What experiment or idea was shown

The key principle

One real-life example

A question you'd ask the scientist

Experimenting — Learn by Doing

Goal: Use hands-on curiosity safely.

Simple at-home or school-friendly investigations:

(Always with adult supervision for heat/electricity!)

Measure **shadow lengths** throughout the day → how does Sun's position affect them?

Drop objects of different weights → observe acceleration (Galileo's principle).

Stretch rubber bands → how does extension change with force? (Hooke's Law idea).

Simple circuit → test what happens when you add bulbs in series vs parallel.

Mini Data Table Template:

Variable changed	Measurement	Observation
Force (N)	Extension (cm)	Linear increase up to 8 N

Listening — Understanding Science Communication

Goal: Develop awareness of how scientists describe ideas in words.

Activities:

Listen to short clips (*BBC Science Focus* podcast, *SciShow Kids*).

Note 3 scientific words + their meanings.

Retell what you heard in your own words to a classmate or parent.

Extension:

Compare a scientist's explanation with how a journalist reports the same idea — how does the style change?

Researching — Science Around You

Goal: Explore connections between theory and real-world use.

Mini Research Projects (choose one per term):

"How do satellites stay in orbit?"

"Why does a rainbow form?"

"How does a wind turbine generate electricity?"

"What makes a car crash dangerous?"

Process:

Find 1 short article + 1 diagram or video.

Write 5 key facts and a 2-sentence summary.

Share findings in class or make a one-slide visual summary.

Student-Led Tasks — Teaching Through Collaboration

Goal: Build teamwork and explanation skills.

Ideas:

Create a “Physics Flash Quiz” with 10 questions for your group.

Choose a topic (e.g. *Energy, Forces, Waves*) and explain it in 60 seconds.

Work in pairs to design a simple experiment demonstration and explain the results to others.

Creative Tasks — Art Meets Science

Goal: Use creativity to visualise physical phenomena.

Projects:

Draw a comic or infographic explaining **Newton’s Three Laws**.

Build a **mini solar oven** (foil + sunlight).

Design a poster: “*Physics in Sports / Music / Space.*”

Make a collage or digital artwork of the **Electromagnetic Spectrum**.

Reflection — Thinking Like a Physicist

Goal: Build awareness of how you think and learn.

At the end of each topic, complete:

“I can now explain...” (1 concept).

“I still wonder...” (1 question).

“This connects to...” (another subject or real-life event).

Example:

“I can now explain how forces act on a parachute. I still wonder how drag is measured. This connects to air resistance in cars.”

Interdisciplinary Connections

Subject	Connection
Maths	Graphs, formulas, units, trigonometry
Chemistry	Energy, states of matter, particle theory
Biology	Light in the eye, sound in the ear, muscles and forces
Geography	Energy resources, climate, weather patterns
Computer Science	Data logging, coding for sensors and robots

Digital & Career Literacy

Goal: Discover how physics links to modern technologies.

Mini Tasks:

Research one career related to physics: *engineer, architect, pilot, meteorologist, radiographer*.

→ Write 50 words: what they do and which physics topics they use.

Explore online simulations:

PhET (*Forces & Motion, Waves on a String, Circuits*)

NASA Kids (*Solar System Explorer*)

Create a digital **Science Journal** with pictures, videos, or reflections on experiments.

Core Topic Map (Cambridge Alignment)

Unit	Theme	Core Concepts
1	General Physics	Measurement, motion, forces, energy
2	Thermal Physics	Heat transfer, temperature, expansion
3	Waves	Sound, light, reflection, refraction
4	Electricity	Circuits, current, voltage, resistance
5	Magnetism & Electromagnetism	Fields, induction, motors
6	Atomic Physics	Structure of atom, radioactivity
7	Energy Resources	Renewable vs non-renewable, efficiency

Weekly Routine (30–40 minutes total)

Day	Task	Outcome
Monday	Read short article	Write 3 facts + 1 question
Tuesday	Watch demo video	4-line summary
Wednesday	Simple measurement / mini experiment	Data table
Thursday	Quiz or flashcards	10 key terms
Friday	Reflection log	3 sentences

Quick Glossary (Starter Physics Vocabulary)

Term	Meaning
Force	A push or pull that changes motion
Mass	Amount of matter in an object (kg)
Weight	Force due to gravity ($N = mg$)
Speed	Distance $\div$ time
Energy	Ability to do work
Work	Force $\times$ distance moved
Power	Work $\div$ time
Current	Flow of charge per second (A)
Voltage	Energy per unit charge (V)
Density	Mass $\div$ volume

Final Thought

“Physics is the poetry of reality.” — Richard Dawkins

The Basic Supercurriculum helps you see the physics that surrounds you — in light, sound, movement, and matter.

It’s not about remembering equations, but about *noticing patterns* and *asking better questions*.

Year 10 Cambridge IGCSE Physics (0625 / 0972) Extended Supercurriculum

Why the Extended Supercurriculum Matters

Physics is the language of the universe.

This extended pathway builds mastery of its grammar — equations, logic, and models — to help students think like scientists, engineers, and innovators.

It transforms curiosity into disciplined problem-solving and links every concept to real-world technology and modern research.

Reading – Deep Understanding Through Context

Objective: Move beyond memorisation to conceptual clarity.

Weekly Advanced Reading (choose 1):

Physics World: “Quantum communication” or “Renewable energy technologies.”

NASA Science News: “Why the sky glows after sunset.”

How Stuff Works / Institute of Physics: “How MRI scans work.”

Task:

Write 150 words: **summary + real-world connection.**

Identify 3 scientific terms and write one sentence definition for each.

Ask a “*how*” or “*why*” question to research next week.

Critical Reading Example Table

Article	Concept	Formula	Real-World Link
“Hydrogen Cars”	Energy conversion	$E = P \times t$	Clean energy systems

Writing – Communicating Physics Precisely

Objective: Express scientific logic using structured argumentation.

Tasks:

Explanation essay (150–200 words): “Why does a spacecraft experience weightlessness?”

Experiment summary: Write step-by-step reasoning for results (not just what happened, but *why*).

Data interpretation paragraph: “Explain the shape of this velocity–time graph.”

Writing Formula (C.A.R.):

Claim – main idea.

Argument – physical principle / equation.

Reasoning – link to data or experiment.

Example:

The rate of fall is constant because air resistance equals weight; net force = 0.

Experimenting – Design, Measure, Analyse

Objective: Think and work like an experimental physicist.

Core Extended Experiments

Hooke's Law Investigation — measure extension vs force; plot graph; calculate spring constant k .

Resistance & Length — measure wire resistance using voltmeter and ammeter; plot $R \propto L$.

Specific Heat Capacity — measure temperature rise of metal block with immersion heater.

Refraction Index — draw normal lines, measure angle of incidence/refraction; calculate $n = \frac{\sin i}{\sin r}$.

Momentum Conservation — use trolleys and motion sensors to confirm $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

Scientific Evaluation

Step	Example
Accuracy	Was the apparatus calibrated?
Reliability	Were 3 or more repeats done?
Validity	Did results match theoretical prediction?
Improvements	Use smaller intervals, remove friction, add digital timers.

Quantitative Physics – From Formulas to Meaning

Objective: Understand what equations represent physically.

Concept	Equation	Interpretation
Motion	$v = u + at$	Acceleration changes velocity over time
Energy	$E_k = \frac{1}{2}mv^2$	Kinetic energy depends on speed squared
Power	$P = \frac{E}{t}$	Rate of doing work
Density	$\rho = \frac{m}{V}$	Mass per unit volume
Pressure	$P = \frac{F}{A}$	Force spread over area
Waves	$v = f\lambda$	Speed = frequency $\times$ wavelength
Electricity	$V = IR$	Potential difference drives current
Efficiency	$\frac{\text{useful}}{\text{total}} \times 100\%$	Energy conservation principle

Advanced Practice:

Rearrange each formula to isolate any variable.

Use correct **units and prefixes** (k, m, μ , n).

Always check **significant figures** and **scientific notation**.

Data Analysis – Interpreting Graphs and Trends

Objective: Read, reason, and calculate from data sets.

Graph-based tasks

Plot displacement–time, velocity–time, and force–extension graphs.

Calculate gradient (rate of change) and area (total quantity).

Describe proportionality (direct / inverse / exponential).

Example:

“The velocity–time graph’s gradient gives acceleration (2 m/s^2). The flat section shows constant velocity.”

Mini challenge:

Analyse a set of experiment data → identify anomalies, suggest physical causes, and predict future results.

Watching – Advanced Applications of Physics

Goal: Link classroom theory with global innovation.

Recommended Channels

Veritasium – thought experiments & paradoxes

MinutePhysics – rapid conceptual animations

BBC Earth Lab – real engineering applications

NASA EDGE – space technology and measurement

Follow-up Task:

Write 5 bullet points on the principle demonstrated, and a 1-line summary of how it affects human life or technology.

Researching – Independent Inquiry

Goal: Explore real scientific challenges and modern discoveries.

Research Questions:

How do renewable energy systems store energy efficiently?

What limits the speed of spacecraft leaving Earth?

How do medical X-rays and MRI scans work?

What are superconductors and why are they important?

How does quantum physics explain LED lights?

Mini Report (250–300 words):

Background

Scientific explanation (laws, equations, energy transfer)

Real-world applications

Reflection: “Why this topic matters for the future”

Student-Led Tasks – Peer Science Communication

Goal: Teach to learn.

Ideas:

Host a 2-minute “Physics Pitch” on your favourite law or device.

Debate: “*Is nuclear power essential for sustainability?*”

Design a digital infographic explaining *Newton’s Laws* or *Wave Properties*.

Pair work: one presents theory, the other demonstrates experiment.

Creative Tasks – Physics in Art & Design

Goal: Combine creativity with analytical skill.

Create a **poster** showing energy flow through a system (e.g. solar panel → battery → motor).

Draw the **life cycle of a star** as a timeline.

Design a **public awareness infographic**: “How physics makes phones possible.”

Model a simple **electric motor** or **bridge structure** using craft materials or CAD.

Reflection – The Physicist’s Mindset

Goal: Think critically about your own learning.

Prompts:

“Which concept felt abstract at first but clicked after experiment?”

“Where did I make a calculation mistake, and why?”

“Which natural event do I now see through physics?”

“How does physics help me make sense of everyday design?”

Keep a **Reflection Journal** (1 paragraph weekly).

Interdisciplinary Links

Subject	Link
Maths	Algebra, geometry, trigonometry, proportionality
Chemistry	Energy transfer, atomic structure, particles
Biology	Biomechanics, sound in hearing, light in vision
Geography	Energy resources, weather, environmental science
Computer Science	Simulations, data logging, coding sensors
Design & Technology	Material properties, structural forces

Digital & Career Literacy

Goal: Connect academic physics to modern careers.

Field	Physics Concepts	Career Example
Engineering	Mechanics, energy systems	Civil, Mechanical, Electrical engineer
Medicine	Radiation, imaging, sound waves	Radiologist, Biomedical engineer
Aerospace	Gravity, motion, materials	Pilot, Astronautics designer
IT & Robotics	Circuits, electromagnetism	Robotics engineer
Renewable Energy	Solar, wind, power transfer	Energy analyst

Mini Task:

Pick one career. Write 100 words: what physics it uses and why it matters.

Capstone Project — “Physics for a Better World”

Duration: 4–6 weeks

Goal: Integrate multiple concepts into one real-world theme.

Choose One:**Energy Systems:** Compare efficiency of wind vs solar power using real data.**Motion Analysis:** Study how athletes use physics to improve performance.**Materials & Forces:** Design a model bridge or tower for maximum load capacity.**Electronics:** Build a working mini alarm or sensor circuit.**Deliverables:**

Written report (500–700 words).

Graphs or diagrams.

2-minute presentation explaining your physics reasoning.

Assessment Criteria:

Criterion	Focus
Scientific accuracy	Equations, data, concepts
Application	Real-world link
Evaluation	Limitations and improvements
Communication	Clear visuals and reasoning

12-Week Extended Planner

Week	Focus	Output
1	Motion & Forces	Velocity–time graph + reflection
2	Energy	Efficiency calculation table
3	Density & Pressure	Experimental sketch + data table
4	Waves	Draw reflection & refraction diagrams
5	Sound	Frequency and amplitude mini project
6	Light	Refractive index graph
7	Electricity	Circuit design + current–voltage graph
8	Magnetism	Electromagnet model
9	Thermal Physics	Conduction/convection comparison
10	Energy Resources	Research poster
11	Atomic Physics	Diagram + radiation safety summary
12	Review	Presentation: “Physics in My Daily Life”

Suggested Resources

Type	Resource	Link
Revision	BBC Bitesize IGCSE Physics	https://www.bbc.co.uk/bitesize/subjects/zpm6fg8
Advanced	HyperPhysics (Georgia State Univ.)	http://hyperphysics.phy-astr.gsu.edu
Simulation	PhET Interactive Simulations	https://phet.colorado.edu
Enrichment	Institute of Physics “I ♥ Physics”	https://www.iop.org

Real Science	NASA Learning / CERN OpenLab	https://science.nasa.gov
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Final Reflection

“Physics teaches us not what to think, but how to think.”

The Extended Supercurriculum turns curiosity into mastery.

You learn not only equations, but the ability to reason, design, and explain.

That is what separates a student of physics from a true physicist.

Cambridge IGCSE Physics (0625 / 0972) – Study Guide

Course Overview

Goal:

To understand how and why physical systems behave as they do — from atomic motion to planetary forces — and to communicate this knowledge through data, reasoning, and mathematical precision.

Key Skills Developed

Experimental investigation

Quantitative problem-solving

Application of laws and models

Graphical interpretation

Clear scientific communication

Paper	Title	Duration	Weight	Focus
Paper 2	Multiple Choice	45 min	30%	Core/Extended factual recall & reasoning
Paper 4	Theory	1 h 15 min	50%	Structured calculation & explanation
Paper 6	Alternative to Practical	1 h	20%	Experimental design, data handling, evaluation

Assessment Objectives (AO)

AO	Description	What It Tests
AO1	Knowledge & understanding	Recall of facts, definitions, laws
AO2	Handling information & problem solving	Applying formulas, interpreting data
AO3	Experimental skills & investigations	Designing, analysing, improving experiments

Paper 2 – Multiple Choice (45 min)

Structure

40 questions (A–D format)

No penalty for wrong answers — attempt every question

Skills Tested

Core concept recall (definitions, units, relationships)

Simple calculations (1-step)

Interpreting diagrams and graphs

Tips

Highlight key terms (*increase/decrease, greater/smaller, same*).

Use unit logic: e.g. $\text{N/kg} = \text{m/s}^2$.

Eliminate impossible options first.

Quick mental estimation before calculator use.

For graphs, check **gradient and area** meaning.

Example:

The gradient of a velocity–time graph represents:

- A) Distance ✓(Wrong)
- B) Displacement ✓(Wrong)
- C) Acceleration ✓(Correct)
- D) Speed ✓(Wrong)

Paper 4 – Theory (1 h 15 min)**Question Types**

Short factual answers (1–2 marks)

Multi-step quantitative problems (3–5 marks)

Graph or diagram interpretation (2–4 marks)

Extended reasoning (4–6 marks)

Marking Focus

Skill	Description
Accuracy	Correct values, units, and working
Reasoning	Logical explanation of relationships
Clarity	Structured paragraphs, labelled diagrams
Consistency	Application of the same principle throughout

Command Words

Word	Action
<i>Define</i>	Give concise scientific meaning
<i>Describe</i>	State what happens
<i>Explain</i>	Use scientific reasoning or law
<i>Calculate</i>	Show working, include units
<i>Sketch</i>	Approximate graph with labelled axes
<i>Deduce</i>	Derive from data or reasoning
<i>Suggest</i>	Propose logical hypothesis

Paper 6 – Alternative to Practical (1 h)**Skills Tested**

Planning investigations

Recording, presenting, and analysing data

Drawing and interpreting graphs

Identifying experimental errors

Suggesting improvements

Question Types

Measurement & Data Table – complete or correct readings.

Graph Plotting – choose suitable scales, draw smooth curve.

Experimental Method – describe how to measure accurately.

Evaluation – state error sources and improvements.

Common Practical Topics

Density by displacement or graph method.

Hooke's Law: force vs extension.

Ohm's Law: current vs voltage.

Specific heat capacity from heating curves.

Light refraction and reflection.

Error Types

Error	Example	Solution
Systematic	Zero error on balance	Check before use
Random	Reaction time delay	Take multiple readings
Parallax	Reading at wrong angle	Eye level alignment
Environmental	Drafts, heat loss	Control environment

Key Equations & Units

Concept	Formula	Unit
Speed	$v = \frac{d}{t}$ or $v = \frac{d}{t}$	m/s
Acceleration	$a = \frac{v - u}{t}$ or $a = \frac{v - u}{t}$	m/s ²
Force	$F = ma$ or $F = ma$	N
Weight	$W = mg$ or $W = mg$	N
Moment	$M = F \times d$ or $M = F \times d$	N m
Density	$\rho = \frac{m}{V}$ or $\rho = \frac{m}{V}$	kg/m ³
Pressure	$P = \frac{F}{A}$ or $P = \frac{F}{A}$	Pa
Work done	$W = Fd$ or $W = Fd$	J
Power	$P = \frac{E}{t}$ or $P = \frac{E}{t}$	W
Energy (kinetic)	$E_k = \frac{1}{2}mv^2$ or $E_k = \frac{1}{2}mv^2$	J
Energy (potential)	$E_p = mgh$ or $E_p = mgh$	J
Wave speed	$v = f\lambda$ or $v = f\lambda$	m/s
Refractive index	$n = \frac{\sin i}{\sin r}$ or $n = \frac{\sin i}{\sin r}$	—
Current	$I = \frac{Q}{t}$ or $I = \frac{Q}{t}$	A
Voltage	$V = IR$ or $V = IR$	V
Power (electrical)	$P = IV$ or $P = IV$	W
Efficiency	$\frac{\text{useful}}{\text{total}} \times 100$ or $\frac{\text{useful}}{\text{total}} \times 100$	%

Constants

Symbol	Meaning	Value
g	gravitational field strength	9.8 N/kg
c	speed of light	3.0 × 10 ⁸ m/s
R	gas constant	not required (beyond scope)

Graph Interpretation Skills

Velocity–Time Graph

Gradient = acceleration

Area under curve = distance travelled

Force–Extension Graph

Linear region = obeys Hooke's Law

Area = work done on spring

Current-Voltage Graph

Straight line = ohmic conductor

Nonlinear = filament lamp (resistance $\uparrow$ with temp)

Best Practice

Always label axes with correct **quantity + unit**.

Use **consistent scale** and smooth curves.

Quote gradient and intercept with calculation.

Experimental Design Framework

Section	Example
Aim	Investigate how resistance varies with wire length
Variables	Independent: length; Dependent: resistance; Control: temperature, wire type
Apparatus	Ammeter, voltmeter, wire, power supply, metre rule
Method	Measure V and I, compute $R = V/I$ for each length
Results	Table with mean values
Graph	Plot R (y) vs L (x); expect straight line
Evaluation	$R \propto L$ confirms theory; mention heat as error source

Topic Overview (Extended)

Unit	Concept Focus	Application
1	Motion, Forces, Energy	Newton's Laws, free fall, efficiency
2	Thermal Physics	Heat transfer, expansion, insulation
3	Waves	Sound, light, reflection, refraction
4	Electricity	Series/parallel circuits, resistance, power
5	Magnetism & Electromagnetism	Magnetic fields, motors, induction
6	Atomic Physics	Structure, radiation, isotopes, safety
7	Energy Resources	Fossil fuels, renewables, environmental balance

Calculation Technique (5 Steps)

Write known quantities with symbols.

Select relevant formula.

Substitute values with correct units.

Calculate using proper significant figures.

Include unit and box the answer.

Example:

A 2 kg object falls 3 m.

$$E_p = mgh = 2 \times 9.8 \times 3 = 58.8 \text{ J} \quad E_p = mgh = 2 \times 9.8 \times 3 = 58.8 \text{ J} \rightarrow \boxed{59 \text{ J}}$$

12-Week Study & Revision Plan

Week	Topic	Key Output
1	Measurement & Motion	Draw graphs, solve 5 speed/acceleration problems
2	Forces	Newton's Laws diagram + 3 worked examples
3	Energy & Work	Efficiency table, energy transfer diagram
4	Density & Pressure	Calculate density for 3 solids/liquids
5	Thermal Physics	Diagram: conduction, convection, radiation
6	Waves	Label reflection/refraction diagrams
7	Light & Sound	Calculate refractive index, echo distance
8	Electricity	Circuit diagrams, I-V graph
9	Magnetism	Sketch field lines + electromagnet design
10	Radioactivity	Half-life graph + safety notes
11	Mixed Calculations	20 mixed past-paper questions
12	Mock & Review	Paper 2 + Paper 4 timed; reflection log

Self-Assessment Rubric

Level	Knowledge	Application	Communication
Emerging	Basic recall	Partial reasoning	Simple phrasing
Developing	Some understanding	Occasional accuracy	Clear layout
Proficient	Accurate, detailed	Logical working	Correct units & structure
Advanced	Applies across topics	Evaluates data	Concise, precise explanations
Expert	Links concepts conceptually	Quantitative mastery	Professional clarity

Common Exam Traps

Trap	Correction
Forgetting units	Always include SI unit after final value
Wrong conversion	1 cm = 0.01 m, 1 kWh = 3.6×10^6 J
Copying data incorrectly	Recheck question values
Mixing mass & weight	Mass = kg, Weight = N
Graph errors	Axes labelled, consistent scales
Rounding errors	Use 2-3 significant figures
Poor evaluation	Always mention "repeats", "control", "precision"

Useful Online Resources

Skill	Resource	Link
Revision	BBC Bitesize – IGCSE Physics	https://www.bbc.co.uk/bitesize/subjects/zpm6fg8
Practice	Save My Exams – Physics Papers	https://www.savemyexams.com
Simulation	PhET Interactive Physics	https://phet.colorado.edu

Concept	HyperPhysics	http://hyperphysics.phy-astr.gsu.edu
Videos	Veritasium / Physics Girl	YouTube
Advanced	Institute of Physics (IOP)	https://www.iop.org

Exam-Day Checklist

- ✓Scientific calculator (check batteries)
- ✓Ruler, pencil, sharpener
- ✓Blue or black pen only
- ✓Eraser (no correction fluid)
- ✓Periodic Table (not usually needed but allowed)
- ✓Label all diagrams
- ✓Show all working — even if unsure

Final Reflection

“Physics is not about memorising formulas; it’s about describing the world with reason.”

By combining mathematical accuracy, experimental curiosity, and logical expression,
you transform from a learner into a problem-solver — capable of explaining how nature truly works.

Year 10 Cambridge IGCSE Mathematics (0580 / 0980) Basic Supercurriculum

Why the Supercurriculum Matters in Mathematics

Mathematics is the universal language of logic and precision.

It teaches you not what to think, but *how to think*.

The Supercurriculum develops curiosity, flexibility, and fluency — the ability to use numbers, symbols, and reasoning to describe the world and solve problems confidently.

Reading – Seeing Maths Everywhere

Goal: Understand that mathematics shapes everything we do.

Weekly Habit: Read or watch one real-life example of maths in action.

Recommended Sources:

BBC Bitesize: Maths in Context

The Guardian Science / Maths section

Numberphile (YouTube)

Khan Academy “Everyday Math”

After each article/video:

Write **3 facts** you learned.

Explain **1 connection** to what you study in class.

Note **1 question** you’d like to explore.

Sample Topics:

How do percentages shape discounts and interest rates?

What is symmetry in art and architecture?

How does geometry help design bridges?

Why is probability essential in weather forecasting?

Writing – Explaining Mathematical Thinking

Goal: Learn to *show working and explain reasoning* clearly.

Activity Template:

Step	What to Include
1	Restate the question in your own words.
2	Write the formula used.
3	Substitute numbers carefully.
4	Show all working.
5	Write the final answer with correct unit.

Mini Task:

Choose one past paper question per week and write a full solution **with explanation sentences**, e.g.

“I divided by 100 because percentage means ‘per hundred’.”

Golden Rule:

If you can explain it aloud, you truly understand it.

Watching – Visualising Patterns and Numbers

Goal: Build intuition through visual and interactive learning.

Recommended Channels / Resources

Numberphile — short mathematical stories

Math Antics — simple conceptual explanations

TED-Ed: Mathematics & Logic — creative insights

Activity:

After watching, summarise:

The main idea

The problem solved

One way this connects to your syllabus

Example:

Topic: “Pythagoras in Real Life” → Connection: Distance on coordinate grids.

Practising – Building Daily Fluency

Goal: Train the mind to think quickly and accurately.

Daily 10-Minute Routine:

5 arithmetic questions (fractions, decimals, percentages)

3 algebra tasks (simplify or solve for x)

2 geometry or data questions (area, angles, mean)

Weekly Challenge:

Create your own “maths story problem” based on something in your week (e.g. travel, money, sports).

“A football travels 24 m in 3 s. What’s its speed?”

Tip: Always check answers *make sense* (estimation).

Listening – Language of Mathematics

Goal: Develop confidence in understanding and speaking maths aloud.

Activities:

Record yourself explaining one solved problem.

Listen to “Maths Stories” podcasts or classroom explanations.

Repeat definitions clearly (e.g. “A polygon is a closed 2D shape with straight sides.”).

Extension:

Pair with a classmate and take turns explaining how to solve one problem in plain English.

Researching – Discovering the Power of Maths

Goal: See maths as a creative and global subject.

Mini Projects (choose one):

“How does geometry shape modern architecture?”

“The maths behind music rhythms.”

“How credit cards calculate interest.”

“Why prime numbers matter in computer security.”

Steps:

Find one reliable article or video.

Write a 5-point summary.

Create one small visual (chart, sketch, infographic).

Student-Led Tasks – Learning by Teaching

Goal: Build confidence and leadership.

Ideas:

Run a 5-minute “Maths Mini Lesson” for your group.

Make a simple quiz on geometry, fractions, or graphs.

Pair up: one student explains, the other checks for clarity.

“Teaching is the best way to learn.”

Creative Tasks – Mathematics in Art and Design

Goal: See patterns as beauty.

Projects:

Design a **tessellation** using polygons.

Create a **fractal drawing** inspired by nature.

Build a **scale model** (room, bridge, or park) using ratio and measurement.

Write a short paragraph: “*Where do I see maths every day?*”

Reflection – Thinking Like a Mathematician

Goal: Develop awareness of your thought process.

End-of-week reflection prompts:

“Which concept challenged me, and how did I overcome it?”

“When did I use logical reasoning outside school?”

“How could I check or prove my answers differently?”

Example:

“I realised I could test my algebra answers by substitution — it worked every time.”

Interdisciplinary Connections

Subject	Connection
Physics	Equations, vectors, units, graphs
Chemistry	Ratios, proportions, data analysis
Biology	Graphs, percentages, rates of change
Geography	Coordinates, statistics, scale
Economics	Graphs, interest, averages
Design & Technology	Measurement, proportion, geometry

Digital & Career Literacy

Goal: Connect mathematics to modern life and technology.

Mini Tasks:

Use **Desmos** or **GeoGebra** to plot lines and curves.

Explore **PhET Simulations – Maths tools** (fractions, area, algebra balance).

Research **3 careers** that use maths daily (architect, data analyst, engineer).

→ Write 50 words about each.

Example:

Data analysts use maths to find trends in large datasets — skills in statistics, averages, and probability are essential.

Core Topic Map (Cambridge Alignment)

Unit	Theme	Key Skills
1	Number	Place value, fractions, percentages, ratio
2	Algebra	Simplifying, expanding, solving equations
3	Geometry	Angles, shapes, symmetry, area, volume
4	Coordinate Geometry	Gradient, line equations
5	Trigonometry	SOH-CAH-TOA, Pythagoras
6	Vectors & Transformations	Direction, magnitude, reflection, rotation
7	Statistics	Data handling, mean, median, mode
8	Probability	Simple outcomes, expected results

Weekly Routine (30–40 minutes total)

Day	Activity	Outcome
Monday	Watch one maths video	Write 3 key ideas
Tuesday	Practice 5 problems	Show full working
Wednesday	Research a real-world application	1 paragraph summary
Thursday	Quiz or flashcards	10 terms revised
Friday	Reflection log	3 sentences about progress

Quick Glossary (Core Mathematical Vocabulary)

Term	Meaning
Equation	Statement showing two equal values
Expression	Combination of numbers and symbols
Mean	Average value
Median	Middle value in data set
Mode	Most common value
Range	Largest minus smallest value
Factor	Number that divides exactly
Multiple	Product of a given number
Perimeter	Distance around a shape
Area	Space inside a shape

Final Thought

“Mathematics is not about numbers, equations, or algorithms — it’s about understanding.” — William Paul Thurston

The **Basic Supercurriculum** helps you build confidence, curiosity, and clarity — transforming maths from a school subject into a *tool for life*.

Year 10 Cambridge IGCSE Mathematics (0580 / 0980) Extended Supercurriculum

Why the Extended Supercurriculum Matters

Mathematics at the Extended level is not about memorising procedures — it's about *thinking structurally*.

It trains precision, argument, and creativity: the same skills used by scientists, engineers, and economists.

The Extended Supercurriculum strengthens conceptual mastery and develops the discipline to analyse, prove, and model complex problems with clarity.

Reading – Understanding Mathematics as a Language

Objective: Read mathematics like a system of logic, not a list of formulas.

Advanced Reading Sources:

NRICH (University of Cambridge) – problem-solving investigations

Plus Maths Magazine – “The maths of music”, “Patterns in nature”, “Chaos theory”

New Scientist – “How algorithms shape the world”

Brilliant.org – concept explanations and challenges

Task Routine (weekly):

Read one short article or problem.

Write a **100-word summary** explaining *how* mathematics was used.

Identify one **formula or pattern** used.

Sketch a quick visual representation (graph, table, or diagram).

Example Table

Article	Concept	Formula / Principle	Real Application
“The maths of risk”	Probability	$P = \frac{\text{favourable}}{\text{total}}$ $P = \frac{\text{total favourable}}{\text{total}}$	Insurance, statistics

Writing – Mathematical Communication and Proof

Objective: Explain and justify every step logically.

Extended Tasks

Proof Paragraph (150 words): Explain *why* the sum of angles in a triangle = 180° .

Reasoned Solution: Solve an algebraic problem, but include an explanation for each manipulation.

“Convince Me” Exercise: Write a short paragraph proving why $0.999... = 1$.

Structure for Written Work

Step	Description
1	Restate the problem clearly.
2	Identify knowns and unknowns.
3	Apply principle or theorem.

4	Write reasoning in full sentences.
5	Conclude logically and verify.

Example:

“The line is perpendicular because its gradient is the negative reciprocal of the other.”

Algebraic Mastery – Modelling and Generalisation

Objective: Move from calculation to *representation*.

Key Focus Areas

Expanding, factorising, and simplifying quadratic and cubic expressions.

Solving simultaneous equations (graphically & algebraically).

Using inequalities to describe regions.

Modelling growth and decay using powers and exponents.

Challenge Tasks

Derive the quadratic formula from $ax^2+bx+c=0$.

Model population growth with $P=Po(1+r)^t$.

Create your own “real-life” quadratic (e.g. height of a thrown ball) and graph it.

Extension:

Research how exponential functions are used in finance and medicine (interest, bacteria growth, half-life).

Geometry & Trigonometry – The Language of Space

Objective: Understand how mathematics describes physical space.

Core Extended Skills

Proofs using congruence, similarity, and parallel lines.

Circle theorems: angle in semicircle, tangent-radius, alternate segment theorem.

Vector geometry: magnitude, direction, scalar product.

Trigonometry:

$\sin, \cos, \tan$ in any triangle.

Sine and cosine rules.

Area of triangle $= \frac{1}{2}ab\sin C = \frac{1}{2}ac\sin B$.

3D trigonometric reasoning.

Applied Project:

Design a simple bridge. Use trigonometry to calculate cable angles and load forces.

Graphs, Functions & Modelling

Objective: Interpret and create mathematical models from data and formulas.

Core Tasks

Plot linear, quadratic, and cubic functions; identify intersections.

Sketch exponential and reciprocal graphs; note asymptotes.

Use transformation rules: *translate, reflect, stretch, invert*.

Model real-life data: speed-time → distance (area under graph).

Mini Investigation:

Measure temperature change of cooling water every 2 min → plot → describe curve → model with exponential decay equation.

Statistics – Data, Probability & Decision-Making

Objective: Apply mathematics to uncertainty and information.

Core Concepts

Data collection & representation (bar chart, histogram, cumulative frequency).

Mean, median, mode, range, interquartile range.

Probability trees & combined events.

Relative frequency → experimental probability.

Expected value and risk analysis.

Mini Project:

Collect data from your classmates (e.g. hours of sleep).

Represent using cumulative frequency graph.

Calculate median, Q_1 , Q_3 .

Interpret what the data tells you about variation.

Vectors, Transformations & Motion

Objective: Link geometry to physics and engineering ideas.

Tasks

Express vectors in column form; add, subtract, multiply by scalar.

Apply transformations to shapes: translation, reflection, rotation, enlargement.

Combine transformations and describe the single equivalent.

Use vector proofs (e.g. show lines are parallel or equal in length).

Applied Extension:

Model a drone's movement using vector addition.

Draw and calculate its resultant displacement.

Problem Solving – Mathematical Reasoning in Action

Objective: Build the habit of tackling unfamiliar questions logically.

Approach (P.R.O.V.E.):

Step	Question
P	What is the <i>Problem</i> asking?
R	What <i>Relationships</i> or formulas connect the data?
O	What <i>Operations</i> are needed?
V	<i>Verify</i> the solution — does it make sense?
E	<i>Extend</i> — how might the problem change?

Weekly Task:

Select one NRICH problem or IGCSE past paper challenge question.

Solve, check, and then explain it aloud.

Identify the key concept behind the problem.

Research & Interdisciplinary Application

Goal: Understand how mathematics supports all other sciences.

Field	Key Mathematical Tool	Real Application
Physics	Algebra, vectors, trigonometry	Motion, energy, waves
Chemistry	Ratios, equations, graphing	Concentration, reaction rates
Biology	Statistics, proportion	Population studies
Economics	Percentage, graph analysis	Inflation, cost-benefit models
Geography	Scale, coordinates	Mapping, land use data
Computer Science	Logic, matrices	Programming, graphics

Mini Research Idea:

Investigate how algorithms use geometry or probability to make recommendations (Netflix, Google, Spotify).

Technology & Digital Tools

Objective: Enhance understanding through mathematical technology.

Tools to Master:

Desmos / GeoGebra: plot functions, transformations, and inequalities.

Excel / Google Sheets: analyse data sets, generate graphs, calculate statistics.

Wolfram Alpha: verify symbolic calculations.

PhET Simulations: experiment with ratios, algebra balances, and trigonometric identities.

Task:

Create a *digital portfolio* with:

2 screenshots of functions you modelled.

1 statistical chart you created.

A 50-word explanation of what each represents.

Student-Led Learning

Goal: Take initiative and develop leadership in problem-solving.

Ideas:

Host a “Maths Masterclass” for younger students.

Design a classroom poster explaining one theorem.

Lead a small-group proof session.

Organise a “Maths Fair” with puzzles and logic games.

Capstone Project – “Mathematics for Real Life”

Duration: 4–6 weeks

Purpose: Connect multiple mathematical topics in one project.

Options:

Architecture & Geometry: Analyse symmetry and ratio in famous buildings.

Business Modelling: Calculate cost, revenue, and profit from given data; create graphs.

Environmental Data: Track temperature or rainfall → analyse using statistics.

Sports Science: Model speed, distance, and acceleration from match data.

Deliverables:

Written report (400–600 words).

At least one graph, one table, and one diagram.

Reflection paragraph: “Which part of maths helped me the most, and why?”

12-Week Extended Planner

Week	Focus	Output
1	Number & Indices	5 calculations with explanation
2	Algebra (Equations)	3 multi-step problems
3	Graphs & Functions	2 plotted graphs + table
4	Geometry & Angles	Proof paragraph
5	Trigonometry	Solve 5 triangle problems
6	Transformations	Draw & describe 3 transformations
7	Statistics	Create histogram + mean/median
8	Probability	Tree diagram + reflection
9	Vectors	Vector diagram & magnitude
10	Measurement	Surface area, volume, unit conversion
11	Mixed Problem Solving	Past paper section + check
12	Review & Project	Summary poster + reflection log

Suggested Resources

Type	Resource	Link
Practice	SaveMyExams – Maths IGCSE	https://www.savemyexams.com
Challenge	NRICH Maths (Cambridge)	https://nrich.maths.org
Explanation	Khan Academy Mathematics	https://www.khanacademy.org
Interactive	GeoGebra / Desmos	https://www.geogebra.org
Enrichment	Plus Maths Magazine	https://plus.maths.org
Simulation	PhET Maths	https://phet.colorado.edu

Final Reflection

“Pure mathematics is, in its way, the poetry of logical ideas.” — Albert Einstein

The **Extended Supercurriculum** builds mastery, creativity, and independence.

It moves beyond memorising formulas — towards *understanding structure, proving truth, and solving the unknown.*

Cambridge IGCSE Mathematics (0580 / 0980) – Study Guide

Course Overview

Exam Board: Cambridge International

Syllabus Codes: 0580 / 0980

Assessment Type: Linear (all exams at end of course)

Aims:

The course develops mathematical reasoning, numerical fluency, and problem-solving across the full range of real-world contexts — from finance to physics.

Paper	Title	Duration	Weight	Level
Paper 2	Short-Answer Questions	1 h 30 min	35%	Core
Paper 4	Extended Structured Questions	2 h	50%	Extended
Paper 6	Investigation & Modelling (Optional)	1 h	15%	Extended Enrichment

Assessment Objectives

AO	Description	Tested Through
AO1	Recall, apply, and interpret mathematical knowledge	Short answers, definitions, formulas
AO2	Solve problems using appropriate methods	Multi-step questions
AO3	Interpret and present information clearly	Graphs, diagrams, reports
AO4	Apply mathematics in real-life situations	Modelling, measurement, estimation

Paper Structure

Paper 2 – Core: Short Answer

56 marks

Mixture of calculation, reasoning, and diagrams

Non-calculator and calculator use combined

Topics Emphasised:

Number, algebra, geometry, mensuration, and data handling

Straightforward application of formulas

Basic graph reading and construction

Strategy

Underline **key values** and **units**.

Write formula *before* substituting.

Keep answers to 3 significant figures.

Double-check rounding and unit conversions.

Paper 4 – Extended: Structured Questions

130 marks

Calculators allowed

Longer questions with real-life contexts and mixed concepts

Common Sections

Multi-step algebra

Geometry and trigonometry proofs

Coordinate geometry (lines, gradients, midpoints)

Graphical and statistical interpretation

Ratio, percentage, and mensuration problems

Marks Awarded For

Type	Marks	Expectation
Method	1–3	Correct formula, logical step
Accuracy	1–2	Correct answer and units
Reasoning	1–2	Clear justification or explanation

Paper 6 – Investigation & Modelling (Extended Optional)

40 marks

Open-ended questions assessing mathematical thinking, not memorisation.

Involves patterns, conjectures, and generalisation.

Example:

Explore how the perimeter of polygons changes when area remains constant.

Skills Tested

Pattern recognition

Hypothesis and algebraic proof

Presentation of logical arguments

Formula & Equation Sheet (Core + Extended)

Number & Algebra

Concept	Formula / Definition
Percentage	$\frac{\text{change}}{\text{original}} \times 100$ or $\frac{\text{original}}{\text{change}} \times 100$
Speed	$\frac{\text{distance}}{\text{time}}$ or $\frac{\text{time}}{\text{distance}}$
Density	$\frac{\text{mass}}{\text{volume}}$ or $\frac{\text{volume}}{\text{mass}}$
Gradient	$\frac{y_2 - y_1}{x_2 - x_1}$ or $\frac{y_2 - y_1}{x_1 - x_2}$
Midpoint	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ or $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Factorisation	$a^2 - b^2 = (a - b)(a + b)$ or $a^2 - b^2 = (a - b)(a + b)$
Simultaneous equations	Solve by elimination or substitution

Geometry & Trigonometry

Concept	Formula / Definition
Pythagoras	$a^2 + b^2 = c^2$ or $a^2 + b^2 = c^2$
Area (triangle)	$\frac{1}{2}ab \sin C$ or $\frac{1}{2}ab \sin C$
Sine Rule	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ or $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
Cosine Rule	$c^2 = a^2 + b^2 - 2ab \cos C$ or $c^2 = a^2 + b^2 - 2ab \cos C$

Circle area	πr^2
Circumference	$2\pi r$
Volume (cylinder)	$\pi r^2 h$
Volume (sphere)	$\frac{4}{3}\pi r^3$
Surface area (sphere)	$4\pi r^2$

Statistics & Probability

Concept	Formula
Mean	$\bar{x} = \frac{\sum x}{n}$
Probability	$P(E) = \frac{\text{favourable}}{\text{total}}$
Expected Value	$\sum (xP(x))$
Cumulative Frequency	Used for estimating median, quartiles
Histogram	Area = frequency density $\times$ class width

Mensuration & Ratio

Concept	Formula
Scale Drawing	Real = Diagram $\times$ Scale Factor
Ratio Simplification	Divide by HCF
Compound Interest	$A = P(1 + \frac{r}{100})^n$
Simple Interest	$I = \frac{PRT}{100}$

Graphing & Modelling Skills

Types of Graphs

Linear: $y = mx + c$ (m = gradient, c = intercept)

Quadratic: $y = ax^2 + bx + c$

Exponential: $y = kax$

Reciprocal: $y = \frac{k}{x}$

Distance-time / speed-time graphs

Key Skills

Identify gradient & intercept

Interpret area under curves

Estimate solutions by graph intersection

Tip:

Always label axes with *variable + unit*, and use consistent scales.

Data Handling and Statistics

Steps for Data Questions

Collect / interpret data table.

Represent data (bar chart, histogram, pie chart).

Find measures of central tendency (mean, median, mode).

Use cumulative frequency to estimate median and interquartile range.

Describe data: compare spread and average.

Describing Data Example

“Group A’s median is higher, but range is smaller — they are more consistent.”

Problem-Solving Framework

Approach (A.I.M.E.)

Step	Action
A	Analyse — what is given and what is asked
I	Identify — which formula, unit, or concept applies
M	Manipulate — calculate and check logic
E	Evaluate — does the answer make real-world sense?

Example:

The surface area of a sphere with $r = 3$ cm
 $= 4\pi r^2 = 4\pi(9) = 113.1 \text{ cm}^2$
 $4\pi r^2 = 4\pi(9) = 113.1 \text{ cm}^2$.
 Check units: area $\rightarrow \text{cm}^2$ ✓

Calculator Skills

Efficiency Tips

Always use **ANS** key for continuation.
 Store variables using **STO** button for reuse.
 Switch between **degrees/radians** correctly (for trig).
 Check **mode** (normal vs scientific notation).
 Round only in the *final answer*.

12-Week Study Plan

Week	Focus	Task
1	Number & Arithmetic	Review fractions, decimals, percentages
2	Ratio & Proportion	Solve mixture and scale problems
3	Algebra Basics	Simplify, expand, and factorise
4	Quadratics & Graphs	Plot parabolas, find roots
5	Simultaneous Equations	Graphical and algebraic solutions
6	Geometry & Angles	Parallel lines, polygons
7	Trigonometry	Solve non-right triangles
8	Mensuration	Volumes, surface areas
9	Statistics	Mean, median, histogram
10	Probability	Tree diagrams, expected value
11	Revision	Mixed past paper problems
12	Mock Exam	Paper 2 + Paper 4 timed practice

Self-Assessment Rubric

Level	Knowledge	Application	Communication
Emerging	Basic recall	Limited problem-solving	Some working shown
Developing	Accurate recall	Applies formulas with help	Logical structure

Proficient	Consistent accuracy	Independent reasoning	Clear steps and units
Advanced	Analytical	Multi-step synthesis	Precise, elegant solutions
Expert	Conceptual mastery	Transfers methods creatively	Mathematically fluent

Common Exam Traps

Trap	Correction
Forgetting units	Always include (cm ² , m ³ , etc.)
Premature rounding	Round at end only
Calculator mode error	Use “DEG” for trigonometry
Missing labels on graphs	Label axes, scales, titles
Formula misuse	Write it before substituting
Time-pressure	Attempt <i>every</i> question — partial marks matter
Skipped steps	Show reasoning → earns method marks

Digital Tools & Practice Platforms

Purpose	Platform	Link
Revision Notes	BBC Bitesize – IGCSE Maths	https://www.bbc.co.uk/bitesize/subjects/zqhs34j
Practice Papers	SaveMyExams	https://www.savemyexams.com
Interactive Graphing	Desmos / GeoGebra	https://www.desmos.com
Problem Solving	NRICH (Cambridge)	https://rich.maths.org
Explainers	Khan Academy Mathematics	https://www.khanacademy.org
Video Tutorials	Maths Genie / ExamSolutions	https://www.mathsgenie.co.uk

Exam-Day Checklist

- ✓Scientific calculator (check mode & batteries)
- ✓Ruler, compass, protractor
- ✓Pencil, eraser, sharpener
- ✓Blue or black pen
- ✓Clear working, boxed final answers
- ✓Label diagrams
- ✓Logical order of solutions

Final Tip:

If stuck → write down the formula and knowns — partial marks are awarded for method.

Final Reflection

“Mathematics is not a spectator sport — you learn by doing.” — George Pólya

The **Study Guide** prepares you to reason, not just calculate.

Every equation is a story of balance, every graph a picture of logic — and every problem an opportunity to think with clarity.

Year 10 Cambridge IGCSE Additional Mathematics (0606 / 4037) Basic Supercurriculum

Why the Supercurriculum Matters in Additional Mathematics

Additional Mathematics builds the bridge between high-school maths and university-level problem-solving.

It's not about speed — it's about *structure*. It teaches how to think abstractly, generalise patterns, and apply algebra and calculus to real situations.

The Basic Supercurriculum lays the foundation for those skills through curiosity, precision, and logic.

Reading – From Numbers to Ideas

Goal: Understand that mathematics is a creative way of describing relationships.

Weekly Habit:

Read one short article or watch a video explaining a concept *beyond the textbook*.

Recommended Sources

NRICH (Cambridge) – algebra and proof challenges

Mathigon – interactive explanations (functions, vectors)

Khan Academy: Algebra II & Pre-Calculus

Plus Maths Magazine – “What is a function?”, “How calculus changed the world”

After reading:

Write 3 key ideas.

Define 1 new term (e.g. *domain*, *derivative*).

Note 1 real-life connection (engineering, design, finance).

Writing – Explaining Mathematical Logic

Goal: Learn to express reasoning clearly and symbolically.

Mini Tasks

Rewrite each step of an algebraic manipulation *in words*.

Keep a “Reasoning Journal”: short explanations of how you solved each problem.

Example entry:

“To solve $2x+3=11$, I subtracted 3 from both sides because equality must stay balanced.”

Style Rule: Always state **why** each step works — not just what you did.

Core Skills Overview

Area	What You'll Learn	Example
Algebraic manipulation	Simplify, expand, factorise, solve	$x^2+5x+6=0$ $x^2+5x+6=0$

	quadratics	
Functions	Understand mappings, domain, range, inverse	$f(x)=2x+3$ $f(x)=2x+3 \rightarrow f^{-1}(x)=x-3$ $2f^{-1}(x)=2x-3$
Coordinate geometry	Equation of line, midpoint, gradient	$y-y_1=m(x-x_1)$ $y-y_1=m(x-x_1)$
Trigonometry	Identities and equations	$\sin^2\theta + \cos^2\theta = 1$ $\sin 2\theta + \cos 2\theta = 1$
Vectors	Magnitude and direction	(
Differentiation	Gradient, maxima/minima	$\frac{d}{dx}(x^2)=2x$ $\frac{d}{dx}(x^2)=2x$
Integration	Area under curve	$\int x^2 dx = \frac{x^3}{3} + C$ $\int x^2 dx = \frac{x^3}{3} + C$

Practising – From Routine to Insight

Goal: Build fluency through short daily exercises.

Routine (10–15 min daily):

- 2 algebraic simplifications
- 2 equations (linear or quadratic)
- 1 trigonometric identity to prove
- 1 derivative or integral (simple power rule)

Weekly Extension:

Create one *real-world problem* that could be solved using algebra or calculus.

“A car’s velocity is $v=3t^2$. Find how far it travels in 4 seconds.”

Watching – Seeing the Abstract in Motion

Goal: Use visual learning to grasp invisible ideas.

Recommended Channels

3Blue1Brown – visual calculus, functions, graphs

Maths Genie – concise IGCSE Add Maths walkthroughs

Khan Academy – *Calculus Basics*

Activity:

After watching, summarise in 3 lines:

The main idea

The visual or graph shown

How it connects to a real-life system (motion, growth, finance)

Experimenting – Discovery through Examples

Goal: Learn by exploring patterns.

Mini Investigations

Plot $y=x^2+2x+1$ and observe its symmetry.

Change coefficients — what happens to the vertex?

Plot $y=2^x$ and $y=\ln x$: how are they related?

Differentiate x^3 , x^2 , x — notice the pattern.

Insight:

Mathematics is discovered, not memorised.

Researching – Linking Mathematics to the Real World

Goal: Connect abstract concepts with technology and science.

Mini Projects

“How derivatives describe the speed of a moving object.”

“How trigonometry is used in GPS and navigation.”

“Why engineers use vectors to model forces.”

“How exponential models describe population or finance.”

Deliverable: 150–200-word paragraph + one diagram or graph.

Student-Led Tasks – Collaboration and Explanation

Goal: Learn through teaching.

Ideas:

Prepare a 2-minute whiteboard explanation of any formula.

Make a short poster explaining *The Power Rule of Differentiation*.

Form study pairs — one solves, the other checks for clarity and logic.

Creative Tasks – The Art of Mathematics

Goal: See the elegance behind equations.

Projects:

Design a “Function Gallery”: draw 5 graphs showing transformations.

Create a visual of $y = \sin x, \cos x, \tan x$ side by side.

Illustrate area under a curve using coloured trapezia (integration idea).

Write a 4-line “math haiku” expressing a concept (e.g. symmetry, infinity).

Reflection – Thinking Like a Mathematician

Goal: Cultivate awareness of learning patterns.

Each week, reflect in your maths journal:

“What rule did I *discover* this week?”

“What error helped me understand a deeper idea?”

“Which concept connects to something outside maths?”

Example: “When I graphed quadratic functions, I realised motion in physics follows the same shape.”

Interdisciplinary Connections

Subject	Connection
Physics	Motion, calculus, vectors
Economics	Growth, decay, optimisation
Biology	Rates of change, population models
Computer Science	Functions, algorithms, binary logic

Geography	Gradient, scale, trigonometric surveying
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Digital & Career Literacy

Goal: Use technology to explore patterns and careers.

Tools:

Desmos / GeoGebra – graphing functions and derivatives

WolframAlpha – verify algebraic steps and differentiation

Excel / Sheets – data tables and regression

Career Links:

Research 3 roles where advanced maths is essential (e.g. engineer, actuary, software developer).

Write 2 lines about what mathematics they use daily.

Core Topic Map (Cambridge Alignment)

Unit	Theme	Key Ideas
1	Algebra	Equations, inequalities, indices, surds
2	Functions	Inverse, composite, domain/range
3	Coordinate Geometry	Gradient, distance, midpoints
4	Trigonometry	Identities, equations, graphs
5	Calculus	Differentiation & integration basics
6	Vectors	Magnitude, direction, scalar products
7	Geometry	Circles, tangents, proofs
8	Application	Modelling, optimisation, kinematics

Weekly Routine (30–40 minutes total)

Day	Task	Outcome
Monday	Read one short article or watch a video	Write 3 facts
Tuesday	Practice 3 algebra or trig problems	Check working
Wednesday	Graph a function on Desmos	Annotate features
Thursday	Research one real-world application	1 paragraph
Friday	Reflection	3-sentence journal entry

Final Thought

“Mathematics is the music of reason.” — James Joseph Sylvester

The **Basic Supercurriculum** for Additional Mathematics is about seeing beyond arithmetic —

it helps you discover patterns, prove relationships, and begin thinking like an engineer, scientist, or analyst.

Year 10 Cambridge IGCSE Additional Mathematics (0606 / 4037) Extended Supercurriculum

Why the Extended Supercurriculum Matters

Additional Mathematics is where maths becomes a *language for science, logic, and prediction*.

Here, students stop memorising and start *reasoning* — learning how mathematical structure underpins everything from orbits and algorithms to architecture and AI.

The Extended Supercurriculum builds mastery, abstraction, and mathematical curiosity for lifelong thinkers.

Reading – Mathematics as Structure and Proof

Objective: Go beyond calculation into conceptual understanding.

Advanced Reading Sources

NRICH – Proof and Reasoning (Cambridge University)

Mathigon: The Mathematics of Motion and Change

Plus Maths Magazine: “Why calculus works”, “Symmetry and the Universe”

3Blue1Brown Essays: “Visualising functions and derivatives”

Weekly Reading Task:

Summarise the main idea in 100 words.

Identify the theorem or property discussed.

Explain how this concept connects to real applications.

Example Table

Reading	Concept	Formula / Theorem	Application
“Curves and change”	Differentiation	$\frac{dy}{dx}$	Predicting velocity, optimisation

Algebra – Abstraction and Generalisation

Objective: Manipulate algebra as a language of patterns and logic.

Extended Concepts

Factor and simplify complex rational expressions.

Solve quadratic and cubic equations (including discriminant).

Apply laws of indices and logarithms.

Work with surds, inequalities, and modulus.

Construct and interpret composite and inverse functions.

Practice Framework

Task	Focus
Derive the quadratic formula	Proof and symbolic reasoning
Solve inequalities graphically	Precision and visual understanding
Simplify $2x^2 - 8x^2 - 4x^2 - 4x^2 - 8$	Factorisation and domain

	awareness
Use $\log_a xy = \log_a x + \log_a y$ $\log_a xy = \log_a x + \log_a y$	Logarithmic manipulation

Exploration:

Investigate how exponential growth equations $y = Ae^{kx}$ describe population, bacteria, or finance models.

Functions and Transformations

Objective: See mathematics as mapping — connecting inputs to outputs.

Key Ideas

Domain, range, inverse, composite functions.

Transformation of graphs: reflection, translation, stretch.

Piecewise and modulus functions.

Sketching functions from algebraic form.

Investigative Task

Use **Desmos** / **GeoGebra** to explore:

$$y = x^2, y = (x-2)^2, y = 2x^2$$

Describe algebraic vs graphical changes.

Extension:

Model “cooling of coffee” as an exponential decay and interpret graph behaviour.

Trigonometry – Beyond Triangles

Objective: Move from geometry to analytic trigonometry.

Advanced Topics

Trigonometric identities:

$$\sin^2 x + \cos^2 x = 1, \tan x = \frac{\sin x}{\cos x}$$

Double-angle formulas:

$$\sin 2x = 2 \sin x \cos x, \cos 2x = 1 - 2 \sin^2 x$$

Solve trigonometric equations graphically and algebraically.

Apply trigonometric modelling in oscillations and waves.

Mini Project:

“Investigate how sine functions describe sound waves and alternating current.”

Challenge:

$$\tan A + \cot A = \frac{2}{\sin 2A}$$

Coordinate Geometry and Vectors

Objective: Link algebra to geometry and motion.

Topics

Equation of a line, perpendicular distance, intersection.

$$(x-h)^2 + (y-k)^2 = r^2$$

Tangents and normals to curves.

Vectors in 2D: magnitude, direction, scalar product, parallel/perpendicular criteria.

Task:

Prove that two lines are perpendicular by showing $m_1 \times m_2 = -1$ or $m_1 \times m_2 = -1$.

Compute the angle between vectors using

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \quad \cos \theta = \frac{|\mathbf{a}| |\mathbf{b}| \cos \theta}{|\mathbf{a}| |\mathbf{b}|}$$

Applied Extension:

Model a drone's or robot's path using vector addition and direction cosines.

Calculus – The Mathematics of Change

Objective: Understand differentiation and integration as inverse processes.

Differentiation

Concept	Formula / Rule	Application
Power Rule	$\frac{d}{dx}(x^n) = nx^{n-1}$ $\frac{d}{dx}(x^n) = nx^{n-1}$	Gradient, rate of change
Product Rule	$(uv)' = u'v + uv'$ $(uv)' = u'v + uv'$	Compound motion, economics
Quotient Rule	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'v^2}{v^2}$ $(\frac{u}{v})' = \frac{u'v - uv'v^2}{v^2}$	Ratio models
Chain Rule	$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$ $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$	Composite functions

Practical Tasks

Find tangents and normals at given points.

Determine turning points (maxima/minima).

Sketch curves showing stationary points.

Application Example:

Optimisation: Find the dimensions of a box with fixed volume that minimises surface area.

Integration

Concept	Formula / Rule	Application
Power Rule	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$ $\int x^n dx = \frac{x^{n+1}}{n+1} + C$	Area under curve
Definite Integral	$\int_a^b f(x) dx$ $\int_a^b f(x) dx$	Total displacement or energy
Reverse Chain Rule	For composite functions	Substitution method
Integration by Parts	$\int u dv = uv - \int v du$ $\int u dv = uv - \int v du$	Product of functions

Mini Investigation:

Compare numerical estimation (trapezium rule) to exact integration

for $y = x^2$ or $y = x^2$.

Real-World Link:

Integration predicts distance from velocity, total charge from current, and energy from power.

Graphs and Modelling

Objective: Turn mathematics into prediction.

Skills

Sketch polynomial, exponential, and logarithmic graphs.

Solve equations by intersection of curves.

Use calculus for curve sketching (stationary points, concavity).

Approximate area and solve optimisation problems.

Challenge Task:

Model the height of a ball:

$$h(t) = -5t^2 + 20t + 1$$

Find time when it hits the ground, its maximum height, and total flight time.

Proof, Reasoning & Generalisation

Objective: Use logical argumentation to justify truth.

Forms of Proof

Direct Proof: Use algebraic manipulation to show equality.

Contradiction: Assume opposite and reach logical impossibility.

Induction (intro): Recognise recursive patterns and generalise.

Examples:

Prove that the sum of two even numbers is even.

Show that $(x+1)^2 - x^2$ is always odd for integer x .

Explore Fibonacci sequences — can you prove ratios approach the golden mean?

Researching – The Real-World Language of Mathematics

Mini Investigations

“How calculus shapes car aerodynamics.”

“Modelling disease spread with exponential functions.”

“The role of complex numbers in engineering.”

“How optimisation underpins modern AI algorithms.”

Output: 300-word summary + one graph or diagram per topic.

Student-Led Projects & Peer Learning

Goal: Collaborate and communicate like mathematicians.

Ideas

Run a “Maths Masterclass” to explain derivatives visually.

Create a poster: “The Journey from Gradient to Curve.”

Debate: “*Should all students learn calculus before university?*”

Work in pairs to design a model connecting two topics (e.g., trigonometry and differentiation).

Creative Mathematics – Beauty Through Logic

Goal: Appreciate symmetry and design in numbers and patterns.

Projects

Draw parametric spirals (e.g. $x = a \cos t, y = a \sin t$).

Use Desmos to animate gradient changes.
 Explore fractals using recursive equations.
 Design an infographic: “Mathematics in Nature – From Shells to Galaxies.”

Reflection – Thinking Like a Mathematician

At the end of each week:

“What was the hardest problem I understood this week?”

“Where did I see structure or symmetry?”

“Which concept connects to another subject?”

“How has calculus changed the way I see the world?”

“In mathematics, every problem is a door to another universe.”

12-Week Extended Planner

Week	Focus	Outcome
1	Algebraic Laws & Proof	Written derivations
2	Quadratics & Functions	Graphs & intersections
3	Logarithms & Exponentials	Exponential model project
4	Trigonometric Identities	Proofs + graphs
5	Trig Equations	Problem set
6	Coordinate Geometry	Circle & tangent problem
7	Vectors	Direction + magnitude investigation
8	Differentiation	Tangents + maxima/minima
9	Integration	Area + accumulation project
10	Kinematics	Motion graphs + velocity models
11	Modelling & Optimisation	Mini report
12	Presentation	Peer explanation + reflection log

Suggested Resources

Type	Resource	Link
Practice	Save My Exams – Add Maths	https://www.savemyexams.com
Advanced Lessons	ExamSolutions – Add Maths	https://www.examsolutions.net
Enrichment	NRICH Advanced Mathematics	https://nrich.maths.org
Visual Understanding	3Blue1Brown – Calculus & Algebra	https://www.3blue1brown.com
Interactive Tools	GeoGebra / Desmos	https://www.geogebra.org
Problem Database	Brilliant.org – Calculus & Vectors	https://brilliant.org

Final Reflection

“Pure mathematics is the world’s best way to learn how to think clearly.”

The **Extended Supercurriculum** transforms mathematics from calculation into creation —

it's where logic meets imagination, where proofs become patterns, and
where *thinking* becomes an art form.

Cambridge IGCSE Additional Mathematics (0606 / 4037) Study Guide

Course Overview

Exam Board: Cambridge International

Syllabus Code: 0606 (IGCSE) / 4037 (O-Level equivalent)

Purpose:

To develop higher-order mathematical thinking, fluent algebraic manipulation, and early calculus skills.

This course is more abstract and theoretical than IGCSE Mathematics, preparing students for A-Level Pure Mathematics, Physics, and Engineering.

Paper	Duration	Marks	Weight	Format
Paper 1	2 hours	80 marks	50%	Short structured questions
Paper 2	2 hours	80 marks	50%	Problem-solving and modelling questions

Assessment Objectives (AO)

AO	Description	Focus
AO1	Knowledge and understanding	Recall, use, and apply mathematical concepts
AO2	Problem solving	Translate situations into mathematical models
AO3	Communication	Interpret, reason, and present solutions clearly
AO4	Use of technology	Employ calculators and graphing tools effectively

Paper Structure

Paper 1 – Short Structured Questions

80 marks, 2 hours

Direct questions testing specific topics and shorter reasoning steps

Each question typically worth 3–8 marks

Common Topics

Algebra, functions, coordinate geometry, trigonometry

Basic differentiation and integration

Vectors and graphs

Advice

Work briskly — about **1.5 minutes per mark**.

Show working for all steps — method marks count.

Always include correct units and rounding.

Paper 2 – Longer, Integrated Problems

80 marks, 2 hours

Multi-step, contextual problems testing links between algebra, calculus, and geometry

Requires structured reasoning and diagram interpretation

Question Examples

Modelling motion using differentiation and integration

Optimisation (maximum/minimum problems)

Curve sketching combining calculus and algebra

Vector geometry and circle theorems

Marking Emphasis

Category	Description
Method Marks	For setting up equations correctly
Accuracy Marks	For correct computation
Communication Marks	For clarity, structure, and justification

Core Topic Overview

Unit	Topic	Key Skills
1	Algebra	Simplification, factorisation, indices, logarithms
2	Functions	Domain, range, inverse, composition
3	Coordinate Geometry	Line, circle, perpendicular distance
4	Trigonometry	Identities, equations, graphs, proofs
5	Calculus	Differentiation, integration, kinematics
6	Vectors	Magnitude, direction, scalar product
7	Inequalities	Linear, quadratic, graphical solution
8	Proof & Modelling	Logic, structure, application

Key Formulae & Theorems

Algebra & Functions

Concept	Formula
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Discriminant	$\Delta = b^2 - 4ac$
Laws of Indices	$a^m \times a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$
Laws of Logarithms	$\log_a(xy) = \log_a x + \log_a y$, $\log_a \frac{x}{y} = \log_a x - \log_a y$
Composite Function	$(f \circ g)(x) = f(g(x))$
Inverse Function	$f^{-1}(x)$: satisfies $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$

Coordinate Geometry

Concept	Formula
Gradient	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Midpoint	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
Equation of Circle	$(x-h)^2 + (y-k)^2 = r^2$
Equation of Tangent	$y - y_1 = m(x - x_1)$

Trigonometry

Concept	Formula
Pythagoras	$a^2 + b^2 = c^2$

Sine Rule	$a \sin A = b \sin B = c \sin C$
Cosine Rule	$c^2 = a^2 + b^2 - 2ab \cos C$
Double Angle	$\sin 2A = 2 \sin A \cos A$; $\cos 2A = 1 - 2 \sin^2 A$
Area of Triangle	$\frac{1}{2} ab \sin C$

Calculus

Concept	Rule	Example
Power Rule	$\frac{d}{dx}(x^n) = nx^{n-1}$	$\frac{d}{dx}(x^3) = 3x^2$
Product Rule	$(uv)' = u'v + uv'$	$y = x \sin x$
Quotient Rule	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$	$y = \frac{\sin x}{x}$
Chain Rule	$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$	$y = \sin(3x)$
Integration (Power)	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$	$\int x^2 dx = \frac{x^3}{3} + C$
Definite Integral	$\int_a^b f(x) dx = F(b) - F(a)$	Area under curve

Vectors

Concept	Formula
Vector Notation	$\mathbf{a} = (x, y)$
Magnitude	$ \mathbf{a} = \sqrt{x^2 + y^2}$
Dot Product	$\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \mathbf{b} \cos \theta$
Unit Vector	$\hat{\mathbf{a}} = \frac{\mathbf{a}}{ \mathbf{a} }$

Graphical & Modelling Skills

Essential Skills

Sketch and interpret polynomial, exponential, and trigonometric functions.

Identify asymptotes, intercepts, stationary points, and concavity.

Model real situations with algebraic or differential equations.

Estimate solutions graphically where algebraic methods are complex.

Applied Example

Given $y = 3x^2 - 12x + 2$:

Gradient = $6x - 12$

Stationary point at $x = 2 \rightarrow$ minimum $y = -10$

Calculus Applications

Differentiation

Tangents and normals

Increasing/decreasing functions

Maxima and minima (second derivative test)

Motion: $v = ds/dt$, $a = dv/dt$, $v = \int a dt$, $s = \int v dt$

Integration

Area under curve between limits

Volume of revolution (introductory)

Mean value of a function over interval $[a, b]$

Displacement and distance in kinematics

Proof, Logic & Reasoning

Proof Strategies

Direct proof – algebraic manipulation.

Contradiction – assume opposite and find impossibility.

Geometric proof – use coordinate or vector reasoning.

Derivation – from first principles (e.g., derivative of x^n).

Mini Challenge:

Prove that $(x+1)^2 - x^2$ is always an odd number for integer x .

Problem-Solving Framework

Method (C.L.E.A.R.)

Step	Description
C	Clarify the problem — list knowns & unknowns
L	Link to formulas or principles
E	Execute calculations systematically
A	Analyse if result is reasonable
R	Reflect — could it be generalised or simplified?

Common Mistakes

Mistake	Correction
Mixing domain & range	Domain $\rightarrow$ input values; Range $\rightarrow$ output
Forgetting +C in integration	Always include constant unless limits are given
Confusing radians/degrees	Convert before using trig functions
Wrong chain rule order	Differentiate inner function last
Rounding early	Round only at final answer
Using wrong log base	$\ln x = \log_e x$, not 10

Study & Revision Techniques

Weekly Practice Routine

2 algebra problems

2 differentiation/integration problems

1 geometry or trigonometry question

1 real-world application question

Resources to Rotate

Cambridge Past Papers (Paper 1 + 2)

SaveMyExams question sets

Desmos for graph sketching
NRICH challenges for proof logic

12-Week Revision Roadmap

Week	Focus	Core Output
1	Algebra & Indices	Simplify, expand, and factorise expressions
2	Quadratic Equations	Discriminant & roots
3	Functions	Composite, inverse, and transformation graphs
4	Trigonometry	Identities and graph modelling
5	Coordinate Geometry	Circle, tangent, gradient
6	Vectors	Direction and magnitude
7	Differentiation	Tangents, stationary points
8	Integration	Area under curve
9	Kinematics	Velocity & acceleration problems
10	Mixed Modelling	Real-life applications
11	Past Paper Practice	Paper 1 timed
12	Exam Simulation	Paper 2 + Reflection log

Calculator & Graphing Tips

- ✓ Use **DEG** mode unless radians specified.
- ✓ Label answers clearly with units.
- ✓ Use **ANS** key for continued calculations.
- ✓ Store constants in **STO** memory.
- ✓ Sketch before calculating → always visualise the curve.

Digital & Learning Resources

Skill	Platform	Link
Revision	BBC Bitesize – Add Maths (UK Equivalents)	https://www.bbc.co.uk/bitesize
Practice	Save My Exams – Add Maths	https://www.savemyexams.com
Lessons	ExamSolutions – Add Maths	https://www.examsolutions.net
Theory	Khan Academy – Pre-Calculus / Calculus	https://www.khanacademy.org
Visuals	3Blue1Brown – Essence of Calculus	https://www.3blue1brown.com
Challenges	NRICH Advanced Mathematics	https://nrich.maths.org

Self-Assessment Rubric

Level	Knowledge	Reasoning	Application
Emerging	Some recall	Limited steps shown	Simple context
Developing	Sound recall	Logical working	Basic modelling
Proficient	Full accuracy	Justified reasoning	Multi-step success
Advanced	Conceptual understanding	Elegant derivation	Cross-topic synthesis

Expert	Proof-level mastery	Creative generalisation	Mathematically fluent
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Exam-Day Checklist

- ✓Scientific calculator (fully charged)
- ✓Ruler, compass, protractor
- ✓Pencil, pen, eraser
- ✓Formula recall sheet (revision use only)
- ✓Show every step → method marks awarded
- ✓Round answers appropriately (3 s.f. or exact form)
- ✓Label diagrams & axes
- ✓Write clear, logical statements

Final Reflection

“Mathematics is not about finding answers; it’s about discovering patterns of truth.”

The **Study Guide** transforms abstract symbols into insight.

Through proof, precision, and creativity, you master the art of reasoning — the skill at the heart of every scientific and technological achievement.

Σκέφτηκε για 7s

Here’s a ready-to-use **Basic Supercurriculum** for **Biology (IGCSE-level, Year 10)**. It’s designed for short, low-prep activities that build secure core knowledge and practical habits. Each item lists the **focus, what to do, materials, time, safety, and success criteria**. Use it as a menu—aim for 3–4 items per week.

Biology — Basic Supercurriculum (Year 10)

Cells & Microscopy

B1.1 Draw it like a scientist

Focus: Cell structure (animal vs plant), scale.

Do: Copy a labelled diagram of an animal and a plant cell; add magnification and scale bars.

Materials: Pencil, ruler, paper. **Time:** 15–20 min.

Success: All major organelles correctly placed; scale bar included.

B1.2 Virtual microscope sketch

Focus: Observational skills.

Do: Use any high-resolution cell image (textbook/online) to sketch what you see; annotate with “low → high power” differences.

Materials: Image, paper. **Time:** 20 min.

Success: Clear differences noted (field of view, detail).

B1.3 Surface area vs volume demo

Focus: Diffusion limits in cells.

Do: Cut 3 cubes (e.g., jelly/gelatine or potatoes). Dye in food colouring for 5–10 min; compare penetration depth.

Materials: Food dye, knife, cubes. **Time:** 25–30 min. **Safety:** Supervision when cutting.

Success: Correctly link SA:V to diffusion speed.

Biological Molecules & Enzymes

B2.1 Food tests at home (mini)

Focus: Starch, sugar, protein, lipid (theory only if reagents unavailable).

Do: Make a results table predicting colours for iodine/benedict/biuret/emulsion tests; check against textbook.

Materials: Paper/table. **Time:** 15 min.

Success: Correct reagents ↔ macromolecule mapping.

B2.2 Enzyme keywords snapshot

Focus: Active site, specificity, denaturation.

Do: Flashcards or a one-page mind map with concise definitions + one example (amylase).

Time: 15–20 min.

Success: Accurate, exam-ready definitions.

Movement of Substances

B3.1 Diffusion in a bag

Focus: Diffusion/osmosis ideas.

Do: Place a tea bag in hot vs cold water; time colour spread; write 3 reasons for difference.

Materials: Two cups, water. **Time:** 15 min. **Safety:** Hot water care.

Success: Temperature linked to particle kinetic energy.

B3.2 Osmosis with potato chips (theory lite)

Focus: Water potential concept.

Do: Plan (not perform) a simple osmosis experiment: variables, method, expected graph.

Time: 15–20 min.

Success: Correct variables & predicted mass change direction.

Plant Nutrition & Transport

B4.1 Leaf structure quick-label

Focus: Palisade/spongy mesophyll, stomata, xylem/phloem.

Do: Label a blank leaf cross-section and add one function for each tissue.

Time: 10–15 min.

Success: Structure–function matched correctly.

B4.2 Stomata count (photo method)

Focus: Gas exchange surfaces.

Do: Photograph the underside of a leaf with phone macro; estimate stomata density (visual estimate).

Time: 20 min.

Success: Reasonable density estimate + one factor affecting it.

Human Nutrition & Digestion

B5.1 Balanced plate audit

Focus: Macronutrients, fibre, water, vitamin/mineral roles.

Do: Sketch today's meals as a "plate"; annotate macronutrient sources; suggest one improvement.

Time: 15–20 min.

Success: Sensible improvement justified (e.g., "add fibre for peristalsis/health").

B5.2 Enzyme locations map

Focus: Amylase, protease, lipase sites & pH.

Do: Draw the gut; place enzymes with substrates → products + optimal pH.

Time: 15 min.

Success: Correct enzyme–site–pH mapping.

Respiration & Gas Exchange

B6.1 Breathing rate mini-investigation

Focus: Exercise & ventilation.

Do: Measure breaths per minute at rest, after 30s steps, after 2 min recovery; plot quick line graph.

Materials: Timer. **Time:** 15–20 min.

Success: Clear increase post-exercise, recovery trend noted.

B6.2 Aerobic vs anaerobic snapshot

Focus: Energy yield, products, uses.

Do: Two-column comparison + one real-life example of each (e.g., sprint vs jog).

Time: 10 min.

Success: Correct products ($\text{CO}_2/\text{H}_2\text{O}$ vs lactate/ethanol in yeast).

Transport in Plants & Humans

B7.1 Xylem vs phloem T-chart

Focus: Direction, content, structure, energy use.

Do: T-chart with four differences, one diagram.

Time: 10–12 min.

Success: "Transpiration stream" and "translocation" correctly used.

B7.2 Heart pathway trace

Focus: Double circulation.

Do: Trace a red blood cell path; label chambers/valves/vessels.

Time: 15–20 min.

Success: Correct sequence and oxygenation changes.

Coordination & Response

B8.1 Reflex arc postcard

Focus: Receptor → sensory → relay → motor → effector.

Do: Draw and caption each step; add why reflexes are fast.

Time: 10–15 min.

Success: Includes synapse and spinal cord role.

B8.2 Eye structure quick-label

Focus: Lens, iris, retina, optic nerve; accommodation & dim/bright reflex.

Do: Label and write two bullet points explaining accommodation.

Time: 10–12 min.

Success: Ciliary muscle/suspensory ligaments correctly explained.

Reproduction, Genetics & Inheritance

B9.1 Mitosis vs meiosis foldable

Focus: Purpose, chromosome number, variation.

Do: Create a foldable/2-column summary with one simple diagram each.

Time: 15–20 min.

Success: Correctly states “diploid → diploid” vs “diploid → haploid”.

B9.2 Punnett square warm-ups

Focus: Monohybrid crosses.

Do: Solve 3 quick genotype → phenotype problems; include % outcomes.

Time: 12–15 min.

Success: Percentages match square; key terms used (allele, dominant, recessive).

Ecology, Energy Flow & Human Impacts

B10.1 Food web sketch

Focus: Producer → consumer levels, arrows as energy transfer.

Do: Draw a 6-organism local food web; label trophic levels.

Time: 15–20 min.

Success: Arrows = “to what eats it”; decomposers included.

B10.2 Carbon cycle story strip

Focus: Processes: photosynthesis, respiration, combustion, decomposition.

Do: 6-frame comic showing a carbon atom’s journey.

Time: 15–20 min.

Success: At least four processes correctly shown.

Practical Skills & Study Habits

B11.1 Variables & graphing drill

Focus: IV/DV, control variables, axes, best-fit lines.

Do: Given any mini-investigation idea, write IV/DV, three controls, and sketch the expected graph shape.

Time: 10–15 min.

Success: Correct axes labels & sensible prediction.

B11.2 10-minute retrieval

Focus: Spaced recall.

Do: From memory, list 10 key terms this week; self-mark with notes in a different colour.

Time: 10 min.

Success: $\geq 8/10$ accurate by week's end.

Low-prep “Watch & Explain” (choose 1 per week)

Focus: Turn passive watching into active learning.

Do: Watch a short clip on a current biology topic (e.g., vaccines, biodiversity). Write **3 bullet takeaways** + **1 exam-style question** you could ask about it.

Time: 15–20 min.

Success: Takeaways use correct terminology; question targets “explain”/“compare”.

How to Use (teacher/student)

Weekly recipe: 1× knowledge builder (diagram/definitions) + 1× mini-practical or data task + 1× retrieval/reflection.

Evidence: Keep pages/photos in a “Bio Basics” folder; title, date, success criteria ticked.

Stretch next: When these are easy, move to **Extended** supercurriculum (longer practicals, data analysis, case studies).

If you want, I can convert this into a printable one-pager or load it into your app's content format (Markdown with tags for **module=super, subject=Biology, tier=basic, skills**).

Είπες:

extended

To ChatGPT είπες:

Here's a **Year 10 Biology — Extended Supercurriculum** set, designed for 45–90 minute tasks that deepen understanding through inquiry, modelling, data-handling, and real-world context. Each task includes **Focus, Brief, Materials, Time, Safety, Deliverables**, and **Success Criteria / Rubric**. Use 1–2 per week alongside classwork.

Biology — Extended Supercurriculum (Year 10)

Cells & Microscopy

E1. Quantifying cell size with scale bars

Focus: Magnification, scale, measurement reliability.

Brief: Using micrographs (textbook/online), calculate real sizes of at least **5 different cells** with scale bars; compare variability and discuss sampling error.

Materials: Printed/onscreen micrographs, ruler, calculator/Sheet. **Time:** 45–60 min.

Deliverables: Table of measurements; graph (box plot or bar with error bars); 150–200-word commentary on error sources.

Success / Rubric: Correct unit conversions; justified error discussion; neat graph with axes/units.

E2. Microscopy method critique

Focus: Method evaluation & improvements.

Brief: Read a simple light-microscope method and **rewrite** it to improve precision (staining, focusing, field of view sampling).

Deliverables: Revised SOP (≤ 1 page) + checklist.

Success: Clear variables/controls; steps minimise bias; safety for stains noted.

Biological Molecules & Enzymes

E3. Enzyme activity data story (virtual or practical)

Focus: Enzyme–temperature or pH relationships.

Brief: Use provided data (or run a safe amylase–starch test if permitted) to plot rate vs temperature **and** identify optimum + denaturation region. Fit a curve or describe trend mathematically.

Materials: Data set or iodine/amylase (if practical). **Time:** 60–90 min.

Deliverables: Graph with best-fit; 200-word explanation linking **active site** + **kinetic energy**.

Success: Correct interpretation of optimum/denaturation; variables/controls acknowledged.

E4. Food tests mini-review

Focus: Specificity & limitations.

Brief: Create a one-page infographic comparing Benedict's, Biuret, Iodine, and Emulsion tests, including **false positives/negatives** and real-world use.

Success: Accurate chemistry rationale; clear, exam-ready visuals.

Movement of Substances

E5. Modelling diffusion & SA:V with math

Focus: Surface area:volume and diffusion time.

Brief: Model three “cells” (1 cm, 2 cm, 3 cm cubes). Calculate SA:V and **predict** relative times for diffusion to the centre. Validate with dyed gelatin (if allowed) or justify theoretically.

Deliverables: Calculations; photo/notes of test; claim–evidence–reasoning paragraph.

Success: Correct SA:V maths; logical link from model → living cells.

E6. Osmosis plan → analysis

Focus: Experimental design & expected data.

Brief: Write a full plan for potato osmosis **with calibration curve** (mass change vs sucrose concentration). Include a sketched **expected** graph and how you'd interpolate plasmolysis point.

Success: Variables, repeats ($n \geq 3$), % change formula, predicted curve shape, interpolation method.

Plant Nutrition & Transport

E7. Photosynthesis rate inquiry (light or CO₂)

Focus: Limiting factors.

Brief: Design an investigation using **Elodea** (or a data set) to test how light intensity or bicarbonate concentration affects bubble rate. Address **control of temperature** and counting error.

Deliverables: Method, results table/graph, conclusion referencing **limiting factor** concept.

Success: Sound controls; valid conclusion; uncertainty considered.

E8. Xylem vs phloem case study

Focus: Structure–function & transport evidence.

Brief: Prepare a 400-word case study explaining how structure supports function in xylem and phloem, including **ringing experiments** or aphid stylet evidence.

Success: Uses evidence and correct terminology (lignin, cohesion-tension, translocation, source-sink).

Human Nutrition & Digestion

E9. Enzyme optimisation for digestion

Focus: pH/temperature optimisation, validity.

Brief: Propose a controlled experiment to find optimal pH for **amylase** using colorimetric end-point (iodine). Include **calibration** and **repeatability** plan; discuss ethical/safety considerations.

Deliverables: Full plan + risk assessment; template results table; how you'd analyse rate.

Success: Operational definition of “rate”; sensible pH range; control of temp and enzyme concentration.

E10. Diet audit & evidence check

Focus: Evaluating nutrition claims.

Brief: Pick one common diet claim (e.g., “high-protein breakfasts improve focus”). Summarise **biological rationale** and limitations in 300 words using at least **two sources**; propose a simple trial.

Success: Balanced reasoning; identifies confounders; clear, testable hypothesis.

Respiration & Gas Exchange

E11. Exercise physiology mini-investigation

Focus: Heart rate, breathing, recovery curves.

Brief: Collect **your own** heart/breathing rate at rest, post-exercise (30–60 s), and during 5-min recovery (1-min intervals). Plot both; compare recovery constants; discuss fitness/confounders.

Deliverables: Dual-axis graph, 200-word analysis.

Success: Clear trend; recovery interpretation; limitations noted (sample size, measurement error).

E12. Yeast respiration data analysis

Focus: Aerobic vs anaerobic, rate inference.

Brief: Given CO₂ production data from yeast in different sugar solutions, identify which substrate yields fastest rate and **explain** in terms of metabolic pathways.

Success: Correct ranking + biochemical reasoning.

Transport in Humans

E13. Blood & circulation modelling

Focus: Pressure/flow relationships.

Brief: Model blood flow using syringes/tubing of differing diameters (or simulate with data). Predict how diameter changes affect flow (Poiseuille idea, qualitatively).

Deliverables: Diagram, predictions, results/discussion.

Success: Correct qualitative link: **smaller radius** → **much lower flow**; relates to atherosclerosis.

E14. Oxygen transport & exercise

Focus: Haemoglobin affinity concept.

Brief: Interpret a standard oxygen dissociation curve and explain how **exercise** shifts oxygen delivery (Bohr effect, conceptual level). Create a one-page explainer.

Success: Correct axes & shift explanation in plain language.

Coordination, Hormones & Homeostasis

E15. Reflexes vs conscious responses

Focus: Neural pathways & speed.

Brief: Measure simple reaction time (apps/ruler drop) across **3 conditions** (no distraction, auditory distraction, visual distraction). Analyse means and variability; discuss synaptic delay.

Deliverables: Data table, graph, 150-word analysis.

Success: Fair testing; mentions synapses and attention.

E16. Blood glucose control storyboard

Focus: Homeostasis, antagonistic hormones.

Brief: Create a 6-panel storyboard showing insulin and glucagon roles, including **negative feedback** and what fails in Type 1 vs Type 2 diabetes.

Success: Accurate pathways; lifestyle/physiology contrast.

Reproduction, Genetics & Inheritance

E17. Modelling inheritance & risk

Focus: Monohybrid + sex-linked traits.

Brief: Build Punnett squares for three traits (one sex-linked). For each, give genotype ratios, phenotype percentages, and a short counselling note to hypothetical parents.

Deliverables: Worked squares + 3× ~80-word notes (sensitive tone).

Success: Correct ratios; clear, empathetic explanations.

E18. DNA → protein mini-model

Focus: Transcription/translation, mutations.

Brief: Use a paper bead/letter code to transcribe/translate a short gene. Introduce a substitution and a frameshift; predict protein impact.

Success: Correct reading frame logic; sensible phenotype predictions.

Ecology & Human Impacts

E19. Quadrat or photo-sampling study (virtual OK)

Focus: Sampling techniques, validity.

Brief: Using **photos** of a small area (or actual quadrats), estimate % cover or species richness at 5 locations; compute mean & range; discuss sampling bias and how to improve.

Deliverables: Map or photo grid; table; mean/range; 200-word evaluation.

Success: Appropriate metrics; honest limitations; improvement plan.

E20. Carbon cycle systems map

Focus: Stores & fluxes; human impact.

Brief: Draw a systems diagram with major **stores** (atmosphere, biosphere, oceans, fossil fuels) and **fluxes** (photosynthesis, respiration, combustion, decomposition). Add **two human interventions** and predict effects.

Success: Correct arrows/labels; plausible intervention effects.

Practical Skills & Data Handling

E21. From messy data to clear graph

Focus: Data cleaning, error bars, best fit.

Brief: Given a “messy” data set (typos, outliers, missing units), clean it; justify outlier handling; plot with **error bars**; write a 150-word “results” paragraph in scientific style.

Success: Transparent cleaning steps; correct units; readable figure; cautious claims.

E22. Design a fair test (proposal)

Focus: Variables, controls, repeatability.

Brief: Propose a small biology investigation you could do at home/school. Include hypothesis, variables, risk assessment, method, **sample size**, and predicted graph.

Success: Feasible, safe, and testable; clear IV/DV; repeats ≥ 3 .

Cross-Unit Applications

E23. Medical case note (integrated)

Focus: Applying multiple topics.

Brief: Write a 300-word case note for a patient with **anaemia** or **asthma**, integrating transport, respiration, and homeostasis. Include symptoms, biology explanation, and lifestyle management ideas.

Success: Accurate mechanisms; realistic advice; scientific tone.

E24. Bioethics position paragraph

Focus: Evidence-based argument.

Brief: Choose a topic (e.g., CRISPR in agriculture, vaccination policies).

Write **250–300 words** presenting **both sides** and a reasoned conclusion supported by biology.

Success: Balanced evidence; clear conclusion; correct key terms.

Marking Mini-Rubric (apply to any Extended task)

A. Biology Accuracy (0–4): Key terms correct; mechanisms explained.

B. Methods & Data (0–4): Variables/controls, repeatability, graph quality.

C. Reasoning (0–4): Claim–evidence–reasoning clear; limitations acknowledged.

D. Communication (0–4): Structure, units, labels, concise style.

E. Reflection (0–4): What went well/next steps; links to prior learning.

(*Excellent* = 16–20; *Secure* = 11–15; *Developing* = 6–10; *Emerging* = 0–5)

Evidence to Upload (for each task)

Photos/scans of work, graphs or spreadsheets, 1–2 minute audio reflection (optional), and the final summary paragraph.

Tips for Use

Aim for **one Extended** task weekly.

Rotate domains (cells → enzymes → transport → genetics → ecology) to spiral knowledge.

Struggle = normal: note **one confusion** and **one fix** in your reflection.

When these become comfortable, step up to **Guide** supercurriculum (mini-investigations, literature snippets, and synoptic reports).

Είπες:

guide

Είπες:

guide

To ChatGPT είπε:

Here's the **Year 10 Cambridge IGCSE Biology — Guide Supercurriculum**, the highest level of the three-tier framework.

These activities extend well beyond the classroom, combining **independent research, inquiry, synthesis, and communication**.

They develop higher-order skills: critical evaluation, experimental design, and scientific literacy — ideal preparation for IGCSE Paper 6 and later A-Level work.

Biology — Guide Supercurriculum (Year 10)

The Living Cell: Microscopic Worlds

G1.1 Research project – “Seeing the Invisible”

Focus: Technological advances in microscopy and their biological impact.

Task: Trace the development from Hooke's light microscope to modern electron and confocal imaging. Explain **how each advance changed our understanding of cell structure or disease**.

Deliverables: 800–1000 word illustrated report or 5-slide presentation.

Success: Accurate timeline, link between instrument and discovery (e.g., mitochondria, viruses, Golgi).

Extension: Interview a lab technician or watch a virtual electron microscopy tour (e.g., HHMI Biointeractive).

G1.2 Independent investigation – “Life at the edge of size”

Focus: Surface area : volume constraints.

Task: Model diffusion limits with your own simulation or spreadsheet; predict maximum sustainable cell sizes for given diffusion coefficients.

Deliverables: Spreadsheet, graph, 300 word interpretation.

Skills: Quantitative modelling, evaluation of biological limits.

Molecules of Life

G2.1 Literature review – Enzymes in industry

Focus: Applications of enzymes (biotechnology, detergents, food processing).

Task: Read at least three scientific or trade sources; summarise enzyme uses, advantages, environmental considerations.

Deliverables: 1000 word article with bibliography (APA).

Success: Balanced evaluation of efficiency, cost, and sustainability.

G2.2 Design thinking – “Your own enzyme lab”

Focus: Experimental design & ethics.

Task: Propose a safe school-level investigation comparing enzyme activity with an accessible indicator (e.g., pineapple protease and gelatin).

Deliverables: Full proposal with variables, risk assessment, expected results, ethical note.

Success: Feasible design with control logic and clear hypothesis.

Transport and Physiology

G3.1 Case study – “The heart under pressure”

Focus: Cardiovascular adaptation and disease.

Task: Compare athlete vs sedentary heart morphology and physiology using credible sources; interpret cardiac output, pulse, and stroke volume data.

Deliverables: 700 word comparative report with one annotated diagram.

Success: Clear data interpretation, use of correct units, and reasoning.

G3.2 Practical extension – Blood viscosity and flow

Focus: Circulatory system mechanics.

Task: Model flow rate through capillary-like tubing using liquids of different viscosities (e.g., water + syrup). Measure time to flow fixed volume; plot rate vs viscosity.

Deliverables: Data table, graph, evaluation.

Safety: Manage spills; eye protection.

Success: Quantitative pattern explained in terms of friction and vessel radius.

Energy and Respiration

G4.1 Data analysis – “Respiration in context”

Focus: Interpreting real biological data.

Task: Use a published dataset (yeast CO₂ production, or pulse oximeter logs) to identify patterns, anomalies, and correlations.

Deliverables: Graphs, summary statistics, 400 word commentary linking to aerobic/anaerobic pathways.

Success: Logical trend analysis; connection to enzyme kinetics and oxygen demand.

G4.2 Extended essay – “The future of cellular energy”

Focus: Mitochondrial efficiency, biotechnology and medicine.

Task: Discuss how understanding cellular respiration informs modern treatments (e.g., mitochondrial therapy, biofuel cells).

Deliverables: 1000 word essay with two cited journal references.

Skills: Research synthesis; scientific communication.

Coordination and Homeostasis

G5.1 Simulation study – “Neural networks: nature vs machines”

Focus: Comparing biological and artificial networks.

Task: Explain the parallels between the nervous system and simple AI neural networks. Create a flow diagram showing inputs, processing, outputs, and feedback.

Deliverables: Diagram + 600 word explanatory brief.

Success: Correct biological analogies; balanced discussion of similarities and limits.

G5.2 Investigation – Temperature regulation challenge

Focus: Homeostasis in action.

Task: Devise a small experiment measuring cooling rates of containers simulating endotherm insulation (fur, feathers, fat). Analyse which variable most reduces heat loss.

Deliverables: Plan, graph, evaluation.

Skills: Hypothesis testing, error analysis, applied physiology.

Reproduction and Genetics

G6.1 Project – “Mapping inheritance”

Focus: Pedigree and genetic counselling.

Task: Construct a hypothetical family pedigree for a single-gene disorder. Calculate genotype probabilities, identify carriers, and discuss ethical issues in screening.

Deliverables: Diagram + 600 word commentary.

Success: Accurate probabilities; ethical awareness.

G6.2 Bioinformatics introduction

Focus: Reading real DNA data.

Task: Use an online DNA translation or BLAST tool to compare a simple gene sequence (e.g., hemoglobin) across species. Summarise evolutionary implications.

Deliverables: Screenshot, short report.

Skills: ICT, data interpretation, evolution concept application.

Ecology and Environment

G7.1 Field project – “Urban biodiversity audit”

Focus: Quantifying biodiversity and human impact.

Task: Select two contrasting local areas (e.g., park vs street). Record species richness or indicator groups (plants, insects, birds) with quadrat or photo-sampling. Calculate a simple biodiversity index and interpret differences.

Deliverables: Map, data tables, index calculation, 400 word discussion.

Success: Sound sampling method; clear link between data and human influence.

G7.2 Policy brief – “Climate, carbon, and you”

Focus: Human impacts & sustainability.

Task: Research one local environmental issue (deforestation, water use, pollution) and propose three evidence-based interventions.

Deliverables: 1-page policy brief (≈500 words) for local council or school board.

Success: Concise, factual, actionable recommendations; correct science.

Bioethics, Society & Communication

G8.1 Opinion editorial – “Should we edit genes?”

Focus: Ethical reasoning and communication.

Task: Write a persuasive 800 word op-ed discussing the promise and risks of gene editing in humans or crops, citing at least two sources.

Success: Balanced tone; evidence from genetics; clear stance.

G8.2 Science communication piece

Focus: Making complex biology accessible.

Task: Create a 3-minute podcast, infographic, or short video explaining a biological idea from the syllabus (e.g., enzymes, immunity) for a younger audience.

Deliverables: Media file + 150 word script summary.

Success: Accurate, engaging, age-appropriate explanation.

Cross-disciplinary & Future Skills

G9.1 STEAM crossover – “Design a bio-solution”

Focus: Applying biology to innovation.

Task: Identify a small community problem (plastic waste, local water quality, food preservation) and propose a biologically informed solution. Include feasibility, cost, and sustainability.

Deliverables: 1-page concept proposal + sketch/model.

Success: Scientific reasoning; creative practicality.

G9.2 Data storytelling – “Numbers behind life”

Focus: Data literacy & communication.

Task: Take a biological dataset (population, disease rates, climate effect on species) and produce an interactive or visual story using graphs or dashboards.

Deliverables: Visualisation + 300 word commentary.

Skills: Data presentation, clarity, narrative thinking.

Reflective Synthesis

G10.1 “Biology and Me” portfolio reflection

Focus: Metacognition and learning narrative.

Task: Compile highlights from Basic + Extended + Guide work. Write a 600 word reflection addressing:

What I learned about how I learn science.

Which topic most changed how I see the world.

Next goals and skills to develop.

Deliverables: Portfolio (PDF or folder) + reflection.

Success: Honest insight; clear self-evaluation; evidence of growth.

Assessment & Teacher Guidance

Criterion	Indicators	Level (0-4)
Scientific accuracy	Correct use of concepts, data, and units	0-4
Inquiry design / reasoning	Hypothesis, evidence, evaluation	0-4
Critical thinking	Links across topics, identifies uncertainty	0-4
Communication	Organisation, clarity, referencing	0-4
Creativity & reflection	Novel approach, self-assessment	0-4

(Excellent = 17-20 pts; Good = 13-16; Developing = 8-12; Emerging < 8)

Implementation Notes

Complete one **Guide** project per half-term or as an **independent study module**.

Encourage use of the **school’s AI Tutor** for research prompts, drafting outlines, and feedback loops.

Combine digital (poster, podcast, report) and practical outputs to build a well-rounded **portfolio for university or scholarship applications**.

Combined Sciences — Basic Supercurriculum (Year 10)

1) The Nature of Science

CS1.1 “What counts as science?”

Focus: Scientific method & evidence.

Task: Pick any everyday claim (e.g. “energy drinks improve focus”). Write a 5-step plan to test it scientifically.

Time: 15–20 min.

Success: Identifies variables, control group, and measurable data.

CS1.2 “Famous failure → discovery”

Focus: The role of error in science.

Task: Research one accidental discovery (e.g. penicillin, X-rays, microwave oven). Summarise what went wrong and what was learned.

Time: 20–25 min.

Success: Clear connection between observation and innovation.

2) Biology Core

CS2.1 “Life under the lens”

Focus: Cell structure & function.

Task: Sketch and label plant and animal cells; colour-code organelles for their roles.

Time: 15–20 min.

Success: At least 5 organelles labelled correctly (e.g. nucleus, cytoplasm, chloroplast).

CS2.2 “Breathe in, breathe out”

Focus: Human gas exchange.

Task: Record your breathing rate before and after 30 seconds of movement; describe the change.

Time: 10–15 min.

Success: Identifies oxygen/carbon dioxide link and body’s need for energy.

CS2.3 “Plants need sunlight”

Focus: Photosynthesis conditions.

Task: Place a potted plant half in shade, half in light for a day; record changes in leaf colour/angle.

Time: 5 min setup + observation.

Safety: Avoid direct sunlight burn.

Success: Recognises light as an essential factor.

3) Chemistry Core

CS3.1 “Matter matters”

Focus: Particle model.

Task: Draw particles in solids, liquids, and gases. Explain how heating changes motion.

Time: 10–15 min.

Success: Correct particle spacing and motion arrows.

CS3.2 “The mystery mixture”

Focus: Separating substances.

Task: Mix salt and sand; describe how to separate them (filter, evaporate).

Time: 15–20 min.

Success: Correct sequence: dissolve → filter → evaporate.

CS3.3 “Acid or alkali?”

Focus: pH basics.

Task: Use universal indicator (or red cabbage extract) to test common liquids.

Time: 20 min.

Safety: Wear gloves, avoid contact with unknown liquids.

Success: Classifies correctly as acid (<7), neutral (=7), base (>7).

4) Physics Core

CS4.1 “Falling faster?”

Focus: Gravity and air resistance.

Task: Drop two paper shapes (flat vs crumpled); note which lands first and why.

Time: 10 min.

Success: Links shape to air resistance.

CS4.2 “Light and shadows”

Focus: Straight-line travel of light.

Task: Shine a torch through a cardboard cut-out; move torch and observe shadow shape changes.

Time: 10–15 min.

Success: Predicts shadow size based on light distance.

CS4.3 “Sound waves visible”

Focus: Vibration and sound transmission.

Task: Place rice on a stretched film over a speaker; play music and observe vibration.

Safety: Low volume only.

Success: Explains how vibrations carry sound energy.

5) Energy & Change

CS5.1 “Where does the energy go?”

Focus: Energy transfer and conservation.

Task: Choose one machine (fan, car, phone). Draw an energy flow diagram showing input → useful output → wasted energy.

Time: 15–20 min.

Success: Identifies energy types accurately (electrical → kinetic + heat).

CS5.2 “Heat on the move”

Focus: Conduction, convection, radiation.

Task: Write one everyday example of each heat transfer type.

Success: Each example matches process (e.g. metal spoon → conduction).

CS5.3 “Renewables in my area”

Focus: Sustainability.

Task: Identify one local renewable energy source (solar panels, wind turbines).

Note one advantage & one limitation.

Success: Balanced evaluation with correct terminology.

6) Earth & Environment

CS6.1 “Water, water everywhere”

Focus: Water cycle processes.

Task: Sketch and label the water cycle; colour condensation, precipitation, evaporation.

Time: 15 min.

Success: All main processes labelled correctly.

CS6.2 “Human impact snapshot”

Focus: Pollution & sustainability.

Task: Take a photo or draw an example of pollution nearby; describe cause, effect, and a possible solution.

Time: 15–20 min.

Success: Logical cause–effect–solution explanation.

CS6.3 “Carbon footprint tracker”

Focus: Energy use awareness.

Task: Estimate your household’s daily energy activities (transport, electricity use). Suggest two ways to reduce carbon output.

Success: Specific, realistic improvements.

7) Thinking Like a Scientist

CS7.1 “Variables challenge”

Focus: Independent, dependent, and control variables.

Task: Choose any mini-experiment idea (e.g., plant growth vs light). Identify all three variable types.

Success: Correctly categorises variables.

CS7.2 “Science in the news”

Focus: Application of scientific literacy.

Task: Read one current science article. Summarise the main finding in 3 sentences and one question it raises.

Success: Uses clear language and relates to a science concept learned.

CS7.3 “Graph it right”

Focus: Graphing and data representation.

Task: Using any numeric data (weather, sports, energy), draw a simple graph with correct axes and labels.

Success: Appropriate scale, labels, and line/columns used.

8) Study Habits & Curiosity

CS8.1 “One question a day”

Focus: Active curiosity.

Task: Write one new science question each day. Try to answer three at week's end using books or videos.

Success: Shows consistent curiosity and follow-up.

CS8.2 "Mini revision flashcards"

Focus: Retrieval practice.

Task: Make 10 question-answer cards (definition, process, formula). Test yourself twice a week.

Success: Improved recall score over time.

CS8.3 "Science & everyday life"

Focus: Transfer of knowledge.

Task: Pick an everyday object (bike, fridge, smartphone). Identify at least three science principles that make it work.

Success: Accurate identification (e.g., energy transfer, circuits, magnetism).

✓Success Rubric (for all Basic tasks)

Level	Indicators
Emerging (1-2)	Attempted task, partial understanding shown.
Developing (3)	Accurate explanation, clear diagram/data, minor errors.
Secure (4)	Correct science, reflection evident.
Excellent (5)	Clear connection to real life or thoughtful extension question.

Implementation Tips

Goal: 3 Basic activities per week (one each from Biology, Chemistry, Physics).

Evidence: Photos, short written notes, or 1-minute video reflections.

Progression: After completing all core sections, move to **Extended**

Supercurriculum, where students plan small investigations and analyse data.

Combined Sciences — Extended Supercurriculum (Year 10)

1) Scientific Thinking & Method

E1.1 Experimental design challenge – “Testable or not?”

Focus: Variables, fair testing, reproducibility.

Task: Write three scientific questions. Decide which are testable and redesign one into a **clear hypothesis** with IV, DV, and controls.

Deliverables: Table of questions, one complete plan.

Success: Identifies measurable variables and control conditions.

E1.2 Data detective

Focus: Data interpretation and uncertainty.

Task: Given a small dataset (e.g., reaction times, temperature change), plot a graph and calculate mean, range, and percentage error.

Deliverables: Graph + 100-word commentary.

Success: Graph correctly labelled; explains reliability limits.

2) Biology Strand

E2.1 Enzyme rate investigation (simulation or practical)

Focus: Effect of temperature or pH on enzyme activity.

Task: Use an online enzyme simulation or safe experiment (e.g., pineapple juice & gelatin) to measure reaction time at different conditions.

Deliverables: Results table, graph, short conclusion.

Success: Recognises optimum and denaturation; controlled variables identified.

E2.2 Heart rate and fitness mini-study

Focus: Human physiology and data handling.

Task: Record resting and post-exercise pulse rates for at least **three volunteers**. Plot recovery curves; compare age, activity level.

Deliverables: Graph + 200-word explanation.

Success: Interprets trends sensibly; mentions oxygen demand and respiration.

E2.3 Ecology at your doorstep

Focus: Sampling and biodiversity.

Task: Choose two nearby habitats (lawn vs paved area). Estimate species variety using quadrats or photos; calculate % cover.

Deliverables: Sketch map, data table, reflection.

Success: Uses fair sampling; suggests reason for difference in biodiversity.

3) Chemistry Strand

E3.1 Reaction speed exploration

Focus: Collision theory.

Task: Use effervescent tablets cut into different sizes or crushed vs whole. Time CO₂ bubble release in water.

Deliverables: Table and graph of time vs surface area.

Success: Correctly explains “more collisions → faster reaction”.

E3.2 Investigating neutralisation (safe setup)

Focus: Acids, bases, indicators.

Task: Test small samples of vinegar + baking soda in varied ratios. Record fizz duration; infer which mixture is closest to neutralisation.

Deliverables: Table, bar chart, 150-word discussion.

Safety: Eye protection; small quantities only.

Success: Links reactant ratio to acid-base balance.

E3.3 Density and materials comparison

Focus: Physical properties and measurement.

Task: Weigh and measure volume of 3–4 household materials (wood, metal, plastic). Calculate density ($\rho = m/v$).

Deliverables: Table + comment on which floats/sinks and why.

Success: Accurate measurements; clear comparison.

4) Physics Strand

E4.1 Stretch and force mini-lab

Focus: Hooke's Law.

Task: Hang small weights from an elastic band; measure extension each time. Plot force vs extension.

Deliverables: Graph and conclusion.

Safety: Don't overstretch; goggles optional.

Success: Recognises proportional region; mentions limit of elasticity.

E4.2 Heat transfer investigation

Focus: Conduction and insulation.

Task: Wrap identical bottles with different materials (foil, cloth, plastic). Measure temperature change over 15 minutes with warm water inside.

Deliverables: Line graph + evaluation.

Success: Identifies best insulator; links material structure to heat flow.

E4.3 Light refraction test

Focus: Refraction and Snell's Law.

Task: Shine light through water or glass block (use phone torch). Measure incident/refraction angles.

Deliverables: Diagram, angle table, ratio $n = \sin i / \sin r$.

Success: Recognises refraction; calculates approximate refractive index.

5) Earth & Environmental Science

E5.1 Water quality survey (safe observation)

Focus: Environmental indicators.

Task: Visit two local water sources (tap, pond, stream). Record observations (clarity, odour, litter) and research what indicates contamination.

Deliverables: Checklist, 200-word summary.

Success: Uses correct indicators (turbidity, smell, presence of algae).

E5.2 Carbon cycle poster

Focus: Energy & matter flow.

Task: Design an annotated poster explaining the carbon cycle with arrows for respiration, photosynthesis, and combustion.

Deliverables: A4 diagram with short notes.

Success: Clear arrows; correct direction and terminology.

E5.3 Renewable energy research

Focus: Energy transformation & sustainability.

Task: Research one renewable energy source used in your region. Create a short fact sheet describing **how it works**, its **advantages**, and one limitation.

Deliverables: 300-word brief.

Success: Balanced evaluation with correct physics principles.

6) Interdisciplinary Tasks

E6.1 “The Science of Sports”

Focus: Applying physics, biology, and chemistry.

Task: Choose a sport (e.g., football). Analyse one aspect from each science:

Physics: motion, force, angle of impact

Biology: respiration and muscle fatigue

Chemistry: energy metabolism (ATP).

Deliverables: 400-word report or poster.

Success: Integrates at least three disciplines coherently.

E6.2 “From atom to ecosystem” concept map

Focus: Systems thinking.

Task: Create a concept map showing how **atoms** → **molecules** → **cells** → **organisms** → **ecosystems** are connected by energy and matter flow.

Deliverables: Visual map + one-paragraph summary.

Success: Logical hierarchy; consistent linking terms.

7) Data Handling & Communication

E7.1 “Analyse the graph”

Focus: Interpreting real data.

Task: Use a real-world dataset (temperature trends, disease rates, or reaction data). Write a short summary explaining correlation, anomalies, and prediction.

Deliverables: Annotated graph + 200-word commentary.

Success: Identifies trend and anomalies; avoids over-generalisation.

E7.2 “Mini science report”

Focus: Communicating scientific findings.

Task: Choose any completed Extended task and write it up using **Introduction–Method–Results–Conclusion** format (IMRC).

Deliverables: 1-page lab report.

Success: Clear headings, concise style, scientific tone.

8) Reflection & Skills Development

E8.1 “Learning from failure”

Focus: Metacognition and scientific resilience.

Task: Pick one investigation that didn't go as planned. Write a 150-word reflection explaining what went wrong and how to improve it.

Success: Honest reflection; evidence of problem-solving.

E8.2 "Career connection"

Focus: Science relevance to future study/work.

Task: Research one STEM career (lab technician, engineer, biologist). Describe what school science topics relate most to that role.

Deliverables: 200-word career note.

Success: Links subject knowledge to real-world application.

✓Mini-Rubric for Extended Tasks

Criterion	Description	Score (0-4)
Scientific accuracy	Uses correct theory and key terms	0-4
Method & data quality	Sound plan, clear measurements, repeatability	0-4
Reasoning & analysis	Interprets results logically	0-4
Communication	Neat tables, labelled graphs, concise writing	0-4
Reflection	Evaluates limitations, suggests improvements	0-4

Excellent: 16-20 | **Secure:** 12-15 | **Developing:** 8-11 | **Emerging:** <8

Implementation Tips

Time: One Extended task per week or every two weeks.

Evidence: Written work, graphs, photos, voice/video reflections.

Feedback: Use rubric above for formative assessment.

Progression: After completing Extended, move to **Guide**

Supercurriculum → student-designed mini research projects and science communication tasks.

Combined Sciences — Guide Supercurriculum (Year 10)

1) Inquiry, Innovation & the Scientific Method

G1.1 Independent Investigation – “Design Your Own Experiment”

Focus: Experimental design from idea to evaluation.

Task: Choose any scientific question (e.g., “How does caffeine affect heart rate?” or “Which material insulates best?”).

Write a hypothesis.

Design fair tests with variables and repeats.

Collect and present data.

Conclude with evaluation of errors and validity.

Deliverables: Full IMRC (Introduction–Method–Results–Conclusion) report.

Success: Original question; measurable data; balanced evaluation.

Stretch: Compare your conclusion with published research.

G1.2 Meta-science Reflection – “What Makes Good Science?”

Focus: Nature and ethics of scientific knowledge.

Task: Analyse one real-world case where science had ethical or societal impact (e.g., gene editing, climate modelling, vaccines).

Deliverables: 600–800 word essay.

Success: Balanced discussion of evidence, ethics, and uncertainty.

2) Biology Investigations

G2.1 Research Project – “The Human Microbiome”

Focus: Symbiosis and human health.

Task: Research how gut bacteria affect immunity, digestion, and mental health. Present as a written summary or video explainer.

Deliverables: 800 word article or 3-min video.

Success: References credible sources; shows cause–effect reasoning.

G2.2 Fieldwork Study – “Biodiversity Index in Urban Spaces”

Focus: Ecosystem health and human influence.

Task: Compare species diversity in two local microhabitats (e.g., grass vs pavement, park vs roadside).

Record species count.

Calculate a simple biodiversity index.

Discuss causes and sustainability links.

Deliverables: Report with tables, graph, and photo evidence.

Success: Accurate sampling; clear link to environmental factors.

G2.3 Bioethics Discussion – “Should We Edit Genes?”

Focus: Modern genetics and ethics.

Task: Debate advantages and dangers of genetic modification in humans or crops.

Deliverables: 800 word balanced essay or 5-min debate recording.

Success: Arguments supported by evidence and ethical reasoning.

3) Chemistry Investigations

G3.1 Green Chemistry Design – “Eco-friendly Reactions”

Focus: Sustainable chemistry principles.

Task: Research how chemists reduce waste, save energy, and replace toxic reactants. Propose a school-level demonstration of a “green” reaction (e.g., baking soda + vinegar as CO₂ source).

Deliverables: 1-page plan and diagram; explanation of “greener” aspects.

Success: Identifies atom economy, safety, energy efficiency.

G3.2 Chemical Patterns Research – “Why Do Elements React?”

Focus: Periodic trends.

Task: Investigate how one property (melting point, density, reactivity) changes down a group or across a period. Use reliable online data, plot a graph, interpret pattern.

Deliverables: Graph + 400 word interpretation.

Success: Explains pattern with reference to atomic structure.

G3.3 Real-world Chemistry – “Materials of the Future”

Focus: Nanotechnology and materials science.

Task: Research a modern material (graphene, aerogel, bioplastics). Explain its properties, uses, and environmental impact.

Deliverables: Poster or one-page infographic.

Success: Correct science + real-world relevance.

4) Physics Investigations

G4.1 Mini Engineering Project – “Energy in Motion”

Focus: Kinetic and potential energy conversion.

Task: Build a small model (e.g., marble ramp, elastic launcher, pendulum).
Measure height and distance.

Calculate energy transfer efficiency.

Deliverables: Photos, data table, calculations, evaluation.

Safety: Eye protection; stable setup.

Success: Quantitative link between energy input and output.

G4.2 The Science of Sound – “Visualising Vibrations”

Focus: Wave properties and frequency.

Task: Use a phone frequency app and vibration setup (speaker + rice on plastic film). Observe how pitch and amplitude affect vibration.

Deliverables: Diagram + short report or video clip.

Success: Relates pitch to frequency and amplitude to loudness.

G4.3 Climate Physics – “Global Energy Balance”

Focus: Energy transfer and greenhouse effect.

Task: Create a simple heat model (light on dark vs light surfaces). Measure temperature change and relate to albedo and absorption.

Deliverables: Data + 300 word discussion.

Success: Explains greenhouse mechanism with evidence.

5) Earth & Environmental Systems

G5.1 Research Project – “Earth’s Changing Atmosphere”

Focus: Historical and modern data interpretation.

Task: Collect CO₂ data from NASA or NOAA. Graph 1960–present; identify trends and possible causes.

Deliverables: Line graph + 400 word summary.

Success: Correctly identifies trend, links to fossil fuels and feedback loops.

G5.2 Practical Design – “Filtering Water Naturally”

Focus: Separation and sustainability.

Task: Design a simple water filter using sand, gravel, and charcoal. Compare clarity before and after filtration.

Deliverables: Sketch, photos, observations, evaluation.

Success: Clear before/after results; correct filtration principle.

G5.3 Sustainability Proposal – “A Greener School”

Focus: Energy conservation & environmental responsibility.

Task: Audit your school or home for waste, water, and electricity use.

Suggest at least **three improvements** with measurable outcomes.

Deliverables: 1-page action plan with data.

Success: Realistic recommendations, quantitative targets.

6) Interdisciplinary Projects

G6.1 “Science of Everyday Life” Investigation

Focus: Integrating physics, chemistry, and biology.

Task: Pick an everyday system (e.g., refrigerator, washing machine, cooking). Explain the **biology**, **chemistry**, and **physics** principles involved.

Deliverables: Poster, 600 word essay, or video explainer.

Success: All three sciences accurately linked and explained.

G6.2 “From Atom to Ecosystem” Extended Concept Map

Focus: Systems thinking and interconnectedness.

Task: Build a large mind map or flowchart showing how energy and matter flow through atoms, molecules, cells, organisms, and ecosystems.

Deliverables: Visual map + 200 word narrative summary.

Success: Logical connections; use of correct key terms.

G6.3 “Science & Society” Debate

Focus: Science literacy and ethics.

Task: Choose a current issue (AI, renewable energy, climate change, medical trials). Research both sides, then conduct or record a structured debate.

Deliverables: Debate transcript or recording, 200 word reflection.

Success: Balanced evidence, respectful reasoning, accurate science.

7) Communication, Reflection, and Leadership

G7.1 “Science Journalism”

Focus: Writing for the public.

Task: Select a new discovery (NASA, WHO, Nature). Rewrite it as a short article for a teen audience with headline, facts, and implications.

Deliverables: 400–600 word article + source list.

Success: Accurate yet accessible writing style.

G7.2 “Create a Mini Documentary”

Focus: Storytelling and science communication.

Task: Produce a 2–4 minute video explaining a science process (e.g., acid rain, circuits, enzymes).

Deliverables: Video + 150 word script outline.

Success: Clear, accurate explanation with visual evidence.

G7.3 “Mentor Moment”

Focus: Peer leadership.

Task: Choose one Extended activity you mastered and design a 10-minute mini lesson to teach it to a younger student.

Deliverables: Lesson plan or slides.

Success: Uses simple language and accurate explanations.

8) Assessment Rubric (for any Guide Project)

Criterion	Indicators	Level (0–4)
Scientific accuracy	Correct use of theory, data, and terminology	0–4
Inquiry & reasoning	Hypothesis, evidence, evaluation	0–4
Critical thinking	Identifies limitations, cross-links disciplines	0–4
Communication	Clarity, organisation, visuals or media	0–4
Reflection & originality	Personal insight, independent thought	0–4

Outstanding: 17–20 | **Strong:** 13–16 | **Developing:** 8–12 | **Emerging:** <8

9) Portfolio Evidence

Students compile:

Photos of experiments / graphs

Written reports or essays

Recorded debates or presentations

Reflection logs (“What I learned” entries)

Peer feedback summaries

At year’s end, this becomes a **Science Exploration Portfolio**, showcasing creativity, inquiry, and literacy — excellent for **Cambridge Checkpoint**, **IGCSE coursework evidence**, or **Duke of Edinburgh-style awards**.

Implementation Guidance

Run **1 major Guide project per term** (3–4 per year).

Each project can integrate **multiple sciences** or focus on one strand.

Teachers act as **mentors**, not directors — learners plan, execute, and reflect.

Encourage **AI Tutor use** for:

Research support & outline drafting

Hypothesis refinement

Feedback on structure & reflection

Simulation of alternative results or “what if” analyses.

Computer Science — Basic Supercurriculum (Year 10)

1) Computational Thinking Basics

CS1.1 “What is an algorithm?”

Focus: Understanding step-by-step logic.

Task: Write an algorithm (in plain English) to make a sandwich or tie your shoes. Number the steps.

Goal: Recognise how clear, ordered instructions avoid ambiguity.

Success: Steps are sequenced, complete, and unambiguous.

CS1.2 “Spot the bug”

Focus: Debugging & logic.

Task: Teacher or peer writes a 5-line pseudocode with one logical error (e.g., wrong comparison or missing step). Identify and fix it.

Success: Corrected version works logically and terminates properly.

CS1.3 “If-Else in real life”

Focus: Conditional logic.

Task: Write 3 real-life examples using IF...THEN...ELSE (e.g., “IF rain THEN umbrella ELSE sunglasses”).

Success: Logical and clear, with both outcomes realistic.

2) Binary & Data Representation

CS2.1 “Binary challenge”

Focus: Conversion between binary and decimal.

Task: Convert 5 decimal numbers (1–31) into binary and back again.

Success: All conversions accurate; shows working.

CS2.2 “Bit, Byte, and Beyond”

Focus: Units of data.

Task: Draw a small chart showing bits, bytes, kilobytes, megabytes, and gigabytes. Add one example (e.g., “photo ≈ 2 MB”).

Success: Correct hierarchy and examples.

CS2.3 “ASCII explorer”

Focus: Character encoding.

Task: Look up ASCII codes for the word “CODE”. Write their binary equivalents.

Success: All codes correct; understands why encoding matters.

3) Hardware & Systems

CS3.1 “Inside a computer”

Focus: Components and their functions.

Task: Label a diagram (CPU, RAM, ROM, HDD/SSD, input, output). Write one sentence about each.

Success: Functions correct (e.g., “CPU processes data”).

CS3.2 “Input or Output?”

Focus: I/O device classification.

Task: Make a two-column table listing 5 input and 5 output devices.

Success: All items correctly classified.

CS3.3 “Storage story”

Focus: Primary vs secondary storage.

Task: Write a short paragraph explaining why computers need both.

Success: Correct distinction (temporary vs permanent memory).

4) Algorithms & Flowcharts

CS4.1 “Flowchart drawing”

Focus: Sequencing.

Task: Draw a simple flowchart for making a cup of tea (start → steps → end).

Success: Proper symbols (start/end ovals, process rectangles, decision diamonds).

CS4.2 “Repeat it!”

Focus: Loops.

Task: Write a pseudocode loop that prints numbers 1 to 10.

Success: Loop logic correct (e.g., FOR i ← 1 TO 10).

CS4.3 “Find the largest”

Focus: Selection + iteration.

Task: Write pseudocode to compare three numbers and print the largest.

Success: Correct conditional structure, runs logically.

5) Programming Concepts (Language-Agnostic)

CS5.1 “Variables & data types”

Focus: Understanding variables.

Task: In pseudocode, declare 3 variables: name (string), age (integer), temperature (real).

Success: Syntax clear; types appropriate.

CS5.2 “Hello, input!”

Focus: Input/output handling.

Task: Write pseudocode to ask for your name and print “Hello, [name]!”.

Success: Both input and output steps correct.

CS5.3 “Simple maths program”

Focus: Arithmetic operations.

Task: Write pseudocode for a program that adds two numbers entered by the user.

Success: Uses INPUT, CALCULATE, OUTPUT sequence correctly.

6) Logic & Boolean Operations

CS6.1 “Logic gate basics”

Focus: AND, OR, NOT gates.

Task: Draw the truth tables for each gate.

Success: All outputs correct.

CS6.2 “Logic gates in real life”

Focus: Application of logic.

Task: Explain one everyday situation that uses AND, OR, or NOT logic (e.g., car ignition = key AND brake pressed).

Success: Correct application of gate logic.

CS6.3 “Build your own gate”

Focus: Combining gates conceptually.

Task: Create a small truth table for (A AND NOT B) and describe when the output is true.

Success: Logical accuracy.

7) Networks & the Internet

CS7.1 “Network mini-glossary”

Focus: Basic terminology.

Task: Define: network, LAN, WAN, server, client.

Success: Clear, concise definitions.

CS7.2 “Wired vs wireless”

Focus: Connection types.

Task: Make a pros/cons list for wired vs Wi-Fi networks.

Success: At least 2 valid points each side.

CS7.3 “Internet safety snapshot”

Focus: Cybersecurity basics.

Task: Write 5 online safety rules for young users.

Success: Practical and correct (e.g., “Use strong passwords”).

8) Cybersecurity & Ethics

CS8.1 “Password strength test”

Focus: Security awareness.

Task: Create 3 sample passwords; evaluate using an online checker (no real passwords).

Success: Recognises what makes a password strong.

CS8.2 “Data protection quick facts”

Focus: Privacy laws and principles.

Task: Summarise 3 rules from the GDPR or a similar framework.

Success: Explains purpose of protecting personal data.

CS8.3 “The ethics of AI”

Focus: Societal implications.

Task: Write a short paragraph on one benefit and one risk of AI.

Success: Balanced argument with examples.

9) Problem-Solving & Thinking Skills

CS9.1 “Decompose it!”

Focus: Breaking big problems into small ones.

Task: Choose a complex problem (e.g., designing a game). Write 4 smaller sub-problems.

Success: Logical breakdown; covers input, process, and output.

CS9.2 “Spot the pattern”

Focus: Pattern recognition.

Task: List three examples of patterns (maths, text, or visuals) computers could use to predict or sort data.

Success: Each example practical and explained.

CS9.3 “Abstraction in action”

Focus: Simplifying systems.

Task: Describe how you could simplify a map, recipe, or timetable to keep only essential info.

Success: Shows understanding of relevant vs irrelevant data.

10) Study & Reflection

CS10.1 “One-minute algorithm”

Focus: Retrieval and clarity.

Task: Choose one process (sorting, looping, logic). Explain it in 1 minute as if teaching a friend.

Success: Clear, simple explanation with correct terms.

CS10.2 “Tech in my world”

Focus: Everyday computing awareness.

Task: Identify 3 everyday systems that use computer science (e.g., elevators, online banking, weather apps).

Success: Realistic examples with brief descriptions.

CS10.3 “Weekly reflection”

Focus: Metacognition.

Task: Write 3 bullet points each week: what I learned, what confused me, one question for next time.

Success: Honest, specific, growth-oriented reflection.

✓Assessment Rubric (for all Basic tasks)

Level	Indicators
Emerging (1–2)	Attempted but incomplete or inaccurate.
Developing (3)	Mostly correct with minor logic or syntax errors.
Secure (4)	Clear understanding, accurate outcomes.
Excellent (5)	Accurate + insightful extension or example.

Implementation

Frequency: 2–3 Basic tasks per week (≈30–45 min total).

Evidence: Screenshots, pseudocode snippets, photos of notes, or short videos.

Progression: After mastering these, students advance to the **Extended**

Supercurriculum (longer algorithmic challenges, real coding practice, and system analysis).

Computer Science — Extended Supercurriculum (Year 10)

1) Algorithms & Problem Solving

E1.1 “Algorithm design challenge”

Focus: Stepwise thinking and optimisation.

Task: Design an algorithm (pseudocode or flowchart) to find the **largest number** in a list of 10.

Extension: Optimise it so it works for any list length.

Deliverables: Algorithm steps + brief explanation.

Success: Correct iteration & comparison logic.

E1.2 “Bubble sort simulation”

Focus: Sorting logic.

Task: Use cards or post-it notes (numbers 1–6 mixed). Simulate bubble sort step-by-step. Record swaps after each pass.

Deliverables: Table of each pass; total swaps counted.

Success: Understands how repeated passes create order.

E1.3 “Flowchart to code”

Focus: Translating logic to syntax.

Task: Take any flowchart (teacher-provided or self-made). Rewrite it in pseudocode and, if possible, in Python or Scratch.

Deliverables: Code or screenshot.

Success: Correct conversion; logical sequence preserved.

2) Programming Practice

E2.1 “Guess the number”

Focus: Input/output, conditionals, and loops.

Task: Write a program that:

Chooses a secret number (1–20).

Lets user guess until correct.

Gives hints (“too high/too low”).

Deliverables: Working program or pseudocode.

Success: Loop + comparison logic correct; counts guesses.

E2.2 “Times table generator”

Focus: Iteration and arithmetic.

Task: Program prints the multiplication table for any number entered (e.g., $7 \times 1-10$).

Success: Proper use of FOR loop, clean formatting.

E2.3 “Average calculator”

Focus: Data processing.

Task: Program reads five numbers and prints their average.

Extension: Modify to work for n numbers until “STOP” entered.

Success: Correct total and count; handles input validation.

3) Computational Thinking

E3.1 “Decompose and conquer”

Focus: Modularisation.

Task: Break a problem (“Create a quiz game”) into subroutines:

```
get_question()  
check_answer()  
update_score()  
display_results()
```

Deliverables: Diagram or pseudocode structure.

Success: Logical separation; reusable subroutines.

E3.2 “Spot the pattern”

Focus: Pattern recognition and efficiency.

Task: Create an algorithm to generate the first 10 even numbers.

Extension: Adjust it to print only numbers divisible by both 2 and 3.

Success: Uses modulus operation correctly.

E3.3 “Debug diary”

Focus: Error analysis.

Task: Run or read a short program with errors (syntax or logic). Record each error, what caused it, and how you fixed it.

Success: At least 3 unique errors correctly explained.

4) Data Representation

E4.1 “Binary conversions table”

Focus: Binary/decimal conversions.

Task: Make a table converting numbers 0–15 into binary, hexadecimal, and decimal.

Success: All correct; clear layout.

E4.2 “ASCII to text converter”

Focus: Character encoding.

Task: Translate the ASCII codes 67, 79, 68, 69 into letters, then back.

Extension: Try to write a simple pseudocode that encodes/decodes text.

Success: Correct understanding of ASCII values.

E4.3 “Image pixels mini-lab”

Focus: Bitmap representation.

Task: Create an 8×8 black-and-white “image” using 0s (white) and 1s (black).

Extension: Colour-code bits using a spreadsheet or graph paper.

Success: Recognises each pixel as one bit of data.

5) Hardware, Systems & Logic

E5.1 “Logic gate builder”

Focus: Boolean logic in circuits.

Task: Draw a logic circuit for (A AND B) OR (NOT C).

Deliverables: Diagram and truth table.

Success: Truth table matches circuit logic.

E5.2 “Fetch–Decode–Execute simulation”

Focus: CPU cycles.

Task: Create a step-by-step comic or diagram explaining what happens when a program runs:

Fetch → Decode → Execute → Store.

Success: Clear explanation of registers and control unit.

E5.3 “Storage comparison chart”

Focus: Types of memory.

Task: Compare RAM, ROM, and secondary storage in a table (function, speed, volatility, cost).

Success: Each characteristic correctly compared.

6) Networks & Security

E6.1 “LAN design”

Focus: Basic network topology.

Task: Sketch a simple LAN for 4 devices. Label router, switch, server, and workstation.

Success: Correct components and connections.

E6.2 “Packet journey story”

Focus: Data transmission.

Task: Write a short narrative (≈150 words) describing how a message travels from one computer to another via IP and packets.

Success: Mentions IP address, packet, router, and reassembly.

E6.3 “Cyber safety case study”

Focus: Digital ethics and data security.

Task: Research one famous cyberattack (e.g., WannaCry). Summarise what happened, how it spread, and how it was stopped.

Deliverables: 300-word case summary.

Success: Balanced account with prevention tips.

7) Databases & Algorithms in Everyday Life

E7.1 “Database vocabulary builder”

Focus: Data organisation.

Task: Define: field, record, table, primary key.

Extension: Design a small table (e.g., school library books).

Success: Accurate definitions and sample structure.

E7.2 “Algorithm everywhere”

Focus: Application of algorithms.

Task: Identify 3 algorithms in daily life (e.g., GPS routing, search engines, online recommendations).

Deliverables: Short paragraph per example.

Success: Correct, concise, real examples.

E7.3 “Data compression mini-demo”

Focus: Storage efficiency.

Task: Take a short text (20–30 words). Apply simple run-length encoding manually (e.g., AAAABCC → 4A1B2C).

Success: Correct compression logic.

8) Real-World Systems & AI

E8.1 “Smart devices audit”

Focus: Embedded systems.

Task: Identify 5 household devices with embedded computers (e.g., washing machine, thermostat). Describe what each controls.

Success: Correctly explains control logic.

E8.2 “AI in everyday life”

Focus: Machine learning awareness.

Task: Write 150–200 words describing one system that uses AI (recommendation engine, chatbot, autopilot).

Success: Describes input → processing → output.

E8.3 “Future of computing”

Focus: Innovation awareness.

Task: Research one emerging trend (quantum computing, edge AI, robotics).

Deliverables: 300-word summary of its potential impact.

Success: Accurate, balanced prediction.

9) Interdisciplinary Applications

E9.1 “Coding meets science”

Focus: Cross-subject integration.

Task: Use pseudocode to model a scientific process (e.g., osmosis, reaction rate, free fall).

Success: Logical steps represent the process correctly.

E9.2 “Maths with loops”

Focus: Programming for maths.

Task: Write pseudocode to calculate the factorial of a number or sum of first n integers.

Success: Uses iteration and variables correctly.

E9.3 “Economics of computing”

Focus: Real-world impact.

Task: Research how computational efficiency affects costs (energy, data storage, bandwidth).

Deliverables: 250-word reflection.

Success: Links efficiency to sustainability and economics.

10) Reflection & Communication

E10.1 “Code journal”

Focus: Learning reflection.

Task: Keep a weekly record: what you coded, what worked, what failed, what to try next.

Success: Honest, specific, continuous reflection.

E10.2 “Explain like a teacher”

Focus: Peer communication.

Task: Teach one concept (binary, loop, variable) to a friend using examples. Ask them to explain it back.

Success: Peer demonstrates understanding.

E10.3 “Mini presentation: My favourite algorithm”

Focus: Science communication.

Task: Create a 2-slide mini-presentation on one algorithm (search, sort, or AI-based).

Success: Concise, accurate, and visually clear.

✓ **Mini-Rubric for Extended Tasks**

Criterion	Indicators	Score (0-4)
Accuracy	Correct syntax, logic, or explanation	0-4
Problem-solving	Reasoning clear, efficient approach used	0-4
Creativity	Original examples or unique applications	0-4
Communication	Clarity, correct format, readability	0-4
Reflection	Evaluation of challenges & improvements	0-4

Excellent: 17–20 | **Secure:** 13–16 | **Developing:** 8–12 | **Emerging:** <8

Implementation Guidance

Frequency: 1 Extended activity per week or 3 per month.

Tools: Pseudocode, Python, or Scratch (language flexible).

Evidence: Screenshots, saved code, flowcharts, or short video demos.

Progression: After completing Extended, students advance to the **Guide**

Supercurriculum, which includes **full coding projects**, **data science mini-labs**, and **computing ethics essays**.

Computer Science — Guide Supercurriculum (Year 10)

1) Independent Project: Design • Code • Reflect

G1.1 “Mini App or Game Project”

Focus: Applying programming to solve a real problem or entertain.

Task:

Choose a purpose (quiz app, flashcard game, currency converter, calculator, or timer).

Design input/output layout (flowchart + pseudocode).

Implement it in Python, JavaScript, or Scratch.

Test, debug, and reflect on usability.

Deliverables:

Code + screenshots.

250–300 word reflection (“What went well / to improve”).

Success:

Runs correctly with input validation.

Demonstrates loops, conditionals, and variables.

Reflection shows problem-solving process.

G1.2 “Automation in my world”

Focus: Control and sensors.

Task:

Investigate a real-world automated system (traffic lights, smart homes, washing machines).

Draw a **control flow diagram** explaining input → process → output.

Suggest one improvement using modern sensors or AI.

Deliverables: 1-page diagram + 200 words explanation.

Success: Correct system model; realistic upgrade suggestion.

2) Algorithms, Efficiency & Data

G2.1 “Algorithm comparison lab”

Focus: Efficiency and Big-O intuition.

Task:

Compare **linear search** and **binary search** for lists of 10, 100, 1000 elements (simulate in pseudocode or spreadsheet).

Count how many steps each takes to find the same item.

Deliverables: Table, graph (steps vs list size), short paragraph.

Success: Recognises binary search scales logarithmically.

G2.2 “Sort race!”

Focus: Algorithmic thinking.

Task:

Code or simulate **bubble sort**, **selection sort**, and **merge sort**.

Measure how many comparisons for a small list.

Deliverables: Table + conclusion on fastest/slowest.

Success: Identifies merge sort as most efficient for large lists.

G2.3 “Real data challenge”

Focus: Applying algorithms to real datasets.

Task:

Download a small CSV file (weather, population, or sports stats).

Write pseudocode or code to compute mean, min, max, or frequency counts.

Deliverables: Code + sample output.

Success: Data processed accurately; code well structured.

3) Data Representation & Cybersecurity

G3.1 “Hidden images”

Focus: Binary and pixels.

Task: Create a black-and-white image (8×8 or 16×16) using 0s and 1s. Encode it as binary and decode it again.

Deliverables: Binary grid + image visualisation.

Success: Correct representation and mapping.

G3.2 “Encryption lab”

Focus: Data protection through simple ciphers.

Task:

Implement a **Caesar cipher** in pseudocode or Python.

Try encrypting and decrypting a short word or sentence.

Explain how it could be improved (frequency analysis, stronger key).

Deliverables: Code snippet + 200 words explanation.

Success: Working encryption; awareness of limitations.

G3.3 “Data privacy case study”

Focus: Ethics and law.

Task: Research one famous data breach (e.g., Facebook–Cambridge Analytica, Equifax).

Deliverables: 400–500 word report explaining what happened, how it broke privacy laws, and how it could have been prevented.

Success: Uses facts; relates to principles of data security and integrity.

4) Hardware & Systems Architecture

G4.1 “Build a virtual computer”

Focus: CPU simulation.

Task:

Create a **diagram or animation** showing the fetch–decode–execute cycle, with registers and buses.

Label each step’s purpose.

Deliverables: Digital diagram or short video.

Success: Clear, sequential explanation; correct terminology (PC, MAR, MDR, ACC).

G4.2 “Storage systems analysis”

Focus: Comparing storage technologies.

Task:

Create a comparison chart for HDD, SSD, Optical, Cloud storage.
Include: capacity, speed, durability, portability, cost, and environmental impact.

Deliverables: Table + summary paragraph.

Success: Complete and evidence-based comparison.

G4.3 “Sustainability of hardware”

Focus: Environmental and ethical issues.

Task: Research the environmental footprint of producing and disposing of computers.

Deliverables: 600-word article “How green is your computer?”

Success: Balanced argument; data or references included.

5) Networks, Internet & Security**G5.1 “Network designer”**

Focus: Topology and configuration.

Task: Design a **school network** for 20 devices. Include routers, switches, wireless access points, server, and printers.

Deliverables: Diagram + labelled description of data flow.

Success: Logical connections; mentions IP addressing and security.

G5.2 “Packet journey simulation”

Focus: Data transmission protocols.

Task: Write a creative narrative from the point of view of a “data packet” travelling from Athens to London, explaining routers, packets, and reassembly.

Deliverables: 200–300 word story.

Success: Technically accurate, imaginative, engaging.

G5.3 “Cyberdefence plan”

Focus: Preventing attacks.

Task: Create a 1-page infographic explaining common cyber threats (malware, phishing, brute-force attacks) and how to defend against each.

Deliverables: Infographic or slide.

Success: Accurate, clear, and visually appealing.

6) Databases & Real Applications**G6.1 “Design a student database”**

Focus: Data modelling.

Task:

Design a database for school students (Name, Age, Grade, House, Points).

Identify fields, primary key, and validation rules.

Deliverables: Table schema diagram.

Success: Logical structure, unique primary key, valid field types.

G6.2 “SQL lite”

Focus: Query logic (theory level).

Task: Write 3 SQL queries:

SELECT all records.

WHERE age > 15.

ORDER BY name ASC.

Deliverables: Written queries + expected results.

Success: Correct syntax and understanding.

G6.3 “Everyday databases”

Focus: Relevance of data systems.

Task: Investigate one real-world system that uses a database (library, hospital, e-commerce).

Deliverables: 300-word explanation.

Success: Correct identification of data flow and purpose.

7) Artificial Intelligence & Emerging Technologies

G7.1 “AI vs Human logic”

Focus: Understanding machine learning principles.

Task:

Compare how humans learn from examples with how ML models are trained.

Write a 400-word essay explaining one benefit and one risk of machine learning in daily life.

Deliverables: Essay.

Success: Balanced reasoning, accurate terminology.

G7.2 “Build a mini AI assistant (text-based)”

Focus: Conditionals and keyword matching.

Task:

Code a chatbot that answers 3–4 simple questions.

Use if–elif–else statements to select responses.

Deliverables: Working code.

Success: Functional program; polite and clear output.

G7.3 “Ethics of automation”

Focus: Social impact of computing.

Task:

Write a 600-word opinion article on “Should robots replace human jobs?”

Deliverables: Essay with examples.

Success: Balanced viewpoint, includes ethical reasoning.

8) Mathematics & Computing Crossover

G8.1 “Randomness & probability in code”

Focus: Algorithmic maths.

Task:

Write a program that rolls two dice 100 times and records how often each total appears.

Plot results in a small bar chart (manual or digital).

Deliverables: Code + graph.

Success: Uses random numbers, loops, and counts correctly.

G8.2 “Binary arithmetic explorer”

Focus: Adding & shifting binary numbers.

Task: Show how binary addition and left shift operations can double a number.

Deliverables: Example table and explanation.

Success: Clear link between binary shift and powers of two.

G8.3 “Fibonacci program”

Focus: Recursion & loops.

Task: Write pseudocode or Python to generate first n Fibonacci numbers.

Deliverables: Code + short reflection on how recursion works.

Success: Correct sequence; identifies base case.

9) Interdisciplinary & Creative Computing

G9.1 “STEAM project: Code meets art”

Focus: Creativity & design.

Task: Use Python Turtle or Scratch to draw geometric patterns.

Deliverables: Screenshot of design + short explanation.

Success: Uses loops and coordinates creatively.

G9.2 “Science simulation”

Focus: Cross-disciplinary modelling.

Task: Create a simulation (e.g., falling object, population growth, chemical reaction rate) using code or spreadsheet formulas.

Deliverables: Code + graph.

Success: Realistic trend; correct variable control.

G9.3 “Computing for good”

Focus: Ethics, sustainability, and innovation.

Task: Research one project using computing to solve a global challenge (e.g., clean water, education access, healthcare AI).

Deliverables: 400-word summary + one visual.

Success: Explains technology + social benefit.

10) Reflection & Portfolio

G10.1 “My learning journey”

Focus: Metacognition and goal setting.

Task: Reflect on your progress across Basic, Extended, and Guide.

What have you learned?

Which skills improved most?

Where do you want to go next (A-Level, AI, game dev, cybersecurity)?

Deliverables: 600-word reflection or 3-min video.

Success: Honest, detailed, future-oriented.

G10.2 “Portfolio compilation”

Focus: Evidence collection.

Task: Collect all code files, screenshots, reports, and reflections into a digital folder or website.

Deliverables: Structured portfolio (title page, projects, reflections).

Success: Organised, complete, visually consistent.

✓Assessment Rubric for Guide Projects

Criterion	Indicators	Score (0-4)
Technical accuracy	Code or logic works correctly	0-4
Design & creativity	Original idea, clear structure	0-4
Problem-solving	Debugging, optimisation, reflection	0-4
Communication	Clarity, documentation, visuals	0-4
Ethical/contextual awareness	Considers impact and purpose	0-4

Outstanding: 17-20 | **Strong:** 13-16 | **Developing:** 8-12 | **Emerging:** <8

Implementation Guidance

Pacing: 3-4 major Guide projects per academic year (≈1 per term).

Tools: Python, Scratch, Replit, or offline pseudocode.

Output: Mix of written reports, presentations, and digital products.

Teacher's role: Mentor, not marker — feedback on logic, ethics, and creativity.

Integration: Ideal for science fairs, digital portfolios, or AI-lab extension.

Art — Basic Supercurriculum (EDEXCEL GCSE / IGCSE)

Daily Drawing Habits

A1.1 5-minute warmups

Focus: Hand–eye coordination.

Do: Continuous line drawings of simple objects (keys, mug, shoe). No erasing.

Success: Bold lines; whole form captured.

A1.2 Blind contour

Focus: Looking, not guessing.

Do: Draw your hand without looking at the page for 2–3 minutes.

Success: Lines track real edges; no tiny, scratchy marks.

A1.3 Tonal ladder

Focus: Value control.

Do: Make a 7-step pencil gradient. Shade a sphere using those values.

Success: Smooth transitions; clear core shadow/highlight.

Observational Studies

A2.1 One object, three views

Focus: Proportion & composition.

Do: Draw the same object from front/side/close-up.

Success: Different viewpoints fill the page; measured proportions.

A2.2 Negative space chair

Focus: Seeing shapes, not symbols.

Do: Draw the space *around* a chair; leave the chair white.

Success: Accurate outer shapes; strong silhouette.

A2.3 Quick still-life

Focus: Tonal contrast.

Do: Arrange 3 objects; 15-minute tonal study (2B/4B).

Success: At least 5 distinct values; cast shadow present.

Elements of Art (Micro Drills)

A3.1 Line library – 20 line types; label feelings/effects.

A3.2 Texture rubbings – Collect 6 textures (coin, leaf, brick) → collage.

A3.3 Colour wheel (12-step) – Mix primaries to secondaries/tertiaries.

A3.4 Warm vs cool – Two mini studies of the same scene; note mood change.

A3.5 Composition thumbnails – 6 tiny boxes testing rule of thirds, symmetry, central focus, diagonal.

Media Tasters

A4.1 Graphite → Ink swap

Draw an object in pencil; trace key lines in brush pen; erase pencil.

Success: Confident line weights.

A4.2 Dry media blend

Soft pastel + charcoal on toned paper; lift highlights with eraser.

Success: Clear light source; edges (hard/soft) varied.

A4.3 Watercolour basics

Flat wash, graded wash, wet-on-wet test squares.

Success: Even coverage; blooms used intentionally.

(Safety: Ventilated space for fixative/solvents; tie hair, apron.)

Photography & Digital

A5.1 Phone composition walk

Focus: Framing.

Do: Shoot 10 photos exploring leading lines/reflections/shadows.

Success: Strong focal point; horizon straight.

A5.2 Light & shadow series

Do: Photograph one object hourly (daylight change).

Success: Notice value shifts; annotate best 3.

A5.3 Edit basics

Crop, rotate, exposure/contrast only. Before/after pair.

Success: Edit serves the subject (not filters for their own sake).

Design & Typography Starters

A6.1 Initials as logo – Sketch 6 variants (mono). Choose best; tidy.

A6.2 Poster hierarchy – Re-layout a school event poster: headline, date, place.

A6.3 Icon set – Draw 6 simple icons (library, lab, sport, music, café, art).

Context in 10 Minutes

(Stick one small reference image per page and respond.)

A7.1 Artist snapshot – One paragraph on an artist's **materials + subject + why it works** (e.g., Hockney, Kahlo, Banksy).

A7.2 Compare & contrast – Two artworks; list 3 similarities / 3 differences (line, colour, mood).

A7.3 “Steal like an artist” – Copy a small section of a chosen artwork, noting technique.

Visual Research & Annotation

A8.1 Mind-map a theme – Choose “*Identity / Nature / Urban / Time*”. Branch to objects, colours, textures, feelings.

A8.2 Page layout triad – Title, a drawing, 5-line annotation answering: *What? How? Why?*

A8.3 Material tests page – Glue small swatches, label brush/pressure, pros/cons.

Quick Creativity Prompts

A9.1 One object, five styles – Realistic, contour, cubist, minimal, collage.

A9.2 30-circle challenge (10 min) – Turn circles into different ideas.

A9.3 Limited palette – 3 colours maximum; paint a simple still-life.

Presentation & Portfolio Habits

A10.1 Page polish – Dates, titles, neat borders, no loose scraps.

A10.2 Before/after pairs – Show progress: first attempt vs improved.

A10.3 Weekly reflection (5 lines) – *What worked? What to fix? Next step?*

Mini-Rubric (use for any Basic task)

Level	Indicators
Emerging (1–2)	Attempted; little control of proportion/tonal range; minimal annotation.
Developing (3)	Mostly accurate shapes; some tonal range; brief notes.
Secure (4)	Confident marks; clear values/composition; useful annotations.
Excellent (5)	Strong design choices; risk-taking; page is presentation-ready.

Materials (cheap & effective)

HB, 2B, 4B pencils; eraser (kneaded + plastic), sharpener

Fineliner / brush pen; soft charcoal; soft pastels

Watercolour set + round brush; glue stick; masking tape

Mixed-media sketchbook (A4) + a few sheets of toned paper

Phone camera; basic editing app

Safety & Studio Etiquette

Ventilate when using sprays/solvents; avoid aerosol near heat.

Tie long hair; cover table; wash hands after charcoal/pastel.

Respect image copyrights when printing references (use for study only).

Weekly Recipe (example)

Mon: A1.1 + A3.5 (10–15')

Wed: A2.3 Still-life (20')

Fri: A7.1 Artist snapshot (10') + A10.3 Reflection (5')

When these feel comfortable, move on to the **Extended Art**

Supercurriculum (longer studies, mixed-media experiments, focused artist research, and small outcome pieces).

Είπε:

guide

To ChatGPT είπε:

Here's the **EDEXCEL GCSE / IGCSE Art — Guide Supercurriculum** (top tier).

It's built to mirror exam practice and push independence. Every task maps to **AO1–AO4**:

AO1: Develop ideas informed by artists/context

AO2: Refine through experiments with media/techniques

AO3: Record observations/ideas (drawing, photos, notes)

AO4: Present a personal, meaningful outcome

Art — Guide Supercurriculum (EDEXCEL GCSE / IGCSE)

How to use this guide

Complete **1 major project per term** (6–8 weeks).

Keep a **rigorous sketchbook**: title, intent, research, experiments, trials, reflection, curation plan.

Each project ends with a **resolved outcome** (A2 board, canvas, digital piece, sculpture) + **500–800 word** artist statement.

Project 1 — Identity: Portrait & Persona (Mixed Media)

Intent: Explore how materials shape identity.

AO1: Research 2–3 artists (e.g., **Frida Kahlo, Kehinde Wiley, Jenny Saville, Barbara Kruger**). Annotate what, how, why.

AO2: Experiments: graphite tonal portraits, acrylic glazing, collage text overlays, gel transfers. Try at least **4 media combos**; test colour palettes and grounds.

AO3: Recordings: 8–12 observational studies (mirror selfies, timed charcoal, photo series under different light).

AO4: Outcome: a resolved portrait mixing paint + collage or paint + text.

Checkpoint: Can a viewer read 3 traits (mood, role, culture) without your text?

Project 2 — Urban Narratives (Photography → Print)

Intent: Tell a visual story about place, time, and movement.

AO1: Study **Henri Cartier-Bresson, Vivian Maier, JR, Shepard Fairey**.

Analyse composition, rhythm, and message.

AO2: Experiments: phone photography (leading lines, reflection), contact sheet curation; convert to high-contrast; make **linocuts** or **monoprints** from chosen frames.

AO3: Recordings: shoot **100+ photos**, select top 12; annotate contact sheets (tick, cross, crop marks).

AO4: Outcome: a print series or photo+print hybrid zine (8–12 pages).

Stretch: Add a simple **AR** layer (QR code linking to a short clip).

Project 3 — Organic → Geometric (Sculpture / 3D)

Intent: Transform natural forms into constructed geometry.

AO1: Research **Barbara Hepworth, Naum Gabo, Ruth Asawa**.

AO2: Experiments: wire armatures, cardboard slot-forms, paper lamination, plaster bandage skins; test balance and negative space.

AO3: Record: contour drawings of plants/shells; shadow studies; scale maquettes.

AO4: Outcome: 3D piece (wire/paper/plaster/laser-cut card), presented with a pedestal or hanging plan and gallery card (title, media, size, statement 100 words).

Safety: Eye protection, ventilate adhesives, cut away from body.

Project 4 — Colour Systems & Emotion (Painting Lab)

Intent: Use colour theory to drive mood.

AO1: Research **Mark Rothko, Hilma af Klint, Hockney** (palette logic).

AO2: Experiments: swatch library (limited palettes, complements, split-complements), glazing vs impasto, underpainting colour tests.

AO3: Record: thumbnails (composition), lighting studies of a chosen interior or landscape at 3 times of day.

AO4: Outcome: A2 painting with a documented palette strategy (swatches stapled in book + notes on why).

Quality bar: 5+ distinct values, intentional edges, colour harmony that matches intent.

Project 5 — Typography & Voice (Graphic Communication)

Intent: Marry image and text to communicate a message.

AO1: Study **Constructivism, Bauhaus, Paula Scher, Barbara Kruger**.

AO2: Experiments: hand-lettering, cut-type, vector layout (Canva/Illustrator), image-type integration; explore hierarchy and kerning.

AO3: Record: poster mockups (rule of thirds, grids), A/B tests of headlines, audience notes.

AO4: Outcome: Campaign set of **3 posters** (or poster + social tiles).

Success check: Message legible at 2 m; consistent visual system across set.

Project 6 — Time & Motion (Sequential Art / Short Video)

Intent: Show change, movement, or process.

AO1: Research **Eadweard Muybridge, Saul Bass, Marjane Satrapi**.

AO2: Experiments: storyboards, frame pacing, limited animation (stop-motion/paper cutout), screen-text overlays.

AO3: Record: 12–24 frame sequence drawings or animatic; notes on rhythm and soundtrack.

AO4: Outcome: 30–90 sec video/animation or 8–12 panel comic.

Deliverable: Export + QR on portfolio page; credit any music.

Project 7 — Material Alchemy (Print / Surface / Collage)

Intent: Turn “ordinary” surfaces into rich images.

AO1: Look at **Rauschenberg, Anselm Kiefer, El Anatsui**.

AO2: Experiments: textured grounds (sand/gel), frottage library, chine-collé, torn-edge collage, stitched elements.

AO3: Record: macro photos of textures; rubbings with notes (source, pressure, effect).

AO4: Outcome: Large mixed-media piece (A2/A1) with layered surfaces.

Evaluation: Does the material choice support your theme (not decoration)?

Project 8 — Nature, Science, Diagram (Info-Art)

Intent: Fuse scientific clarity with artistic interpretation.

AO1: Research **Ernst Haeckel**, **Agnes Denes**, contemporary data-visual work.

AO2: Experiments: diagrammatic drawing, data-driven shapes, ink washes + fineliner overlays.

AO3: Record: field sketches, measured studies, collected data set (e.g., leaf sizes, tide times).

AO4: Outcome: Poster or triptych combining observation + diagram + key.

Stretch: Publish a small **fold-out guide** (risograph/printable PDF).

Project 9 — Curate & Present (Mini Exhibition)

Intent: Practice AO4 presentation and critical selection.

Steps:

Select **10–15** best pages/works across projects.

Write a **curatorial statement** (150–200 words): theme, journey, choices.

Plan a wall layout (scale diagram), captions, QR links to process videos.

Mount and present in a corridor/class gallery or digital slideshow.

Assessment: Cohesion, sequencing, captions, professionalism.

Research & Annotation Prompts (use across all projects)

WHAT am I exploring? (subject, theme)

HOW did this artist achieve the effect? (media, process, composition)

WHY does it matter? (meaning, context, audience)

SO WHAT for my work? (what I will borrow, adapt, reject)

NEXT: Concrete action (test palette X; try ground Y; scale up).

Evidence Checklist (per project)

☐ Title & intent (1–2 sentences)

☐ **AO1:** 2–3 artist/context pages with analysis (not biography)

☐ **AO2:** At least 4 structured experiments (label tools, timings, pros/cons)

☐ **AO3:** 8–12 primary recordings (drawings/photos) + notes

☐ **AO4:** Mockups → final outcome + installation/printing plan

☐ Reflection: 150–300 words on decisions, risks, what to improve

Marking Mini-Rubric (Guide level)

Band	Indicators (AO1–AO4 woven together)
Outstanding	Research genuinely informs original decisions; media control is confident and purposeful; recordings are extensive and from life; outcome is coherent, resolved, and communicates intent; curation professional.
Strong	Clear links to research; good range of experiments; mostly secure recording; resolved outcome with minor weaknesses; thoughtful reflection.
Developing	Surface-level research; experiments limited; recording thin; outcome

	under-developed; reflection descriptive more than analytical.
Emerging	Minimal research/experiments; little evidence of development; outcome unclear; weak presentation.

Materials & Practicalities

Sketchbook A4/A3, mixed-media paper; pencils (HB–8B), fineliners, brushes
Acrylics/watercolours, inks; charcoal/pastel; glue, tape, craft knife, cutting mat
Lino + cutters / gel plate (optional); wire, cardboard, plaster bandage (3D)
Phone camera/tripod; basic editing/typography software (Pixlr, Canva, Photopea)

Safety: Ventilate sprays/solvents; cut away from body; eye protection for lino/wire; cover surfaces.

Term Planner (example)

Weeks 1–2: Intent + AO1 research, first AO2 media tests

Weeks 3–4: AO3 recording, more AO2 experiments, mini mockups

Weeks 5–6: Final outcome build (AO4), mount/photograph

Week 7: Curate, title cards, statement, critique circle

Critique Circles (peer review prompts)

What's the **intent**? Do the choices support it?

Which **experiment** should be pushed further?

Where does **composition** lead the eye?

What would you **remove** to strengthen the piece?

What is the **next, concrete step**?

If you'd like, I can convert this into a **print-ready checklist & planner** (one-pager) or into your app's content format (tagged by **module=Art**, **tier=guide**, with AO1–AO4 fields).

Είπε:

Spanish basic EDEXCEL GCSE / IGCSE

To ChatGPT είπε:

Perfect 🌟 — here's a **Basic Supercurriculum** for EDEXCEL GCSE / IGCSE

Spanish, designed to complement the main syllabus while developing confidence, vocabulary, pronunciation, and cultural awareness.

Each Basic task takes **10–25 minutes** and targets all four skills: **listening, speaking, reading, and writing**, plus cultural context.

Aim: 3 short activities per week + 1 reflection.

Spanish — Basic Supercurriculum (EDEXCEL GCSE / IGCSE)

📌 1) Listening Foundations

S1.1 “Spanish Sound Check”

Focus: Pronunciation & phonics.

Task: Watch a 5-min video on Spanish alphabet & sounds (e.g., BBC Bitesize).

Do: Repeat 10 words aloud; record yourself and compare.

Success: Accurate vowel and “r/rr” sounds.

S1.2 “Music & Meaning”

Focus: Listening for gist.

Task: Pick a Spanish pop song (Shakira, Rosalía, Morat). Write down 5 words you understand.

Success: Identifies cognates and repeated phrases.

S1.3 “Mini podcast”

Focus: Key info recognition.

Task: Listen to a 2–3 minute beginner podcast (e.g., Coffee Break Spanish, Ep.1–3). Note 3 greetings or phrases.

Success: Recognises context and main topic.

2) Speaking Confidence

S2.1 “Daily greeting challenge”

Focus: Routine expressions.

Task: Use 5 greetings aloud each day — *Hola, ¿Qué tal?, Buenos días, Buenas tardes, Adiós.*

Success: Natural pronunciation and tone.

S2.2 “Mirror talk”

Focus: Fluency.

Task: Speak for 30 seconds introducing yourself (*Me llamo..., Tengo... años, Soy de...*).

Success: Clear structure without reading.

S2.3 “Mini role-play”

Focus: Real-life exchanges.

Task: Practice short dialogues from textbook or online (at a café, asking directions).

Success: Uses correct intonation and question form.

3) Reading Practice

S3.1 “Cognate hunt”

Focus: Vocabulary recognition.

Task: Read a short paragraph in Spanish. Highlight 10 words that look like English. Translate.

Success: Correct meaning in context.

S3.2 “Headline scan”

Focus: Skimming for gist.

Task: Visit a Spanish news site (e.g., *El País Niños, BBC Mundo*). Find one story title you understand.

Success: Identify topic + one key word.

S3.3 “Menu mystery”

Focus: Authentic text decoding.

Task: Find an online Spanish menu. List 5 food words you can recognise.

Success: Basic food vocabulary identified.

📝 4) Writing & Vocabulary Building

S4.1 “10-word diary”

Focus: Simple sentences.

Task: Write 10 words per day from class topics (family, school, hobbies).

Success: Reuses them correctly in short phrases.

S4.2 “Mini postcard”

Focus: Controlled writing.

Task: Write 3 sentences to a friend: where you are, what you do, what the weather is like.

Success: Uses *estoy en*, *hace sol*, *me gusta...* correctly.

S4.3 “Emoji story”

Focus: Creativity.

Task: Use 5 emojis to write a 5-line story (e.g., 🐼🐼🐼🐼🐼).

Success: Logical, grammatical sentences.

🌍 5) Culture & Context

S5.1 “Country snapshot”

Focus: Cultural awareness.

Task: Pick a Spanish-speaking country. Write 3 facts (capital, flag colours, food).

Success: Correct and concise info.

S5.2 “Festival spotlight”

Focus: Tradition.

Task: Watch a short clip about *La Tomatina* or *Día de los Muertos*. Write 3 notes about what happens.

Success: Identifies when, where, what.

S5.3 “Famous person focus”

Focus: Real-world connection.

Task: Choose one Spanish-speaking figure (e.g., Messi, Frida Kahlo). Write 3 sentences about them.

Success: Uses *es*, *tiene*, *vive en...* correctly.

📖 6) Grammar & Structure Mini Tasks

S6.1 “Verb match”

Focus: -AR/-ER/-IR endings.

Task: Conjugate *hablar*, *comer*, *vivir* for *yo*, *tú*, *él/ella*.

Success: Correct endings and subject agreement.

S6.2 “Gender swap”

Focus: Noun agreement.

Task: Write 5 pairs: *el chico* / *la chica*, *el profesor* / *la profesora*, etc.

Success: All articles + endings match.

S6.3 “Sentence builder”

Focus: Word order.

Task: Arrange scrambled sentence cards (e.g., *gusta / me / el / fútbol* → *Me gusta el fútbol*).

Success: Word order correct; accents accurate.

7) Visual & Audio Culture

S7.1 “Film scene freeze”

Focus: Listening & emotion.

Task: Watch a 3-min clip from a Spanish cartoon or film. Pause and describe (in English) what you think they’re saying.

Success: Correct tone & gist.

S7.2 “Advertisement decode”

Focus: Authentic text.

Task: Find a Spanish poster/ad online. Identify 3 key words and translate.

Success: Recognises message type (offer, event, etc.).

S7.3 “Music & identity”

Focus: Cultural exploration.

Task: Choose a Spanish or Latin American artist. Listen and describe rhythm, theme, instruments (in English).

Success: Mentions 2 cultural aspects.

8) Revision & Reflection

S8.1 “5 words review”

Focus: Retrieval.

Task: Each week, pick 5 words you forgot. Write a mini-sentence for each.

Success: All in context, correctly spelled.

S8.2 “One-minute talk”

Focus: Spontaneous recall.

Task: Speak for 1 minute on *Mi familia* or *Mi colegio*. Record and replay.

Success: Continuous flow; no long pauses.

S8.3 “Reflection journal”

Focus: Metacognition.

Task: End of week: 3 sentences —

“I learned...”

“I found difficult...”

“Next week I will...”

Success: Honest and specific self-review.

Mini-Rubric (for any Basic Task)

Level	Description
Emerging (1–2)	Few words correct, unclear pronunciation, little structure.
Developing (3)	Some correct vocabulary and grammar, limited flow.

Secure (4)	Accurate phrases and pronunciation, clear meaning.
Excellent (5)	Fluent, expressive, culturally aware use of Spanish.

Suggested Weekly Routine

Day	Activity	Skill
Monday	Song + 5 new words	Listening/Vocab
Wednesday	Mini dialogue	Speaking
Friday	Country or festival	Culture
Weekend	10-word diary + reflection	Writing/Review

Simple Resources

- 🔗 **Audio:** BBC Bitesize Spanish, Coffee Break Spanish
- 🔗 **Videos:** *Dreaming Spanish*, *Señor Jordan*, *Easy Spanish* (YouTube)
- 🔗 **Reading:** *BBC Mundo Niños*, *El País Educación*, *News in Slow Spanish*
- 🔗 **Apps:** Duolingo, Linguno, Memrise, Conjuguemos

Spanish — Extended Supercurriculum
(EDEXCEL GCSE / IGCSE)

1) Speaking & Conversation Practice

E1.1 “My World in 60 Seconds”

Focus: Spontaneous speaking (Theme: *Identity & culture*).

Task: Record a 1-minute video talking about your daily routine or hobbies.

Must include:

One reflexive verb (*me levanto, me acuesto*).

One opinion phrase (*me encanta, odio*).

One connector (*además, sin embargo*).

Deliverable: Script + recording.

Success: Continuous flow, correct tense & connectors.

E1.2 “Role-play rehearsal”

Focus: Functional speaking (Theme: *Local area, travel & tourism*).

Task: Write and perform a short dialogue: booking a hotel room, ordering food, or asking for directions.

Deliverable: 6–8 lines per speaker.

Success: Correct question forms + polite expressions (*¿Podría...?, por favor*).

E1.3 “Interview a classmate”

Focus: Exchange of information (Theme: *School or work*).

Task: Interview your partner (in Spanish) about studies, subjects, and future jobs. Record both roles.

Extension: Write 5 follow-up questions.

Success: Uses *¿Qué?, ¿Dónde?, ¿Por qué?* accurately; answers in full sentences.

2) Writing: Expression & Control

E2.1 “Letter from the past”

Focus: Narrative writing (Theme: *Past holidays or experiences*).

Task: Write 100–120 words describing your favourite holiday — where, with whom, what you did, and how you felt.

Include:

pretérito verbs (*fui, hice, comí, visité*)

One opinion + reason (*Fue divertido porque...*)

Success: Logical order, correct tense agreement.

E2.2 “My dream future”

Focus: Future tense.

Task: Write about your plans and ambitions for the next five years.

Estudiaré, viajaré, trabajaré, tendré, viviré...

Deliverable: 80–100 words.

Success: At least 5 accurate *futuro simple* forms.

E2.3 “Diary of a busy week”

Focus: Using three tenses.

Task: Write a 7-line diary (Mon–Sun) mixing present, past, and future actions.

Success: Correct tense use; connects ideas with *y, luego, después, mañana*.

3) Reading for Depth & Inference

E3.1 “Teen blogs in Spanish”

Focus: Reading authentic texts.

Task: Read 2 Spanish teen blogs about daily life or hobbies (BBC Mundo / Linguno).

Answer: 5 short questions in English.

Success: Correctly identifies opinions and reasons.

E3.2 “Spot the opinion”

Focus: Reading comprehension & tone.

Task: Highlight all opinion words in a short article (*pienso que, creo que, me gusta, no soporto*).

Success: Finds at least 6 accurate opinion markers.

E3.3 “Cultural article summary”

Focus: Cultural reading.

Task: Read an article on a Spanish-speaking festival or region (e.g., Andalucía, Día de los Muertos).

Write: 5 key facts + a 3-line English summary.

Success: Summarises key ideas, not full translation.

4) Listening & Pronunciation Skills

E4.1 “News snippet decode”

Focus: Authentic listening (Theme: *Current affairs*).

Task: Listen to a short Spanish news clip (1–2 mins). Identify:

The topic

3 key words

1 number or date

Success: Captures gist + one detail accurately.

E4.2 “Pronunciation mirror”

Focus: Accuracy & rhythm.

Task: Read a 6-line paragraph aloud twice. Record and compare to native audio (YouGlish).

Focus on: vowels (a/e/i/o/u), r/rr, and accent marks.

Success: Improved stress and clarity on replay.

E4.3 “Lyrics gap-fill”

Focus: Active listening.

Task: Use a simple Spanish song lyric sheet (remove 10 words). Fill in missing words as you listen.

Success: 70%+ accuracy and rhythm awareness.

5) Grammar & Structure Mastery

E5.1 “Tense grid challenge”

Focus: Core verb conjugation.

Task: Create a grid for *hablar, comer, vivir* in *present, preterite, future*.

Success: All 9 forms correct.

E5.2 “Error detective”

Focus: Proofreading.

Task: Correct 5 grammar errors in a short paragraph (teacher-provided).

Success: Fixes gender, tense, and adjective agreement.

E5.3 “Sentence expansion”

Focus: Style & variety.

Task: Take a short simple sentence (*Me gusta el fútbol.*). Expand 3 times:

Add adjective (*Me gusta el fútbol emocionante.*)

Add reason (*Me gusta el fútbol porque es emocionante.*)

Add connective + tense change (*Aunque es difícil, jugué ayer.*)

Success: Clear structure; wide range of linkers.

6) Cultural Enrichment

E6.1 “Festival research project”

Focus: Understanding tradition.

Task: Research one Spanish/Latin American festival (*Las Fallas, Carnaval, Día de los Muertos*).

Deliverable: 8-slide mini-presentation or 300-word report in English/Spanish mix.

Success: Includes location, date, purpose, and traditions.

E6.2 “Food from Spain & Latin America”

Focus: Cultural vocabulary.

Task: Create a “menu page” with 5 dishes from 3 countries (Spain, Mexico, Argentina).

Success: Correct spelling + origin of each dish.

E6.3 “Virtual tour”

Focus: Real-world context.

Task: Take a Google Street View “walk” through a Spanish city (Barcelona, Seville, Madrid).

Deliverable: 5 screenshots + short captions in Spanish (e.g., *Aquí hay una plaza grande.*).

Success: Describes accurately with basic structures.

7) Translation & Interpretation

E7.1 “Spot the false friend”

Focus: Vocabulary precision.

Task: Make a list of 10 false friends (e.g., *embarazada* ≠ *embarrassed*).

Success: Correct translations; example sentences for each.

E7.2 “Translate the dialogue”

Focus: Two-way translation.

Task: Translate a short English dialogue into Spanish and back.

Success: Meaning preserved both ways.

E7.3 “Mini paragraph translation”

Focus: Grammar in context.

Task: Translate a 60-word Spanish paragraph into English, then back into Spanish from memory.

Success: Maintains original structure with correct tenses.

8) Intercultural Communication & Reflection

E8.1 “Pen-pal message”

Focus: Real communication.

Task: Write an informal email to a Spanish-speaking pen-pal (real or simulated).

Introduce yourself, family, hobbies.

Ask 3 questions about them.

Success: Natural flow; no English interference.

E8.2 “Compare my country”

Focus: Cultural comparison.

Task: Write 100 words comparing your town with one Spanish-speaking city (*Mi ciudad es más pequeña que Madrid.*).

Success: Correct comparatives/superlatives.

E8.3 “My Spanish learning journey”

Focus: Self-awareness.

Task: Reflect (in English or Spanish) on what you’ve learned so far and which skills improved most.

Success: Identifies real progress and next goals.

✓Mini Rubric (for Extended Tasks)

Level	Indicators
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Emerging (1–2)	Uses limited vocabulary and short sentences; frequent grammar errors.
Developing (3)	Understands task; some errors but meaning clear.
Secure (4)	Mostly accurate grammar; clear communication; uses connectives.
Excellent (5)	Fluent, confident, accurate; opinions, tenses, and cultural awareness integrated.

Suggested Weekly Plan

Week	Focus	Output
1	Speaking (Role-play)	Record conversation
2	Writing (Past holiday)	100-word paragraph
3	Reading (Festival article)	Summary + vocab list
4	Listening (Song gap-fill)	Fill sheet + reflection
5	Culture (Food/country)	Poster or presentation

Recommended Tools

- 🔗 **Listening:** *Coffee Break Spanish, Notes in Spanish, Easy Spanish YouTube*
- 🔗 **Practice:** *Quizlet, Memrise, Conjuguemos, FluentU*
- 🔗 **Reading:** *BBC Mundo Educación, News in Slow Spanish, Linguno.com*
- 🔗 **Culture:** *Netflix Kids Spanish shows, El Internado, Hilda (ES dub)*

Spanish — Guide Supercurriculum (EDEXCEL GCSE / IGCSE)

1) Independent Speaking & Debate

G1.1 “Mini TED Talk: Who am I?”

Focus: Fluency, confidence, and description (Theme: *Identity and culture*).

Task: Prepare a 2-minute recorded talk introducing yourself — interests, origins, and aspirations.

Language goals:

3 tenses (past, present, future)

Connectors (*además, por eso, sin embargo*)

Opinion + justification (*pienso que..., porque...*)

Deliverables: Written script (100–150 words) + video/audio.

Success: Fluent delivery, accurate grammar, varied vocabulary.

G1.2 “Formal debate: Technology — ¿Aliada o enemiga?”

Focus: Argumentation and register.

Task: Conduct a classroom or online debate in Spanish on whether technology improves or harms our lives.

Deliverables: 5 arguments per side, summary paragraph (100 words).

Success: Logical structure, discourse markers (*por un lado..., por otro lado...*), confident rebuttal.

G1.3 “Interview project”

Focus: Real interaction.

Task: Interview a Spanish speaker (teacher, exchange student, or online partner) about their routine and city.

Deliverables: 6-question script + transcript or notes.

Success: Natural question forms, correct tense use, polite tone.

2) Extended Writing & Creative Expression

G2.1 “Travel blog”

Focus: Narrative writing.

Task: Write a 200-word blog post describing your dream trip to a Spanish-speaking country.

Include 3 tenses (past experience, current impressions, future plan).

Add at least 5 cultural words (food, places, festivals).

Success: Logical flow, stylistic flair, correct tenses.

G2.2 “Opinion article: The environment”

Focus: Argumentative writing (Theme: *Global issues*).

Task: Write a 180-word newspaper column titled “*Mi planeta, mi responsabilidad.*”

Mention 3 global problems (contaminación, deforestación, pobreza).

Propose 3 solutions using *debería* / *se debe* / *es necesario que* + *subjunctive*.

Success: Advanced structures, cohesive flow, persuasive tone.

G2.3 “Letter to my future self”

Focus: Reflective writing.

Task: Compose a 150-word letter about your goals and dreams, using **future and conditional tenses**.

Viajaré, estudiaré, me gustaría, podría...

Success: Emotional, authentic tone; grammatical accuracy.

3) Advanced Listening & Shadowing

G3.1 “News summary challenge”

Focus: Listening comprehension and summarisation.

Task: Listen to a 2-minute *BBC Mundo* or *News in Slow Spanish* clip.

Note 10 key words + 3 facts.

Summarise in Spanish (50 words).

Success: Captures gist and one supporting detail.

G3.2 “Shadow reading”

Focus: Pronunciation and rhythm.

Task: Read aloud with a native speaker recording, matching intonation and speed.

Choose any 100-word authentic text (article, ad, dialogue).

Success: Improved accent accuracy and pacing.

G3.3 “Podcast review”

Focus: Listening for opinion and tone.

Task: Listen to a Spanish podcast episode on youth culture or music.

Write a 120-word review in English: topic, host tone, vocabulary learned.

Success: Identifies tone, audience, and main ideas.

4) Translation & Linguistic Precision

G4.1 “Creative translation: short story”

Focus: Accuracy and nuance.

Task: Translate a short English story (100 words) into Spanish, preserving tone and rhythm.

Extension: Compare with AI translation — note 5 differences.

Success: Faithful yet natural rendering.

G4.2 “Reverse translation”

Focus: Grammar awareness.

Task: Translate a Spanish news paragraph into English, then back to Spanish without looking.

Success: Recovers 80%+ of original phrasing correctly.

G4.3 “Proverbs & idioms challenge”

Focus: Figurative language.

Task: Research 10 Spanish idioms and match them with English equivalents.

Example: *Más vale tarde que nunca* → *Better late than never*.

Success: Correct meaning and context example for each.

5) Culture, Society & Global Awareness

G5.1 “Festival documentary”

Focus: Cultural analysis.

Task: Research a Spanish or Latin American festival (*San Fermín, Día de los Muertos, Inti Raymi*).

Create a 5-slide digital presentation in Spanish (who, where, when, why, symbolism).

Success: Authentic information, visual accuracy, full sentences in target language.

G5.2 “Artist study: Frida Kahlo / Pablo Neruda / Rosalía”

Focus: Literature and arts.

Task: Research one artist or musician.

Write 150 words about their work and influence.

Extension: Include one quote or lyric in Spanish and comment on its meaning.

Success: Analytical, not biographical.

G5.3 “Social issue investigation”

Focus: Critical thinking.

Task: Choose a theme (*la pobreza, la igualdad, el medio ambiente*).

Read 1–2 Spanish sources.

Summarise cause, impact, and possible solutions in 150 words.

Success: Balanced discussion with factual support.

6) Interdisciplinary Projects

G6.1 “STEAM in Spanish”

Focus: Cross-subject vocabulary.

Task: Create a bilingual mini poster explaining a science concept in Spanish (e.g., *el cambio climático, la energía solar*).

Success: Correct terminology + creative visuals.

G6.2 “My bilingual podcast”

Focus: Speaking & media.

Task: Record a 3-minute bilingual podcast episode (Spanish + short English segments) about school life or technology.

Success: Smooth transitions between languages; clear pronunciation.

G6.3 “Cultural exchange e-journal”

Focus: Reflection.

Task: Keep a 2-week diary (in Spanish) imagining you live in Spain or Mexico.
Note school, meals, customs, and surprises.

Success: 7+ entries; realistic language; evidence of cultural understanding.

7) Literature & Film (Enrichment)

G7.1 “Short story reading”

Focus: Literary understanding.

Task: Read a short story or extract (*El principito*, *Cuentos cortos para niños*).
Write 5 comprehension questions + 5-line summary.

Success: Summarises accurately and identifies tone.

G7.2 “Film study: Coco / Voces Inocentes / Volver”

Focus: Cultural cinema.

Task: Watch a Spanish-language film.

Write a 200-word review (plot, characters, message, personal opinion).

Success: Uses vocabulary of emotion and evaluation.

G7.3 “Poetry reading”

Focus: Literary rhythm and vocabulary.

Task: Choose a short Spanish poem (Neruda, Lorca).

Read aloud and annotate 5 literary devices (metaphor, imagery).

Success: Clear understanding of tone and meaning.

8) Reflection & Independent Learning

G8.1 “My language portfolio”

Focus: Evidence compilation.

Task: Collect recordings, essays, translations, reflections, and projects into one bilingual portfolio.

Success: Organised, labelled by skill area.

G8.2 “My learning strategy”

Focus: Metacognition.

Task: Write 150 words on how you study Spanish effectively — tools, routines, challenges.

Success: Specific examples; actionable plan.

G8.3 “Beyond GCSE: My next steps”

Focus: Lifelong learning.

Task: Research post-GCSE opportunities (A-Level, IB, DELE, exchange). Create a personal roadmap in English/Spanish mix.

Success: Realistic goals, practical steps.

✓Mini Rubric for Guide Tasks

Level	Indicators
Outstanding (17–20)	Fluent, coherent, accurate use of complex language; integrates cultural context confidently.
Strong (13–16)	Good range of structures and vocabulary; minor errors; shows independence and cultural insight.
Developing (8–12)	Task achieved; some inconsistency in grammar or expression; limited depth.
Emerging (<8)	Minimal detail; vocabulary gaps; weak structure or accuracy.

Implementation Guidance

Frequency	Output	Suggested Tools
1 project every 2–3 weeks	Written or multimedia	Canva, Padlet, Flipgrid, Google Docs, Notion
Each term	1 speaking + 1 writing + 1 cultural piece	
At year's end	"Mi Portafolio de Español" (digital or printed)	

Recommended Authentic Resources

📖 **Reading:** *El País Educación, BBC Mundo, Revista Eñe*

🎧 **Listening:** *Radio Nacional de España, Notes in Spanish, News in Slow Spanish*

📺 **Viewing:** *Netflix – La Casa de Papel, Coco, Las Chicas del Cable*

📱 **Apps:** *LingQ, Anki, Clozemaster, FluentU*

📖 **Literature:** *Pablo Neruda – Poemas de amor, Laura Esquivel – Como agua para chocolate*

Physical Education — Basic Supercurriculum (EDEXCEL GCSE / IGCSE)

1) Fitness Foundations

PE1.1 “Know your components”

Focus: Components of fitness.

Task: Write a definition and example for: *strength, endurance, flexibility, speed, coordination, balance*.

Example: “Flexibility = range of motion at a joint → gymnast performing splits.”

Success: All six definitions accurate and linked to a sport.

PE1.2 “Mini fitness test”

Focus: Self-assessment.

Task: Choose 3 quick tests:

Sit-ups in 30 sec (muscular endurance)

Sit-and-reach (flexibility)

Sprint 20m (speed)

Record your results + compare next month.

Success: Accurate recording, correct link between test and component.

PE1.3 “Heart rate check”

Focus: Monitoring exercise intensity.

Task: Take resting pulse, then jog for 2 min, retake. Calculate % increase.

Success: Understands that higher effort = higher heart rate.

2) Anatomy & Physiology Basics

PE2.1 “Label the body”

Focus: Skeleton system.

Task: Label 10 key bones on a simple diagram (e.g., skull, femur, tibia, humerus, pelvis).

Success: 8+ correct placements.

PE2.2 “Joint action in sport”

Focus: Movement terminology.

Task: Match 6 actions to joints (e.g., flexion – bicep curl, extension – football kick).

Success: Correct movement-term pairing.

PE2.3 “Muscle match”

Focus: Muscular system.

Task: Draw arrows showing which muscles work in a push-up and in a squat. Label agonist/antagonist.

Success: Identifies major pairs (biceps/triceps, quadriceps/hamstrings).

3) Health, Diet & Lifestyle

PE3.1 “Food diary”

Focus: Nutrition awareness.

Task: Record everything eaten in one day. Mark which meals included **carbs, protein, fat, fibre, water**.

Success: Correct identification; 1–2 improvement suggestions.

PE3.2 “Hydration quiz”

Focus: Fluid balance.

Task: Research how dehydration affects performance. Write 3 key symptoms + 3 prevention tips.

Success: Clear physiological links (fatigue, slower reaction, cramps).

PE3.3 “Sleep and sport”

Focus: Recovery & lifestyle.

Task: Reflect: how many hours do you sleep per night? Write 3 sentences on how rest affects performance.

Success: Relates fatigue to performance outcomes.

4) Skills, Technique & Performance

PE4.1 “Skill vs ability”

Focus: Key definitions.

Task: Define both and give one example per sport.

Example: “Skill = learned action (tennis serve); ability = natural coordination.”

Success: Distinction clear; both linked to sport.

PE4.2 “Types of skill”

Focus: Open/closed, simple/complex skills.

Task: Write two examples per type.

Open = pass in football

Closed = basketball free throw

Success: Accurate categorisation.

PE4.3 “Video breakdown”

Focus: Observation & feedback.

Task: Watch a 30-sec clip of a professional sport. Identify one skill performed well and one improvement area.

Success: Uses technical vocabulary (timing, control, coordination).

5) Exercise, Training & Improvement

PE5.1 “FITT principle quick plan”

Focus: Training design.

Task: Create a short plan to improve one area of fitness using FITT (Frequency, Intensity, Time, Type).

Example: Jog 3x/week, 20 min, moderate pace.

Success: All four elements addressed.

PE5.2 “SMART goals”

Focus: Motivation & planning.

Task: Write one goal for your sport using the SMART framework.

Specific, Measurable, Achievable, Realistic, Time-bound.

Success: Clear measurable detail (e.g., “Run 5km under 30 minutes in 4 weeks”).

PE5.3 “Warm-up design”

Focus: Safety & preparation.

Task: Create a 3-part warm-up: pulse raiser, stretches, skill rehearsal.

Success: Logical sequence, sport-specific.

7) Psychology & Motivation in Sport

PE6.1 “Why do I play?”

Focus: Intrinsic vs extrinsic motivation.

Task: Write a paragraph explaining what motivates you to exercise.

Success: Uses correct terms (enjoyment vs rewards).

PE6.2 “Visualisation exercise”

Focus: Mental rehearsal.

Task: Close eyes and imagine completing a skill successfully. Write 3 sentences on how it felt and how it may improve performance.

Success: Connects mental focus with confidence.

PE6.3 “Team spirit reflection”

Focus: Team dynamics.

Task: After a group activity, write one example of **communication**, **leadership**, or **cooperation** that improved play.

Success: Identifies positive behaviour linked to performance.

7) Sports, Society & Culture

PE7.1 “Global sports map”

Focus: Sporting geography.

Task: On a blank world map, label 10 countries and one famous sport or event from each.

Success: Accurate country-sport matching (e.g., Brazil–Football, Japan–Judo).

PE7.2 “Olympic values”

Focus: Ethics & fair play.

Task: List the 6 Olympic values. Choose one athlete who represents them and explain why.

Success: Clear link between value and real example.

PE7.3 “Media & sport”

Focus: Influence of media.

Task: Find one recent sports headline. Explain in 3 lines how media can help or harm athletes.

Success: Balanced reasoning.

8) Reflection & Self-Evaluation

PE8.1 “Weekly reflection”

Focus: Self-assessment.

Task: Write 3 bullet points:

What I improved

What challenged me

What I’ll try next week

Success: Honest and specific reflection.

PE8.2 “Fitness tracking”

Focus: Consistency.

Task: Record one fitness measure weekly for 4 weeks (time, reps, HR). Plot progress.

Success: Trend visible; data accurate.

PE8.3 “My health triangle”

Focus: Physical, mental, social balance.

Task: Draw a triangle labelled *physical* / *mental* / *social health*. Add one behaviour under each that helps you stay balanced.

Success: All dimensions connected to daily life.

✓Mini-Rubric for Basic PE Tasks

Level	Description
Emerging (1–2)	Shows basic effort but limited understanding of concepts.
Developing (3)	Demonstrates sound knowledge, some minor inaccuracies.
Secure (4)	Clear understanding and accurate explanations; applies to own experience.
Excellent (5)	Insightful, reflective, applies theory to practice confidently.

Suggested Weekly Structure

Day	Focus	Example
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Monday	Fitness & anatomy	Mini test or drawing
Wednesday	Skills & motivation	20-min activity + reflection
Friday	Health & culture	Short research + summary

Recommended Quick Resources

- 🔗 BBC Bitesize GCSE PE
- 🔗 YouTube: *EverLearner PE*, *My PE Exam*
- 🔗 Apps: *MapMyRun*, *Nike Training Club*, *Strava*
- 🔗 Sports news: *BBC Sport*, *The Guardian Sport*
- 🔗 Anatomy visualiser: *InnerBody.com* (for muscles & bones)

Physical Education — Extended Supercurriculum (EDEXCEL GCSE / IGCSE)

1) Fitness & Training Design

E1.1 “Personal Fitness Plan”

Focus: Applying the FITT and SMART principles.

Task: Design a 4-week plan targeting one component (e.g. stamina, flexibility, strength).

Specify frequency, intensity, time, type.

Add SMART goal (e.g., “Improve plank hold from 30s → 60s by Week 4”).

Deliverables: Table + weekly reflection log.

Success: Plan realistic, measurable, and progressive.

E1.2 “Target heart rate zones”

Focus: Calculating and using training zones.

Task:

Calculate your maximum HR ($220 - \text{age}$).

Determine aerobic (60–80%) and anaerobic (80–90%) zones.

Record HR during 2 different workouts; plot graph.

Success: Data accurate; correct analysis of training intensity.

E1.3 “Circuit training challenge”

Focus: Exercise selection and evaluation.

Task:

Create a 6-station circuit (bodyweight only).

Test once a week; record performance each round.

Reflect on improvements and limiting factors.

Success: Logical progression; applies principles of overload and recovery.

2) Anatomy & Movement Application

E2.1 “The movement match-up”

Focus: Joint actions and muscles in performance.

Task: Choose 3 sporting skills (e.g., football kick, basketball jump, tennis serve).

For each, identify:

Main joint action

Agonist & antagonist muscles

Type of contraction (isotonic concentric/eccentric or isometric).

Deliverables: Table with annotated diagrams.

Success: Accurate pairings; links to movement phases.

E2.2 “Energy system snapshot”

Focus: ATP production and energy pathways.

Task: Pick 3 sports and state which energy system dominates (ATP-PC, lactic acid, aerobic).

Justify with duration/intensity data.

Deliverables: Chart or infographic.

Success: Correct matching; valid physiological reasoning.

E2.3 “The circulatory loop”

Focus: Understanding oxygen transport.

Task: Draw and label a diagram of the double circulatory system.

Trace blood flow before, during, and after exercise.

Include arrows, valves, oxygenated/deoxygenated blood.

Success: Diagram labelled, colour-coded, accurate flow direction.

3) Health, Nutrition & Wellbeing

E3.1 “Macronutrient breakdown”

Focus: Balanced diet and sport performance.

Task: Research recommended daily intakes for athletes vs non-athletes.

Present as a bar chart.

Reflect in 100 words: How does nutrition affect recovery?

Success: Data correct, reflection links diet to performance.

E3.2 “Dietary analysis challenge”

Focus: Applying knowledge.

Task: Keep a 3-day food diary; categorise foods into carbs, fats, proteins, micronutrients.

Compare intake to athlete guidelines.

Suggest 2 improvements.

Success: Accurate categorisation; realistic nutrition advice.

E3.3 “Lifestyle evaluation”

Focus: Health choices.

Task: Assess your daily habits under **Physical / Social / Emotional** headings.

Give one behaviour per category that helps and one that hinders wellbeing.

Success: Balanced self-assessment and improvement plan.

4) Skill, Technique & Performance

E4.1 “Biomechanical breakdown”

Focus: Technique analysis.

Task: Record yourself performing a skill (e.g., free throw, serve, sprint start).

Use slow motion to identify phases: preparation, action, recovery.

Annotate with forces, balance, and follow-through.

Success: Identifies mechanical principles (force, leverage, stability).

E4.2 “Skill classification case study”

Focus: Theory in practice.

Task: Choose 2 contrasting skills.

Classify each as **open/closed, gross/fine, self-paced/externally paced.**

Explain how this affects training design.

Success: Correct classifications with sport-specific reasoning.

E4.3 “Feedback in action”

Focus: Coaching and self-improvement.

Task:

Ask a partner/teacher to give verbal and visual feedback.

Record both and reflect on which improved performance most.

Success: Shows understanding of intrinsic vs extrinsic feedback.

❏ 5) Exercise Physiology & Training Effects

E5.1 “Short-term effects experiment”

Focus: Immediate physiological responses.

Task: Record HR, breathing rate, and perceived exertion pre-, during-, and post-exercise.

Plot changes on a line graph.

Success: Clear data trend; correct physiological interpretation.

E5.2 “Long-term training adaptations”

Focus: Aerobic vs anaerobic training outcomes.

Task: Research 5 physiological changes after 6 weeks of consistent training (e.g., increased stroke volume, capillarisation).

Link each to improved performance.

Success: All adaptations correctly explained with sport examples.

E5.3 “Recovery & injury prevention”

Focus: Safety and rehabilitation.

Task: Design a 10-minute post-session recovery routine including stretching, hydration, and nutrition.

Success: Balanced plan; connects recovery to reduced injury risk.

❏ 6) Sport Psychology in Context

E6.1 “Personality and performance”

Focus: Psychological profiles.

Task: Research Type A / Type B personalities.

Write 100 words on how personality affects performance in team vs individual sports.

Success: Clear connection between traits and behaviour.

E6.2 “Arousal curve study”

Focus: Yerkes–Dodson law.

Task: Draw arousal–performance graph; explain in your own words.

Apply to one athlete example (archer vs sprinter).

Success: Accurate curve and logical explanation.

E6.3 “Goal setting case”

Focus: Motivation and confidence.

Task: Interview a peer about their training goals; categorise them (process/performance/outcome).

Give one suggestion for improvement.

Success: Uses correct terminology and constructive feedback.

7) Socio-Cultural Influences

E7.1 “Sport for all?”

Focus: Access and equality.

Task: Research one barrier to participation (e.g., gender, cost, disability).

Propose 2 solutions for inclusion.

Success: Balanced argument with practical ideas.

E7.2 “Role model analysis”

Focus: Media & representation.

Task: Choose one athlete. Explain how they inspire participation or controversy.

Include one quote or statistic.

Success: Explains media impact clearly.

E7.3 “Ethics in sport”

Focus: Fair play, drugs, and conduct.

Task: Write a 200-word essay: “Should doping bans be lifelong?”

Use evidence and sporting examples.

Success: Coherent argument and evaluation.

8) Data & Evaluation Skills

E8.1 “Performance data table”

Focus: Collecting and analysing data.

Task: Track two measurable variables (e.g., 1km run time, push-ups) for 4 weeks.

Create a graph; interpret trends.

Success: Accurate data representation and realistic conclusions.

E8.2 “Observation schedule”

Focus: Systematic analysis.

Task: Observe a teammate in one sport. Create a checklist for success criteria (e.g., accuracy, decision-making).

Success: Uses data to justify feedback.

E8.3 “Performance review poster”

Focus: Evaluation & presentation.

Task: Combine visual data, photos, and comments into a performance summary (before/after training).

Success: Clear structure; uses evidence to support progress.

9) Reflection & Personal Growth

E9.1 “My PE journey”

Focus: Metacognitive awareness.

Task: Reflect on what you’ve improved most this term (skill, fitness, confidence).

Deliverable: 150-word reflection.

Success: Honest, detailed self-analysis.

E9.2 “Mind–body balance”

Focus: Holistic health.

Task: Keep a 7-day log of physical and mental wellbeing (mood, energy, stress).
Identify 2 links between exercise and emotional state.

Success: Shows understanding of mental health benefits.

E9.3 “Next steps in sport”

Focus: Lifelong participation.

Task: Research 3 ways to stay active beyond school (local clubs, volunteering, online fitness).

Success: Feasible plan connected to personal interests.

✓Mini-Rubric for Extended Tasks

Level	Indicators
Emerging (1–2)	Basic attempt; limited link between theory and application.
Developing (3)	Some understanding; partial accuracy in data or reasoning.
Secure (4)	Clear application; logical, accurate explanations with examples.
Excellent (5)	Analytical, evidence-based response; connects multiple PE concepts confidently.

Suggested Term Plan

Week	Focus Area	Output
1–2	Fitness & Training	4-week plan + HR log

3-4	Anatomy & Movement	Diagrams + analysis
5-6	Psychology & Motivation	Reflection essay
7-8	Data & Evaluation	Report or presentation

Resources

📖 **Textbook:** EDEXCEL GCSE (9-1) PE Student Book

📺 **Video:** *The EverLearner PE, My PE Exam*

🌐 **Sites:** BBC Bitesize PE, TeachPE.com

📱 **Apps:** Strava, MyFitnessPal, Fitbod (for training data)

📖 **Reference frameworks:** FITT, SMART, RAMP, Yerkes-Dodson, Principles of Training

Physical Education — Guide Supercurriculum (EDEXCEL GCSE / IGCSE)

1) Personal Training Project

Focus: Apply physiological, biomechanical, and psychological knowledge to design and evaluate your own programme.

Task:

Select **one area for improvement** (e.g. cardiovascular endurance, reaction time, strength).

Design a **6-week personal training plan** following:

FITT principles (Frequency, Intensity, Time, Type)

SMART targets (Specific, Measurable, Achievable, Realistic, Time-bound)

Include warm-up, cool-down, progression, and recovery.

Deliverables:

Training log with HR data and self-assessment.

300-word written evaluation on effectiveness and limitations.

Success:

Clear physiological reasoning.

Evidence-based adjustments.

Demonstrates understanding of *overload*, *specificity*, and *reversibility*.

2) Applied Anatomy & Physiology Analysis

Focus: Deep anatomical understanding and biomechanical linkage.

Task:

Choose **three sports skills** (e.g. sprint start, volleyball spike, shot put).

Analyse each using:

Bones & joints involved

Agonist/antagonist muscles

Types of contraction

Planes and axes of movement

Energy systems in use

Deliverables:

Detailed labelled diagrams and written commentary (400–500 words total).

Annotate with **sport images or stills**.

Success:

Precise anatomical language.

Links mechanics to performance outcome.

Connects movement to energy pathways logically.

3) Performance Analysis Investigation

Focus: Objective analysis and evaluation of live performance.

Task:

Record or observe a full performance in your main sport.

Analyse using these steps:

Identify strengths and weaknesses (skills, fitness, tactics).

Apply **data collection methods** (observation schedule, match stats, timing).

Link performance outcomes to **fitness components** and **training principles**.

Write a **500-word evaluation** with suggestions for improvement.

Success:

Structured, evidence-based commentary.

Links data directly to theory.

Includes justified, specific improvement plan.

❓ 4) Sports Psychology & Motivation Portfolio

Focus: Psychological influences on elite performance.

Task:

Choose **one psychological factor**: arousal, anxiety, motivation, or confidence.

Produce a **research poster** explaining:

Theoretical background (e.g. Yerkes-Dodson Law).

One athlete case study.

Two techniques for management or improvement (goal-setting, relaxation, self-talk).

Reflect in 150 words on your own mental approach in PE or sport.

Success:

Accurate theory; applied to real or personal context.

Professional layout and clarity of analysis.

❓ 5) Health & Lifestyle Case Study

Focus: Nutrition, rest, and holistic performance.

Task:

Research a **professional athlete's diet and recovery regime**.

Create a 1-page comparison with your own:

Macronutrients, hydration, sleep, mental health strategies.

Write 200 words analysing *what you could adapt* to improve performance and wellbeing.

Success:

Balanced evaluation using sport science.

Practical recommendations supported by reasoning.

🏆 6) Leadership & Coaching in Practice

Focus: Planning, communication, and reflection.

Task:

Plan and lead a **10-minute warm-up or skills session** for peers or younger pupils.

Include learning objective, equipment list, safety checks.

Gather peer feedback.

Reflect:

What went well?

What would you improve?

How did communication affect motivation and performance?

Success:

Leadership clarity, safety awareness, adaptive feedback.

Written reflection (200–300 words) demonstrating insight and responsibility.

7) Socio-Cultural & Ethical Debate

Focus: Evaluate sport's role in society.

Task Options (choose one):

Essay (300 words) – *“Technology in sport enhances fairness.” Discuss.*

Podcast (2–3 mins) – *“The media's role in creating or destroying sporting heroes.”*

Panel debate – *“Should trans athletes compete in open categories?”*

Deliverables:

Structured argument with evidence from at least 2 sports.

Evaluation of ethical, legal, and social perspectives.

Success:

Balanced argument; supported by research.

Demonstrates understanding of equality, access, and moral reasoning.

8) Data Handling & Evaluation Skills

Focus: Quantitative analysis and interpretation.

Task:

Collect performance data (e.g., reaction time, vertical jump, 12-min run).

Display using graphs or tables.

Calculate % improvement and standard deviation.

Write 150 words analysing reliability and validity of results.

Success:

Accurate calculations.

Evaluates test limitations (e.g., environment, motivation).

9) Long-Term Athlete Development Pathway

Focus: Linking theory to lifelong participation.

Task:

Research one athlete's career from grassroots to elite.

Identify key milestones: foundation → participation → performance → excellence.

Note support systems (clubs, coaches, facilities).

Compare to your own opportunities.

Suggest how schools/communities can better support talent.

Deliverables:

Timeline diagram + 300-word reflection.

One practical improvement suggestion.

Success:

Real-world understanding of development pathways.

Evidence of critical thinking.

10) Reflection & Goal Evaluation

Focus: Self-regulation and independent learning.

Task:

Review your term's work using these prompts:

What have I improved physically?

What have I learned about the science of performance?

Which psychological strategy works best for me?

What's my next measurable goal?

Deliverables:

250-word final reflection or short video log.

Optional peer feedback form.

Success:

Insightful, self-aware, action-oriented reflection.

Mini-Rubric for Guide Tasks

Level	Indicators
Outstanding (17–20)	Deep analysis, advanced scientific and evaluative thinking, real evidence used.
Strong (13–16)	Solid understanding, logical application, supported with accurate theory.
Developing (8–12)	Some analytical insight; partial evidence or limited evaluation.
Emerging (<8)	Basic description without depth or clear linkage.

Suggested Guide-Level Project Plan

Term	Project	Outcome
Autumn	Personal Training Plan + Anatomy Analysis	6-week portfolio + diagrams
Spring	Performance Analysis + Psychology Poster	Report + visual presentation
Summer	Coaching Session + Ethical Debate	Leadership & Evaluation Essay

Recommended Resources

Books:

Edexcel GCSE (9–1) PE Student Book

Sport and Exercise Science: An Introduction for All Levels (Hodges)

Websites:

BBC Bitesize PE

TeachPE.com

Sports Coach UK

Videos:

The EverLearner PE, PE with Mr Duda, CrashCourse Anatomy & Physiology

Apps/Tools:

Strava (endurance data)

MyFitnessPal (diet tracking)

Coach's Eye (video analysis)

Canva / Padlet (poster or reflection boards)

📋 Evidence Portfolio Template (recommended)

Section	Description
1. Project title	e.g. "Personal Fitness Plan"
2. Objective	What I aimed to improve
3. Methods	Training type, schedule, principles
4. Data collected	Results, HR, performance stats
5. Analysis	Graphs, comparison, interpretation
6. Reflection	What worked / next steps

Chinese — Basic Supercurriculum

(EDEXCEL GCSE / IGCSE — Mandarin, simplified characters; traditional optional)

1) Pronunciation & Tones (Foundation)

C1.1 Tone gym (5 mins/day)

Say 5 minimal pairs aloud: 妈 mā / 麻 má / 马 mǎ / 骂 mà / 吗 ma.

Record yourself; check consistency.

C1.2 Pinyin quick map

Create a chart of finals you find hard (e.g., ü, -iong, -uan). Add 2 example words each.

C1.3 Tone sandhi snapshot

Practice 3→2 rule with 你好 nǐ hǎo → ní hǎo; 很好 hěn hǎo → hén hǎo.

Read 5 phrases slowly, then at natural speed.

2) Characters & Radicals (Little & Often)

C2.1 Stroke order mini-drill

Write 我、你、他、她、们 ×3 with correct stroke order.

Circle any inconsistent strokes.

C2.2 Radical hunt

Make a 6-box

page: 口 (mouth) 氵 (water) 女 (female) 心 (heart) 讠 (speech) 木 (wood) .

Find 2 characters per radical (e.g., 喝、问 for 口) and note meanings.

C2.3 Spaced copy

Copy 8 high-frequency words (HSK1/GCSE list) with pinyin + meaning:

中国、学生、老师、学校、朋友、喜欢、吃、看.

3) Speaking Starters (Set phrases → short turns)

C3.1 Greetings chain

Drill: 你好！你好吗？我很好，你呢？再见！

Goal: smooth intonation, no English filler.

C3.2 Self-intro 4 lines

我叫...；我...岁；我是...人；我住在...。 Record 20–30 secs.

C3.3 Café role-play (2–3 lines)

请问...？我要...；多少钱？谢谢！

Swap roles; keep it brisk.

4) Listening (Gist → Detail)

C4.1 Number sprint

Listen to numbers 1–100 (any beginner clip). Write what you hear for 5 items: ages, prices, times.

C4.2 Keyword bingo

Make a 3×3 grid of topic words

(e.g., 学校、老师、作业、喜欢、朋友、运动、电脑、手机、电影).

Play a beginner podcast 2–3 mins; tick words you catch.

C4.3 Date & time decode

Hear 今天星期几？现在几点？ Practice answers: 星期三；三点半.

5) Reading (Fast recognition)

C5.1 Sign spotting

Collect 6 screenshots/photos of real signs (e.g., 出口、入口、厕所、危险).

Translate in 1 line each.

C5.2 Menu mini

From a Chinese menu, list 5 dishes you recognise

(e.g., 饺子、米饭、面条、鸡、牛肉) + guess ingredients.

C5.3 Micro-texts

Read a 40–60 character profile (textbook/blog). Underline **time words** (今天、昨天、明天) and **measure words** (个、杯、张).

6) Writing (Controlled → Creative micro-outputs)

C6.1 Sentence bricks

Build 5 SVO sentences with 也/很/都:

我也喜欢运动。她很忙。我们都在学校。

C6.2 Postcard 3 lines

Topic My weekend: 我星期六...；我跟...一起...；因为...所以....

C6.3 Emoji story (5 lines)

Use emojis (🐼🐼🐼🐼) to write 5 simple lines in Chinese; add pinyin under tough words.

7) Grammar Micro-tasks (Core exam structures)

C7.1 是 / 有 / 在

Make 6 contrasts: 这里是教室 / 这里有电脑 / 我在教室 (meaning changes!).

C7.2 Questions: 吗 / 呢 / 多少 / 几

Convert 4 statements to questions: 你是英国人。→ 你是英国人吗？

你呢？ follow-up practice.

C7.3 了 / 过 taster

Write 3 past experience sentences with 过: 我去过中国 / 我吃过北京烤鸭.

Write 3 completed actions with 了: 我做了作业.

8) Culture & Real-world Context

C8.1 Country snapshot

3 facts about one place: 北京/上海/台北/香港 (food, landmark, transport).

C8.2 Festival flash

5 bullets for 春节/中秋节 (date, food, symbol, greeting 新年快乐 / 中秋快乐).

C8.3 Name card

Design your Chinese name card: 中文名 + 拼音 + 英文名; explain meaning if known.

9) GCSE/IGCSE Topic Boosters (pick one per week)

Identity & culture: family words (爸爸、妈妈、哥哥、妹妹); 会/能/要 for abilities/plans.

Local area/travel: places (银行、饭店、地铁), directions (左/右/往前), tickets/prices.

School: subjects (数学、历史、体育), timetable (第一节...), opinions with 因为/所以.

Future & work: jobs (医生、老师、工程师), 想/打算.

Media & tech: 手机、电脑、上网、发短信; pros/cons with 但是/虽然...可是....

10) Study Habits & Retrieval

C10.1 10-word deck

Make 10 flashcards/week (character → pinyin/meaning → example). Review day 1/3/7.

C10.2 One-minute talk

Record 60s on a topic (family/school/weekend). Aim for **3 set phrases + 1 connector**.

C10.3 Weekly reflection (3 bullets)

New words learned; still tricky; next goal.

✓Mini-Rubric (any Basic task)

Level	Indicators
Emerging (1–2)	Some recall; tone/character errors block meaning.
Developing (3)	Simple phrases mostly accurate; occasional tone/word-order slips.
Secure (4)	Clear meaning; correct tones in set phrases; characters/pinyin aligned.
Excellent (5)	Fluent short turns; accurate tones; neat characters; uses connectors/opinions.

Suggested Weekly Routine (≈40–60 mins total)

Mon: Tones + characters (C1.1 + C2.1, 10–15')

Wed: Listening or speaking (C4.x or C3.x, 15–20')

Fri: Reading or writing (C5.x or C6.x, 15–20')

Weekend: Culture + reflection (C8.x + C10.3, 10')

Handy Core Lists (starter)

Numbers 0–100 ; Days 星期一–星期日 ; Months 一月–十二月 ; Time 点/分/半/刻

Measure words: 个、位、张、杯、本

Connectors: 和、但是、因为...所以...、虽然...但是...、如果...就... (recognition at Basic)

Recommended Beginner Resources

Dictionaries/Tools: Pleco app; MDBG; Hanzi stroke order (YellowBridge/ArchChinese)

Reading: Du Chinese (Beginner); The Chairman's Bao (Foundation); Decodable graded readers

Listening: ChinesePod (Newbie), Coffee Break Chinese (early episodes), Chillchat (slow)

Video: Yoyo Chinese (pinyin/tones), Mandarin Corner (slow chats), Easy Mandarin (street interviews with subs)

Typing: Install Chinese keyboard; Practice pinyin-to-character selection for 5 mins/day

When these basics feel comfortable, we can move to the **Extended Supercurriculum** for Chinese (longer speaking turns, short essays, listening with note-taking, character goals, and mini cultural projects).

Chinese — Extended Supercurriculum
(EDEXCEL GCSE / IGCSE — Mandarin, simplified characters)

1) Extended Speaking Practice

E1.1 “My Life in Three Tenses”

Focus: Fluency & time reference.

Task: Prepare a 1-minute talk about your life:

过去 (past): 我去过北京。

现在 (present): 我每天学习中文。

将来 (future): 我想去中国旅行。

Success: Uses all 3 tenses with connectives (因为...所以..., 然后, 但是).

Extension: Record and self-assess pronunciation.

E1.2 “Role-play scenarios”

Focus: Real-world communication.

Task: Write and act two short dialogues (6–8 lines each):

Ordering food in a restaurant

Buying a train ticket

Grammar focus: Use question forms (多少钱? 怎么走?) and polite phrases (请、谢谢、不客气).

Success: Natural tone and interactional flow.

E1.3 “Mini interview project”

Focus: Interpersonal speaking.

Task: Pair up or use a voice note exchange. Ask 5 questions about hobbies, school, and daily life.

Use 吗 / 呢 / 几 / 多长时间 question types.

Deliverable: 2-minute recording + short transcript.

Success: No English; smooth question-answer rhythm.

2) Writing: Structure & Creativity

E2.1 “My Ideal Day” (理想的一天)

Focus: Connectors and adverbs of time.

Task: Write 80–100 characters describing your perfect day.

Include 早上、中午、下午、晚上 sequence.

Add one reason and opinion.

Success: Coherent flow; 3+ connectors used naturally.

E2.2 “Letter to a Pen Pal” (电子邮件)

Focus: Informal communication & paragraphing.

Task: Write an email introducing yourself and your school life.

Mention 3 school subjects, 1 hobby, and a question to your partner.

End with 祝好 or 再见.

Success: Polite tone, balanced structure, varied vocabulary.

E2.3 “Short story: The Lost Phone” (丢了手机)

Focus: Narrative writing & sequencing.

Task: Write 100–120 characters about losing and finding your phone.

Use 3–4 verbs in 了 form.

End with reflection: “我学到了...”

Success: Clear past sequence; correct use of 了 and 因为...所以....

2 3) Listening & Note-taking

E3.1 “News clip summary”

Focus: Extracting key information.

Task: Listen to a 2–3 min beginner/intermediate podcast (e.g., *ChinesePod*, *Slow Chinese*).

Write down 5 nouns, 3 verbs, and 1 adjective you understand.

Summarise in 3 lines (English or Chinese).

Success: Captures gist; uses real examples.

E3.2 “Character sound recognition”

Focus: Linking sound to written form.

Task: Choose 10 words you can say but not write. Listen → identify → write in characters.

Success: At least 7/10 correctly identified.

E3.3 “Mini documentary watch”

Focus: Cultural comprehension.

Task: Watch a 3–5 min video (e.g., *Mandarin Corner* “A Day in Beijing”).

List 5 new words + 2 facts about local life.

Success: Cultural note-taking + 50% vocabulary recall.

2 4) Reading & Interpretation

E4.1 “Weibo post decoding”

Focus: Authentic short texts.

Task: Find 2–3 social media posts (screenshots or textbook examples).

Translate key information (time, emotion, activity).

Success: Correctly identifies key words; conveys tone.

E4.2 “Short article comprehension”

Focus: Skimming & scanning.

Task: Read a 120-character article about school or travel.

Answer 5 questions (who, where, when, what, why).

Success: 4/5 correct; short written summary.

E4.3 “Character families”

Focus: Recognition and meaning-building.

Task: Group words by root meaning:

吃、喝、饭、菜、米、面 (food)

学、校、书、文、笔、字 (education).

Success: Can explain each meaning group in 1–2 lines.

5) Grammar in Action

E5.1 “Measure word matrix”

Focus: Accuracy and variety.

Task: Create a 3-column table: 个、张、本、杯、件、位 + 2 nouns per word.

Extension: Use 4 in full sentences.

Success: No mismatched measure words.

E5.2 “把 & 被 structures”

Focus: Sentence pattern mastery.

Task: Write 4 sentences each using 把 (active) and 被 (passive).

Ex: 我把书放在桌子上。 / 我的手机被朋友拿走了。

Success: Correct word order, particle placement.

E5.3 “Comparisons (比 / 没有 / 跟...一样)”

Focus: Comparative language.

Task: Write 5 comparison sentences (e.g., *Beijing is bigger than London*).

北京比伦敦大。 / 我的家没有学校远。 / 他跟我一样高。

Success: All structures correct and meaningful.

6) Cultural Exploration

E6.1 “Festival poster”

Focus: Cultural knowledge.

Task: Create a bilingual mini-poster about one Chinese festival (春节, 端午节, 中秋节).

Include: date, customs, food, and 3 key vocabulary.

Success: Accurate info + creative presentation.

E6.2 “Famous Chinese people”

Focus: Research + communication.

Task: Choose a figure (Confucius, Jackie Chan, Yao Ming, etc.).

Write 5–6 sentences introducing them.

Use 是/有/在/因为...所以....

Success: Grammatically correct, informative.

E6.3 “Travel brochure project”

Focus: Functional writing.

Task: Design a 1-page Chinese-English travel guide for a city in China.
Include 3 attractions, 2 foods, 1 travel tip.

Success: Clear layout + accurate vocab for location and activities.

7) Vocabulary & Translation

E7.1 “Synonym challenge”

Focus: Word range.

Task: Find 3 synonyms each for *big* (大), *good* (好), *beautiful* (漂亮), and *happy* (高兴).

Success: Correct meanings + context sentences.

E7.2 “Sentence translation drill”

Focus: Grammar accuracy.

Task: Translate 10 English sentences into Chinese (using current topics).

Focus on **word order, particles, and tense**.

Success: 80% grammatical accuracy.

E7.3 “Error correction notebook”

Focus: Conscious learning.

Task: Each week collect 3 mistakes (tone, structure, word). Rewrite correctly with explanation.

Success: Consistent self-correction habit.

8) Reflection & Communication

E8.1 “My Chinese learning diary”

Focus: Metacognition & self-assessment.

Task: Write 4–5 lines per week:

“This week I learned...”

“I found ... difficult.”

“Next week I will...”

Success: Reflective and specific.

E8.2 “Culture comparison essay”

Focus: Intercultural communication.

Task: Write 100–150 characters comparing your culture and Chinese culture (food, family, festivals).

Use 虽然...但是... and 不同.

Success: Balanced comparison and correct conjunctions.

E8.3 “One-minute presentation”

Focus: Confidence & clarity.

Task: Deliver a 60–90s talk about one topic: school, travel, food, or family.

Must include: 1 reason, 1 connector, 1 opinion.

Success: Natural tone, correct grammar, and confident delivery.

✓Mini-Rubric (for Extended Level Tasks)

Level	Description
Emerging (1–2)	Limited vocabulary; simple sentence patterns; some tone/character confusion.
Developing (3)	Clear message; mostly correct grammar; emerging flow in speech and writing.
Secure (4)	Confident, mostly accurate; combines connectives and tenses effectively.
Excellent (5)	Fluent, culturally aware, and grammatically precise; good intonation and style.

Suggested Monthly Cycle (example)

Week	Focus	Core Output
1	Speaking (E1)	1-min talk recording
2	Writing (E2)	100-character paragraph
3	Listening (E3) + Reading (E4)	Notes + summary
4	Culture (E6) + Reflection (E8)	Poster + diary entry

Recommended Resources

- 📖 **Listening:** *Slow Chinese, Mandarin Bean, ChinesePod (Elementary)*
- 📖 **Reading:** *The Chairman's Bao (HSK1–3), Du Chinese (graded)*
- 🗣️ **Speaking:** *HelloTalk, Tandem, Speechling (for feedback)*
- ✍️ **Writing/Character:** *Skritter, ArchChinese, Hanping Pro*
- 📺 **Culture:** *Mandarin Corner, Easy Mandarin, BBC Mandarin Learner Series*
- 📖 **Reference Textbook:** *EDEXCEL GCSE Chinese Student Book (Pearson)*

Chinese — Guide Supercurriculum
(EDEXCEL GCSE / IGCSE — Mandarin, simplified characters)

1) Extended Oral Communication & Debate

G1.1 “My Identity and Future” (我的身份和未来)

Focus: Expressing complex ideas and aspirations.

Task: Record a 2-minute talk (≈180–200 characters script) on who you are and what you want to do in the future.

Must include **three tenses** (我小时候... / 现在... / 将来...)

At least **one connector per sentence**

group (因为...所以..., 虽然...但是..., 如果...就...).

Deliverables: Script + video/audio.

Success: Fluent tone flow, clear structure, confident delivery.

G1.2 “Formal discussion: Technology in life” (科技对生活的影响)

Focus: Argumentation & register.

Task: Conduct a 3-minute recorded discussion with a partner:

Positive & negative impacts of technology.

Express opinion → support → counterargument.

Language goals: 认为、觉得、影响、方便、危险 + opinion linkers
(一方面...另一方面...).

Success: Balanced debate; no English; turn-taking smooth.

G1.3 “Interview project: Voices of China” (中国人的生活)

Focus: Authentic interaction.

Task: Interview (live or online) a native speaker about daily life, festivals, or education.

Prepare 8 questions + short transcript.

Reflect in 100 English words on what you learned culturally.

Success: Real engagement; correct register; intercultural understanding.

2) Advanced Writing & Reflection

G2.1 “My Dream City” (我理想的城市)

Focus: Descriptive & persuasive writing.

Task: Write 150–180 characters describing your ideal city — appearance, environment, lifestyle.

Include 5 adjectives, 3 comparative structures (比、没有、跟...一样).

Success: Logical paragraphing; uses advanced connectors
(不仅...而且..., 除了...以外...还...).

G2.2 “Social issue article: 环保从我做起”

Focus: Opinion essay.

Task: Write 180–200 characters about environmental protection or social responsibility.

Use **modal verbs + passive voice** (应该 / 必须 / 被...).

Include at least 1 rhetorical question.

Success: Strong argument, accurate formal register, natural transition words.

G2.3 “Creative journal: 我的中文之旅”

Focus: Personal reflection in narrative style.

Task: Keep a 1-week diary (60–80 characters/day) describing how you Practice Chinese and what frustrates/excites you.

Success: Shows self-expression, evolving vocabulary, and tone control.

3) Listening & Analysis

G3.1 “Podcast critique” (播客评论)

Focus: Detailed listening + evaluation.

Task: Listen to one *Slow Chinese* episode (3–5 min).

Note 10 new words + main idea + one opinion about delivery.

Success: Captures gist & details; interprets tone and bias.

G3.2 “Movie dialogue dictation” (电影听写)

Focus: Intensive listening accuracy.

Task: Choose a 30–45 sec clip from 哪吒, 流浪地球, or Coco 中文配音.

Pause → transcribe → check accuracy → translate.

Success: ≥80% transcription accuracy; smooth syntax awareness.

G3.3 “Mini news report” (新闻摘要)

Focus: Real-world comprehension.

Task: Listen to a Mandarin news summary (2–3 mins).

Note topic, 3 key facts, and 2 numbers/dates.

Present a 40–50 character written summary.

Success: Correct data recall, concise rephrasing.

4) Reading for Inference & Culture

G4.1 “Short story annotation” (短文分析)

Focus: Literary understanding.

Task: Read a 200–250-character story (e.g., 狐狸和乌鸦, 愚公移山).

Identify: 3 new words, 1 moral, 1 cultural value.

Success: Accurately explains moral lesson in English or Chinese.

G4.2 “Opinion reading journal” (读后感)

Focus: Expressing reaction.

Task: Read a blog or article on 学生压力 / 家庭关系 / 健康.

Write a 100-character reflection using 我同意 / 不同意, 因为....

Success: Demonstrates understanding and personal stance.

G4.3 “Character etymology study” (字的故事)

Focus: Cultural linguistics.

Task: Choose 5 Chinese characters and research their origins (pictographic, ideographic, etc.).

Write meanings + origin diagrams.

Success: Explains symbol–meaning link correctly.

5) Grammar & Syntax Mastery

G5.1 “Advanced connectors”

Focus: Flow & coherence.

Task: Create a connector sheet and use 10 in new sentences:

虽然...但是..., 只要...就..., 不但...而且..., 为了..., 由于...因此....

Success: Sentences natural and varied.

G5.2 “Complex sentence weaving”

Focus: Subordinate structures.

Task: Combine two short sentences into one using

*因为、所以、虽然、如果、但是、除了...还.... *

Example: 我喜欢看电影。我没有时间。→

虽然我喜欢看电影，但是我没有时间。

Success: All logical relationships correct.

G5.3 “Aspect mastery: 了, 过, 着”

Focus: Verbal aspect precision.

Task: Write 3 sentences each with 了, 过, 着.

Identify time context for each.

Success: Uses aspect particles with correct time logic.

6) Cultural & Research Projects

G6.1 “Chinese festival vlog” (中国节日视频)

Focus: Cultural presentation.

Task: Produce a 2–3 min bilingual video about a festival you’ve researched or celebrated.

Include key vocabulary + visuals (lanterns, food, music).

Success: Accurate info, expressive delivery, confident tone.

G6.2 “Modern China mini-report” (现代中国报告)

Focus: Social awareness & global connection.

Task: Research one modern issue: environment, tech, or education reform.

150–180 characters + infographic.

Success: Balanced analysis; academic style language.

G6.3 “Chinese art & symbolism” (中国艺术与象征)

Focus: Cultural aesthetics.

Task: Study one Chinese art form (calligraphy, Peking opera, painting, or architecture).

Create 5 bilingual slides (origin, features, vocabulary, symbolism).

Success: Correct terminology, engaging visual explanation.

7) Translation & Comparative Thinking

G7.1 “Translate the message” (双语翻译)

Focus: Meaning equivalence.

Task: Translate 5 short Chinese texts into natural English; then rewrite one in more authentic English tone.

Success: Maintains both content & nuance.

G7.2 “Cultural phrasebook” (文化短语本)

Focus: Idiomatic language.

Task: Collect 10 idioms or proverbs (成语) with English meaning + literal meaning + usage.

e.g., 对牛弹琴 — “playing the lute to a cow” → *talking to the wrong audience*.

Success: Accurate translation + context sentence.

G7.3 “English–Chinese comparison essay”

Focus: Language reflection.

Task: Write 120–150 characters comparing English and Chinese grammar or idioms.

Example: *Word order, tones, characters vs alphabet.*

Success: Clear linguistic insight; comparative reasoning.

8) Independent Projects & Portfolios

G8.1 “HSK3–4 Vocabulary Sprint”

Focus: Lexical expansion.

Task: Create digital flashcards (Anki/Quizlet) for 300–400 new words across 10 topics.

Success: Weekly 85% recall accuracy in self-tests.

G8.2 “Language & culture blog” (双语博客)

Focus: Public expression.

Task: Maintain a bilingual online or print blog: one 150-word post per month (Chinese + English).

Topics: learning tips, travel, music, cultural reflections.

Success: Consistent output; bilingual clarity.

G8.3 “Final portfolio presentation”

Focus: Cumulative mastery.

Task: Compile speaking recordings, essays, reflections, and one cultural project.

Present orally in Chinese (2–3 mins).

Success: Shows progress, fluency, and self-awareness.

✓Mini-Rubric for Guide Tasks

Level	Indicators
Outstanding (17–20)	Fluent, accurate, creative; integrates complex grammar and cultural context confidently.
Strong (13–16)	Very good accuracy; clear argument or description; minor errors.
Developing (8–12)	Understands main ideas; some complexity but limited accuracy.
Emerging (<8)	Basic communication; frequent grammar or tone slips; limited range.

📅 Suggested Year-Long Structure

Term	Focus	Output
Autumn	Personal & social topics	Oral talk + short essay
Spring	Global issues & research	Debate + article + vlog
Summer	Literature & portfolio	Story analysis + presentation

📖 Recommended Advanced Resources

Listening: *ChinesePod Upper Beginner, Slow Chinese (HSK3–4)*

Reading: *The Chairman’s Bao (HSK3+), Du Chinese Intermediate*

Grammar Reference: *Chinese Grammar Wiki (AllSet Learning)*

Vocabulary: *HSK Online App, Pleco flashcards*

Culture: *BBC Mandarin Learner, CGTN Culture, SupChina Podcasts*

Video Practice: *Mandarin Corner (intermediate), Yoyo Chinese Grammar Series*

📁 Suggested Portfolio Layout

Section	Content	Evidence Type
1. Speaking	Debates, monologues	Audio/video links
2. Writing	Essays, reflections, creative pieces	Text files or screenshots
3. Listening	Notes + transcriptions	PDFs / charts
4. Culture	Research or festival project	Slides/posters
5. Reflection	Weekly journals + goals	Word doc / Notion board

Greek — Basic Supercurriculum
(EDEXCEL GCSE / IGCSE — Modern Greek)

1) Speaking & Pronunciation Foundations

G1.1 “Alphabet challenge”

Focus: Sound recognition and reading fluency.

Task: Read aloud all 24 Greek letters + 5 example words (e.g., α – άνθρωπος, β – βιβλίο).

Success: Clear pronunciation; no confusion between similar letters (η/ι/υ, θ/δ).

G1.2 “Daily greetings” (Καλημέρα – Καλησπέρα – Καληνύχτα)

Focus: Everyday expressions.

Task: Practice a short greeting dialogue:

- Καλημέρα! Τι κάνεις;
- Καλά, ευχαριστώ! Εσύ;

Success: Correct tone and natural rhythm.

G1.3 “Name and origin” (Πώς σε λένε;)

Focus: Introducing oneself.

Task: Learn and repeat:

- Με λένε... (My name is ...)
- Είμαι από... (I'm from ...)
- Μένω στην... (I live in ...)

Success: Smooth delivery without hesitation.

2) Writing & Spelling Basics

G2.1 “Alphabet practice”

Focus: Correct letter formation.

Task: Write all uppercase and lowercase letters twice.

Success: Legible handwriting; consistent accent placement (').

G2.2 “Ten key phrases”

Focus: Essential survival phrases.

Task: Write 10 sentences (with translation):

- Γεια σου / Καλημέρα / Ευχαριστώ / Παρακαλώ / Συγγνώμη / Πόσο κοστίζει; / Τι κάνεις; / Ναι / Όχι / Εντάξει.

Success: Accurate spelling and diacritics.

G2.3 “My family” (Η οικογένειά μου)

Focus: Vocabulary and structure.

Task: Write 5 short sentences about your family:

- Έχω έναν αδελφό και μία αδελφή.
- Η μητέρα μου δουλεύει σε σχολείο.

Success: Correct article–noun agreement.

🔍 3) Listening Practice

G3.1 “Number quiz” (Αριθμοί)

Focus: Listening comprehension.

Task: Listen to Greek numbers 1–20; repeat and write down 5 random ones.

Success: Correct recognition and spelling.

G3.2 “Days & months” (Ημέρες & μήνες)

Focus: Familiarisation with time words.

Task: Listen to a recording of days of the week; write them in order.

Δευτέρα, Τρίτη, Τετάρτη, Πέμπτη, Παρασκευή, Σάββατο, Κυριακή.

Success: Full sequence memorised.

G3.3 “Mini dialogue dictation”

Focus: Everyday listening.

Task: Listen to a short 2-line dialogue and fill in missing words.

Example: – Τι κάνεις; – ... (Καλά, ευχαριστώ.)

Success: 80% transcription accuracy.

🔍 4) Reading for Meaning

G4.1 “Cognate hunt”

Focus: Recognising Greek–English similarities.

Task: From a short text, find 10 words that look or sound similar to English (τηλέφωνο, μουσική, πρόβλημα).

Success: Correct identification + English meaning.

G4.2 “Signs around us”

Focus: Everyday literacy.

Task: Search online or photograph 5 public Greek signs (Έξοδος, Είσοδος, Φαρμακείο, Ταμείο, Τουαλέτα) and translate.

Success: Meaning correctly interpreted.

G4.3 “Simple biography”

Focus: Comprehension.

Task: Read a 50–80 word text about a person (Ο Γιάννης είναι μαθητής. Μένει στην Πάτρα...).

Answer 3 questions: Who? Where? What?

Success: 3/3 answers correct.

🔍 5) Everyday Conversation Practice

G5.1 “In the café” (Στο καφέ)

Focus: Ordering & politeness.

Task: Practice:

Θα ήθελα έναν καφέ, παρακαλώ.

Πόσο κάνει;

Ορίστε, ευχαριστώ.

Success: Fluent use of polite forms (*θα ήθελα, παρακαλώ, ευχαριστώ*).

G5.2 “At school” (Στο σχολείο)

Focus: School vocabulary.

Task: Learn 10 words: *μαθητής, δάσκαλος, τάξη, πίνακας, βιβλίο, θρανίο, μάθημα, διάλειμμα, τετράδιο, σχολείο.*

Extension: Write 3 sentences using them.

G5.3 “My daily routine” (Η καθημερινή μου ζωή)

Focus: Simple present tense.

Task: Write or say 5 lines describing your day:

Ξυπνάω στις επτά. Πηγαίνω στο σχολείο. Τρώω το μεσημέρι...

Success: All verbs correctly conjugated in the present.

6) Grammar Foundations

G6.1 “Articles & gender”

Focus: Article–noun agreement.

Task: Group 6 nouns by gender:

ο πατέρας, η μητέρα, το παιδί.

Success: 6/6 correct + colour-coded chart.

G6.2 “Verb endings -ω / -εις / -ει”

Focus: Conjugation of 1st group verbs.

Task: Practice *μιλάω / γράφω / τρώω / παίζω* for *εγώ, εσύ, αυτός/αυτή.*

Success: All forms correct.

G6.3 “Adjectives practice”

Focus: Agreement & use.

Task: Match adjectives with nouns:

ο μεγάλος σκύλος, η όμορφη πόλη, το μικρό παιδί.

Success: Gender and endings accurate.

7) Greek Culture & Everyday Life

G7.1 “Greek map quiz”

Focus: Geography vocabulary.

Task: Label: *Αθήνα, Πάτρα, Θεσσαλονίκη, Κρήτη, Ρόδος, Σαντορίνη.*

Success: Correct spelling and region recognition.

G7.2 “Food favourites” (Ελληνική κουζίνα)

Focus: Cultural vocabulary.

Task: List 5 traditional dishes: *σουβλάκι, μουςακάς, χωριάτικη, σπανακόπιτα, παστίτσιο.*

Write 2–3 lines: *Μου αρέσει το σουβλάκι γιατί είναι νόστιμο.*
Success: Correct food words + use of *μου αρέσει*.

G7.3 “Festival facts” (Ελληνικά έθιμα)

Focus: Cultural awareness.

Task: Research 3 Greek holidays (Πάσχα, Καθαρά Δευτέρα, 25η Μαρτίου).
Write 1 sentence about each.

Success: Accurate and meaningful facts.

8) Vocabulary Building

G8.1 “Word families”

Focus: Expanding vocabulary logically.

Task: From *γράφω*, find *γράμμα*, *γραφείο*, *συγγραφέας*.

Success: Identifies pattern and root meaning.

G8.2 “Opposites” (Αντίθετα)

Focus: Adjectives in contrast.

Task: Match pairs: *μεγάλος–μικρός*, *όμορφος–άσχημος*, *εύκολος–δύσκολος*.

Success: 6 correct pairs.

G8.3 “Verb of the week”

Focus: Repetition & fluency.

Task: Pick 1 verb (π.χ. *πηγαίνω*). Write 3 sentences in present, past, and future.

Success: All tense markers correct (*πήγα*, *θα πάω*).

9) Reflection & Self-Assessment

G9.1 “My word diary”

Write 10 new words per week with translation and one example sentence.

G9.2 “Weekly reflection”

3 prompts:

“This week I learned...”

“This was difficult...”

“Next week I will...”

G9.3 “Mini quiz”

Create 5 short Q&A in Greek with a classmate or parent:

Πώς σε λένε; Από πού είσαι; Τι ώρα είναι; Πού είναι το σχολείο; Πόσο κοστίζει;

✓Mini-Rubric for Basic Tasks

Level	Indicators
Emerging (1–2)	Knows a few words and sounds; struggles with alphabet or accent.
Developing (3)	Can form simple sentences; some errors with endings or tones.

Secure (4)	Accurate basic grammar; clear pronunciation; short conversations possible.
Excellent (5)	Confident in everyday topics; uses polite forms and simple tenses naturally.

Suggested Weekly Plan

Day	Focus	Example
Monday	Alphabet & pronunciation	G1.1 + G2.1
Wednesday	Listening & reading	G3.1 + G4.3
Friday	Speaking or writing	G5.x or G6.x
Weekend	Culture & reflection	G7.x + G9.x

Recommended Beginner Resources

- 🔗 **Listening:** BBC Languages Greek, *GreekPod101*
- 🔗 **Reading:** *LingQ Greek Library*, *Easy Greek (YouTube)*
- 🔗 **Apps:** *Duolingo Greek*, *Memrise*, *Anki* (for flashcards)
- 🔗 **Video:** *Easy Greek Street Interviews (YouTube)*
- 🔗 **Grammar:** *Greek123 Beginner Grammar Book* or *Oxford Greek Grammar & Workbook*

Greek — Extended Supercurriculum
(EDEXCEL GCSE / IGCSE — Modern Greek)

1) Speaking for Meaning & Confidence

E1.1 “My daily life” (Η καθημερινή μου ζωή)

Focus: Expressing habits & routines in the present tense.

Task: Record or present 6–8 sentences describing your day.

Example: Ξυπνάω στις επτά, πηγαίνω στο σχολείο, διαβάζω το απόγευμα.

Success: Uses verbs correctly; smooth pronunciation and rhythm.

E1.2 “Future plans” (Τα σχέδιά μου για το μέλλον)

Focus: Using θα + present verb to describe future actions.

Task: Write or say 5 lines about what you’ll do next weekend or next year.

Το Σάββατο θα πάω στον κινηματογράφο. Του χρόνου θα ταξιδέψω στην Κύπρο.

Success: Clear tense control and logical time sequence.

E1.3 “A short interview” (Μικρή συνέντευξη)

Focus: Asking & answering questions naturally.

Task: Create 5 Q&A with a partner:

Πού μένεις; Τι σου αρέσει να κάνεις; Πόσο χρονών είσαι;

Success: Questions correct; natural conversational flow.

2) Writing: Building Structure & Expression

E2.1 “Describe your house” (Το σπίτι μου)

Focus: Adjectives, prepositions, and structure.

Task: Write 80–100 words describing your home and room.

Το σπίτι μου είναι μεγάλο και φωτεινό. Στο δωμάτιό μου υπάρχει ένα γραφείο και μια καρέκλα.

Success: 5+ adjectives, correct prepositions (σε, πάνω σε, δίπλα σε).

E2.2 “My weekend” (Το Σαββατοκύριακό μου)

Focus: Past tense introduction (έκανα, πήγα, έφαγα, είδα).

Task: Write 80–120 words recounting your last weekend.

Include time words: το πρωί, μετά, το βράδυ.

Success: At least 3 verbs correctly in the past tense.

E2.3 “Email to a friend” (Γράμμα σε φίλο)

Focus: Informal writing conventions.

Task: Write 5–6 sentences about school life, friends, and hobbies.

Use greetings: Γεια σου Μαρία! Ελπίζω να είσαι καλά...

End: Φιλιά, Νίκος.

Success: Cohesive and natural structure.

3) Listening & Comprehension

E3.1 “Daily conversation challenge”

Focus: Understanding natural speech.

Task: Listen to a 1-minute dialogue (e.g., *Easy Greek*).

Note down 5 key words and 2 main ideas.

Success: Accurate gist capture.

E3.2 “Number dictation”

Focus: Listening for accuracy.

Task: Listen to random prices, phone numbers, and times.

Example: Δώστε μου δύο κιλά μήλα, κοστίζουν τέσσερα ευρώ και πενήντα λεπτά.

Success: Correctly writes $\geq 80\%$ of numbers.

E3.3 “Story sequence”

Focus: Listening for order and detail.

Task: Listen to a short story (≈ 1 min).

Reorder 4–5 sentences that summarise it.

Success: Logical sequencing; correct time words identified.

4) Reading: Comprehension & Vocabulary

E4.1 “Short biography” (Βιογραφία)

Focus: Identifying personal details.

Task: Read 100-word text about a person (*Η Μαρία μένει στην Αθήνα, δουλεύει σε σχολείο...*)

Answer 5 WH-questions.

Success: $\geq 4/5$ correct.

E4.2 “Text message decoding”

Focus: Informal written Greek.

Task: Read 3 short SMS/email examples.

Translate them into English or summarise in Greek.

Success: Accurate meaning capture, including slang/abbreviations.

E4.3 “Menu reading” (Στο εστιατόριο)

Focus: Real-world literacy.

Task: Read a menu and note 5 dishes and 3 drinks.

Write one short dialogue ordering something.

Success: Recognises vocabulary for food and cost.

5) Grammar & Structure in Use

E5.1 “Verb endings practice”

Focus: 3 common tenses (present, past, future).

Task: Conjugate γράφω, διαβάζω, τρώω in **εγώ, εσύ, αυτός/αυτή, εμείς, αυτοί.**

Success: 90% correct.

E5.2 “Prepositions drill”

Focus: Spatial relationships.

Task: Write 5 sentences using: *σε, πάνω σε, κάτω από, δίπλα σε, μπροστά από.*

Example: *Η γάτα είναι πάνω στο τραπέζι.*

Success: Prepositions and articles agree.

E5.3 “Comparisons”

Focus: Using *πιο... από* and *όσο... τόσο.*

Task: Write 3 comparison sentences:

Η Μαρία είναι πιο ψηλή από τον Νίκο.

Όσο διαβάζεις, τόσο μαθαίνεις.

Success: Correct structure and agreement.

6) Greek Culture & Real Life

E6.1 “Greek island project”

Focus: Geography and tourism vocabulary.

Task: Choose one island (Κρήτη, Σαντορίνη, Ρόδος).

Write 5 sentences about it: location, weather, food, what to do.

Success: Accurate cultural and vocabulary use.

E6.2 “Traditional food guide”

Focus: Descriptive language.

Task: Describe your favourite Greek dish (ingredients, flavour, when it's eaten).

Μου αρέσει η χωριάτικη σαλάτα. Έχει ντομάτα, αγγούρι, φέτα και ελιές.

Success: At least 5 correct food words; one opinion phrase.

E6.3 “Festivals & customs” (Έθιμα και γιορτές)

Focus: Cultural awareness.

Task: Research one Greek celebration (Πάσχα, Απόκριες, 15 Αυγούστου).

Write 5 sentences about how people celebrate.

Success: Describes 3+ traditions correctly.

7) Vocabulary & Expressions

E7.1 “Topic cards”

Focus: Vocabulary recall.

Task: Create 10 cards per topic (family, school, food, weather).

One side: Greek word; other side: meaning + example sentence.

Success: 90% correct recall.

E7.2 “Connectors in action”

Focus: Sentence cohesion.

Task: Write 5 sentences using *και, αλλά, γιατί, όταν, μετά, λοιπόν*.

Μου αρέσει το καλοκαίρι γιατί κάνει ζέστη.

Success: Logical connectors, correct syntax.

E7.3 “Common mistakes log”

Focus: Error awareness.

Task: Each week, note 3 mistakes and correct them.

Λάθος: Εγώ πάω στην σχολείο → ✓Εγώ πάω στο σχολείο.

Success: Self-corrects consistently.

8) Reflection & Progress

E8.1 “Weekly progress tracker”

Fill 3 lines:

“This week I learned...”

“A phrase I can use confidently...”

“Something I found difficult...”

E8.2 “Mini presentation”

Topic: *Η οικογένειά μου / Το σχολείο μου / Τα χόμπι μου.*

Speak for 1–2 minutes from notes only.

Success: Uses 3 tenses and opinions.

E8.3 “Self-assessment quiz”

Skill	Emerging	Developing	Secure	Excellent
Vocabulary	☐	☐	☐	☐
Grammar	☐	☐	☐	☐
Pronunciation	☐	☐	☐	☐
Listening	☐	☐	☐	☐
Culture	☐	☐	☐	☐

✓Mini-Rubric for Extended Level

Level	Indicators
Emerging (1–2)	Short, memorised phrases only; limited understanding.
Developing (3)	Understands and produces simple sentences; can write short texts with support.
Secure (4)	Clear pronunciation; accurate basic grammar; can describe, narrate, and ask questions.
Excellent (5)	Fluent, spontaneous responses; consistent use of tenses and connectors.

Suggested Monthly Cycle

Week	Focus	Output
1	Speaking & Writing	Self-intro + short paragraph
2	Grammar & Listening	Verb practice + dictation
3	Reading & Culture	Biography + food/festival
4	Review & Reflection	Presentation + quiz

Recommended Resources

- 📖 **Textbooks:** *Edexcel GCSE Greek Student Book*, *Greek123 – Level 2/3*
- 📺 **Listening:** *Easy Greek (YouTube)*, *GreekPod101 (Intermediate)*
- 📺 **Video:** *ERT Learn Greek*, *Lifo Podcast* (slow Greek speech)
- 📱 **Apps:** *Memrise Greek*, *Clozemaster*, *Anki decks for GCSE Greek*
- 🌐 **Websites:** *BBC Languages Greek*, *Filoglossia+*, *LearnGreekFree.com*
- 📖 **Grammar Reference:** *Greek Grammar in Practice* by D. Holton et al.

Greek — Guide Supercurriculum
(EDEXCEL GCSE / IGCSE — Modern Greek)

1) Extended Oral Expression & Debate

G1.1 “Who am I?” (Ποιος είμαι εγώ;)

Focus: Self-expression across tenses.

Task: Deliver a 2-minute speech (≈180 words) covering:

Your background, hobbies, ambitions, and cultural identity.

Use *παρελθόν-παρόν-μέλλον*: Όταν ήμουν μικρός..., Τώρα..., Στο μέλλον θα...

Success: Clear structure, accurate tense control, natural rhythm.

G1.2 “Debate: Technology helps or harms?” (Η τεχνολογία βοηθά ή βλάπτει;)

Focus: Formal argument and spontaneous response.

Task: Pair debate (2–3 min each).

Each student: 2 arguments + 1 counter-argument.

Use *πιστεύω ότι / διαφωνώ γιατί / από τη μία... από την άλλη...*

Success: Logical reasoning and register control.

G1.3 “Interview a Greek speaker”

Focus: Authentic interaction & intercultural listening.

Task: Record a short interview (6–8 Qs) with a Greek speaker (teacher, family, online).

Topic: *Ελληνική ζωή, εκπαίδευση ή φαγητό.*

Write a short reflection: *Τι έμαθα από αυτή τη συζήτηση.*

Success: Smooth turn-taking; cultural awareness shown.

2) Advanced Writing: Structure & Style

G2.1 “A day that changed my life” (Μια μέρα που άλλαξε τη ζωή μου)

Focus: Past narrative with emotion.

Task: 150–180 words using narrative connectors (*όταν, μετά, ξαφνικά, τελικά*).

Success: Coherent plot; varied vocabulary; consistent past tense.

G2.2 “Modern life and stress” (Το άγχος στη σύγχρονη ζωή)

Focus: Opinion essay (argumentative).

Task: 150–200 words exploring causes and solutions.

Use *πρέπει να / δεν χρειάζεται να / είναι σημαντικό να*.

Include at least 1 contrast (*αντίθετα, παρόλο που*).

Success: Persuasive tone; logical paragraphing.

G2.3 “Letter to the Mayor” (Επιστολή στον Δήμαρχο)

Focus: Formal writing register.

Task: Write 180 words about improving your neighbourhood.

Structure: greeting, problem, solution, closing.

Use *θα ήταν καλό αν...*, *προτείνω να...*

Success: Polite tone, accurate syntax, civic vocabulary.

3) Listening for Gist, Detail & Inference

G3.1 “Podcast summary”

Focus: Comprehending authentic spoken Greek.

Task: Listen to a 3–5 min Greek podcast (*ERT, Lifo, Easy Greek*).

Summarise main ideas in 5 bullet points.

Success: 4/5 points accurate; notes in Greek or English.

G3.2 “News report breakdown”

Focus: Understanding journalistic style.

Task: Listen to a 1–2 min TV/radio news clip.

Identify: Who? Where? What happened? When? Why?

Success: All five questions answered concisely.

G3.3 “Song analysis”

Focus: Listening for metaphor & culture.

Task: Choose a Greek song (e.g., Χατζιδάκις, Θεοδωράκης, Μηδενιστής).

Translate key lyrics; identify 3 cultural or emotional ideas.

Success: Explains imagery; connects to Greek values.

4) Reading & Interpretation

G4.1 “Opinion article” (Άρθρο γνώμης)

Focus: Identifying arguments and tone.

Task: Read 150–200-word article (π.χ. για το περιβάλλον ή τα social media).

Highlight connectives and write a 3-line summary.

Success: Detects main opinion + at least 2 supporting points.

G4.2 “Short story analysis” (Ανάλυση διηγήματος)

Focus: Literary appreciation.

Task: Read a short Greek story (e.g., Παπαδιαμάντης excerpt).

Note 3 unknown words, 2 characters, 1 theme.

Write a 100-word response in English or Greek.

Success: Explains theme clearly and personally.

G4.3 “Cultural news report”

Focus: Reading current culture.

Task: Read one cultural news piece (*συναυλία, φεστιβάλ, θέατρο*).

Identify date, place, artists, and main idea.

Success: Full factual accuracy; correct cultural context.

5) Grammar & Syntax Mastery

G5.1 “Complex sentences”

Focus: Using conjunctions for fluency.

Task: Combine 5 short sentences with *επειδή, παρόλο που, όταν, αν, μέχρι να*.
Δεν βγήκα, επειδή έβρεχε.

Success: Perfect conjunction use; natural flow.

G5.2 “Subjunctive mood”

Focus: *να + verb* structures.

Task: Write 5 examples showing intention or wish:

Θέλω να μάθω ελληνικά.

Ελπίζω να πάω διακοπές.

Success: All subjunctive uses correct.

G5.3 “Passive voice & participles”

Focus: Formal and descriptive Greek.

Task: Transform 4 active → passive sentences.

Ο δάσκαλος διδάσκει τα παιδιά → Τα παιδιά διδάσκονται από τον δάσκαλο.

Success: Accurate endings and meaning retention.

6) Cultural Research & Connection

G6.1 “Greek film review”

Focus: Cultural analysis through media.

Task: Watch a Greek film (*Πολίτικη Κουζίνα, Ακαδημία Πλάτωνος, Ελιές της Οργής*).

Write 150-word review: plot, characters, message.

Success: Analytical yet personal response; accurate film vocabulary.

G6.2 “Tradition and modernity” essay

Focus: Cultural evolution.

Task: Write 180 words: “Η Ελλάδα σήμερα: Παράδοση και πρόοδος.”

Compare past and present; mention food, family, society.

Success: Balanced argument; shows cultural depth.

G6.3 “Greek innovation project”

Focus: Contemporary achievements.

Task: Research one Greek inventor, artist, or company (e.g., Corallia, Nanophos, Yannis Behrakis).

Present short bilingual slide summary.

Success: Concise, accurate, culturally insightful.

7) Translation & Comparative Thinking

G7.1 “Translate a newspaper headline”

Focus: Conciseness & syntax.

Task: Translate 5 Greek headlines → English. Discuss literal vs idiomatic meaning.

Success: 4/5 idioms accurately captured.

G7.2 “English → Greek translation practice”

Focus: Word order & register.

Task: Translate 5 English sentences on family, school, and society.

Pay attention to articles and prepositions.

Success: >85% accuracy.

G7.3 “Idioms & sayings collection” (Παροιμίες)

Focus: Figurative language.

Task: Collect 10 idioms or proverbs + English meaning.

Ὅποιος βιάζεται σκοντάφτει → “Haste makes waste.”

Success: Correct translation and appropriate usage example.

8) Independent Projects & Reflection

G8.1 “My bilingual journal”

Focus: Reflection & metacognition.

Task: Keep a weekly journal (100 words): part Greek, part English.

Reflect on what you learned, cultural observations, goals.

Success: Consistent, honest, linguistically mixed reflection.

G8.2 “Personal portfolio”

Focus: Evidence of achievement.

Task: Compile 5 samples (speaking, writing, reading, listening, culture).

Add short commentary on growth and targets.

Success: Clear progress narrative; ready for teacher review.

G8.3 “Final oral presentation”

Focus: Summative fluency & confidence.

Task: 3–4 min presentation on one chosen theme:

Η εκπαίδευση, τα ταξίδια, η οικογένεια, το περιβάλλον, η μουσική.

Include visuals and audience questions.

Success: Natural delivery, correct structures, cultural references.

✓Mini-Rubric for Guide Level

Level	Indicators
Outstanding (17–20)	Fluent, idiomatic, and culturally rich language with precise grammar.
Strong (13–16)	Very good control; natural connectors; minor errors only.
Developing (8–12)	Clear meaning but limited range or tense consistency.
Emerging (<8)	Simple ideas; frequent structural mistakes.

☐ Suggested Annual Plan

Term	Focus	Project Output
Autumn	Identity, Society, Communication	Speech + opinion essay
Spring	Culture, Media, Environment	Debate + film review
Summer	Reflection, Future, Global links	Portfolio + oral presentation

☐ Recommended Advanced Resources

Textbooks: *Edexcel GCSE Greek Student Book* (Pearson)

Listening: *ERT Podcasts, Easy Greek Intermediate, Lifo Radio*

Reading: *Ta Nea* (simplified), *Greek Reporter Education Section*

Culture: *Cultural Foundation of the National Bank, Greek Film Centre*

Apps: *Clozemaster, Anki HSK-style Greek Deck, HelloTalk (Greek)*

Grammar Reference: *Greek Grammar in Practice* – Holton, Mackridge & Philippaki-Warbuton